PARTICLE TECHNOLOGY SERIES

THE SCIENCE AND ENGINEERING OF GRANULATION PROCESSES

Jim Litster and Bryan Ennis

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The Science and Engineering of Granulation Processes

Particle Technology Series

Series Editor

Professor Brian Scarlett
Technical University of Delft

The Kluwer Particle Technology Series of books is the successor to the Chapman and Hall Powder Technology Series. The aims are the same, the scope is wider. The particles involved may be solid or they may be droplets. The size range may be granular, powder or nano-scale. The accent may be on materials or on equipment, it may be practical or theoretical. Each book can add one brick to a fascinating and vital technology. Such a vast field cannot be covered by a carefully organised tome or encyclopaedia. Any author who has a view or experience to contribute is welcome. The subject of particle technology is set to enjoy its golden times at the start of the new millennium and I expect that the growth of this series of books will reflect that trend.

The Science and Engineering of Granulation Processes

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Dedication

To my wife, Annette, and my children, Sam, Rachel and Angus,
for their unfailing love and support.

JL

Preface

This book had its origins in a meeting between two (relatively) young particle technology researchers on Rehobeth Beach in Delaware in 1992 near the holiday house of Reg Davies (then Director of the Particle Science and Technology Research Center in Dupont). As we played in the sand, we shared an excitement for developments in particle technology, especially particle characterization, that would lead operations such as granulation to be placed on a sound scientific and engineering footing. The immediate outcome from this interaction was the development of new industry short courses in granulation and related topics which we taught together both in Australia and North America. This book follows closely the structure and approaches developed in these courses, particularly the emphasis on particle design in granulation, where the impact of both formulation properties and process variables on product attributes needs to be understood and quantified.

The book has been a long time in the making. We have been actively preparing the book for at least five years. Although the chapters have relatively good bibliographies, this book is not a review of the field. Rather it is an attempt by the authors to present a comprehensive engineering approach to granulator design, scale up and operation. It is exciting for us to see the explosion of research interest around the world in this area in the last five to seven years. Some of the most recent work will have to find its way into the second edition.

The book is written primarily as a tool for engineers and technologists in industrial R&D to use directly in the engineering design and troubleshooting. It should also be a useful text or reference for honours and graduate student level courses in Particle Technology. Indeed, drafts of some chapters have already been tested on students in Chemical Engineering at the Universities of Queensland and Delaware. Chapter 1 gives a good overview of the approach taken and advice for readers on how to make best use of the book.

The book relies heavily on work of current and former research students at The University of Queensland and we owe a great debt of gratitude to these people – Anthony Adetayo, Lian Liu, Rakhman Sarwono, Simon Iveson, Karen Hapgood, Hans Wildeboer and Rachel Smith. We also want acknowledge a wide range of research colleagues from industry and academia who have contributed strongly in the research development through discussion and active collaborations: Ian Cameron, Ted White, Tony Howes and Fu Yang Wang (University of Queensland), Neil Page, Simon Iveson and Simon Biggs (University of Newcastle), Flip Wauters, Gabrie Meesters and Brian Scarlett (Delft Technical University) Jim Michaels (Merck and Co), Paul Mort (P&G), Greg Sunshine (Dupont), Gabriel Tardos (City College, New York), Mike Hounslow (Cambridge and Sheffield Universities), Sarah Forrest and John Bridgwater (Cambridge University), Jonathan Seville (Surrey and Birmingham Universities). JL would particularly like to thank Lian Liu (Research Fellow) for her constructive criticisms of the manuscript and enormous assistance in editorial work and Vicki Thompson, without whose dedicated efforts in preparation of the manuscript this book would never have seen the light of day.

Jim Litster
Bryan Ennis
September 2003

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THE SCIENCE AND ENGINEERING OF GRANULATION

INTRODUCTION

Fine powders are a pain. They are difficult to handle and to process. Granulation converts fine powders to granular materials with controlled physical properties. Granulation and related processes cover a wide range of techniques used to form agglomerates that range in size from ~100µm to 20mm and varying in structure from loose aggregates to dense compacts. Sometimes even the fine powder feeds are avoided with granules formed from liquid (solution, slurry, melt) feed. Particle size enlargement techniques are used in every processing industry which handles particulate feeds, intermediates or products. This in itself is testimony to the value of the operation. Table 1-1 summarises the many reasons for granulating fine powders.

Table 1.1. Why granulate?

Reason	*Typical Application*
To produce useful structural forms	powder metallurgy
To provide a defined quantity for dispensing and metering	agricultural chemical granules, pharmaceutical tablets
To eliminate dust handling hazards or losses	briquetting of waste fines
To improve product appearance	food products
To reduce caking and lump formation	fertilisers
To improve flow properties for further processing	pharmaceuticals, ceramics
To increase bulk density for storage	detergents
To control dispersion and solubility	instant food products
To control porosity and surface-to-volume ratio	catalyst supports
To improve permeability for further processing	ore smelting
To create non-segregating blends of powder ingredients	ore smelting, agricultural chemicals, pharmaceuticals

A word about linguistics. Many terms are used for the variety of industrial size enlargement processes. These include *agglomeration, granulation, pelletisation* and *balling*. Terminology is very industry specific, so let's be clear. In this book we divide size enlargement processes into two categories (figure 1-1):

- *Granulation* uses a liquid binder to form interparticle bonds and agitates the powder-liquid mass to promote liquid dispersion and granule growth. This class of processes include fluidised beds, tumbling drums and pans, and mixer granulators.
- *Compression* processes use pressure to promote interparticle bonds. These processes are often performed dry. Where liquid is added, it is to enhance powder flow. This class of processes include roll pressing, extrusion (dry or wet), tabletting and briquetting.

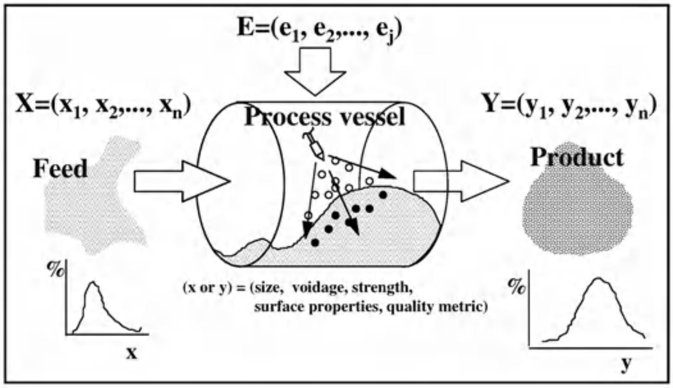
We will call the agglomerated product from a granulation process *granules*. The *formulation* is the combination of powder and liquid feed materials to the granulator.

Table 1.2. Size-Enlargement Methods and Application

Method	*Typical applications*
Tumbling granulators	
• Drums • Discs	Fertilisers, iron ore, non-ferrous ore, agricultural chemicals
Mixer and Planatary granulators	
• Continuous high shear	Chemicals, detergents, carbon black
• Batch high shear	Pharmaceuticals, ceramics
Fluidized granulators	
• Fluidized beds • Spouted beds • Wurster coaters	Continuous: fertilisers, inorganic salts, detergents Batch: pharmaceuticals, agricultural chemicals, nuclear wastes
Centrifugal granulators	Pharmaceuticals, agricultural chemicals
Spray methods	
• Spray drying	Instant foods, detergents, ceramics
• Prilling	Urea, ammonium nitrate
Compression agglomeration	
• Extrusion • Roll press • Tablet press • Molding press • Pellet mill	Pharmaceuticals, catalysts, inorganic chemicals, organic chemicals, plastic preforms, metal parts, ceramics, clays, minerals, animal feeds

This book focuses primarily on *granulation.* Table 1.2 lists the key size enlargement processes and some typical applications. Chapter 7 gives more details on the range of particle size enlargement processes, their applications and idiosyncrasies. Within a process plant, the size enlargement circuit also includes a number of peripheral operations such as milling, blending, drying and classification (figure 1-2). In general, the size enlargement operation cannot be considered in isolation. Interactions with the other processes must be taken into account.

Historically, granulation circuits have been designed and run based on no science, a little bit of empirical engineering and a lot of art. However, our understanding of powder mechanics is greatly improved. Sophisticated techniques to characterise formulations and granules are available both in the laboratory and in the plant. This book is devoted to developing a sound scientific and engineering approach to design, scale up and operation of size enlargement processes. The remainder of this chapter introduces the conceptual framework that we will use to achieve this.



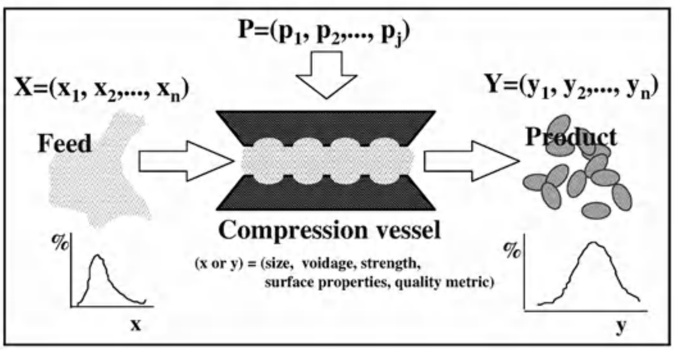


Figure 1.1. The unit-operations of (a) granulation and (b) compression agglomeration. X and Y represent a set of particle attributes, such as size and porosity. E represents additional process streams such as spray rates of moisture or slurry, steam, or heated air. P represents a description of the applied compression stress, such as maximum applied load and the rate and duration of load

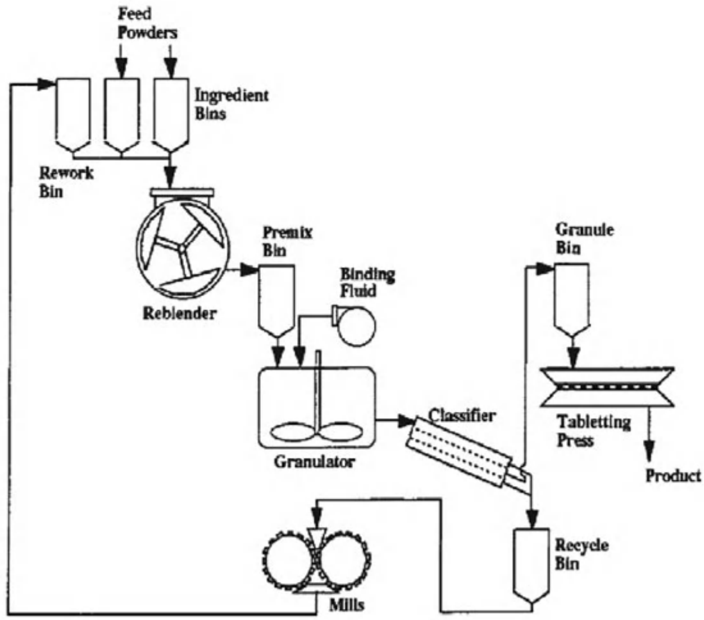


Figure 1.2. A typical size enlargement circuit for pharmaceutical or agricultural chemicals involving both granulation and compression techniques

1.1. Granulation Rate Processes

Granulation is a complex process with several competing physical phenomena occurring in the granulation vessel, which lead ultimately to the formation of the granules. We will divide these phenomena into three groups of rate processes (figure 1-3):

1. wetting, nucleation and binder distribution
2. consolidation and growth
3. attrition and breakage

The first stage of granulation is the addition and distribution of the binder to give nuclei granules. This occurs within the liquid spray zone in the granulator. The liquid binder is usually sprayed onto the moving powder bed. Ideally, each drop will imbibe into the powder bed, engulfing particles to form a single granule nucleus. If the drop does not easily wet the powder, or the rate of imbibition is slow, large wet agglomerates will form at the powder surface. Sometimes shear forces within the powder bed are large enough to break up these clumps of wet material to further distribute the liquid.

Nucleation gives a distribution of loosely packed granule nuclei. Under good wetting conditions, the distribution will be narrow and correlate closely with the liquid drop size distribution. Poor wetting and nucleation causes a broad nuclei size distribution and often poor final granule properties. The nucleation process is strongly influenced by the characteristics of the liquid spray, the flux of powder through the spray zone and the wetting properties of the formulation.

Granule nuclei will consolidate through collisions with other granules, and granulator. The extent of consolidation depends on the intensity of agitation in the granulator and resistance of the granule to deformation. Granules made from fine powders or with viscous liquid binders resist deformation on impact and consolidate slowly. Granules made from coarse powders deform a lot on impact and quickly reach their minimum porosity. Granule consolidation controls the final granule porosity which influences many other granule properties.

When two granules collide they may stick together to form a single large granule. This is growth by coalescence. For successful coalescence (a) the energy of impact must be absorbed during collision so that the granules do not rebound; and (b) a strong bond must form at the contact between the colliding granules. The presence of liquid at the surface of the granule is important for growth by coalescence and coalescence rate is very sensitive to liquid content. Consolidation and coalescence are closely related and are considered together in this book.

Granules formed through a combination of nucleation and growth processes are eventually dried. The dry granules may break to give several fragments or attrit to give a fine powder. These processes are controlled by the fracture mechanics of the granule. Granules resist attrition if their fracture toughness is high (related to formulation properties) and the number and size of flaws is small (related to granule porosity).

These rate processes are at the heart of the granulation process and determine the final granule properties. It is important to know which rate processes dominate in any particular application. For example, if nucleation is the dominant rate process, adjusting the liquid spray nozzles will have a profound effect on granule size distribution. If coalescence rates are high, granule size will be very sensitive to liquid content and granule residence time.

We therefore spend a lot of time examining each of these processes in detail. We will:

- examine the underlying physics behind each rate process
- define the controlling formulation properties and process parameters for each rate process
- use regime maps to establish the operating regime for the granulator
- provide quantitative relationships to predict the effect of changing operating parameters and formulation properties
- provide troubleshooting and scale up guidelines

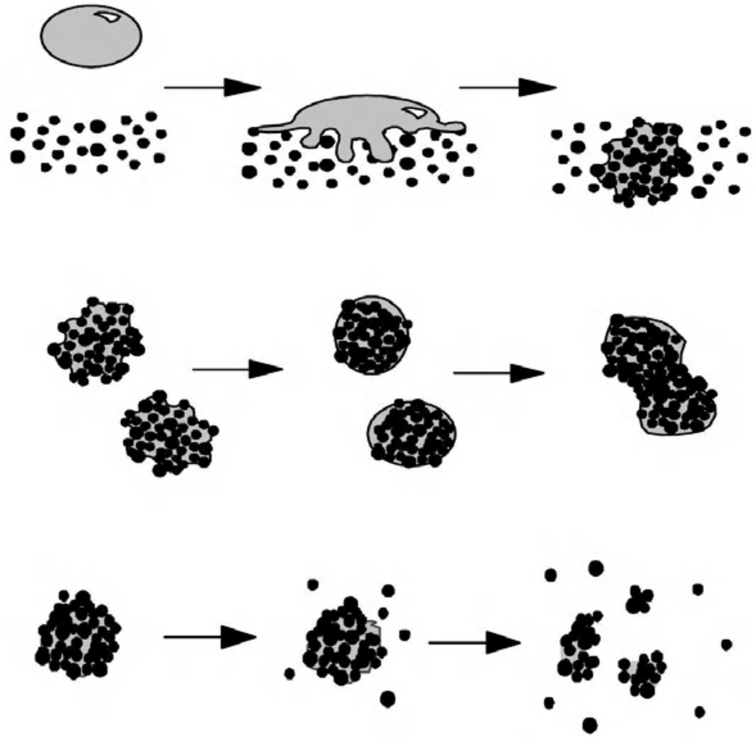


Figure 1.3. Rate Processes in Granulation (a) wetting and nucleation; (b) consolidation and growth; (c) breakage and attrition

1.2. *Considering a Granulator as a Reactor*

There is an analogy between a granulation process and a chemical reactor. A chemical reactor converts feed materials to different chemical species via a series of chemical reactions. A granulator converts fine powders to granules via a series of physical rate processes. Chemical reactor engineering is very well established. We can apply many of the methologies from reaction engineering to approach granulator design. There are four natural scales of analysis for a granulator or reactor (figure 1-4):

1. Particle scale: A granule is an agglomerate of many small particles. The granulation process depends ultimately on the particle-particle and particle-binder physicochemical interactions at a small scale. This is analogous to examining chemical reactions at the molecular level.
2. Volume of powder scale: Within some volume element in the granulator, we define the physical rate processes that are occurring (nucleation, coalescence, etc.). Kinetic expressions are written for rate process which give the rate of change of granule properties eg. size distribution. This is analogous to writing rate expressions for several chemical reactions in a reactor.
3. Granulator scale: A series of balance equations is written to describe the evolution of the granule property distributions in the granulator. These equations incorporate rate

expressions for each of the physical rate processes, residence time distributions and mixing patterns in the granulator. The mass, energy and population balances for a granulator are equivalent to the total mass, energy and individual component balances for a reactor.

4. Granulation circuit scale: The performance of a chemical reactor cannot be considered in isolation. Products need to be separated after reaction. Unreacted feed needs to be recycled. In the same way, design and operation of granulation circuits requires consideration of all the unit operations and their interactions. Note that most granulation plants are either batch operations or continuous operations with large recycle. Both cases provide special problems for process control.

Good engineering requires investigation at **all four scales**. Furthermore, knowledge and models at all scales of analysis should be linked. Rate expressions for physical rate processes (volume of powder scale) are of limited value unless they are sensibly derived from underlying phenomena at the particle scale. Granulator performance may be dominated by the properties of recycle materials to the reactor. While this may seem commonsense, it is rarely applied in practice.

In this book we will use the reactor analogy and reaction engineering methodology extensively. A lot of emphasis is placed on linking the four scales of analysis.

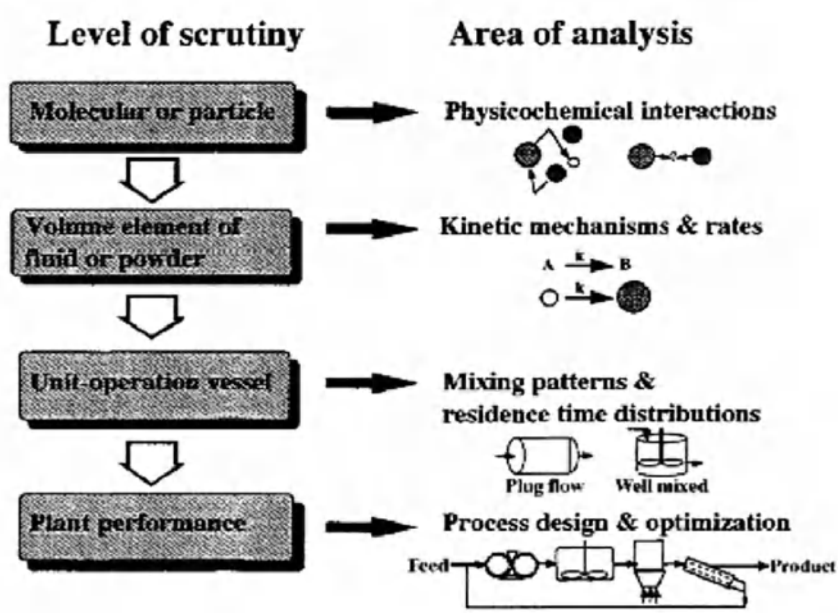


Figure 1.4. Comparisons of scales of analysis of chemical reaction and size-enlargement processes

1.3. Granulation as Particle Design

Granulation should be used to produce granules with defined properties to meet specific end use requirements. Key primary properties are granule size distribution and porosity. These two properties which dominate a wide range of secondary properties may also be important, depending on the application. Achieving the desired *designer particle* is a combination of process engineering and product engineering. Process engineers choose processes and manipulate process variables to change the granule properties. Product engineers manipulate the ingredients or *formulation* to achieve the same effect.

Let's illustrate the particle design concept with an example. Herbicide and pesticide powders must be dispersed in water by the farmer before spraying onto crops. Small particle size is needed so the particles don't settle out in the tank (of order a few microns). However, fine powders are hard to disperse in the water. They are dusty, cake in the bag and can be inhaled by the farmer. The designer particle must therefore have the following properties:

- easy to handle
- non-dusty
- strong enough to avoid attrition during handling
- disperses in water to a stable colloidal suspension within seconds

Several companies now produce these designer particles commonly known as water dispersible granules (WDGs). The granules are made from finely milled powder (for final suspension properties). The fine powder is granulated with water to produce porous free flowing granules. The large granules will break the surface of water in the tank easily and disperse to their unit particles as water enters the granule.

To achieve the desired properties, clays and surfactants are added to the formulation to help granules disperse once they are wet. Additional binders may be added to give the granules dry strength during transportation. Several different granulation techniques have been used to produce WDGs. It is critical in processing to control the granule porosity. If the porosity is too low, the granules will not disperse in water; too high, and granules attrit to produce dust during handling. Other process issues affect the granule properties eg. if the residence time during drying is too long, the surfactant will age leading to poor dispersion. Control of both process variables and formulation properties is necessary to achieve the desired WDG properties.

In any granulation process, process variables and formulation properties *interact* to give the final particle design. Often, different groups in a company are responsible for formulation design and process design/scale up. If these two groups *do not* interact, problems are likely to occur! This lack of interaction has also occurred in the scientific literature on granulation.

Good particle design requires four things:

1. Clear definition of the required granule properties.
2. Good characterisation of formulation ingredients and granules.
3. Understanding the impact of formulation properties on granule properties.
4. Understanding the impact of process choice and operating variables on granule properties.

In this book, granulation is approached from a particle design perspective. There is a heavy emphasis on (a) defining and measuring both formulation and granule properties, and (b) how granule properties are evolved from the interaction between formulation properties and process variables.

1.4. How To Use This Book

The aim of this book is to provide a scientific and engineering basis for design and operation of granulation processes. It is divided into two parts:

- Part A: Granulation Process Fundamentals
- Part B: Equipment Selection, Design and Operation

In Part A we examine the underlying phenomena that control the granulation process at the particle and volume of powder scale of investigation. Chapter 2 defines important properties of particles and particle assemblies. Chapters 3 to 5 examine each class of rate process for granulation.. Each chapter describes the physics that control the rate process and relevant characterisation techniques for formulation and granule properties. Rate expressions are developed in terms of equipment independent process parameters. Examples from different types of equipment are used to illustrate the concepts and a summary table presented on ways to control the rate process. Part A finishes with the mathematical description of rate processes to predict the evolution of granule property distributions in the granulator (Chapter 6).

Part B analyses granulation at the granulator and granulation circuit scale. Chapter 7 classifies and compares different granulation and compression agglomeration techniques. This chapter gives a rationale for equipment selection. The following chapters each focus on an individual technique, giving brief descriptions of equipment and looking at equipment specific design, scale up and operation. These chapters refer heavily to the understanding developed in Part A to describe and quantify the rate processes for the technique being studied.

Readers can begin with either Part A or Part B. Researchers and formulation designers may wish to start at Chapter 2 and gradually build up a fundamental understanding of the rate processes and underlying science. Process engineers and technologists in manufacturing who want to get a broad overview or focus on one type of equipment only, may start with Chapter 7 and refer back to Part A only as indicated in the equipment specific chapters.

CHAPTER 2

PARTICLE AND GRANULE MORPHOLOGY

"Measure Properties, don't perform rituals"[1]

Good particle design means understanding particle-particle and particle-fluid interactions and using this knowledge to properly design processes and products. It is the properties of the particle that dictate these interactions so particle characterisation is at the heart of particle design. Good characterisation of both the feed powders and the product granules is essential.

However, particle characterisation presents special challenges that don't exist for the measurement of fluid properties:

- The discrete nature of a particulate system means that it is heterogeneous. Two particles in the same system don't have the same properties. We need to measure **property distributions** and sampling is a major issue.
- The properties of particulate systems can't be tabulated like the thermodynamic properties of fluids. The properties of the system depend not only on the chemical composition, but also the *particle morphology.* You can't look up surface area of sand in The Handbook of Chemistry and Physics! You have to measure it.
- The properties are not always easy to define. The simplest example of this is particle size (see below).
- Historically, many particle properties have been difficult and time consuming to measure.

These challenges have led many practising engineers and technologists to "perform rituals" rather than measure properties. These rituals measure some empirical parameter that may bear no relationship to any fundamental property, but involve tests that can be easily and routinely performed. Such rituals are enshrined in company and industry standards. **The need for these rituals is gone**. Effective on-line and off-line measurement of real properties is now possible with modern particle characterisation techniques.

Figure 2.1 illustrates some important particle properties. The particle properties determine the macroscopic properties of the particle assembly or *bulk solid.* We cannot yet predict completely bulk solid properties from knowledge of particle properties so that ability to define and measure bulk solids properties is also important.

Each chapter in the first part of this book defines important particle or granule properties and discusses their measurement. This chapter focuses on the morphological properties - size, shape, density and porosity. All other granule attributes are a strong function of these properties so they need to be defined clearly at the start. We first define the properties for any particle. At the end of the chapter we look at special issues associated with granule morphology.

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[1] From a speech by Roland Clift to the 1st International Particle Technology Forum, Denver 1994.

Particle characterisation is a book in itself and that book has already been written (Allen, 1997). You should refer to Terry Allen's excellent book for a more detailed discussion of the topic.

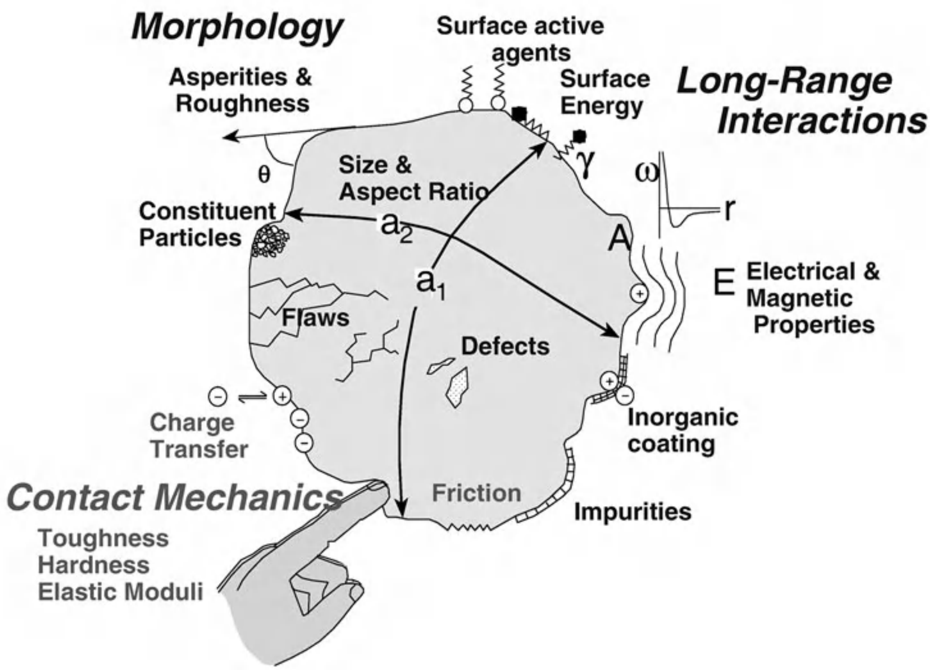


Figure 2.1. Particle properties affecting particle and assembly behavior

2.1. Particle Size, Shape and Size Distribution

2.1.1. Particle Size Definitions

Size is the single most important property of a particle in a formulation. It defines the basic building block of the granule. Particle size strongly influences other important particle and granule properties (Table 2.1). One reason for this is that size controls the balance between surface forces and body forces with the surface forces becoming increasingly dominant as size becomes small. So knowledge of the particle size of a system is essential.

However, real life is not that simple. There is no clear definition of particle size for real (irregularly shaped) particles. In practice, there are *many* definitions of particle size. Some of the most important are "equivalent diameters". The equivalent diameter is the diameter of a sphere with the same property as the particle eg. the volume equivalent diameter is the diameter of a sphere with the same volume as the particle:

$$V_p = \frac{\pi}{6} d_v^3 \tag{2-1}$$

Similarly, the specific surface area equivalent diameter is defined by:

$$(S/V)_p = 6/d_{sv} \tag{2-2}$$

Often, size is not measured directly. Instead, a size dependent property is measured and the size is inferred. For example, settling velocity may be measured and the size calculated from Stokes Law. In this case the measure of size is the *Stokes diameter* d_{St}, ie. the diameter of a sphere with the same settling velocity as the particle. Table 2.2 gives ten examples of different measures of particle size. Ideally, the choice of particle size measurement should match the application. There is no clear choice for granulation formulations and volume equivalent diameter and sieve diameter are the most commonly used. The specific surface diameter (eqn. 2-2) is also an important measure, especially for fluidised bed granulators and driers.

2.1.2. Particle Shape

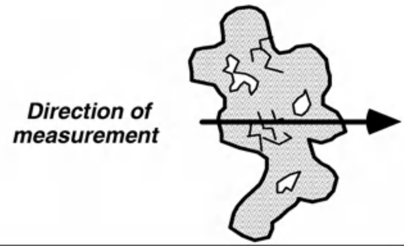
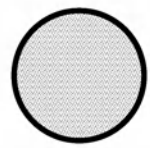
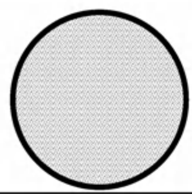
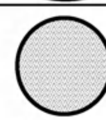
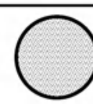
Particle shape affects flowability, packing density and particle-fluid interactions. Shape is more difficult to define and measure than particle size. Qualitative descriptions are available but are not useful for quantitative calculations. One useful quantitative measure of shape is the shape factor. The shape factor relates some property of the particle to a measure of particle size. For example, the volume shape factor, surface ar ea shape factor and specific surface shape factor is defined by the equation:

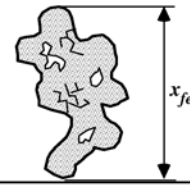
$$V_p = \alpha_v x^3; S_p = \alpha_s x^2; (S/V)_p = \frac{\alpha_{sv}}{x} \tag{2-3}$$

Table 2.1. Properties of particles, granules and particle assemblies that depend on particle size

Property	*Trend with decreasing size*	*Other relevant particle properties*
Properties of single particles		
1. Homogeneity	Increasing	
2. Elastic-plastic behavior (ductility)	Increasing	Elastic moduli
Probability of breakage	Decreasing	Toughness
Particle strength	Increasing	Hardness
Wear behavior	Decreasing	Flaw distribution
3. Properties resulting from competition between volume and surface forces, e.g.: adhesion, agglomeration, suspendability	Increasing	Surface energy Charge distribution Surface groups Impurities Hardness Surface asperities Density Shape Interparticle forces
Properties of particle assemblies and granules		
1. Bulk density	Decreasing	Density, shape, friction, interparticle forces
2. Rheological behavior: elasticity, yield point,	Mostly increasing	Size distribution, density, shape, interparticle forces, friction
3. Wetting	Decreasing	Surface energy, contact angle
4. Strength of agglomerates & briquettes	Increasing	Fracture toughness, hardness, elasticity voidage, flaws
5. Powder mechanics: shear stress, unconfined yield stress, cohesive stress, wall friction	Increasing	Surface energy, friction, asperities, hardness, elasticity
6. Fluid-mechanics: permeability, fluidizability	Usually decreasing	Density, interparticle forces
7. Ignitability, explosive behavior	Increasing	Chemical and surface properties
8. Reactivity, solubility (e.g. taste, activity)	Increasing	Surface energy Flaw distribution

Table 2.2. Nine typical measures of size or length scale for irregularly shaped particles

Particle under study		
Volume diameter	Diameter of a sphere having the same volume as the particle $V_p = \frac{\pi}{6} d_v^3$	
Surface diameter	Diameter of a sphere having the same surface area as the particle $S_p = \pi d_s^2$	
Stokes' (or drag) diameters	Diameter of a sphere having the same terminal velocity as the particle $u_t = \frac{g(\rho_p - \rho) d_{St}^2}{18\mu}$	
Specific surface diameter	Diameter of a sphere having the same specific surface area as the particle $(S/V)_p = 6/d_{sv}$	
Particle sieve diameter	Width of the minimum square aperture through which the particle will pass	
Projected area diameter	Diameter of a sphere having the same projected area as the particle $A_p = \pi d_a^2$	

Feret's diameter	Projection of the particle's outline onto a line perpendicular to the direction of measurement	
Martin's diameter	Chord parallel to the direction of measurement which bisects the projected area	
Scattering diameter	Diameter of a sphere scattering light at the same intensity	

Note that the numerical value of the shape factor depends not only on the particle shape *but also on the definition of particle size x.* One shape factor deserves special mention. It is the particle sphericity ψ which is defined as:

$$\psi = \frac{\mathit{surfaceareaof\ aspherewithsamevolume}}{\mathit{surfaceareaof\ the\,particle}} \tag{2-4}$$

As a sphere is the shape with minimum surface area, the maximum value of ψ is 1. Low values of ψ indicate large deviations of particle shape from spheres eg. needles, fibres and plates. More complex definitions for particle shape include the fractal dimension of the particle. This is particularly useful for flocs and loose particle aggregates.

We can estimate the relationships between different particle size definitions if the particle shape is known. From the definition in eqn. 2-4:

$$d_V = \psi^{0.5} d_s \tag{2-5}$$

A very useful approximate relationship between sieve size and equivalent volume size is:

$$d_{sieve} = \psi \ d_v \tag{2-6}$$

Particle shape is measured directly by microscopy combined with image analysis, indirectly by measuring particle size by two different techniques or by light scattering techniques.

2.1.3. Basic Particle Size Distribution Definitions

Particle size (and other properties) present a further problem. They do not have single values. Instead there is a distribution of values of the property and we need to be able to define and describe these distributions. There are two ways to represent the particle size distribution :

- a **cumulative** distribution: The cumulative distribution, *N(x)*, is the number of particles less than size *x*.
- a **frequency** distribution: The frequency distribution, *n(x)*, is defined such that *n(x)dx* is the number of particles between sizes *x* and *x+dx*

Clearly, *N(x)* and *n(x)* are related. From their definitions:

$$N(x) = \int_0^x n(x')dx'; \quad n(x) = \frac{d}{dx} N(x) \tag{2-7}$$

Figure 2.2 shows the relationship between the frequency and the cumulative size distributions. From the definition of the frequency size distribution, it follows that the total area under the $n(x)$ curve must be N_T, the total number of particles:

$$\int_0^\infty n(x)dx = N_T \tag{2-8}$$

We often express the cumulative and frequency distributions as *normalised* distributions (see figure 2.2):

$$f(x) = \frac{n(x)}{N_T}; \quad F(x) = \frac{N(x)}{N_T} \tag{2-9}$$

Note that the frequency distribution *f (x)* has the dimensions ***L^{-1}*** e.g. units of μm^{-1}. The fraction of particles in a size range is represented by ***the area under the curve*** of the distribution. We can't define the number of particles of size *x*, only the number between two sizes, *x* and *x+dx*. ***When viewing size distribution data, always plot the true frequency distribution, not the fraction in a size interval against size.***

Equations 2-7 to 2-9 are written for a continuous distribution. In practice, we usually measure size distributions divided into sections or intervals. Each size interval is defined by the largest size in the interval (x_i) and the width of the interval ($\Delta x_i = x_i - x_{i-1}$). Within each interval, we assume constant values for particle properties e.g. the particle frequency (see figure 2.3). We can rewrite all the size distribution definitions for a sectional distribution. Table 2.3 summarises the relationships between the standard ways of representing particle size distributions.

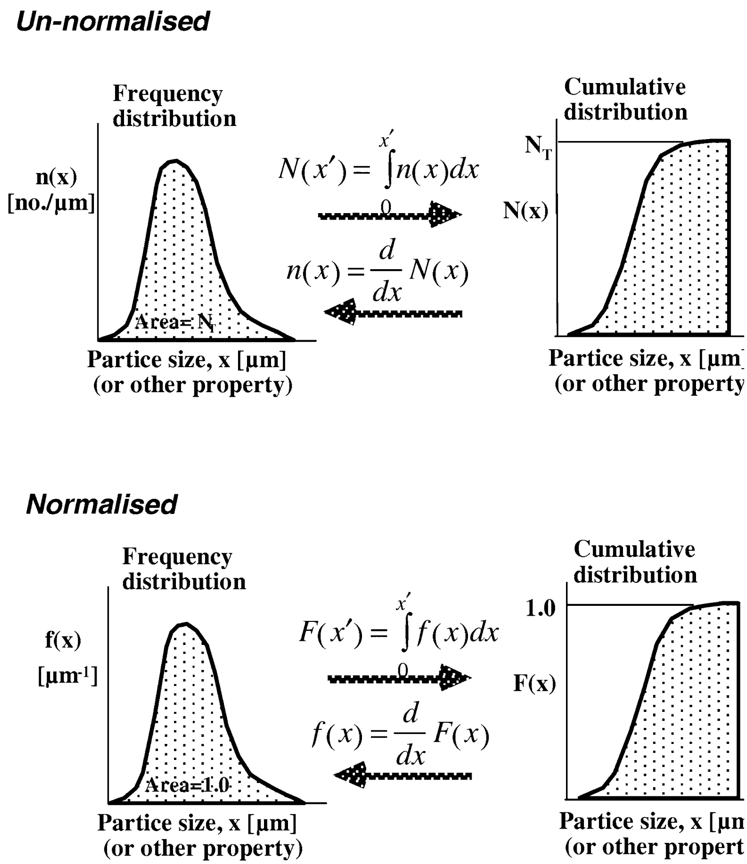


Figure 2.2. Cumulative and frequency size distributions

Table 2.3. Summary of size distribution definitions

Type of distribution	*Definition*	*Relationships*
<u>Un-normalised/continuous</u> $N(x)$ $n(x)dx$	no. of particles less than size x no. of particles between size x and $x+dx$	$N(x) = \int_0^x n(x')dx'$ $n(x) = \frac{d}{dx}N(x)$; $\int_0^\infty n(x)dx = N_T$
<u>Normalised/continuous</u> $F(x)$ $f(x)dx$	no. fraction of particles less than size x no. fraction of particles between size x and $x+dx$	$F(x) = \int_0^x f(x')dx'$ $f(x) = \frac{d}{dx}F(x)$ $\int_0^\infty f(x)dx = 1$ $f(x) = \frac{n(x)}{N_T}; F(x) = \frac{N(x)}{N_T}$
<u>Un-normalised/sectional</u> x_i $\Delta x_i = x_i - x_{i-1}$ N_i $n_i^* = n_i \Delta x_i$	maximum size of particles in size interval i width of size interval i no. of particles in all intervals up to and including i no. of particles in size interval i	$N_i = \sum_{j=1}^{i} n_j^* = \sum_{j=1}^{i} n_j \Delta x_j$ $n_i \Delta x_i = n_i^* = N_i - N_{i-1}$ $\sum_{i=1}^{\infty} n_i \Delta x_i = N_T$ $n_i = \frac{\int_{x_{i-1}}^{x_i} n(x)dx}{\Delta x_i}; N_i = N(x_i)$

Normalised/sectional F_i $y_i = f_i \Delta x_i$	no. fraction of particles in all intervals up to and including i no. fraction of particles in size interval i	$F_i = \sum_{j=1}^{i} y_j = \sum_{j=1}^{i} f_j \Delta x_j$ $f_i \Delta x_i = y_i = F_i - F_{i-1}$ $\sum_{i=1}^{\infty} f_i \Delta x_i = 1$ $f_i = \frac{\int_{x_{i-1}}^{x_i} f(x)dx}{\Delta x_i}; F_i = F(x_i)$

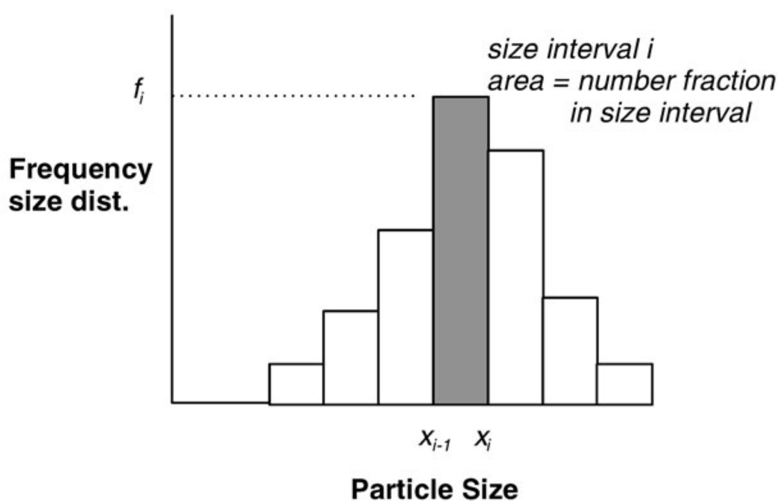


Figure 2.3. Frequency histogram for a discrete size distribution

2.1.4. Different Ways to Represent Particle Size Distributions

So far, we have looked at the distribution of *number* of particles with *size x*. However, we may generalise the analysis and represent a distribution of any measure of ***quantity*** of particles with any ***property*** . We write a generalised distribution as $f_\alpha(\xi)$ where ξ is the particle property of interest and the subscript α denotes the measure of quantity of particles. $f_\alpha(\xi)$ is defined such that the fraction of the amount of α between ξ and

$\xi + d\xi$ is $f_\alpha(\xi)d\xi$. When plotting the distributions, the *quantity* is the y-axis variable and the *property* is the x-axis variable (see figure 2.4).

When the quantity α is the number of particles we usually omit the subscript. The most common alternative quantity used is mass (volume) because many sizing techniques measure the mass of particles, rather than count the number.

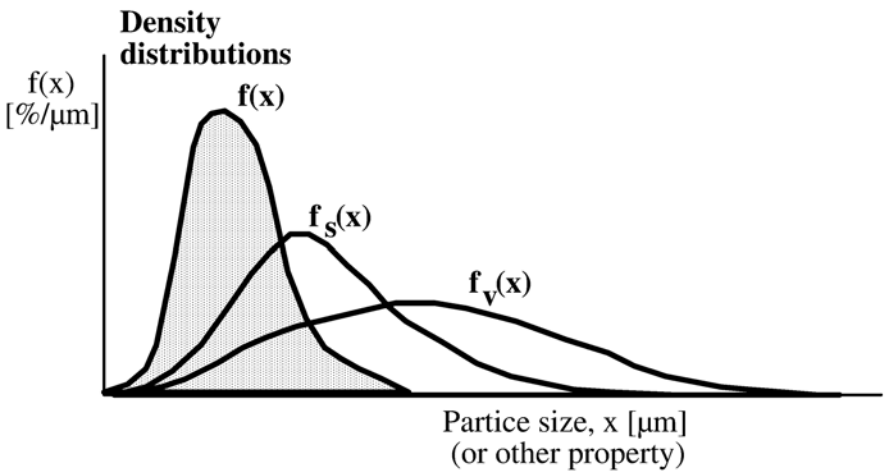


Figure 2.4. Examples of different frequency size distributions

Apart from the linear size x, other choices for the property ξ are particle volume and log of particle size. The logarithmic frequency distribution is very useful where data is collected for geometric size intervals ie. $x_i = kx_{i-1}$. Common values of k are 2, $\sqrt{2}$ and $\sqrt[4]{2}$.

In principle, we can readily convert from one type of distribution to another. Table 2.4 gives common conversions to change either the property or the quantity of the distribution. These relationships can be derived directly from the size distribution definitions (see Allen, 1997). Despite the fact that converting distributions from one form to another is mathematically straightforward, great caution should be shown in manipulating the measure of quantity for real data. The reason is clear from figure 2.4. Volume distributions give much greater weighting to the large particle end of the distribution where there are very small numbers of particles and the error in the number size distribution is very large. The errors in converting real broad size distribution data from a volume (mass) basis to number basis can be extremely large.

A quick look at Tables 2.3 and 2.4 show an impressive list of different symbols and definitions. In fact, most of the difficulties in handling size distributions come from confusion in definitions and nomenclature.

Table 2.4. Converting between different distributions

Property (ξ)	*Quantity* (α)	*Continuous*	*Discrete*
size	number	$f(x)$	f_i
size	surface area	$f_s(\mathrm{x}) = \dfrac{\mathrm{x}^2 f(\mathrm{x})}{\int_0^\infty \mathrm{x}^2 f(\mathrm{x})d\mathrm{x}}$ $f(\mathrm{x}) = \dfrac{\mathrm{x}^{-2} f_s(\mathrm{x})}{\int_0^\infty \mathrm{x}^{-2} f(\mathrm{x})d\mathrm{x}}$	$f_{si} = \dfrac{\bar{\mathrm{x}}_i^2 f_i}{\sum_j \bar{\mathrm{x}}_j^2 f_j \Delta\mathrm{x}_j}$ $f_i = \dfrac{\bar{\mathrm{x}}_i^2 f_{si}}{\sum_j \bar{\mathrm{x}}_j^{-2} f_{sj} \Delta\mathrm{x}_j}$
size	volume or mass	$f_v(\mathrm{x}) = \dfrac{\mathrm{x}^3 f(\mathrm{x})}{\int_0^\infty \mathrm{x}^3 f(\mathrm{x})d\mathrm{x}}$ $f(\mathrm{x}) = \dfrac{\mathrm{x}^{-3} f_v(\mathrm{x})}{\int_0^\infty \mathrm{x}^{-3} f_v(\mathrm{x})d\mathrm{x}}$	$f_{vi} = \dfrac{\bar{\mathrm{x}}^3_i f_i}{\sum_j \bar{\mathrm{x}}_j^3 f_j \Delta\mathrm{x}_j}$ $f_i = \dfrac{\bar{\mathrm{x}}^{-3}_i f_{vi}}{\sum_j \bar{\mathrm{x}}_j^{-3} f_{vj} \Delta\mathrm{x}_j}$
volume	number	$f(v) = \dfrac{f(x)}{3\alpha_v \mathrm{x}^2}$	$f_i^v = \dfrac{f_i}{3\alpha_v \bar{\mathrm{x}}_i^2}$
ln(size)	number	$f(\ln \mathrm{x}) = xf(\mathrm{x})$	$f_i^{\ln x} = \bar{x}_i f_i$

2.1.5. Measures of Mean Size and Spread of the Size Distribution

A full size distribution is a lot of information to carry around and manipulate. It is useful to define a limited set of parameters that characterise the whole distribution. We start by defining the un-normalised moments of the number-size distribution:

$$\mu'_k = \int_0^\infty x^k n(x)dx = \sum_{i=1}^\infty \bar{x}_i^k n_i \Delta x_i \tag{2-10}$$

The total number and volume of particles are related to the zeroth and third moment:

$$N_T = \mu_0' \tag{2-11}$$

$$V_T = \alpha_v \mu_3' \tag{2-12}$$

A general definition of "mean size" is also written in terms of the moments:

$$\bar{x}_{m,n} = \left(\frac{\mu_m'}{\mu_n'} \right)^{\frac{1}{m-n}} \tag{2-13}$$

There is no unique mean size that reveals everything about the distribution. Important examples of mean sizes based on this definition are:

- $\bar{x}_{1,0} = \mu_1' / \mu_0'$
- $\bar{x}_{3,2} = \mu_3' / \mu_2'$ (the specific surface or Sauter mean)
- $\bar{x}_{4,3} = \mu_4' / \mu_3'$ (the mass-moment or volume-moment mean)

The key point is to use the mean size relevant to your application. For example, the specific surface mean $\bar{x}_{32}$ should be used to calculate the surface area per unit volume of the powder:

$$(S/V) = \frac{\alpha_{sv}}{\bar{x}_{32}} \tag{2-14}$$

Any other mean size used in eqn. 2-14 will give the wrong answer. Different measures of mean size will vary widely for a broad size distribution.

An alternative approach is to define measures of average size and spread of the size distribution in terms of values related to the cumulative distribution:

- the median size x_{50} is the 50% passing size ie. $F(x_{50}) = 0.5$.
- the range of the distribution is the difference between the 95% passing size and the 5% passing size ($x_{95} - x_{05}$).

2.1.6. Particle Size Measurement

A wide range of techniques are used for particle size analysis. The choice of technique depends on the application. Some techniques measure particle size directly, while others measure a property that depends on particle size. Table 2.5 shows a number of commonly used particle sizing methods (see Allen for more exhaustive descriptions). For particles above 0.1 mm in size, sieving is easily the most popular technique. For finer sizes there is no single best methods although light diffraction techniques are common because of the speed and ease of the measurement.

Measuring particle size distribution correctly takes some skill. There are some important points to remember:

1. Getting a representative sample is vital. Without a representative sample, correct sizing is impossible. Often, much more effort is required in getting the sample than in doing the size analysis.
2. Use extreme caution when comparing size distributions measured by different techniques as they may be measuring different particle sizes. If possible use a technique that measures the "size" you want to know e.g. microscopy is a good technique for sizing paint pigment because projected area is the property of interest.
3. If fine dust is present in the sample, dry sizing techniques will give a different size distribution to wet sizing techniques because the dust adheres to the larger particles. In particular, beware of dust blinding screens for dry sieve analysis. It may be necessary to "deslime" the sample over a fine screen first.
4. If you need the number size distribution use a technique that counts particles. If you want the mass size distribution use a technique that measures mass directly. Very large errors can occur from converting from one form of the size distribution to the other.

Never accept size analysis data on face value. Always view the data critically in light of the comments made above.

2.1.7. Particle Size Summary

Particle size is the single most important property of a particulate system, hence the detail it has received in this chapter. Size is also a good case study for common issues and problems associated with any particle property. In summary:

- There is no single definition of particle size. Know the common definitions and be sure you know what "size" you are talking about.
- Size distributions can be represented as cumulative or frequency distributions. Be comfortable with the mathematics of size distributions and common transformations eg. converting from mass to number size distributions.
- There are many definitions of mean size. Choose the definition appropriate to your application.
- There are many ways to measure size and size distributions. They measure different types of particle "size" and different types of size distribution. Correct sampling and sample preparation are very important. Never take particle sizing data at face value. View it with a critical eye.

Table 2.5. Commonly employed methods of particle size analysis

Method	Size range [µm]	Condition	Particle size measured	Type of distribution	Cost range [$] On-line capability(+)
Microscopy					
Optical	1-500	Dry or wet	A range of	Number	100-2000
Electron (e.g.TEM,SEM)	.01-100	Dry	geometric		30,000-250,000
Image analysis		Dry or wet	diameters		3,000-150,000 (+)
Sieving					
Wire woven	37-4000	Dry or wet	Sieve diameter	Mass	100-1000 (+)
Electro-formed	5-120	Dry or wet			
Sedimentation					
Micromerograph	5-75	Dry/gravity	Stoke's diameter	Mass	15,000
Pipette extraction	2-75	Wet/gravity		Mass	60
Photosedimentation	2-75 (.05-5 centrifugal)	or		Surface	1000-5000 (25,000
X-ray sedimentation	.1-75 (.05-5 centrifugal)	centrifugal		Mass	centr.)
Sediment balance	2-75	" "		Mass	10,000-20,000
Elutriation/cyclone tech.	2-100	" "		Mass	3000-8000 (+)
		Dry or wet			
Sensing zones					
Electrical (e.g Coulter)	.5-1000	Wet	Volume	Number	10,000-20,000
Optical:		Wet or dry	diameter		
Scattering	.1-100			Number or	20,000-50,000(+)
Diffraction (e.g.	1-3500		Scattering diam.	vol.	50,000-90,000 (+)
Sympatec Helos or			Scattering diam.	Volume	
Malvern)					
Obscuration	1-9000				50,000-80,000 (+)
			Proj. area diam.	Number	
Surface methods					
Surface permeametry	.1-75	Dry	Surface-volume	Surface	500-10,000
Gas adsorption		Dry	diameter		5000-50,000
Thermal conductivity		Dry			
Adsorption from solution		Wet			
Heats of solution/wetting		Wet			
Miscellaneous					
Ultrasonic attenuation	1-500	Wet	Volume	Volume	30,000-50,000 (+)
Photon correlation spect.	.003-3 µm	Wet	diameter	Number	(+)
Scanning IR laser (e.g.			Scattering diam.		
Lasentec Labtec	3-100 µm	Wet		Number	30,000-50,000 (+)
100)			Chord diameter		

2.2. Density, Porosity and Pore Size Distribution

2.2.1. Density and Porosity Definitions

Powder density is an important property in determining requirements for granulation. In fact, granulation may be primarily for the purposes of *either* increasing *or* decreasing powder density. For any particle assembly, there are three important densities to define for a given mass of solid M (see figure 2.5). The difference is in the reference volume used. The three densities are:

- bulk density: which relates to the volume occupied by a bulk solid including all void space.
- particle (apparent) density: which relates to the volume occupied by a single particle including internal porosity. The volume is defined by an imaginary envelope around the particle.
- skeletal (true) density: this is the true solid density of the material.

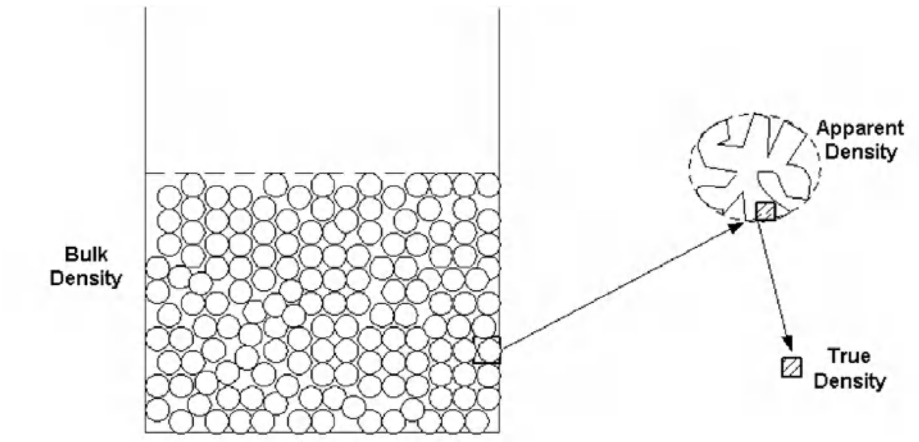


Figure 2.5. Different types of solid density

Only the third of these is a property of the material that can be tabulated. Bulk density and particle density are strong functions of particle morphology (size, shape, porosity). Bulk density is an important property in determining bulk solid storage and handling requirements. Particle density is important in determining the interactions between particles and fluids.

The three densities are related by the fraction of the volume occupied by the fluid. We define the voidage ε as the fraction of bulk solid volume present as voids *between* particles. The porosity ε_{pore} is the fraction of particle volume present as pores *within* the particle. Table 2.6 summarises all the density and porosity definitions. The densities, voidage and porosity are related as follows:

$$
\begin{aligned}
\rho_p &= \rho_b (1-\varepsilon) \\
\rho_s &= \rho_p (1-\varepsilon_{pore})
\end{aligned}
\tag{2-15}
$$

Table 2.6. Density, porosity and voidage definitions

Property	*Definition*
bulk density	$\rho_b = \dfrac{M}{V_s + V_v + V_p}$
particle (apparent) density	$\rho_p = \dfrac{M}{V_s + V_p}$
skeletal (true) density	$\rho_s = \dfrac{M}{V_s}$
voidage	$\varepsilon = \dfrac{V_v}{V_s + V_v + V_p}$
porosity	$\varepsilon_p = \dfrac{V_p}{V_s + V_p}$

M = solid mass; V_s = solid volume; V_v = void volume in particles; V_p = pore volume

2.2.2. Density measurement

The general principle in density measurement is to measure the volume occupied (or fluid volume displaced) by a known mass of solid. **Solids preparation is critical.** The measured density is a strong function of the way the sample is prepared.

Bulk density is relatively easy to measure by simply observing the volume taken up by a known mass of bulk solid. The loose filled bulk density is measured immediately after pouring into the container. This is not a very reliable measurement. The tapped bulk density measures the volume occupied by the bulk solid after tapping the container mechanically till it reaches it maximum settled density. For fine solids, bulk density may increase significantly under pressure. The compressibility of fine solids will be considered further in the section on bulk solids characterisation.

Skeletal (true) density is measured by the volume of a liquid or gas displaced by the solid. The density is measured with a liquid or gas pycnometer. Standard techniques are available and well documented eg. BS 3483, Part B8 (1978). In both cases, sample preparation is critical. For a liquid pycnometer it is important the liquid chosen completely wets the solid. The solid is first ground finely and evacuated before adding liquid. These preparation steps are aimed at ensuring that all gas in the sample is completely displaced by the liquid.

In gas pycnometry, an inert gas (Argon or Helium) is used. After evacuating the sample, the volume of gas taken up by the sample at several different pressures is measured. The solid volume can be inferred from these measurements.

Particle density is the most difficult density to measure accurately. The problem is that only an imaginary envelope separates the internal pores from the external voids and the displacement fluid is not smart enough to know the difference! Techniques generally involve three steps:

1. Fill all particle pores with a liquid eg. soaking in boiling water.
2. Drain particles and surface dry.
3. Measure volume of water displaced by the wet particles.

It is difficult to separate the fluid in the pores from the surface fluid. Thus, the techniques are very inaccurate for particles less than 1mm. For these particles, the particle density must be inferred from measurements eg. combining measurement of particle porosity and skeletal density in eqn. 2-15.

2.2.3. Porosity and Pore Size Distribution Measurement

Total porosity can be calculated from independent measurements of skeletal and particle densities (eqn. 2-15). Porosity and pore size distribution are measured directly by mercury porisimetry. This technique relies on the fact that pressure must be exerted to force mercury into the pores of a particle because the contact angle between mercury and all practical solids is greater than 90º. A known mass of powder is first evacuated then exposed to a pool of mercury. As the pressure is increased, the mercury is forced into pores of smaller and smaller size. Assuming cylindrical pores, the pore size is related to the applied pressure by the Laplace-Young equation:

$$P = \frac{4\gamma_{lv} \cos\theta}{d_{pore}} \tag{2-16}$$

The volume of mercury that penetrates into the powder is measured for each pressure increment. Thus the method measures directly cumulative volume basis pore size distribution $V(d_{pore})$ ie. the cumulative volume of pores per unit mass of solid greater than a size d_{pore}. The total pore volume is $V(d_{min})$ where d_{min} is the size of a pore penetrated at the maximum applied pressure. The pore volume is related to porosity as defined above by:

$$\varepsilon_p = V(d_{\min})\rho_p; \ or \ \varepsilon_p = \frac{V(d_{\min})\rho_s}{1+V(d_{\min})\rho_s} \tag{2-17}$$

The cumulative pore size distribution can be converted to a frequency distribution if desired. The same rules for manipulation and correct representation of distributions developed for particle size distributions above also apply for pore size distributions. Figure 2.6 shows an example of mercury porosimetry data for a series of granules of the same feed formulation granulated under different conditions.

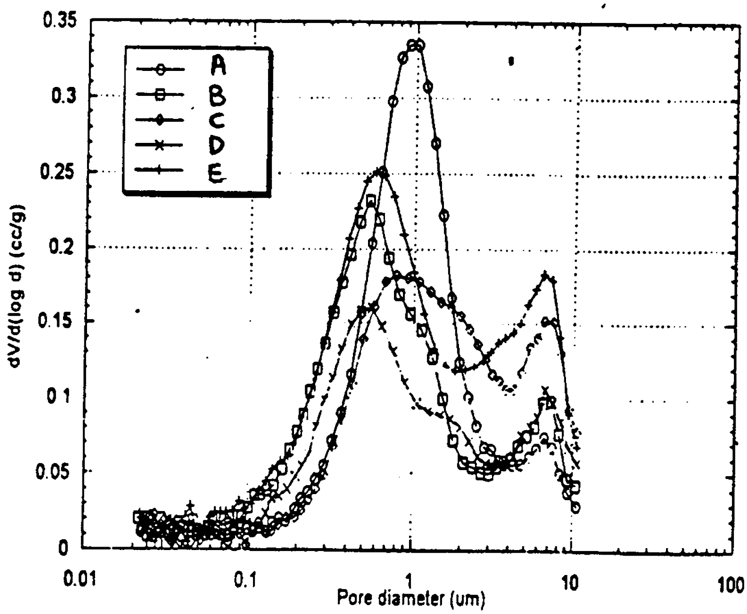


Figure 2.6. Typical Mercury Porosimetry Data

Some of the limitations of the mercury porosimetry technique are:

1. The minimum pore size is limited by the maximum pressure applied. Typically, this is of order a few nanometers. Mercury porosimetry is not applicable for characterising microporous materials.
2. The pores are not cylindrical and are connected. This causes hysteresis in porosimetry results.
3. For fine powders or coarse powders with large macropores, it may be difficult to distinguish between mercury intruding into interparticle voids and intraparticle pores.
4. Powders have heterogenous surfaces so there may be a distribution of surface energies (and contact angles) as well as of pore size.

2.3. Granule Properties

The key properties of any granule are its size and porosity. Virtually all other granule properties of importance depend directly on one or the other. Figure 2.7 shows the strong relationship between granule strength and porosity. Other important morphology dependent properties include bulk density, flowability, reactivity, permeability, dispersibility and dissolution.

Granules are a particular type of particle and the techniques and definitions listed above are all applicable to granules. However, sometimes there are special issues associated with analysing granules and some of these are described below.

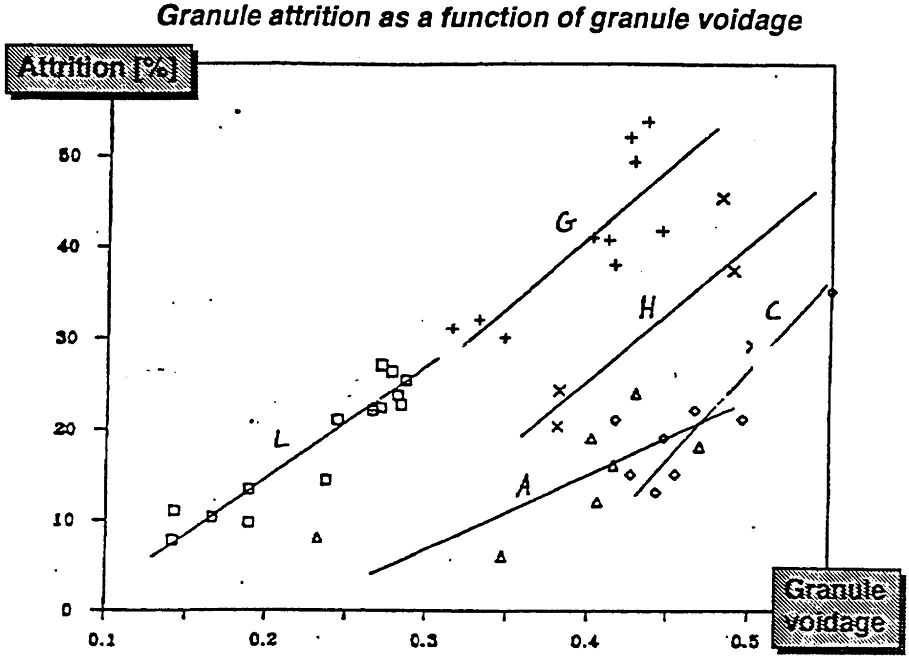


Figure 2.7. Relationship between granule porosity and fracture toughness for herbicide granules

2.3.1. Granule Size Distribution

The desired size range for granules is always in the sizing range for sieve analysis. Wet granules, however, present special problems for sieving:

- they are sensitive to handling and may either granulate further or break up before their size distribution is measured
- wet granules may cake on the screens or blind the screen cloth
- it is sometimes difficult to define what is a granule and what is a set of loosely connected particles.

Sample preparation and handling is more important for granules than for "standard" particles. One useful approach in the laboratory is to freeze the wet granules with liquid nitrogen [Hinkley et al., 1994]. The liquid nitrogen can be poured directly into the granulator to freeze the granules *in situ* or a sample of granules is poured into a stirred container of liquid nitrogen. The frozen granules are immediately sieved for one or two minutes in a precooled set of sieves. Granules greater than 0.25 mm can be easily sieved by this technique but smaller granules are problematic.

The main preparation alternative is to dry the granules before sieving but granules may cake together during tray drying or attrit and break during fluid bed drying. Care should be taken to check if dry granules break or attrit during the sieving operation.

The greatest problems occur when the final granule size distribution overlaps initial particle size distribution, the material is only partially granulated with a very broad size distribution. In all cases, the sample preparation technique used needs to be well documented and repeatable.

2.3.2. Porosity and Granule Density

The porosity of dry granules can be measured by mercury porisimetry as described above (eg. Sastry et al., 1977). Porous, friable granules may deform or crush under the pressure exerted by the mercury. Thus, a minimum dry granule strength is needed for mercury porosimetry to be applicable.

As an alternative, the apparent density of wet granules has been measured successfully by displacing a non-wetting fluid with a sample of granules (Capes and Danckwerts, 1965, Hinkley et al., 1994, Iveson, 1997). The granule porosity can then be calculated as:

$$\varepsilon_{pore} = 1 - \frac{\rho_{gran}}{\rho_s (1+w)} \tag{2-18}$$

Kerosene is the non-wetting fluid of choice. Performed with care, this technique gives errors in porosity of ±2% relative error. However, the technique is only applicable to well formed granules and does not work well for very porous granules or loose aggregates Iveson, 1997.

X-ray microtomography is a new technique which has great potential to give information on the internal structure of granules (Golchert *et al.*, 2002). This approach uses technology borrowed from medical imaging with bench top equipment now available for paticle analysis. Figure 2.8 gives an example of tomography analysis of a singe granule. Note that the pore structure is very non-uniform with large macropores or flaws in some places. This technique gives potential to study in detail how the structure of granules change with processing and variations in formulation.

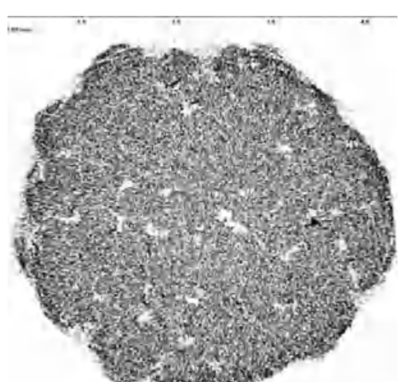


Figure 2.8. X-ray tomography image of a slice through a 2mm diameter granule made from 50µm glass ballotini with salt solution as binder

2.3.3. Wet Granule Morphology

During the wet granulation process, the granule is a three phase system consisting of particles, liquid binder and air. The granulation process is very sensitive to the liquid content, more particularly to the level of liquid saturation in the granule which depends both on the amount of liquid present and the granule porosity. We define the liquid saturation as:

$$s = \frac{w\rho_s\left(1-\varepsilon_{pore}\right)}{\rho\varepsilon_{pore}} \tag{2-19}$$

As the granule saturation increases, the structure of the liquid phase inside the granule changes (Newitt and Conway-Jones, 1959) (see figure 2.8). At low saturations, the liquid exists as discrete pendular bridges. As saturation increases some regions of the granule will have liquid filling capillary networks. At saturation a little below 100%, the granule will be in the capillary state ie. entirely saturated in the centre of the granule but surface dry with the liquid sucked into the entrances of the surface capillaries. For s > 100%, the granule have a liquid layer on the outside. At higher liquid contents, the granules will collapse into a slurry.

Granule saturation cannot be measured directly. Instead, it must be inferred from independent measurements of granule porosity and moisture content.

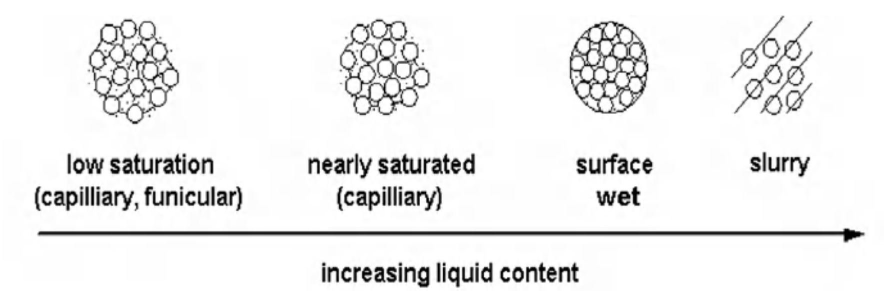


Figure 2.9. The structure of granules with different liquid saturations

2.3.4. Granule Performance Properties

Depending on the desired application, there are many possible performance properties that may be measured. It is in this sphere that rituals abound. Rituals may have a place in quality control but have no place in granule design and troubleshooting. For these purposes, we need to measure properties.

"Attrition resistance" is a good example to illustrate the point. Empirical techniques abound by which the size distribution of granules are measured before and after simulated materials handling eg. in a pneumatic conveying circuit, in a fluid bed, in a tumbling drum. The amount of fines generated is an index of attrition. Such tests are rituals because:

1. They give no information on why the batch had good or poor resistance to attrition.
2. Results are only useful for handling conditions very similar to the test conditions and cannot predict behaviour in other handling environments.
3. Results from different attrition index tests correlate poorly.

An alternative approach is to measure the granule properties that control its resistance to attrition eg. fracture toughness, porosity and pore size distribution. Changes to either the formulation or the process can then be targeted to overcome the problem eg. increasing residence time or process intensity will reduce granule porosity.

Table 2.7 lists a number of granule performance areas and associated important particle or assembly properties. Many of these properties are discussed in detail in the ensuing chapters.

Table 2.7. Relevant properties for granule quality

Performance Area	*Relevant Properties*	*Measurement Techniques*
Strength, attrition resistance	Fracture toughness Hardness Porosity	Uniaxial compression Indentation, Notched bar Mercury Porosimetry
Bulk Solids Flow	Internal angle of friction Wall angle of friction Unconfined yield stress	Shear cell
Packaging and Storage	Bulk density Size distribution	
Segregation	Size distribution	
Caking	Moisture content Hydroscopic tendency Phase transitions	Drying Differential Scanning Calorimetry Dilatometry
Dispersion and dissolution	Contact angle Porosity, pore size distribution Dissolution rate	Sessile drop, Washburn, Flotation, IGC
Resistance to fluid flow	Permeability Size distribution	Permeameter
Appearance	Shape Colour	Image analysis

2.4. Summary

A scientific understanding of granulation requires good characterisation of both feed formulation and product granules. This should be achieved by measuring particle properties rather than empirical indicies (rituals).

Particle granule size and size distribution are always key properties. Take care with the definitions and measurement of particle size, correct presentation of the size distributions, and careful choice of distribution parameters. For granule sizing, special care is needed in sample preparation through drying or freezing the granules.

Next to size, granule density and porosity are the most important properties. These can be measured, with care, using a combination of fluid displacement and porosimetry techniques described in this chapter. The fractional saturation of wet granules can also be inferred from these measurements.

Powder and granule morphological properties are important in all aspects of granulation and compression processes. Further important properties are defined in the ensuing chapters.

A final cautionary note on sampling - the most careful property measurement is useless unless it is carried out on a representative sample. Correct sampling and sample preparation are sometimes difficult, but always essential. See Allen (1997) or Gy (1979) for detailed discussion of sampling techniques.

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2.6. Nomenclature

a	arbitrary property of particles
c	cohesion of bulk solid (Nm^{-2})
d_v, d_s, d_{sv}	equivalent diameters - volume, surface area, specific surface area respectively (m).
f_c	unconfined yield stress (Nm^{-2})
$f(x)$	normalised frequency size distribution (m^{-1})
$f_\alpha(\xi)$	normalised frequency distribution of property ξ with respect to quantity α
$F(x)$	normalised cumulative size distribution
M	total mass
M_s	sample mass
$n(x)$	un-normalised frequency distribution (no. m^{-1})
n^*_i	number of particles in size interval i
$N(x)$	un-normalised cumulative size distribution
N_T	total number of particles
P	pressure (Pa)
p	probability
Q	number of sample increments
r	pore radius
s	particle surface area (m^2)
S	surface area (m^2)
SE	sampling error
t	time
$t_{0.025}$	t statistic
v	particle volume (m^3)
V	volume (m^3)
x	particle size (m)
x_i	maximum size of particle in interval i (m)
$\bar{x}_i$	mean size of interval i (m)
Δx_i	width of size interval i (m)
$\bar{x}_{m,n}$	the "m,n" mean size
K	permeability
y_i	fraction of particles in size interval i
α_v, α_s, α_{sv}	shape factors -volume, surface area, specific surface area respectively.
$\gamma_{lv}, \gamma_{ls}, \gamma_{sv}$	surface energies - liquid/vapour, liquid/solid, solid/vapour respectively (Nm^{-1})
δ	effective angle of friction
ε	voidage (volume fraction fluid)
ρ	density (kg m^{-3})
ρ_b, ρ_p, ρ_s	bulk, particle, skeletal densities respectively (kg m^{-3})
σ^2	variance of a distribution
σ	major consolidating stress
τ	shear stress

ϕ_w	wall angle of friction
ϕ	angle of internal friction
θ	contact angle
μ_k	kth moment of a distribution

CHAPTER 3

WETTING, NUCLEATION AND BINDER DISTRIBUTION

In Chapter 1, we divided granulation rate processes into three classes: wetting and nucleation, consolidation and coalescence, and breakage and attrition. Chapters 3 to 5 address each of these classes in turn beginning with wetting and nucleation.

The first stage in any wet granulation process is the distribution of the liquid binder through the feed powder. Two extremes can be considered: (1) The liquid drop size is large compared to unit particle size; and (2) The particle size is large compared to the drop size.

When the drop size is larger than the particle size, wetting the powder with the liquid gives a distribution of seed granules or ***nuclei***. The nucleation process can be divided into four stages (see figure 3.1):

1. Droplets are formed at the spray nozzle at some size distribution and frequency.
2. Binder droplets impact on the powder surface. Drops may coalesce at the powder bed surface and increase the effective drop size.
3. The drop spreads across the bed surface and penetrates into the bed by capillary action to give a loosely packed nuclei granule.
4. Shear forces within the bed may break up large wet clumps and nucleate into smaller entities.

These processes combine to define the nuclei size distribution produced as the powder passes through the spray zone of the granulator.

When the drop size is small compared to the unit particle size, the liquid will ***coat*** the particles. The coating is produced by collision between the drop and the particle followed by spreading of the liquid over the particle surface. If the particle is porous, then liquid will also suck into the pores by capillary action (see figure 3.2). The wetting dynamics control the distribution of coating material which has a strong influence on the later stages of growth.

The initial wetting and liquid distribution to produce nuclei or coat particles is important for a variety of reasons:

- poor wetting leads to very broad nuclei size distributions and in extreme cases a mixture of overwet and ungranulated material.
- often the granulation retains a memory of the nucleation stage with broad nuclei size distributions leading to broad granule size distributions.
- preferential distribution of liquid between individual ingredients can cause component segregation with granule size.
- Wetting phenomena also influence downstream granule processes such as drying and redispersion in fluids.

Both the ***rate*** and ***extent*** of wetting are important for understanding nucleation and binder distribution.

There are relatively few studies in the literature on nucleation and wetting phenomena. Table 3.1 summarises some of the key experimental studies. Wetting and binder distribution have been variously blamed for changes in granule mean size, breadth of the granule size distribution, defluidisation and the presence of large clumps or agglomerates. Many variables, both formulation properties and process parameters, have been found to influence the quality of nucleation. In many cases, the experimental data are difficult to interpret because nucleation, growth and perhaps breakage are occurring simultaneously. One thing is clear, however. Poor nucleation and binder distribution almost always leads to poor control of granule properties with broad granule size distributions, possible wet quenching in fluid beds and clump formation in mixers and tumbling granulators.

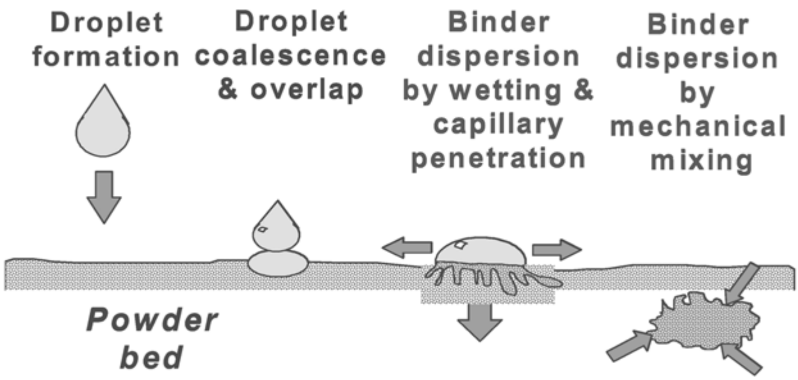


Figure 3.1. Nucleation during granulation of fine particles

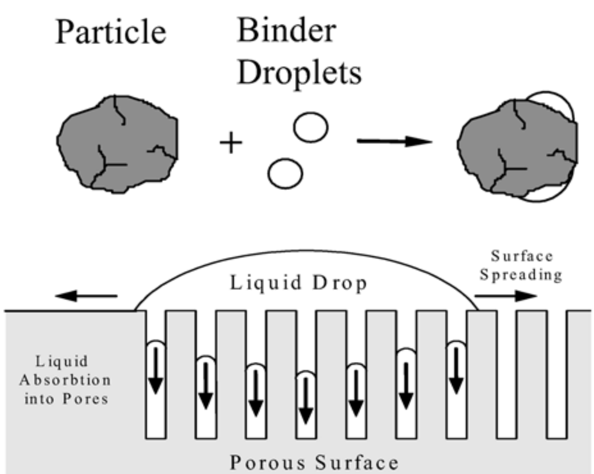


Figure 3.2. Wetting and liquid spreading in systems where the particles are larger than the droplets

Table 3.1. Summary of research into wetting and granulation (Hapgood, 2000)

Worker	Materials	Equipment	Investigated
Schæfer and Wørts (1978)	Lactose and maize starch powder with gelatine, PVP and MCC binders	Fluid bed	Drop size
Schaafsma et al (1999)	Lactose powder with water and 3-8% PVP solution binders	Packed bed in petrie dish	Drop size
Waldie (1991)	Lactose and ballotini powders with water and 5% PVP solution	Fluid bed	Drop size
Watano et al (1994)	Lactose and cornstarch with water	Hybrid fluid bed with agitator	Drop size, binder distribution
Holm et al. (1983, 1984)	Lactose and calcium hydrogen phosphate with water and 10-15% PVP solution	High shear mixer	Drop size, binder distribution and shear forces
Aulton and Banks (1979)	Lactose and salicylic acid powder with water	Fluid bed	Contact angle
Gluba et al (1990)	Talcum, chalk, and kaolin powder combinations with water	Drum	Contact angle
Jaiyeoba and Spring (1980)	Lactose, boric acid, kaolin, suphanilamide and salicylic acid powders with water.	Fluid bed	Contact angle
Krycer and Pope (1983)	Paracetamol powder with HPMC, PVP and water	Fluid bed	Spreading coefficients
Zajic and Buckton (1990)	Methycellulose powder with HPMC, PVP and water	Mixer granulator	Spreading coefficients
Rankell et al (1964)	Aluminium hydroxide and sucrose powder with water	Fluid bed	Binder addition rate, powder flux, spray zone
Heim and Antkowiak (1989)	Talc powder and water	Drum	Binder addition rate
Crooks and Schade (1978)	5% phenylbutazone in lactose powder mix with 10% solution	Fluid bed	Binder addition rate
Kristensen and Schæfer (1994)	Lactose powder with PEG300 and PEG6000	High shear	Viscosity, binder dispersion, shear forces,
Schæfer et al. (1992)	Lactose powder with PEG300 and PEG6000	High shear	Viscosity, binder dispersion, shear forces, powder flux
Cartensen et al (1976)	Lactose, sucrose and starch powder mix with water	High shear	Binder dispersion
Kokubo et al (1996)	Lactose-cornstarch-MCC with HPMC 3cP	High speed mixer	Powder flux
Davies and Gloor (1971)	Lactose, cornstarch, magnesium stearate, gelatin and benzene powder with gelatin solution	Fluid bed	Spray zone

In this chapter, we take a systematic approach to nucleation and binder distribution, starting by defining important powder surface properties and how they can be measured. We then quantify nucleation phenomena in terms of some controlling groups and discuss how wetting and nucleation problems can be identified and overcome by changes to process conditions or formulation properties.

3.1. Powder Surface Properties

3.1.1. Some Definitions

Molecules at the surface of a phase lack near neighbours. This leads to an excess free energy associated with the surface, or **surface energy**. When a drop of liquid sits on a solid surface there are three surface (interfacial) energies to consider (see figure 3.3):

- the liquid-solid interfacial energy γ_{sl}
- the liquid-vapour interfacial energy γ_{lv}
- the solid-vapour interfacial energy γ_{sv}

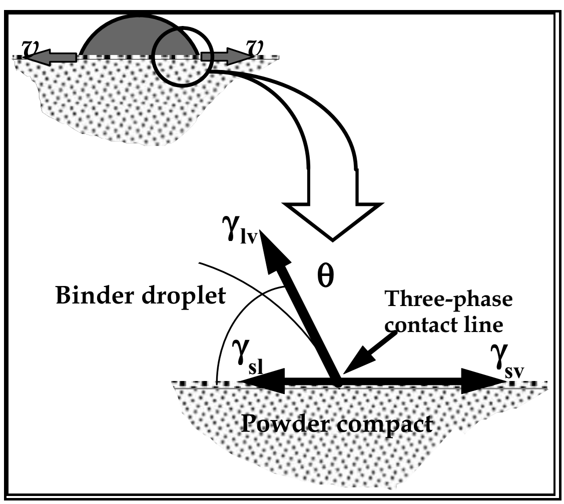


Figure 3.3. Contact angle on a powder surface, where γ_{sv}, γ_{sl}, γ_{lv}, are the solid-vapor, solid-liquid, and liquid-vapor interfacial energies, and θ is the contact angle

Surface energy has units of Jm^{-2} (energy per unit area of surface). Surface energy is often also called surface tension, especially when referring to the liquid-vapour interfacial energy. If a drop is static on the solid surface, the angle the drop makes with the surface, its **contact angle θ**, is such that the horizontal components of the three surface energies balance (see figure 3-3). The contact angle is a measure of the affinity of the fluid for the solid as given by the Young-Dupré equation:

$$\gamma_{sv} - \gamma_{sl} = \gamma_{lv} \cos\theta \tag{3-1}$$

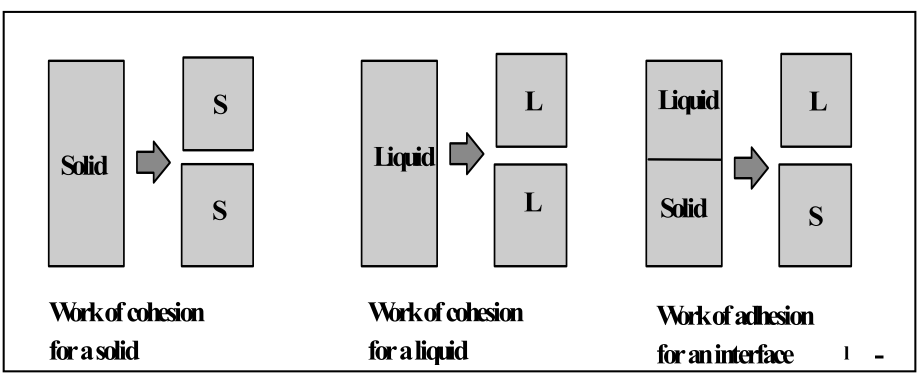
When the solid-vapor interfacial energy exceeds the solid-liquid energy, the fluid wets the solid with a contact angle less than 90°. In the limit of $\gamma_{sv} - \gamma_{sl} \geq \gamma_{lv}$, the contact angle equals 0° and the fluid **spreads** on the solid. The extent of wetting is controlled by the group $\gamma_{lv} \cos\theta$ which is referred to as **adhesion tension**.

Work is required to create new surface. We can define the work of cohesion as the work required to separate a unit cross sectional area from itself and the work of adhesion as the work required to separate a unit area of cross section at the interface between two phases (see figure 3.4). The work of adhesion and cohesion can be written in terms of surface energies:

$$W_{cs} = 2\gamma_{sv} \tag{3-2a}$$

$$W_{cl} = 2\gamma_{lv} \tag{3-2b}$$

$$W_a = \gamma_{sv} + \gamma_{lv} - \gamma_{sl} = \gamma_{lv}(1 + \cos\theta) \tag{3-2c}$$



$W_{cs} = 2\gamma_{sv}$ $\quad W_{cl} = 2\gamma_{lv}$ $\quad W_a = \gamma_{sv} + \gamma_{lv} - \gamma_{sl}$

Figure 3.4. Diagram showing cohesion and adhesion in solid and liquid systems (adapted from York (1995))

We should emphasise that surfaces are not homogeneous or pure (see figure 3.5). Both hydrophobic and hydrophilic sites may be present on the same particle, as well as ions and other adsorbed species eg. impurities and surfactants. To a first approximation, we can consider the total surface energy of the powder surface to be made up of a ***dispersive*** and a ***polar*** component:

$$\gamma_{sv} = \gamma_{sv}^P + \gamma_{sv}^D \tag{3-3}$$

These components reflect the chemical character of the interface, with the polar component due to hydrogen bonding and other polar interactions and the dispersive component due to van der Waals interactions.

The surface energy of the liquid binder can also be arbitrarily divided into polar and dispersion components. When solid and liquid surfaces are brought together, the dispersion and polar components of the two surface energies will interact to reduce the overall solid-liquid interfacial energy. Various models exist to predict this interaction quantitatively (York, 1995). The two most common models are the geometric mean and harmonic mean models:

$$\gamma_{sl} = \gamma_{sv} + \gamma_{lv} - 2\left(\sqrt{\gamma_{sv}^D \gamma_{lv}^D} + \sqrt{\gamma_{sv}^P \gamma_{lv}^P}\right) \tag{3-4a}$$

$$\gamma_{sl} = \gamma_{sv} + \gamma_{lv} - 4\left(\frac{\gamma_{sv}^D \gamma_{lv}^D}{\gamma_{sv}^D + \gamma_{lv}^D} + \frac{\gamma_{sv}^P \gamma_{lv}^P}{\gamma_{sv}^P + \gamma_{lv}^P}\right) \tag{3-4b}$$

Equations 3-1 and 3-4a can be combined to give the contact angle as a function of the polar and dispersive interactions between the powder and the binder eg. using the geometric mean:

$$\cos\theta = \frac{\gamma_{sv} - \gamma_{sl}}{\gamma_{lv}} = -1 + \frac{2\left(\sqrt{\gamma_{sv}^D \gamma_{lv}^D} + \sqrt{\gamma_{sv}^P \gamma_{lv}^P}\right)}{\gamma_{lv}} \tag{3-5}$$

These models are semi-empirical rather than fundamentally based. The polar interaction, in particular, is difficult to predict. In discussion below, we will use the geometric mean model (eqns. 3-4a and 3-5) but caution is advised.

This is a brief summary of key surface properties. For more detailed discussion of powder surface properties, wetting and dispersion see Parfitt (1986) and Ayala (1985).

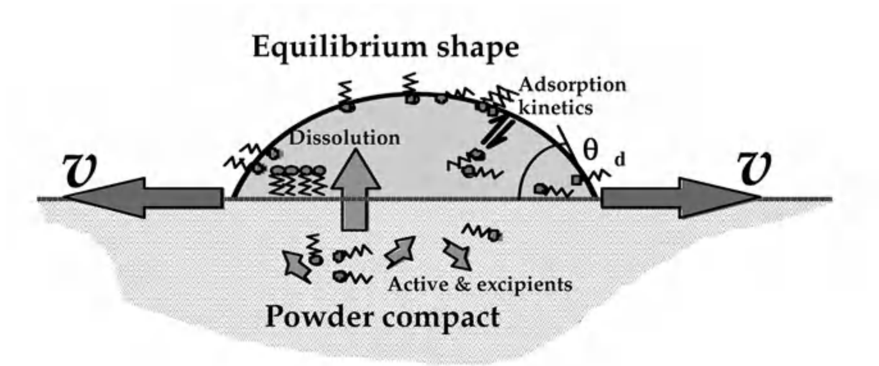


Figure 3.5. The heterogeneous nature of powder-liquid interfaces

3.1.2. Methods of Measurement

There is no single ideal method to characterise the surface properties of powders. The four possible approaches are summarised in Table 3.2. Each of the methods is discussed below.

Sessile Drop Studies. Sessile drop studies to measure contact angle can be performed on powder compacts in the same way as on planar surfaces. A drop is placed on the compact surface and the contact angle measured in one of three ways:

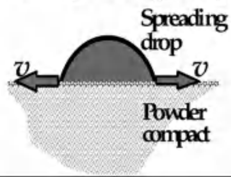
1) direct measurement of the contact angle from the tangent to the air-binder interface,
2) solution of the Laplace-Young equation involving the contact angle as a boundary condition, or
3) indirect calculations of the contact angle from measurements of drop height.

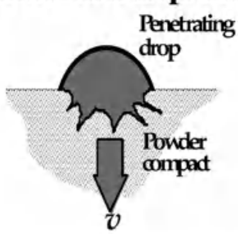
If the compact is dry, the drop will simultaneously spread and sink into the compact. Therefore, to measure static contact angle, the compact must first be saturated with fluid. However, experiments with dry compacts to give useful dynamic measurements may be made through a computer imaging goniometer as depicted in figure 3.6 (Pan *et al.*, 1995).

Figure 3.7 gives an example of dynamics of drop penetration and spreading on a compact of herbicide granules. The short time dynamic data is found to be more sensitive to the differences between the different batches than the longer time data and that spreading velocity decreases with time. These dynamics are affected by the rate of ingredient dissolution and adsorption/desorption kinetics of surface active agents.

The main weakness in sessile drop techniques for powders is that it is very difficult to get reproducible data. Results are extremely sensitive to how the powder compact is prepared, especially to porosity at the compact surface. Preparing the compact may also break particles or granules, changing their surface properties.

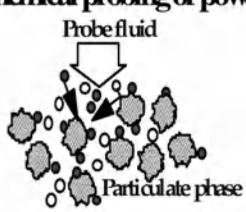
Table 3.2. Methods of Characterizing Powder Surface Properties

<table>
<tr><th>Mechanism of wetting</th><th>Characterization method</th></tr>
<tr><td>Spreading of drops on powder surface
</td><td>Contact angle goniometer
Contact angle
Drop height or volume
Spreading velocity

References: Kossen & Heertjes, Chem Eng. Sci., 20, 593 (1965).</td></tr>
<tr><td>Penetration of drops into powder bed
</td><td>Washburn test
Rate of penetration by height or volume

Bartell cell
Capillary pressure difference

References: Parfitt (ed.), Dispersion of Powders in Liquids, 1986. Washburn, Phys. Rev., 17, 273 (1921). Bartell & Osterhof, Ind. Eng. Chem, 19, 1277 (1927).</td></tr>
<tr><td>Penetration of particles into fluid
</td><td>Flotation tests
Penetration time
Sediment height
Critical solid surface energy distribution

References: R. Ayala, Ph.D. Thesis, Chem Eng, Carnegie Mellon Univ. (1985). Fuerstaneau et al., Colloids & Surfaces, 60, 127 (1991).</td></tr>
<tr><td>Chemical probing of powder
</td><td>Inverse gas chromatography
Preferential adsorption with probe gases
Electrokinetics
Zeta potential and charge
Surfactant adsorption
Preferential adsorption with probe surfactants</td></tr>
</table>

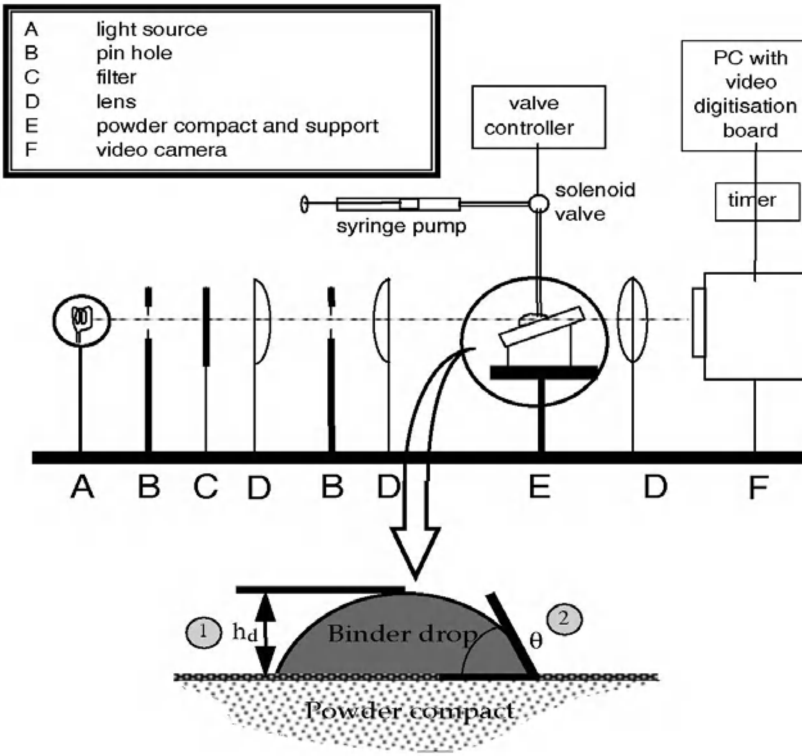


Figure 3.6. A contact angle goniometer

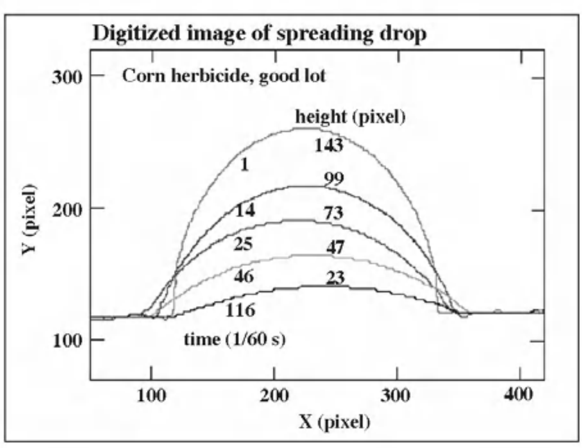


Figure 3.7. Dynamic drop penetration and spreading for herbicide formulation using contact angle goniometry equipment as shown in figure 3.6

Penetration of Liquid into a Powder Bed. The second approach to characterize wetting measures the extent and rate of fluid rise by capillary suction into a column of powder. The **Washburn test** is the best known of this class of technique**.** Consider the powder to consist of capillaries of effective radius R, as illustrated in figure 3.8. The equilibrium height of rise h_e is given by the balance between capillary and gravity forces acting on the liquid in the capillary:

$$h_e = \frac{2\gamma_{lv}\cos\theta}{\Delta\rho g R} \tag{3-6}$$

When the fluid first starts to rise up the capillary, it will be resisted by viscous losses. From the Hagen-Poiselle equation, the viscous pressure drop at some height h is:

$$\frac{\Delta P}{h} = \frac{8\mu\bar{v}}{R^2} = \frac{8\mu(dh/dt)}{R^2} \tag{3-7}$$

If we ignore gravity ($h << h_e$), then this pressure drop is equal to the available capillary pressure difference:

$$\Delta P = \frac{2\gamma_{lv}\cos\theta}{R} \tag{3-8}$$

Substituting into eqn. 3-7 and integrating gives the height of liquid in the capillary as a function of time:

$$h = \sqrt{\left[\frac{R\gamma_{lv}\cos\theta}{2\mu}\right]t} \tag{3-9}$$

The contact angle can be calculated from the slope of a plot of h^2 *vs* t provided R, μ and γ_{lv} are known.

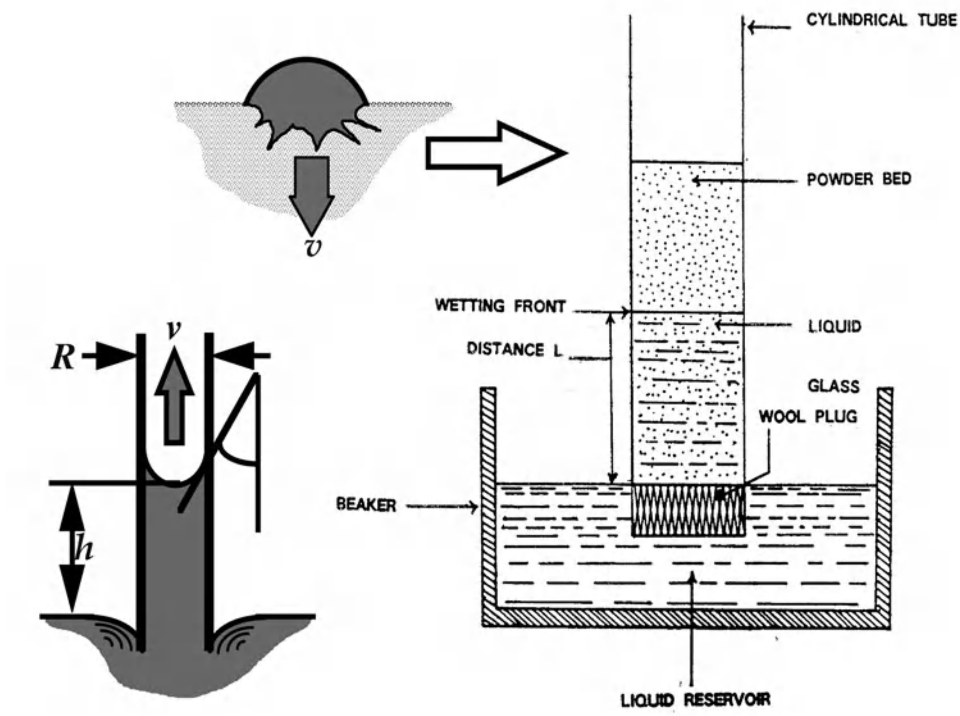


Figure 3.8. The Washburn measurement of wetting dynamics involving fluid Rise driven by capillary pore pressure. Here h is height of penetration, R is tortuosity or pre radius, v is the rate of rise (dh/dt), and θ is binder contact angle with the particulate phase

The Washburn test measures the penetration profile of a liquid in a powder bed as a function of time. First, the penetration profile for a liquid that perfectly wets the powder (zero contact angle) is measured to determine the effective capillary radius *R*. Then the contact angle of any other fluid can be measured from the penetration profile for that liquid through a similarly prepared powder bed. The test can also be used to investigate the influence of powder preparation (e.g. the effects of milling, surface modification, or surfactant concentration) on penetration rates for a given binder fluid. Figure 3.9 gives an example of Washburn test data for two formulations with heptane and decane as fluids. Differences in wetting dynamics are clearly seen.

The **Bartell cell** is related to the Washburn test except that adhesion tension is determined by a variable gas pressure which opposes penetration (Bartell and Osterhof, 1927).

The Washburn test and its variants have the advantage that they mimic the actual wetting process in granulation. A drop sucking into a powder bed by capillary action is equivalent to liquid rising into a powder column. The required equipment is simple and cheap. There are disadvantages, however. It is difficult to separate the effects of contact angle θ and bed pore size in the Washburn test and results are very sensitive to the way the powder bed is prepared. In addition, there is no way to assure that the best wetting fluid has been chosen for a assumed contact angle of zero. Non-zero intercepts on the h^2 *vs t* may occur with surfactant solutions due to the kinetics of surfactant sorption at the air-water and solid surfaces. For coarse materials, the gravity force should be included in eqns. 3-9 (Parfitt, 1986).

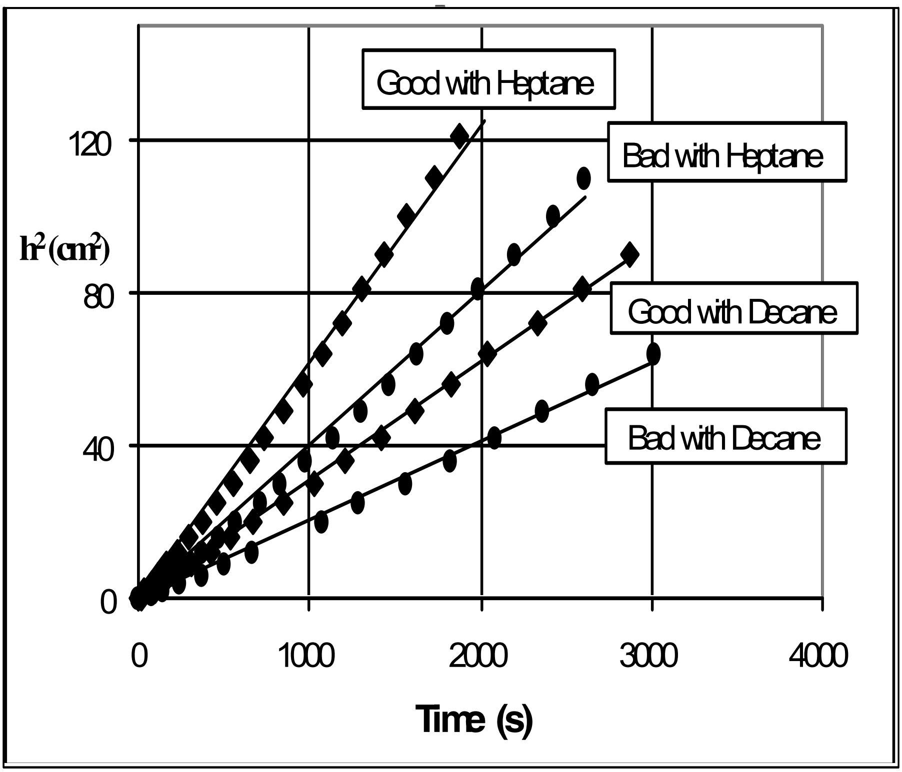


Figure 3.9. Washburn test data for two formulation with different wetting characteristics

Penetration of Particles into Fluids. A third approach to characterize wetting measures the penetration of particles into a series of reference fluids of varying surface tension (Ayala, 1985; Fuerstanau *et al.*, 1991). Particles will float on a liquid of high surface tension and penetrate and sink into a liquid of low surface tension. The liquid surface tension that just allows the solid to completely penetrate is defined as the critical surface energy of the solid. We can relate this measurement to our previously defined properties by examining the wetting process (Parfitt et al, 1986).

For a particle to penetrate into a fluid it must complete three stages of wetting (see figure 3.10) and each step involves a change in the surface energy of the system:

- Adhesional wetting: The particle adheres to the liquid surface replacing air-particle and air-liquid interfaces by a particle-liquid interface. The work of adhesion is given by eqn. 3-2c.

- Immersional wetting: The particle pushes through the liquid surface replacing particle-air interfaces with particle-liquid interfaces.

$$W_i = \gamma_{sl} - \gamma_{lv} = -\gamma_{lv}\cos\theta \tag{3-10}$$

- Spreading wetting: The liquid must spread over the top of the particle, replacing a particle-air interface with a liquid-air and particle-liquid interface.

$$W_s = (\gamma_{sl} + \gamma_{lv}) - \gamma_{sv} = -\gamma_{lv}(\cos\theta - 1) \tag{3-11}$$

The total change in free energy is:

$$W_T = W_a + W_i + W_s = -\gamma_{lv}\cos\theta \tag{3-12}$$

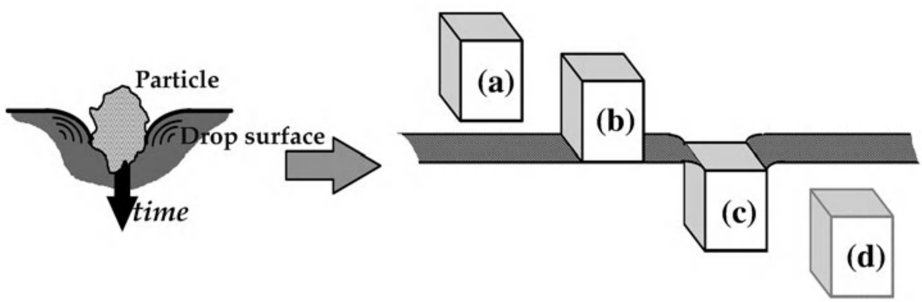


Figure 3.10. Particle Penetration into a Fluid - The Three Stages of Wetting (Parfitt, 1986): (a)→ (b) adhesional wetting; (b)→(c) immersional wetting; (c)→(d) spreading wetting

The complete wetting process is only spontaneous if the required work for each step is negative. Although the total work W_T is negative if $\theta \leq 90^\circ$ (eqn.3-12), the work of spreading will only be negative if $\theta \leq 0^\circ$ (eqn. 3-11). Thus, the critical surface energy should correspond to a contact angle of 0° (Fuerstenau et al., 1991). However, complete immersion could occur for 0°<θ<90° if gravity or mechanical mixing act to overcome the work of spreading (Ayala, 1985).

It is not possible to calculate the contact angle with other solvents from this measurement. However, the technique can give fast information about the interaction between the powder and real binder liquids if the binder liquid is used rather than reference solvents. Furthermore, this technique is the only one that can measure a distribution of surface properties, rather than an average value. As real powders may be heterogeneous with a distribution of surface energies, this is the key advantage of this technique. Figure 3.11 shows an example of the distribution of surface energy of coal particles measured by the powder immersion technique.

The critical surface energy may also be determined from the variation of sediment height with the surface tension of the solvent (Vargha-Butler, 1989).

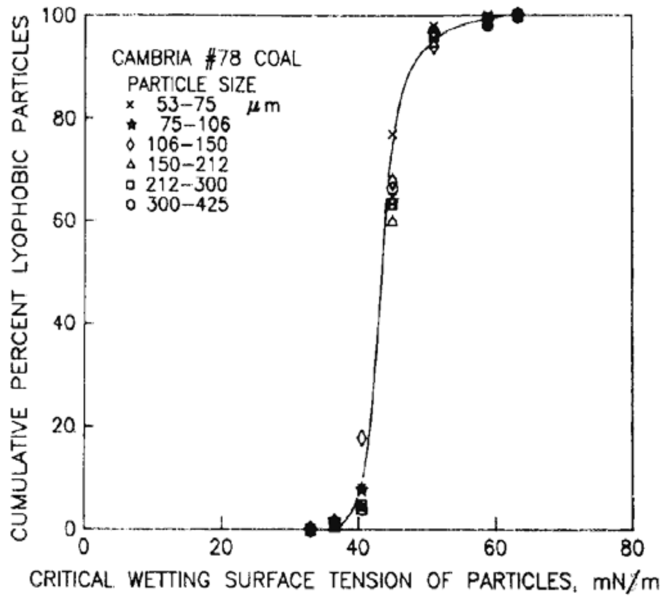


Figure 3.11. Distribution of Critical Solid Surface Energy for a Coal Sample Based on Penetration Tests with Aqueous Methanol Solutions (Fuerstenau et al., 1991)

Chemical Probing of the Powder Surface. Vapour adsorption techniques (organic sorption balance and inverse gas chromatography (IGC)) can be used to give information about powder surface properties without wetting the powder with any liquid. By probing the powder with different solvents, the dispersive and polar contributions of the powder surface energy are extracted.

The IGC technique uses standard gas chromatography equipment with the powder of interest as the packing. The retention time is measured for a number of non-polar and polar vapours. For each probe, the free energy of desorption is:

$$\Delta G = RT \ln V_N \tag{3-13}$$

This free energy is related to the work of adhesion which is further broken up into dispersive and polar interactions (eqn. 3-4a):

$$\Delta G = RT \ln V_N \propto W_a = \sqrt{\gamma_{sv}^D \gamma_{lv}^D} + \sqrt{\gamma_{sv}^P \gamma_{lv}^P} \tag{3-14}$$

For an alkane (non-polar) probe, the dispersive surface energy will be given by:

$$\Delta G \propto \sqrt{\gamma_{sv}^D \gamma_{lv}^D} + c \tag{3-15}$$

A plot of ΔG against $\sqrt{\gamma_{lv}}$ for series of alkane probes will give a straight line. The dispersive component is calculated from the slope of the line using eqn. 3-15. If polar probes are used, the measured ΔG will fall above the reference line for alkanes and distance above the line is the polar free energy contribution. Thus, the polar contribution to the solid surface energy is calculated as:

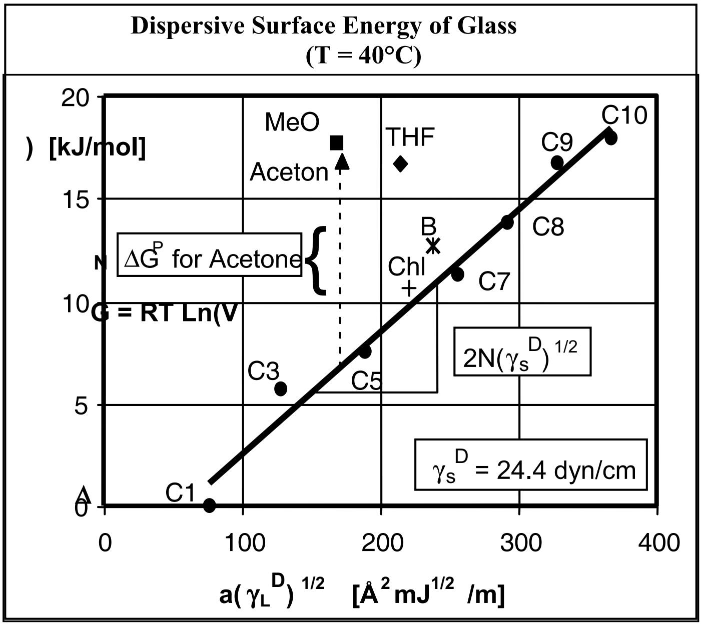
$$\Delta G_P = \Delta G - \Delta G_D \propto \sqrt{\gamma_{sv}^P \gamma_{lv}^P} \tag{3-16}$$

Figure 3.12 gives an example of calculating the dispersive and polar surface energy components for a glass powder using IGC. Having calculated γ_{sv}^D and γ_{sv}^P, the contact angle and solid-liquid surface energy for the powder with any other solvent can be calculated from eqns. 3-4a and 3-5.

This technique is potentially very powerful in matching powders and binders. It does rely heavily on the mixing rule for the polar and non-polar interactions (eqns. 3-4a and 3-14). The non-polar term has a fundamental basis. However, the polar interaction is entirely empirical. Nevertheless, the work of adhesion with standard polar and non-polar probes provides a fast and reproducible method of ranking powders and identifying batch to batch variability in surface properties.

The polar and non-polar surface energies can also be obtained from a series of Washburn tests using a variety of polar and non-polar solvents (Zisman, 1964). This approach is much more tedious, however, and cannot be used if the powder dissolves in some of the reference solvents.

Charges may also exist at interfaces. In the case of solid-fluid interfaces, these may be characterized by **electrokinetic** studies (Shaw, 1983).



Liqui	Notatio	γ (mJ/m^2)	γ^D (mJ/m^2)	γ^P (mJ/m^2)	α (Å^2)	A (-)	DN (-)	Polar Character
Pentan	C	16.0	16.0	0	46.	-	-	Neutra
Heptan	C	20.	20.	0	5	-	-	
Octan	C	21.6	21.6	0	62.	-	-	
Nonan	C	22.	22.	0	68.	-	-	
Decan	C1	23.8	23.8	0	75.	-	-	
Benzen	B	28.	26.	2.	4	8.	0.	Acid
Chloroform	Chl	28.	26.	1.	4	8.	0	
THF	THF	26.	22.	4	4	8	2	Bas
Methano	MeO	22.4	1	4.4	26.	41.	1	Amphoteric
Aceton	Ac	2	2	4	28.	12.	1	

Figure 3.12. Measurement of Interfacial Properties of a Glass Powder by IGC

3.2. Wetting and Nucleation Regimes for Granulation

Section 3.1 established techniques for characterizing the wetting behaviour of powders. To use this information for our granulation processes, we must first identify conditions that will ensure good nucleation. Consider the nucleation process described at the start of the chapter and illustrated in figure 3.1. We arbitrarily break down the nucleation possibilities into three regimes:

1. *Droplet controlled*: Each individual drop wets completely and quickly into the powder bed to form a single nuclei granule. The nuclei size distribution is essentially controlled by the drop size distribution.
2. *Shear controlled*: Liquid pooling or caking occurs where the spray meets the bed. Binder distribution occurs only by breakage of lumps or granules due to shear forces

within the powder bed and the "nuclei" size distribution is independent of the drop size distribution.

3. *Intermediate:* This regime is intermediate between droplet controlled and shear controlled. Some agglomeration does occur in or near the spray zone without complete caking or pooling. The nuclei size distribution will be sensitive to many formulation properties and operating parameters.

Wetting thermodynamics, wetting kinetics and the ratio of powder to liquid fluxes in the spray zone will all influence the nucleation regime. Ideally, we would like to operate in the drop controlled regime. It is much easier to control the size (and size distribution) of drops from a spray nozzle than it is to mechanically disperse the liquid through the bed. Only mixer granulators have any chance of achieving good mechanical dispersion.

To achieve drop controlled nucleation, two conditions are required:

1. Drops penetrate into the bed quickly and do not roll on the bed surface and contact other drops.
2. Drops overall on the powder surface is minimal.

Drop penetration rate is set by wetting thermodynamics and kinetics and primarily influenced by formulation properties. Drop overlap is related to the flux of drops hitting the powder surface and is set by operating parameters. We will consider each of these phenomena in turn.

3.2.1. Drop penetration into powder beds

Theoretical prediction of drop penetration time. Consider a drop hitting the loosely packed, moving powder bed surface in a tumbling or mixer granulator. How long will this drop take to penetrate into the powder bed and what is the effect of formulation properties on penetration time?

We can imagine the porous powder bed as consisting of a series of capillary pores. For a drop to penetrate the pores, the contact angle between the liquid and the powder must be less than 90º (wetting thermodynamics). If this is so, drop penetration is driven by capillary suction and the rate at which liquid penetrates the pore is given by the Washburn equation (see eqns. 3-7 to 3-9 and figure 3.8).

Consider a drop of volume V_d hitting the powder surface. The drop will have a circular footprint on the surface of radius:

$$r_d = \left(\frac{3V_d}{4\pi}\right)^{1/3} \tag{3-16}$$

The rate at which the liquid flows from the drop is:

$$Q = \frac{dV}{dt} = \pi r_d^2 \varepsilon \bar{v} \tag{3-17}$$

where ε is the powder bed voidage. The average velocity in the pore $\bar{v}$ is given by the differential form of the Washburn equation:

$$\bar{v} = \frac{dh}{dt} = \sqrt{\frac{R\gamma_{lv}\cos\theta}{8\mu t}} \tag{3-18}$$

Combining eqns. 3-17 and 3-18 and integrating gives the penetration time ie. the time for the total volume of drop to penetrate the bed:

$$t_p = 1.35\frac{V_d^{2/3}}{\varepsilon^2 R}\frac{\mu}{\gamma_{lv}\cos\theta} \tag{3-19}$$

This equation was derived separately by Middleman (1995) and Denesuk *et al.* (1993). There are many assumptions made in this derivation but the key one is assuming the total voidage of the loosely packed powder bed is present as uniform cylindrical pores. With this assumption, the pore radius can be the bed voidage and the specific surface mean particle size:

$$R = \frac{\varphi d_{32}}{3}\frac{\varepsilon}{1-\varepsilon} \tag{3-20}$$

This Kozeny model for the bed voidage is reasonable for a tapped powder bed in random close packing. However, a loosely packed bed of fine powder has very heterogeneous voidage distribution. Hapgood *et al.* (2003) divided the voidage into two parts: micropores and macrovoids (see figure 3.13). Liquid will flow through the micropores, but there is no capillary driving force for the liquid to flow into the expanding macrovoids. In effect, the liquid does not see the macrovoids. Using the Kozeny approach will overestimate both the voidage and the pore size seen by the penetrating fluid. Hapgood introduced an effective voidage ε_{eff}. If we assume all the voidage above the tapped bed voidage is present as macrovoids, then the effective bed voidage for capillary driven flow is:

$$\varepsilon_{eff} = \varepsilon_{tap}(1-\varepsilon+\varepsilon_{tap}) \tag{3-21}$$

The effective pore size seen by the liquid is the average micropore size:

$$R_{eff} = \frac{\varphi d_{32}}{3}\frac{\varepsilon_{eff}}{1-\varepsilon_{eff}} \tag{3-22}$$

and the drop penetration time is:

$$t_p = 1.35 \frac{V_d^{2/3}}{\varepsilon_{eff}{}^2 R_{eff}} \frac{\mu}{\gamma_{lv} \cos\theta} \quad (3\text{-}23)$$

Eqns. 3-21, 3-22 and 3-23 allow us to calculate the drop penetration time from measurable properties of the powder and liquid binder.

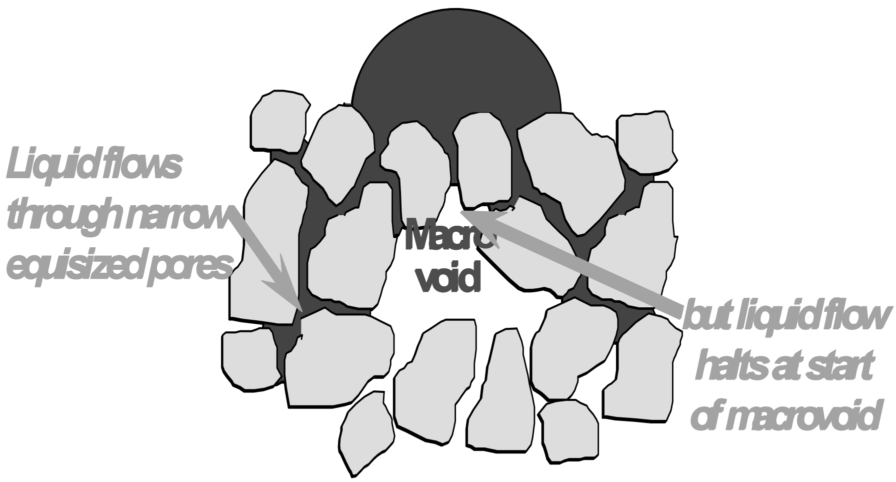
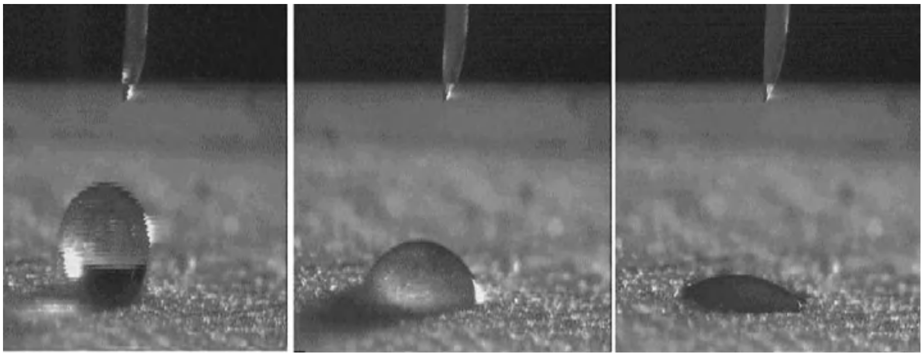


Figure 3.13. Heterogeneous packing of particles in a loose powder bed. A drop on the powder surface will penetrate the micropores by capillary action but stops where the pore expands into a macrorvoid

Experimental measurement of drop penetration time. Hapgood *et al.* (2002) did an extensive study of drop penetration into loosely packed powder beds. It is a relatively easy experiment to perform, similar to the contact angle goniometry (figure 3.6). A carefully metered single drop is placed on a carefully prepared powder surface and the time for complete penetration of the drop is measured (see figure 3.14).



(a) impact (b) 2 seconds (c) 5.4 seconds

Figure 3.14. Drop penetration time measurement: A single drop of polyethyleneglycol solution (PEG200) penetrates into a bed of glass ballotini

Figure 3.15 shows the effect of liquid properties on drop penetration time. For a range of different powder beds, the penetration time varies linearly with the group $\mu / \gamma_{lv} \cos\theta$ as predicted by eqn.3-23. Penetration time is proportional to liquid viscosity and inversely proportional to adhesion tension. This emphasizes the importance of both wetting thermodynamics and kinetics. The contact angle needs to be less than 90º to ensure penetration. Provided this is the case, the dominant liquid parameter is the viscosity, which can vary at least two orders of magnitude for typical liquid binders used in granulation.

Figure 3.16 shows an example of the effect of powder properties. The penetration time decreases sharply as the specific surface mean particle size is increased. Note also that penetration time is different for broad size distribution powders because the bed voidage is a function of the spread of the size distribution.

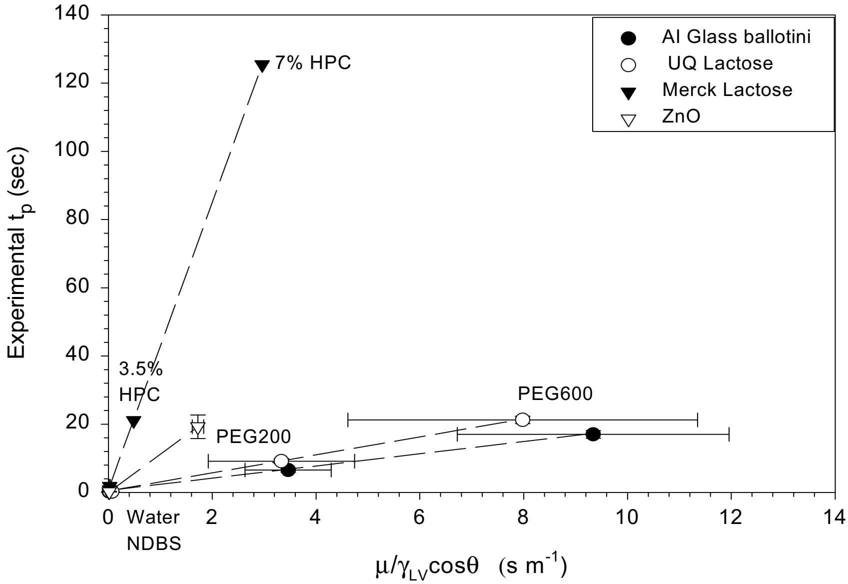


Figure 3.15. Effect of liquid properties on drop penetration time In a rang of powder beds (Hapgood et al., 2002)

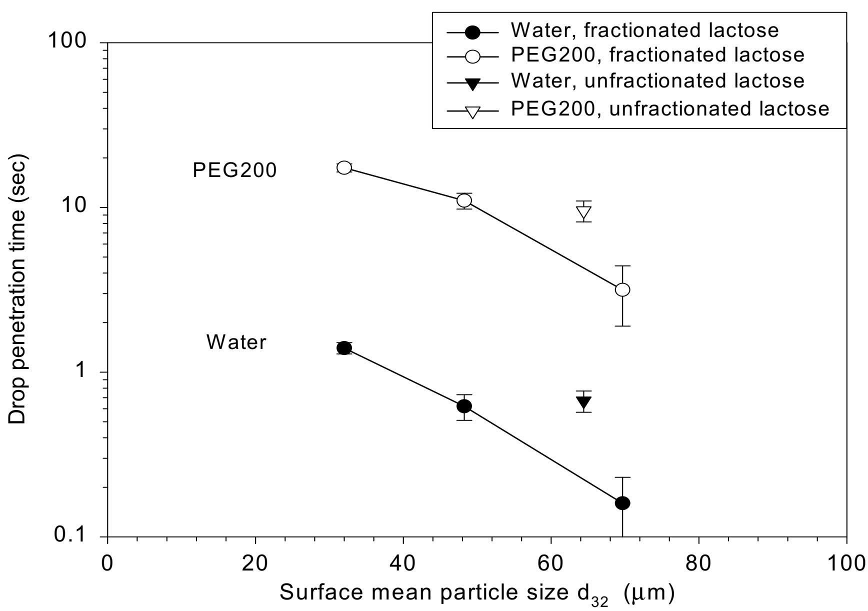


Figure 3.16. Effect of particle size on the penetration time of water and PEG200 drops into lactose powder beds beds (Hapgood et al., 2002)

Figure 3.17 compares the penetration time predictions from the Middleman/Denusek model (eqns. 3-19 and 3-20) with the experimental results for a wide range of powder/liquid combinations. If the model is correct the data should be scattered around the solid equality line or at least within the dashed lines which represent ± 1 second. The Middleman/Denusak model with the Kozeny approach to pore size works well for glass ballotini powders which are free flowing and have very low Hausner ratios. However, the existing methods perform very poorly on lactose, ZnO and TiO_2 powders. The experimental penetration times are an order of magnitude larger than the predictions from eqn.3-19.

Figure 3.18 compares the predictions from the modified theory (eqns. 3-21 to 3-23) with the experimental penetration times. Most of the data is scattered around the equality line and within ±1 second of the penetration time. The improved theory is much better at predicting penetration times on loosely packed powders, and allows t_p to be predicted within an order of magnitude for all powders. The powder structure of the fine, cohesive powders (ZnO, TiO_2, fine lactose and Merck lactose) is not completely described by the effective porosity. Eqns. 3-21 to 3-23 are fairly crude estimates of effective porosity and pore size in loosely packed beds and there is much room for improvement. Nevertheless, this relatively simple model is very useful for estimating drop penetration time, and the effect of liquid and powder properties for all but the finest powders.

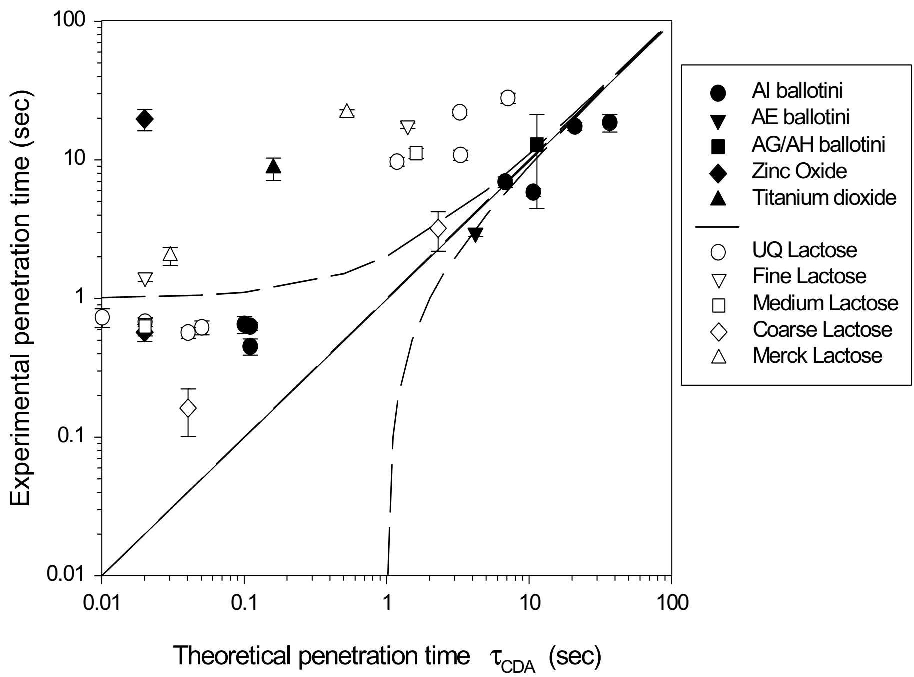


Figure 3.17. Comparison of experimental penetration times with those predicted by the Middleman/Denesuk model theoretical penetration times. Solid line is the equality line and the dashed lines show ±1 second (Hapgood et al., 2002)

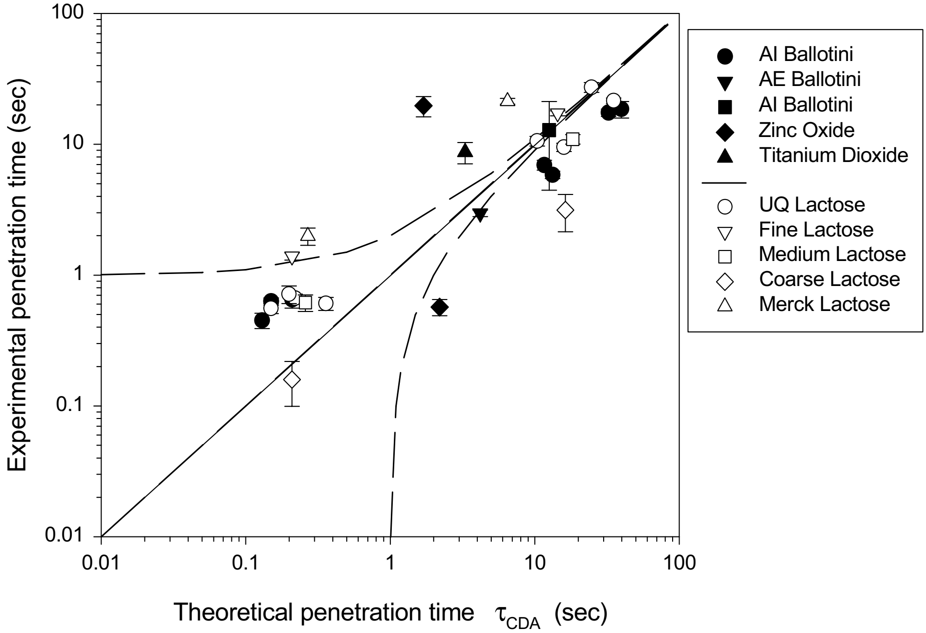


Figure 3.18. Experimental drop penetration times compared to those predicted by the improved theory with ε_{eff} and R_{eff} from eqns. 3-21 and 3-22. Solid line is the equality line and the dashed lines show ±1 second (Hapgood et al., 2002)

3.2.2. Analysis of the spray zone in granulators – the dimensionless spray flux

Consider a powder surface traversing through a spray zone with a flux $\dot{A}$. If a flat spray is used, the powder flux through the spray zone is simply given by (see figure 3.19):

$$\dot{A} = vW \tag{3-24}$$

Each drop hitting the powder surface will leave a footprint as it wets into the powder. If a second drop overlaps this footprint, a doublet will form. The number of drops hitting the powder surface per unit time is:

$$\dot{n}_d = \dot{V}\left(\frac{6}{\pi d_d^3}\right) \tag{3-25}$$

Thus, the rate of production of covered area in the spray zone is:

$$\dot{a} = \dot{V}\left(\frac{6}{\pi d_d^3}\right)\left(\frac{\pi d_d^2}{4}\right) = \frac{3\dot{V}}{2d_d} \tag{3-26}$$

Let us define the dimensionless spray flux as the ratio of the rate at which wetted area is covered by the droplets to the area flux of powder through the spray zone:

$$\psi_a = \frac{\dot{a}}{\dot{A}} = \frac{3\dot{V}}{2\dot{A}d_d} \tag{3-27}$$

The dimensionless spray flux is a measure of the density of drops falling on the powder surface. At low spray flux ($\psi_a << 1$) drop footprints will not overlap and each drop will form a separate nucleus granule. At high spray flux ($\psi_a \approx 1$) there will significant overlap of drops hitting the powder bed. Nuclei granules formed will be much larger and bear little relationship to original drop size. The process is illustrated schematically in figure 3.20. It is exactly analogous to watering the lawn with a garden hose.

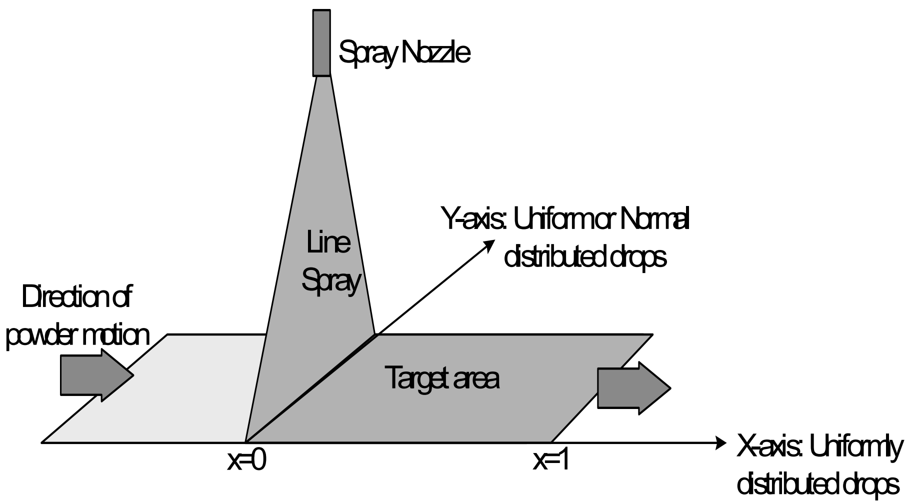


Figure 3.19. An idealized spray zone in a granulator using a flat spray

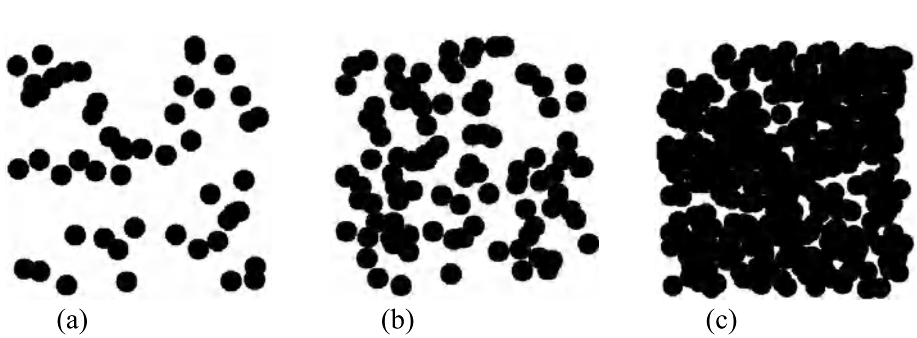


Figure 3.20. Monte-Carlo simulations of drop on the powder bed after the spray zone: (a) 50 discs Ψ_a=0.29, $f_{covered}$=0.26; (b) 100 discs Ψ_a=0.59, $f_{covered}$=0.45; (c) 400 discs Ψ_a=2.4, $f_{covered}$=0.91. Image 500x500 pixels. Disc radius 20 pixels. (Hapgood, 2000)

We can quantify this analysis. If the assumption of *complete spatial randomness* (CSR) is invoked, spatial statistics can be used to derive an analytical solution for both the fraction surface coverage and fraction agglomerates. Under these conditions, it follows that the drops landing randomly on the target area describe a Poisson distribution. It follows that the fraction surface coverage is given by (Hapgood, 2000):

$$f_{covered} = 1 - exp\,(-\,\Psi_a) \tag{3-28}$$

Similarly, we can calculate the number of single drops, not overlapping with any other drops, and by difference, the number of agglomerates:

$$f_{single} = exp(-4\Psi_a) \tag{3-29}$$

$$f_{agglom} = 1 - f_{single} = 1 - exp(-4\Psi_a) \tag{3-29a}$$

The impact of ψ_a on nuclei formation can be studied in ex-granulator nucleation experiments where the nuclei size distribution is analysed after a single pass of the powder through the spray zone (Litster et al., 2001). Figure 3.21 illustrates the dramatic effect of ψ_a on the nuclei size distribution. At low spray flux (v=1.36 m/s; ψ_a= 0.22) the nuclei size distribution is quite narrow. As spray flux increases, the distribution broadens as agglomerates begin to form. At the highest spray flux (v=0.25 m/s; ψ_a = 1.2) the spray zone has become a continuous cake and the nuclei distribution bears no resemblance to the drop distribution. However, when the spray flux is low and we are in the drop controlled regime, changes to the spray drop distribution are directly mapped onto the nuclei size distribution (see figure 3.22).

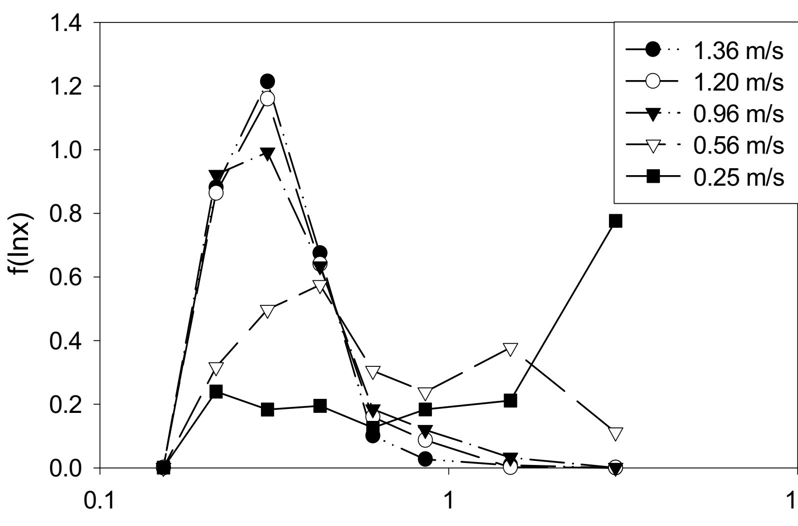


Figure 3.21. Effect of powder velocity on nuclei size distribution in ex-granulator nucleation experiments: lactose powder with water spray (Litster et al., 2002)

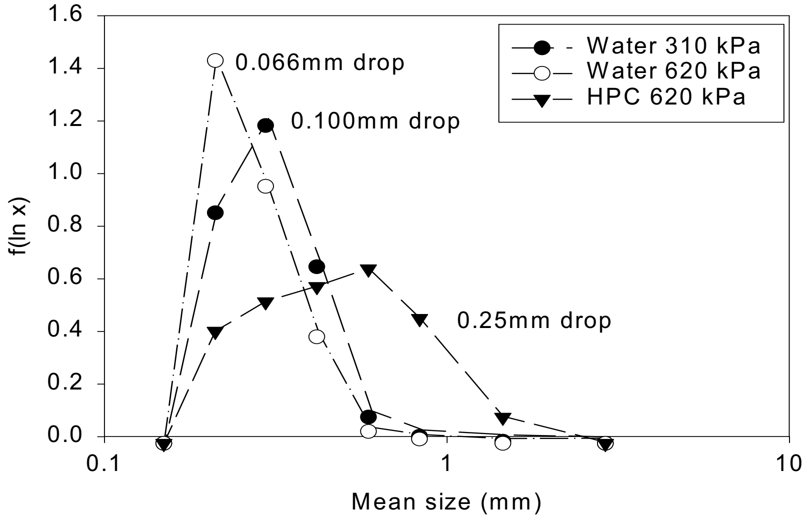


Figure 3.2. Effect of spray drop size distribution on nuclei size distribution (v=1.36 m/s; $\psi_a = 0.22$). Data for water and HPC solutions varying nozzle pressure (Litster et al., 2002)

The dimensionless spray flux characterizes the equipment operating parameters with respect to nucleation in a single dimensionless group. Figure 3.23 shows that the fraction of agglomerates formed in ex-granulator and in-granulator experiments is predicted extremely well by eqn.3-29a with spray flux as the only parameter. The dimensionless spray flux provides a good basis for equipment scale up to maintain good nucleation.

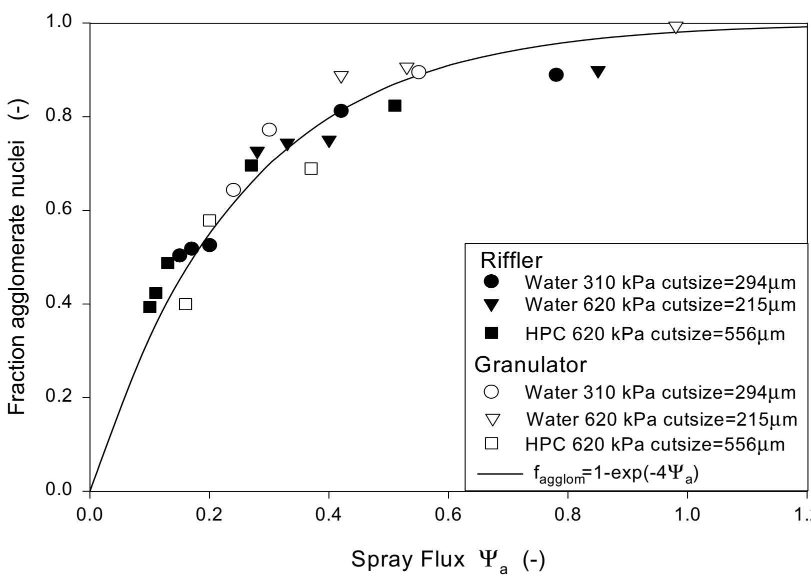


Figure 3.23. Agglomerate formation as a function of spray flux for in-granulator and ex-granulator nucleation experiments using lactose powder with water and HPC solution binders (Hapgood, 2000)

3.2.3. A nucleation regime map

We have defined three regimes for nucleation: *drop controlled, intermediate and shear controlled*, and analysed the factors which dictate the operating regime in any particular system. It is useful to try and summarise in a nucleation regime map. Drop controlled nucleation should occur when there is both :

1. Low Ψ_a - relatively few drops overlap; and

2. Fast t_p - the drop must wet into the bed completely before bed mixing brings it into contact with another partially absorbed drop on the bed surface.

If *either* criterion is not met, powder mixing characteristics will dominate: this is the *mechanical dispersion regime*. Viscous or poorly wetting binders are slow to flow through the powder pores and form nuclei. Drop coalescence on the powder surface (also known as "pooling") may occur and create a very broad nuclei size distribution. In the mechanical dispersion regime, nucleation and binder dispersion can only occur by mechanical mixing and agitation, and the solution delivery method (drop size, nozzle height etc.) has a minimal effect on the nuclei properties.

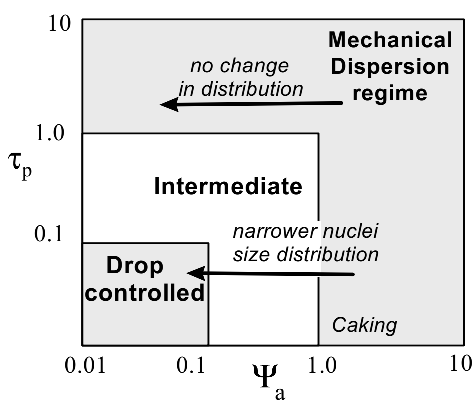


Figure 3.24. Proposed Nucleation Regime Map. For ideal nucleation in the drop controlled regime, must have (i) low Ψ_a and (ii) low t_p . In the mechanical dispersion regime, one or both of these conditions are not met, and good binder dispersion requires good mechanical mixing

Figure 3.24 summarises these concepts in a regime map for nucleation. The map centres around the drop controlled regime, where one drop makes one nuclei, provided the drop penetrates fast enough and the drops are well separated from each other. The regime limit lines are indications only. The horizontal axis is the dimensionless spray flux Ψ_a, which describes the spray pattern and multiple drop behaviour. On the vertical axis is dimensionless drop penetration time τ_p:

$$\tau_p = \frac{t_p}{t_c} \tag{3-30}$$

where t_p is the penetration time of the spray drops and t_c is the circulation time, which is the time taken for a packet of powder to return to the spray zone. The circulation time is a function of powder flow patterns and the amount of material in the granulator and is equipment dependent. However, typical values in industrial granulators will be the order of seconds implying the drop penetration time needs to be of order 0.1s to ensure drop controlled behaviour.

We can validate the regime map by carrying out granulation experiments varying both equipment parameters and formulation conditions and observing the changes to the granule size distribution (Hapgood, 2000). Figures 3.25 and 3.26 show examples of such tests in a 10l batch mixer granulator. Drop penetration time is varied by changing formaultion properties (the viscosity of the binder liquid). Spray flux is varied by changing operating parameters (the spray rate through a single nozzle). Figure 3.25 shows the data after a very short granulation time. Results should be dominated by nucleation only. The spread of the granule size distribution (represented by the parameter δ) increases with both τ_p and Ψ_a. The narrowest granule size distributions occur only in the bottom left hand corner of the map – the drop controlled regime. A similar effect is seen at longer spray times and larger liquid contents (figure 3.26) even though nuclei rewetting and granule growth will now also influence the granule size distribution.

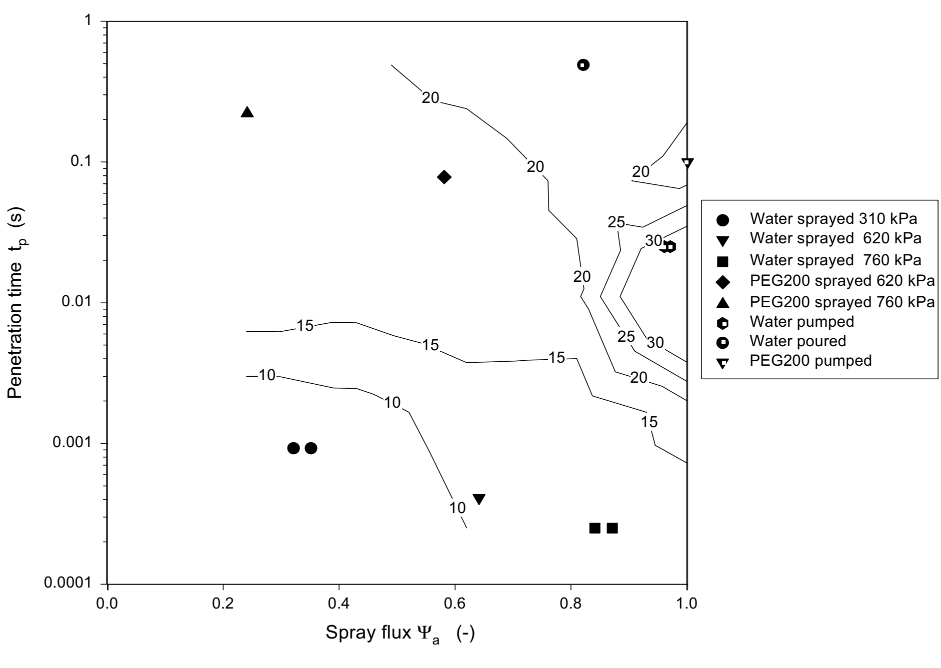


Figure 3.25. Nucleation regime map in 6L Hobart mixer after 10 seconds. Contour lines estimate δ. UQ lactose with water and PEG200 (Hapgood, 2000)

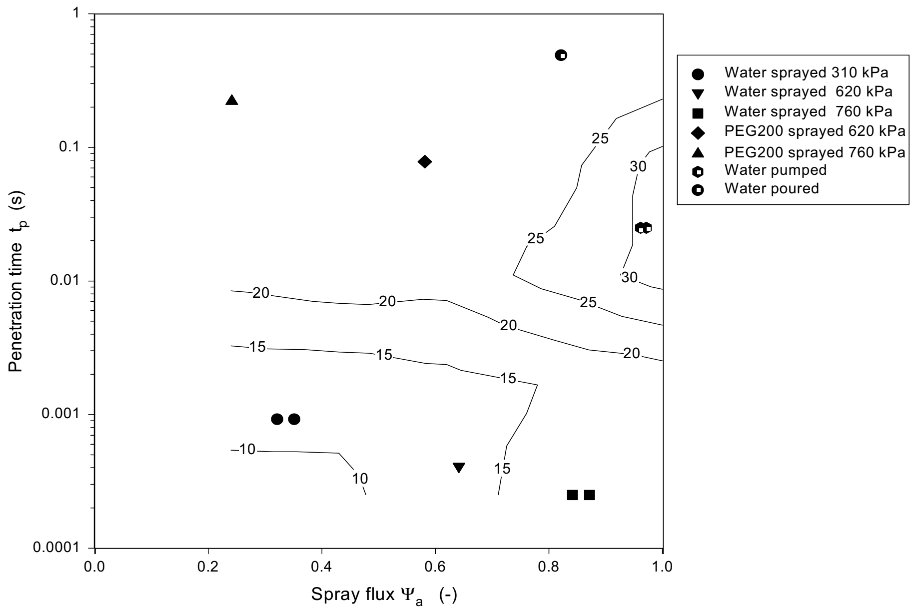


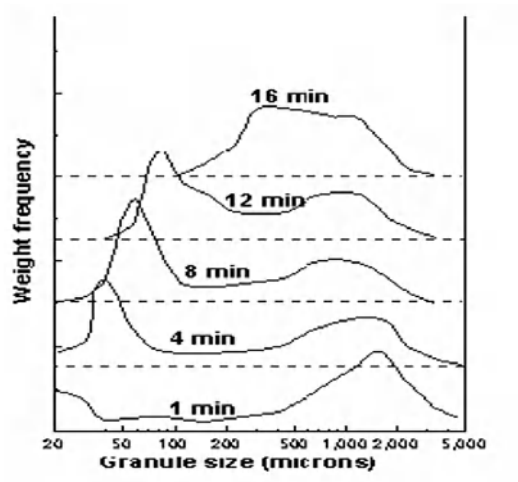
Figure 3.26. Nucleation regime map in 6L Hobart mixer at 3% liquid content. Contour lines estimate δ. UQ lactose with water and PEG200 (Hapgood, 2000)

The regime map is a helpful tool to focus trouble shooting of wetting and nucleation problems, eg. If the drop penetration time is large, then making adjustments to spray rates and nozzle positioning will not lead to narrower granule size distributions because the system will remain in the mechanical dispersion regime (see figure 3.24). Significant changes to wetting and nucleation will only occur ***if changes take the system across a regime boundary.*** This can occur is an undesirable way if processes are not scaled with due attention to remaining in the drop controlled regime (see section 3.3 and later chapters on granulation equipment).

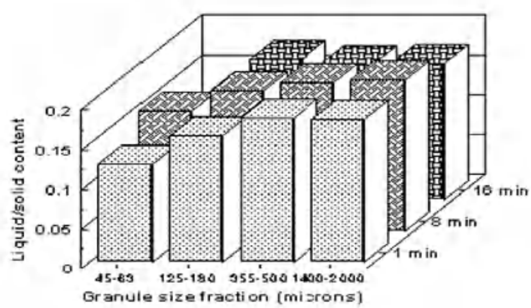
3.2.4. Related studies on wetting and nucleation

Iveson *et al.* (2001) gives a comprehensive review of the literature related to wetting and nucleation in wet granulation. Much of the experimentally reported phenomena fit into the framework we have developed here.

When binder is poured into a granulator, the fraction of coarse granules increases (Knight et al. 1998a,b) in comparison to atomised binder. Adding the binder by pouring creates local patches of high moisture content and preferential growth. The bimodal granule distribution can persist for some time and that the largest granules were the most saturated (see figures 3.27).



(a)



(b)

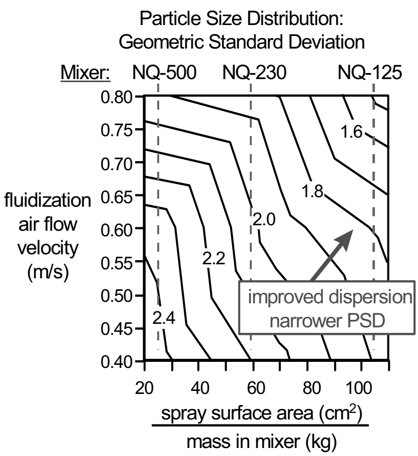
Figure 3.27. Effect of poor binder distribution (a) persistance of bimodal granule size distribution over time (b) larger granules contain higher liquid binder content. (Knight et al., 1998a,b)

A controlled spray droplet size distribution leads to a more controlled granule size distribution (Butensky et al. 1971; Nienow 1995), as the size and distribution of the droplets determine the nuclei size distribution. Several authors (Schaafsma et al. 1998; Schæfer et al. 1977b; 1978; 1991; Waldie et al. 1987) have found a marked correlation between the drop size and nuclei size distributions in fluidised bed granulators. Waldie (1991; 1987) found a direct correlation between measured individual drop sizes and the granule sizes in a fluidised bed granulator. One drop tended to form one granule according to the relation:

$$d_g \propto d_d^n \tag{3-21}$$

For fluidized bed granulators, operation outside the drop controlled regime is unstable, leading to defluidisation, sometimes called quenching (see chapter 9). Therefore, it is not surprising to see close correlation between drop size and granule size in fluid bed granulators.

A number of studies have shown the granule size distribution is related to the relative flux of liquid in the spray zone (Akermans *et al,* 1998; Schaafsma *et al.*, 1999,2000; Tardos *et al.*, 1997). Figure 3.28 shows how the granule size distribution in a fhybrid mixer/fluidized granulator narrows as the fluidization air rate increases (increase in powder flux) and the spray surface area increases (decrease in liquid area flux). Both these effects will decrease the dimensionless spray flux, pushing the system towards the drop controlled regime. The flux number developed by Akermans *et al.* to assist in fluidized bed granulator scale up is in fact a particular case of the dimensionless spray flux defined ineqn.3-27 (Hapgood, 2000).



Agglomeration in an Agitated Fluidized Bed

	Mixer Type		
	NQ-125	NQ-230	NQ-500
Powder feed (kg)	0.36	2.23	22.9
Spray area (cm^2)	38.5	132.7	594
Spray area / Feed	107	60	26

Figure 3.28. Geometric standard deviation of granule size in an agitated fluid-bed granulator as a function of gas fluidisation velocity and binder dispersion (measured using spray surface area to mass in mixer). Figure from Mort et al. (1998). Data from Tardos et al. (1997)

Apart from Hapgood's work, the best available data to demonstrate the influence of wetting kinetics on nucleation is that of Gluba et al. (1990). Their data for the effect of fluid penetration rate and the extent of penetration on granule size distribution for drum granulation is illustrated in figure 3.29. In particular, note that increasing the penetration rate decreases the variance (spread) of the granule size distribution.

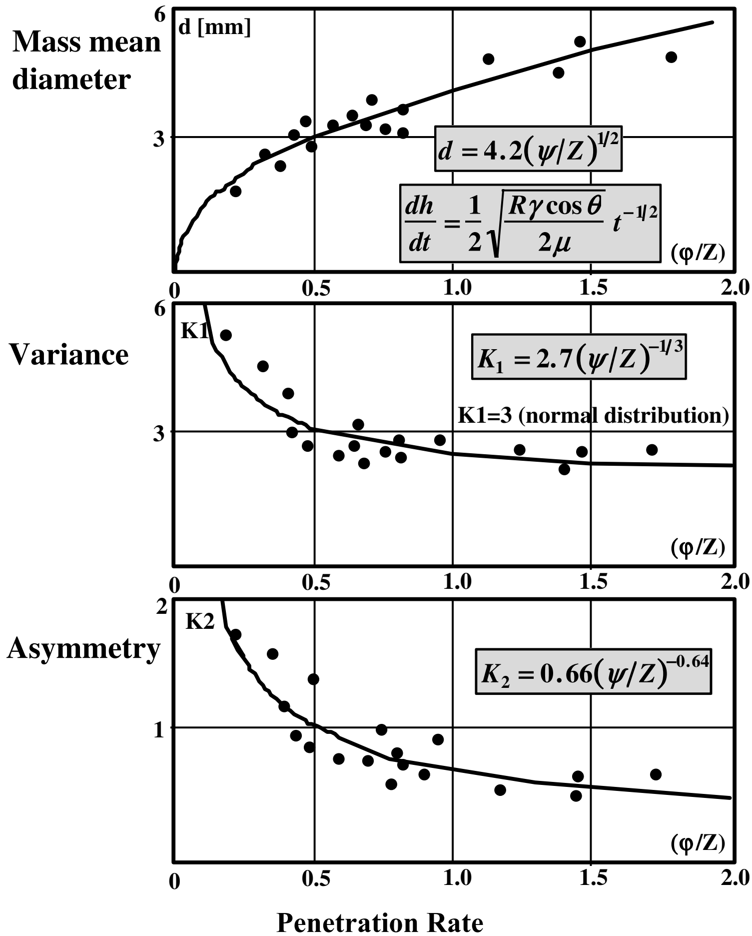


Figure 3.29. The influence of penetration rate as determined by capillary rise on granule size distribution for drum granulation of talcum, chalk, and kaolin powder mixtures. [Gluba et al., (1990).]

3.3. Control of Wetting and Nucleation for Particle Design

Table 3.3 provides a summary of the typical changes in material and operating variables which are necessary to improve wetting uniformity. Also listed are appropriate routes to achieve these changes in a given variable through changes in either the formulation or in processing. It is important to note that the wetting regime analysis above gives the ***quantitative*** trends of the effect of material and process variables on both wetting dynamics and wetting thermodynamics. This avoids the need for a "suck it and see" approach to avoiding wetting problems. The recommended approach for most situations is to ensure the system is operating in the drop control regime and choose a spray system to control the nuclei size as desired. This approach has rarely been used in practice.

3.3.1. Adhesion tension control

In general, adhesion tension $\gamma_{lv} \cos\theta$ should be maximized from the point of view of increasing the rate of drop penetration (eqn. 3-23). While other variables can be manipulated to improve wetting kinetics, only adhesion tension is available to improve wetting thermodynamics. A contact angle less than 90º is essential and a value close to 0º is very desirable.

Maximizing adhesion tension is achieved by minimizing contact angle and maximizing surface tension of the binding solution. These two aspects work against one another as surfactant is added to a binding fluid, and there is an optimum surfactant concentration which must be determined for the formulation (Ayala, 1985). In addition, surfactant type influences adsorption and desorption kinetics at the three-phase contact line. An inappropriate choice of surfactant can lead to Marangoni interfacial stresses which slow the dynamics of wetting (Pan *et al.* , 1995). Adhesion tension can also be varied by changing the granulating solvent eg. alcohol/water mixtures. Additives for other purposes also change the surface activity of the solution including 1) polymeric binders, and 2) dissolution of the powder into the liquid binder.

If the powder and potential liquid binders are sufficiently well characterised, eqns. 3-4 and 3-5can be used to tailor a liquid-powder combination for formulation. The gas absorption/desorption techniques used to achieve such data are still developmental, and the mixing models need further validation. However, the approach is potentially very powerful and likely to be used more regularly in the future.

3.3.2. Other powder and liquid properties

The main influence of other powder and liquid properties is to vary the wetting kinetics. The penetration time t_p is proportional to binder viscosity and inversely proportional to the particle size of the powder (eqns. 3-21 to 3.23). It is also very sensitive to powder bed voidage.

The prime control over the viscosity of the binding solution is through binder concentration. Therefore, liquid loading and drying conditions strongly influence binder viscosity. For processes *without* simultaneous drying (tumbling and mixer granulation), binder viscosity generally decreases with increasing temperature. For processes *with* simultaneous drying, however, the dominant observed effect is that lowering temperature lowers binder viscosity and enhances wetting due to decreased rates of drying and increased liquid loading. Decreasing binder viscosity will also decrease the drop size from a given nozzle and this also contributes strongly to reducing liquid penetration time.

Changes in particle size distribution effect the pore distribution of the powder. Large pores between particles enhance the *rate* of binder penetration. Eqns. 3.21 to 3.23 enable a quantitative evaluation of the effect of changes powder size and packing, even for relatively fine powders.

3.3.3. Spray distribution

The drop distribution and spray rate of binder fluid have a major influence on wetting. Decreasing drop size *increases* ψ_a at constant spray rate (eqn. 3-27) but *decreases* t_p. Therefore, it is important to understand the reason for wetting problems before attempting to vary drop size (see figure 3-24). If the system is in the drop controlled regime, nuclei size is directly related to drop size and this presents the possibility of very close control of final granule size distribution.

The effect of spray rate is straightforward. Decreasing spray rate will always reduce ψ_a and move the system towards the drop controlled regime. Increasing the *width* of the spray zone perpendicular to direction of powder flow will increase $\dot{A}$ and therefore also reduce ψ_a, as will distributing the same spray flow over several spaced nozzles in a large granulator. Note that spray area *per se* is not the important variable. Increasing the spray area by increasing the *length* of the spray zone is no help whatsoever. Thus, nozzles which give a fan shaped spray are often an advantage, provided they are properly aligned with respect to the powder flow. Increasing the height of the nozzle above the bed will increase $\dot{A}$.

Improved spray distribution is also aided by good housekeeping measures such as avoiding spray entrainment, dripping nozzles, or powder caking on process walls.

3.3.4. Agitation Intensity

Good bed mixing correlates closely with increasing the powder flux in the spray zone and is therefore desirable. Complete bed turn over also ensures that the spray is being distributed over the whole bed contents. Mixing patterns in granulators are very equipment specific. In general, improved mixing is achieved by increasing agitation intensity ie. mixer impeller or chopper speed, drum and disc rotation rate, or fluidization gas velocity.

Increased agitation intensity will also break up lumps and aid binder distribution if operating in the shear controlled regime (Holm et al., 1983, 1984; Mort et al., 1997). However, it is very likely that many of the agitation intensity effects attributed to lump breakage may in fact be due to better mixing moving the system from the shear controlled regime towards the drop controlled regime ie. forming less lumps in the first place. The deformation and breakage of wet granules in the granulator will be considered in later chapters.

3.3.5. Simultaneous Drying

If drying occurs simultaneously with wetting, the effect of drying can substantially modify the expected impact of a given process variable and this should not be overlooked. In addition, simultaneously drying often implies that the *dynamics* of wetting are far more important than the *extent*.

Table 3.3. Controlling Wetting In Granulation Processes

Typical changes in material or operating variables which improve wetting uniformity	**Appropriate routes to alter variable through formulation changes**	**Appropriate routes to alter variable through process changes**
Increase adhesion tension Maximize surface tension Minimize contact angle	Alter surfactant concentration or type to maximize adhesion tension and minimize Marangoni effects. Precoat powder with wettable monolayers, e.g. coatings or steam.	Control impurity levels in particle formation. Alter crystal habit in particle formation. Minimize surface roughness in milling.
Decrease binder viscosity	Lower binder concentration. Change binder. Decrease any diluents and polymers which act as thickeners.	Raise temperature for processes without simultaneous drying. Lower temperature for processes with simultaneous drying since binder concentration will decrease due to increased liquid loading. This effect generally offsets inverse relationship between viscosity and temperature.
Increase pore size to increase rate of fluid penetration	Modify particle size distribution of feed ingredients.	Alter milling, classification or formation conditions of feed if appropriate to modify particle size distribution.
Improve spray distribution	Improve atomization by lowering binder fluid viscosity.	Increase wetted area of the bed per unit mass per unit time by increasing the number of spray nozzles, lowering spray rate, increase air pressure or flow rate of two fluid nozzles.
Increase solids mixing	Improve powder flowability of feed.	Increase agitation intensity (e.g. impeller speed, fluidization gas velocity, or rotation speed.)
Minimize moisture buildup and losses	Avoid formulations which exhibit adhesive characteristics with respect to process walls.	Maintain spray nozzles to avoid caking and nozzle drip. Avoid spray entrainment in process air streams, and spraying process walls.

3.3.6. Concluding Comments

In any granulation process the first aim should be to ensure good wetting and nucleation, thus removing binder distribution problems from the picture and allow the engineer to concentrate on other issues. The characterisation procedures and regime analysis presented in this chapter provide the tools for quantitative analysis and design. Some examples of how these tools can be applied to scale up and operation in different types of granulators are given in chapters 8 to 10.

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3.5. Nomenclature

$\dot{a}$ Area flux of drops hitting powder surface (m^2s^{-1})
$\dot{A}$ Area flux of powder through the spray zone (m^2s^{-1})
d_d liquid drop size (m)
d_g nuclei granule size (m)
d_{32} specific surface mean particle size (m)
f_{agglom} fraction of overlapping drops on powder surface (-)
$f_{covered}$ fraction of powder surface covered by drops (-)
f_{single} fraction of single drops on powder surface (-)
g gravitational acceleration (N/m^2)

h penetration depth (m)
h_e equilibrium height of rise (m)
$\dot{n}_d$ number of drops hitting the powder surface per unit time (no. s^{-1})
Q flow rate of liquid into powder bed (m^3s^{-1})
r_d projected area of drop on powder surface (m^2)
R effective pore radius (m)
t time (s)
t_c powder circulation time (s)
t_p drop penetration time (s)
$\bar{v}$ liquid penetration velocity (m s^{-1})
$\dot{V}$ volumetric spray rate (m^3s^{-1})
V_d drop volume (m^3)
V_N molar volume
W_a work of adhesion (N m^{-1})
W_{cl} work of cohesion for liquid (N m^{-1})
W_{cs} work of cohesion for solid (N m^{-1})
W_i surface energy in immersion wetting (N m^{-1})
W_s surface energy in spreading wetting (N m^{-1})
W_T total work of wetting (N m^{-1})
γ_{lv} liquid-vapour interfacial energy (N m^{-1})
γ_{sl} liquid-solid interfacial energy (N m^{-1})
γ_{sv} solid-vapour interfacial energy (N m^{-1})
ΔG free energy of desorption (J mol^{-1})
ΔP pressure drop (Pa)
ε loose packed powder bed voidage (-)
ε_{ett} effective powder bed voidage (-)
ε_{tap} tapped powder bed voidage (-)
θ contact angle (°)
ψ_a dimensionless spray flux (-)
φ sphericity (-)
τ_p dimensionless penetration time (-)
μ Viscosity (Pa.s)
ρ fluid density (kg m^{-3})

Superscripts

D dispersive component of surface energy
P polar component of surface energy

CHAPTER 4

CONSOLIDATION, COALESCENCE AND GROWTH

If wetting and nucleation are effectively controlled, particles leaving the spray zone will be in one of two forms:

1. Individual particles coated with a layer of liquid binder (when the drop size is small compared to the particle size); or
2. Loose, partially saturated nuclei granules (where the drop size is large compared to the particle size).

From this point, granules may increase in size (granule growth) and density (granule consolidation). In many cases, final granule properties are determined by the extent of growth and consolidation.

Table 4.1 summarises the many different types of growth behaviour that have been observed. A corresponding large number of macroscopic mechanisms to describe these behaviours have been proposed and classified (Sastry and Feurstenau (1975), Capes and Dankwerts (1965), Kapur (1978)). However, all the behaviours can be understood by answering the following simple question. Will two granules which collide in the granulator stick together (coalesce) or rebound? Answering this question involves examining granule mechanics and understanding how energy is absorbed on impact. This examination leads us to predict a wide variety of granulation behaviour because granules are complex three phase entities consisting of a particle assembly partially saturated with a binding liquid whose properties vary as the granule consolidates with time.

In this chapter we begin by examining granule mechanics and defining macroscopic properties that characterise the behaviour of a granule on impact. We then quantify consolidation and growth phenomena in terms of controlling groups and set guidelines for particle design through changes to formulation design or process conditions.

4.1. Granule Mechanics

A granule is a complex particle assembly partially saturated with a binder liquid (see section 2.3.3). Three mechanisms contribute to the yield strength of the granule and its resistance to deformation on impact:

- Surface energy associated with pendular bridges of liquid between particles and liquid necks in capillaries at the granule surface;
- Friction between particle surfaces within the granule; and
- Viscous dissipation in the liquid phase between particles within the granule

In his pioneering work, Rumpf (1962) derived an expression for the tensile strength of a particle assembly considering surface forces only:

$$\sigma_T = s\frac{1-\varepsilon}{\varepsilon}\frac{\gamma_{lv}\cos\theta}{d} \qquad (4\text{-}1)$$

However, in reality interparticle friction and viscous dissipation are important mechanisms that cannot be neglected. This has been demonstrated experimentally (Kristesen et al, 1985) and through discrete element simulations of granule behaviour on impact (Adams et al, 1994). Figure 4.1 shows that as little as 5% of energy absorbed on collision of a deformable granule may be due to rupture of capillary bridges.

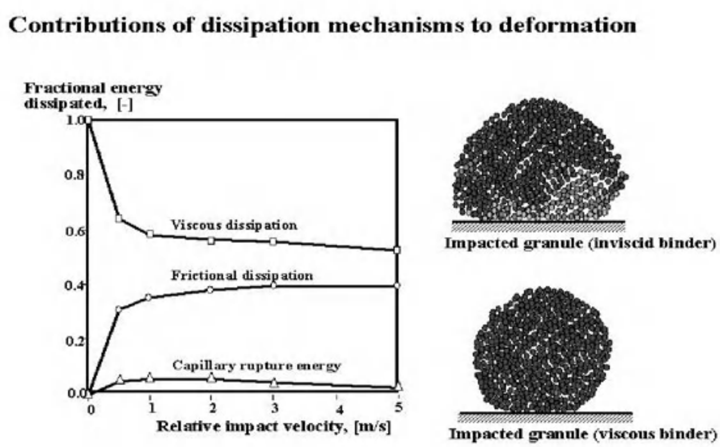
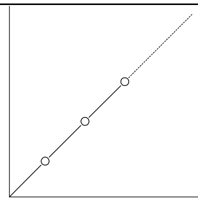
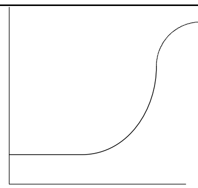
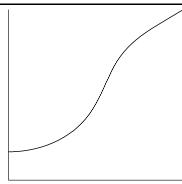
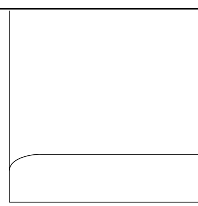
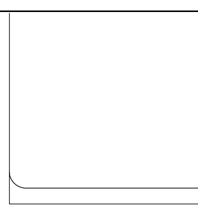
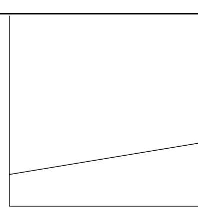


Figure 4.1. Discrete element simulations of granule deformation on impact (Adams et al., 1994).

In general, we cannot yet predict the assembly mechanical behaviour quantitatively from all the particle-particle and particle-fluid interactions in the granule. Instead, we define some macroscopic constitutive model for the granule behaviour. For example, consider the granule as an elastic-plastic solid as shown in Figure 4.2a. + When the assembly is compressed, energy is first stored by elastic deformation. Once the dynamic yield stress is reached, the assembly deforms plastically at constant stress if the assembly is perfectly plastic. When the stress is released, the elastic portion of the deformation is recovered and the elastically stored energy is released.

Table 4.1. Reported granule growth behaviour

Description of Growth Behaviour	*Granule size vs time*	*Examples Systems*	*References*
Fast growth with no limit to granule size		Coarse particles with low viscosity binders in drums; coarse particles with medium viscosity binders in mixers	Newitt and Conway-Jones (1958), Capes and Dankwerts (1965), Linkson et al. (1973), Kenningley et al. (1997), Iveson and Litster (1998a,b)
An induction time with no growth followed by sudden onset of rapid growth		Fine powders and/or high viscosity binders in drums and mixers	Linkson et al. (1973), Sastry et al. (1977), Kapur (1978), Hoornaert et al. (1994), Iveson and Litster (1998a,b)
Growth to a maximum equilibrium granule size		Various fluid bed granulations, some mixer and drum granulations	Nienow and Rowe (1985), Ennis et al. (1991), Schaefer et al. (1990).
Granules formed by nucleation that grow no further		Fluid bed granulation of fine powders, drum granulation of fine powders at low liquid contents	Waldie (1991), Schaefer et al., (1977), Iveson and Litster (1998a,b)
Weak crumb material that does not grow at all		Coarse particles with weak binders in drums and high shear mixers	Capes and Dankwerts (1965), Kenningly et al., (1997)
Slow granule growth by layering of melt, solution, slurry or fine powder onto seed particles		Fluid bed urea granulation, fluidised drum granulation, spouted bed coating and powder layering	Litster and Sarwono (1996), Liu and Litster (1993)

The key granule properties that determine its behaviour on impact is its elastic modulus E and particularly the dynamic plastic yield stress Y. Techniques to measure these properties are described below.

There are several likely complications to this very simple model. Firstly, the dynamic yield stress is probably strain rate dependent (Figure 4.2b). Certainly, any contribution of viscous dissipation in the binder liquid must be proportional to strain rate. Ideally, the dynamic yield stress should be measured as at varying strain rates, or at least at strain rates typical of the process of interest.

Secondly, the assembly may strain harden or strain weaken (Figure 4.2c). Strain hardening is likely to occur where there is compaction of the assembly over time.

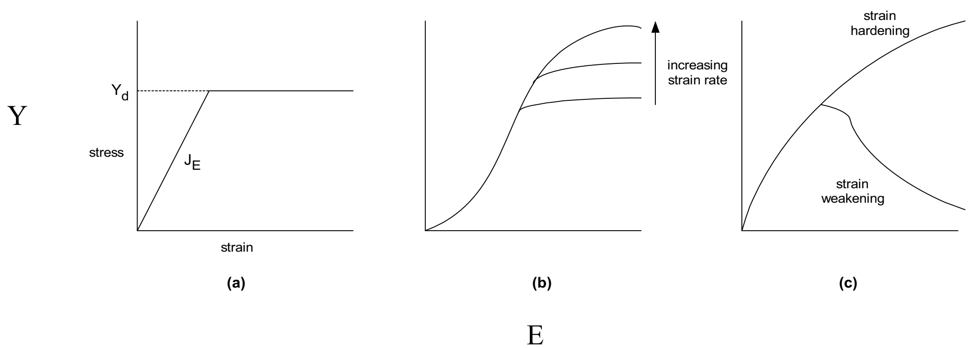


Figure 4.2. Possible granule stress-strain behaviours (a) Simple elastic-plastic solid, (b) with strain rate dependent yield stress, (c) with strain hardening or strain weakening

4.1.1. Measuring the Dynamic Yield Stress of a Particle Assembly

The yield stress of a powder compact can be measured in standard uniaxial or triaxial testing machines. Tests of this kind have been performed on partially saturated particle assemblies by Schubert et al. (1975) and Holm et al. (1985). Figure 4.3 shows measured stress-strain relationship for dicalcium phosphate granules in uniaxial compression across the diameter of a cylindrical compact. As liquid saturation is increased the yield stress decreases and the compact supports an increasing amount of plastic deformation before final failure.

The problem with these studies is that very low strain rates are used and the tests are essential static. The *dynamic* yield stress of a granule can be measured at strain rates similar to impacts in a wet granulation process in a high-speed hydraulic load frame capable of reaching speeds of up to 15 cm/s (strain rate 10 s^{-1}) (Iveson *et al.*, 2002). A wet pellet of known porosity and liquid content is formed in a hand press and then compressed in the load frame. Figure 4.4 shows an example of the stress-strain response from a series of tests at different strain rates. In fact, the wet pellet rheology is much more complicated than a simple elastic-plastic solid and clearly is strain rate dependent. In all cases, the pellet stress goes through a maximum stress as strain increases. We will take this peak flow stress σ_D as a measure of the the dynamic yield stress of the powder. ie. $Y \approx \sigma_D$.

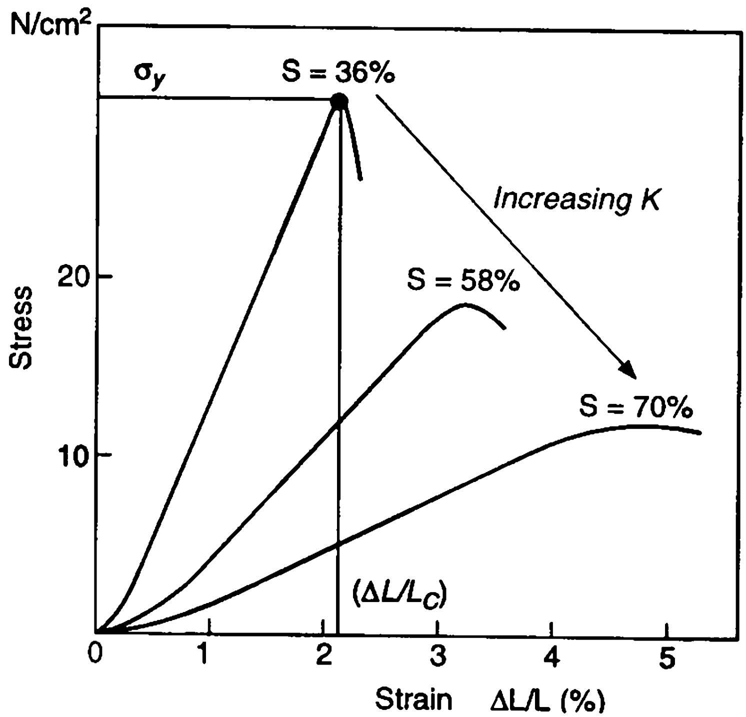


Figure 4.3. The influence of moisture as a percentage of sample saturation S on granule deformability. Dicalcium phosphate with a 15 wt % binding solution of PVP/PVA Kollidon® VA64. (Holm et al., 1985.)

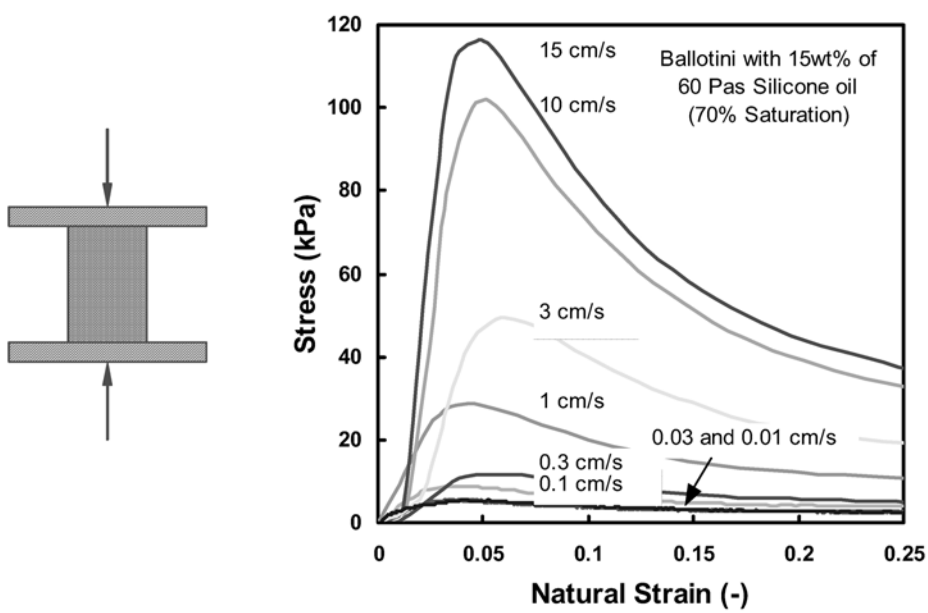


Figure 4.4. Typical data for the fast compression of an unconfined wet pellet in an load frame with cross head speeds ranging from 0.1 to 15 cm/s.

Figure 4.5 shows the peak flow stress versus strain rate for glass ballotini with water, glycerol and a range of silicone oil binders. There is an initial region where flow stress is independent of strain rate and then a second region where the flow stress increases with increasing strain rate. This suggests that the capillary and inter-particle friction forces are dominant at low strain rates and that viscous forces become significant at high strain rates.

This hypothesis is confirmed by the fact that these results, which cover a range of viscosities from 0.001 to 60 Pa·s and surface tensions from 0.025 to 0.072 N/m, all collapse onto one curve when plotted in terms of two dimensionless groups (see figure 4.6):

$$\frac{\sigma_D d_p}{\gamma} = 5.0 + 320\left(\frac{\mu \dot{\varepsilon} d_p}{\gamma}\right)^{0.64} \tag{4-2}$$

where d_p is the powder surface mean size (m), γ is liquid surface tension (N/m), μ is liquid viscosity (Pa·s) and $\dot{\varepsilon}$ is the strain rate (s^{-1}). The group on the left-hand side is the dimensionless strength, *Str**, which is the ratio of peak flow stress to capillary forces. The group on the right-hand side is the capillary number, *Ca*, which is the ratio of viscous to capillary forces.

This trend is not unique to spherical glass ballotini. Results with different sized powders show similar strain-rate dependent behaviour. Non-spherical powders have higher low strain rate strength (presumably due to increased particle-particle contact) but also show identical dependence on capcilliary number. For $Ca<10^{-4}$, $Str^* \neq f(Ca)$ while for $Ca>10^{-4}$, $Str^* \;\alpha\; Ca^{0.64}$.

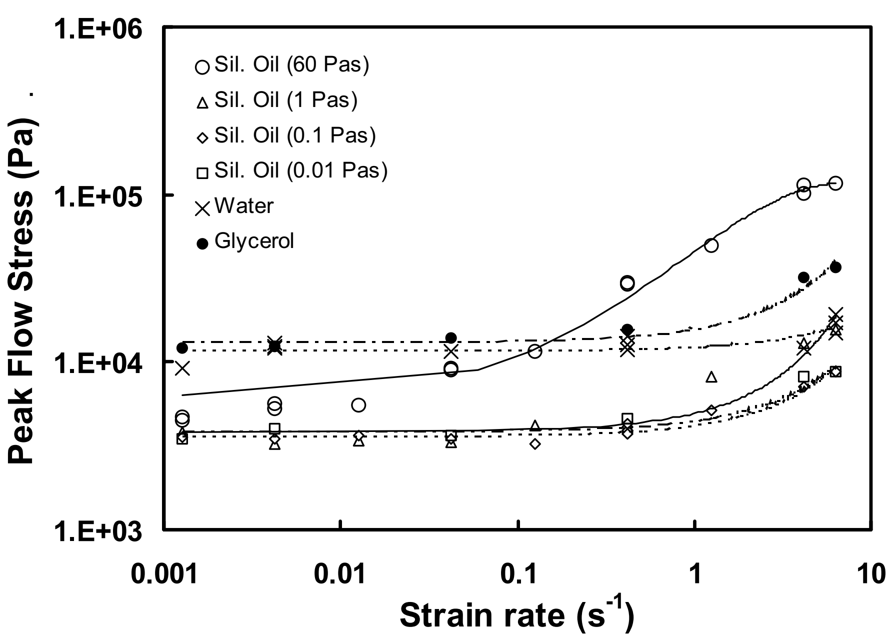


Figure 4.5. Peak flow stress versus strain rate for pellets made with glass ballotini (x_{32} = 34.5 μm) and water, glycerol or silicone oil binders. Pellets 25 mm long and 20 mm diameter (Iveson et al, 2002).

A very significant feature of figure 4.5 is not only that the peak flow stress is a function of the strain-rate, but that the ***ranking*** of flow stress for different systems is strain-rate sensitive. At low strain rates the water and glycerol bound granules have similar peak flow stress and both are stronger than the silicone-oil bound granules. However, at high strain rates the glycerol-bound granules and high-viscosity silicone-oil bound granules become stronger than the water-bound ones.

This result is of profound importance to the understanding of granulation phenomena, since it means that the strength of systems measured at low strain rates may not, ***even qualitatively***, give any indication of how different materials will behave at the much higher strain rates encountered in most granulators. This means that traditional quasi-static methods of measuring granule strength are misleading when used to model granule growth behaviour. This effect is unlikely to be significant in applications involving low-viscosity binders and gentle agitation, such as the drum granulators used in the minerals and fertiliser industries, where viscous effects are minor. However, it will become significant for high binder viscosities at high strain rates, such as in the detergents and pharmaceutical industries, and these are precisely the same industries that are most in need of *a priori* predictive methods for determining granulation behaviour, since they have such frequently changing formulations.

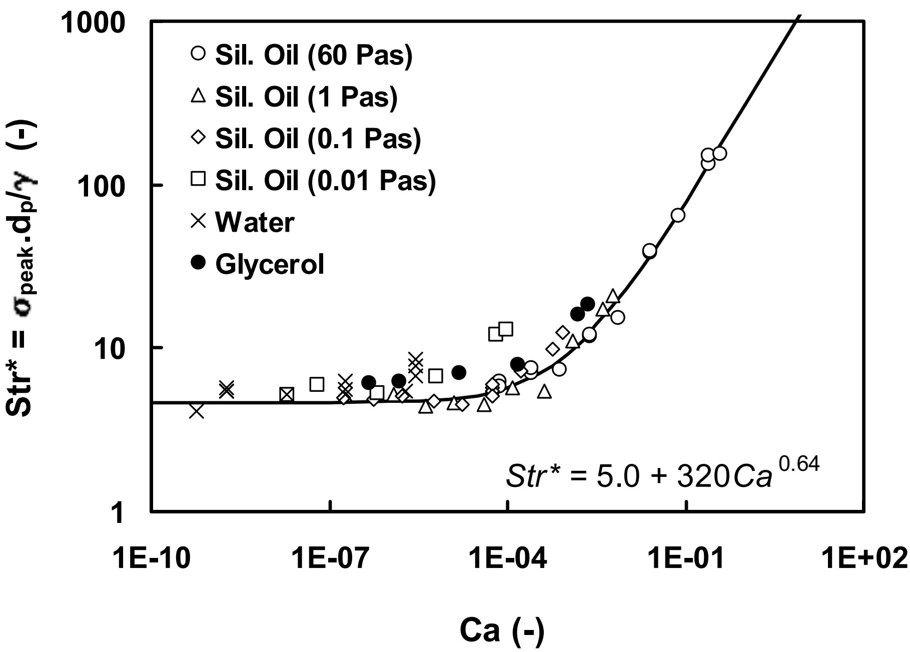


Figure 4.6. Dimensionless strength versus capillary number for pellets made with glass ballotini (x_{32} = 34.5 μm) and water, glycerol or silicone oil binders. Pellets 25 mm long and 20 mm diameter (Iveson et al, 2002).

Figure 4.6 and eqn.4.2 can be used to predict the effect of formulation changes on granule strength. For example, decreasing the particle size making up the pellet or granule s increase its strength. Binder viscosity will have a strong effect, but only when $Ca<10^{-4}$. Thus, a limited number of dynamic yield stress measurements provide a powerful tool in formulation design.

Not only the peak flow stress, but also the mode of failure vary with strain rate. Figure 4.7 shows schematically the crack behaviour observed in pellets as a function of capillary number. At low *Ca*, pellets fail by macroscopic crack propogation – brittle failure. At high *Ca*, pellets fail by plastic flow. This interesting and unusual result has important consequences for wet granule breakage.

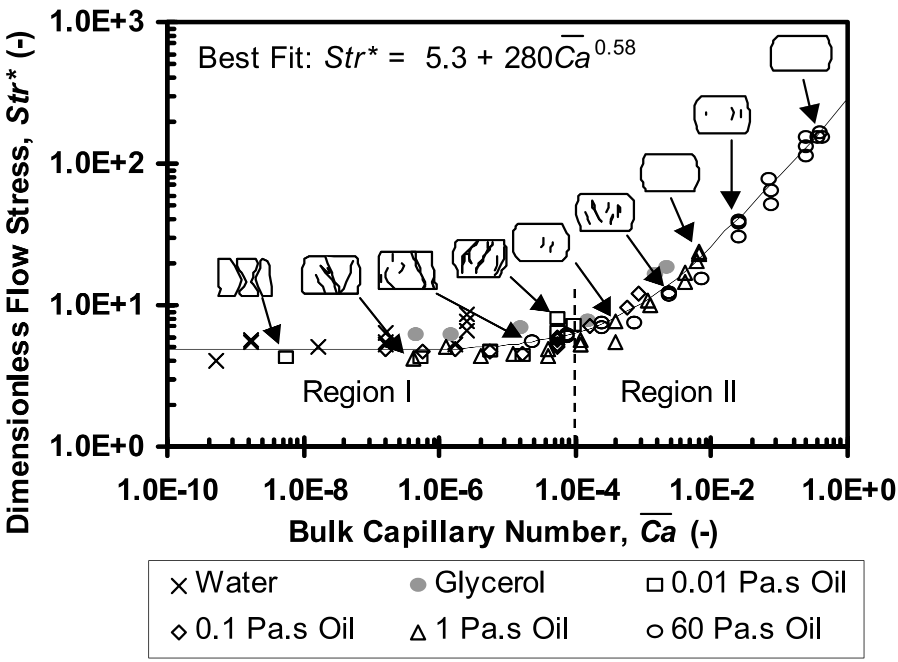


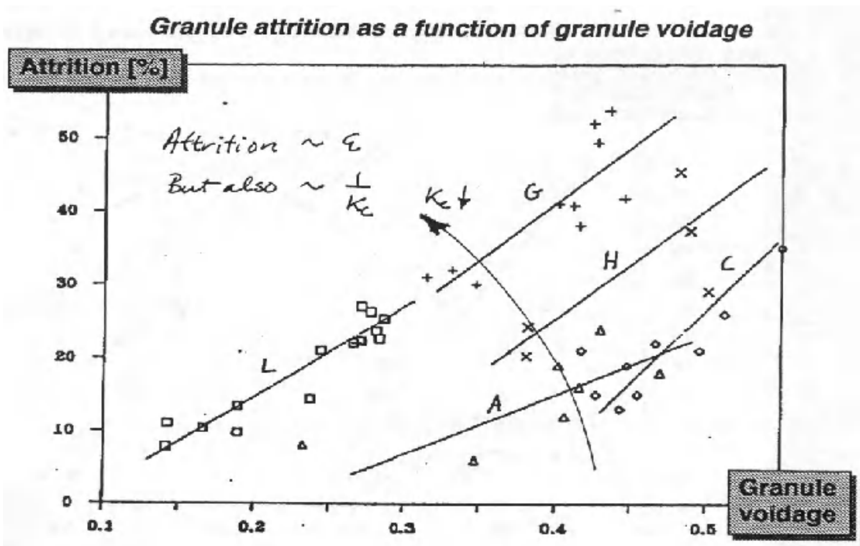
Figure 4.7. Granule strength correlation showing schematically how mode of failure changes with Ca (Iveson et el., 2002)

4.2. Granule Consolidation

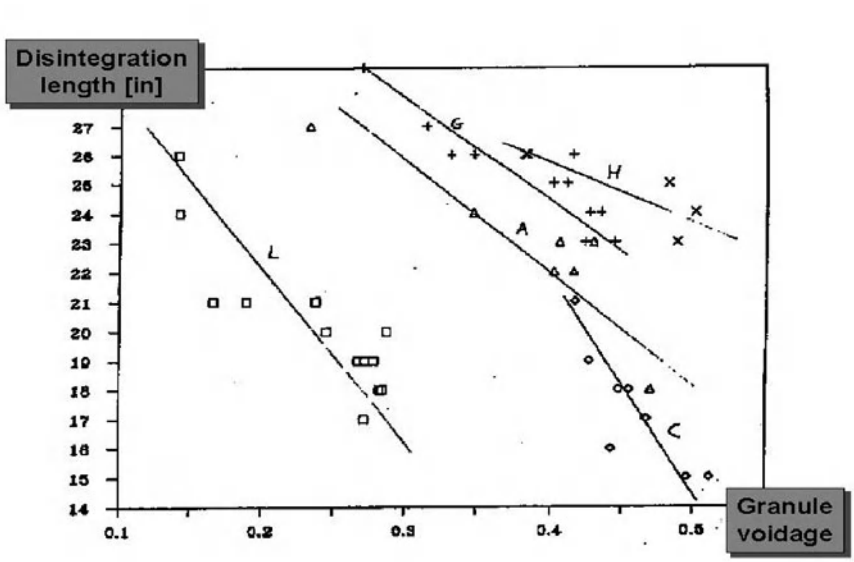
As granules are agitated in a granulator they experience many collisions with other granules and with walls of the equipment. These collisions cause *consolidation* of the granules (also called densification or compaction) which reduces their porosity and size, squeezes out entrapped air and possibly even squeezes liquid binder to the granule surface.

The consolidation process often determines the granule porosity. Granule porosity controls granule strength, an important product property. Porous granules are weak, friable and disperse quickly in solution, whereas low porosity granules are strong and slow to disperse. Figure 4.8 shows an example of the importance of porosity in determining granule properties. For a range of formulations, both attrition resistance and dispersibility are strongly related to granule porosity.

Granule porosity also controls granule deformability and liquid saturation which strongly influence granule growth mechanisms. Hence, consolidation affects both the granule product properties and the growth mechanisms that form them.



(a)



(b)

Figure 4.8. The effect of granule porosity (voidage) on product attributes for five herbicide formulations (a) attrition resistance, and (b) dispersibility in water.

. A Mechanistic Understanding of Granulation

Iveson and Litster (1998b) give a good overview of the consolidation process. Consolidation results from many granule impacts with other granules, the walls of the granulator and granulator internals eg. the impeller in a mixer granulator. Granules may both dilate and compact during these impacts (Kapur, 1978). Three mechanisms control this behaviour: inter-particle friction, viscous dissipation in the liquid and capillary (surface energy) considerations. Inter-particle friction and viscous forces are dissipative i.e. they resist both consolidation and dilation of the particle matrix. In contrast, capillary forces are conservative i.e. they always act to pull particles together and thus aid consolidation and resist dilation.

As granules consolidate, the inter-particle friction and viscous resistance increases (due to a greater number of inter-particle contacts and smaller gap distances). Capillary forces also increase until the granule becomes saturated. Hence, granules consolidate until they become strong enough to resist any further consolidation. At this minimum porosity ε_{min}, there is a dynamic equilibrium between their tendency to dilate and consolidate during impact.

A number of workers have observed the process of consolidation to minimum porosity in drum and mixer granulation (Newitt and Conway-Jones, 1958; Sastry *et al.*, 1977; Riatala *et al.*, 1986; Iveson and Litster, 1998b). We can represent this process by a simple empirical exponential decay model (Iveson and Litster, 1998b):

$$\frac{\varepsilon - \varepsilon_{\min}}{\varepsilon_0 - \varepsilon_{\min}} = \exp(-kN) \tag{4-3}$$

Figure 4.9 shows examples of this model fitted to laboratory drum granulation data. Although this model is purely empirical, we can relate the model parameters ε_{min} and k to formulation properties and process conditions. We will use this model to quantify and discuss consolidation data.

Consolidation is the sum of many granule impacts. Therefore, the coalescence rate constant k will be proportional to frequency of granule impacts β and the amount of deformation that occurs at each impact. The deformation of a granule during an impact in the granulator will be a function of the granule yield stress compared to the kinetic energy absorbed plastically during the collision. This ratio is called the Stokes deformation number St_{def} (Tardos *et al.*, 1997; Iveson and Litster, 1998c):

$$St_{def} = \frac{\rho_g U_c^2}{2Y} \tag{4-4}$$

where U_c is a representative collision velocity for the granulator. Thus, we expect that granule consolidation rate will vary with formulation and process parameters as:

$$k = \beta_c f(St_{def}) \tag{4-5}$$

Figure 4.10 shows the relationship between k and Y for drum granulation of glass powders of varying size with liquid binders of varying viscosity and surface tension. k decreases exponentially with Y suggesting the following form for equation. 4-5:

$$k = \beta_c \exp(a.St_{def}) \tag{4-5a}$$

Figure 4.10 is significant because it shows that the effect of a wide range of formulation properties on consolidation rate can be interpreted through a single assembly property, the dynamic yield stress and a single dimensionless group, the Stokes deformation number. We will return to this approach later when studying coalescence of deformable granules.

It is not so easy to predict ε_{min} because the minimum porosity is a balance between dilation and compaction during impact. Interparticle friction and viscous friction are dissipative mechanisms which resist both compaction and dilation. Capillary forces are conservative and always act to resist dilation.

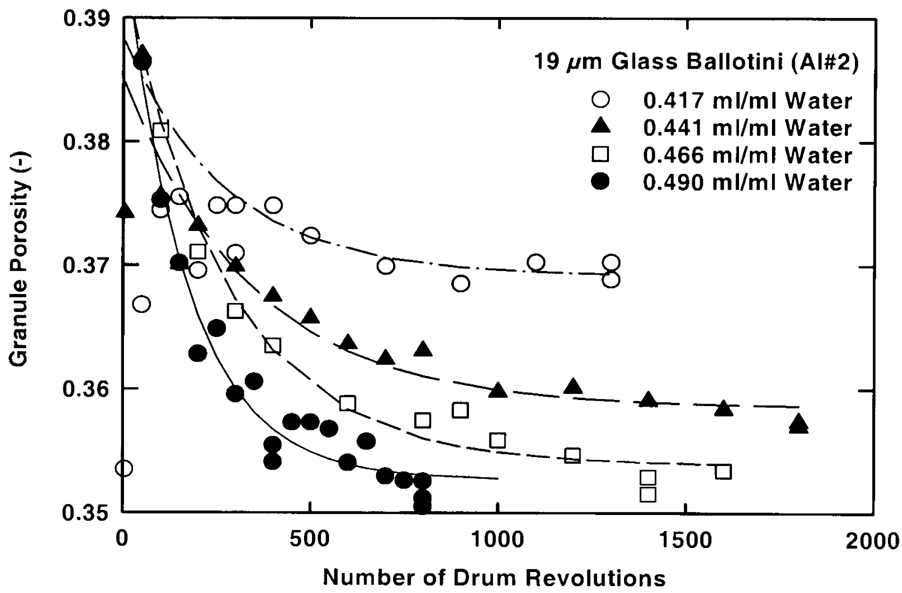


Figure 4.9. Granule consolidation to minimum porosity during laboratory drum granulation (Iveson et al., 1996)

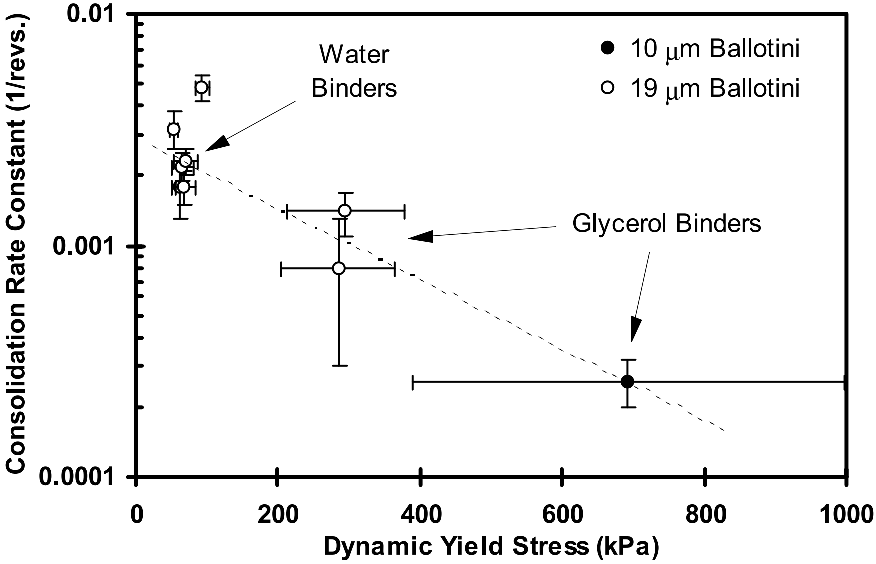


Figure 4.10. The relationship between dynamic yield stress and consolidation rate constant (drum granulation in a 0.3m diameter drum at 30 rpm) (Iveson and Litster, 1998b)

4.2.2. Effect of Formulation Properties and Process Conditions on Consolidation

There is limited data available on granule consolidation when compared to granule growth. Iveson and coworkers (Iveson et al., 1996; Iveson and Litster, 1998b) give the only comprehensive study of the effect of formulation properties on consolidation in tumbling granulators. There is no comprehensive study of the effect of process scale and intensity.

The effect of formulation properties on k mirrors their effect on the granule dynamic yield stress Y (see section 4.1.2). Strong granules consolidate slowly. Thus, decreasing particle size strongly reduces the consolidation rate with k roughly proportional to the specific surface mean particle size (see figure 4.11a). Similarly, k decreases with increasing binder viscosity and increasing binder surface tension (figure 4.11b,c).

The effect of liquid binder content on k is harder to interpret, reflecting competing effects of interparticle friction and viscous dissipation on dynamic yield stress. Mostly, k increases by only a small amount as binder content increases.

Granules made from fine powders have a higher equilibrium porosity ε_{min} than those made from coarse powders. The effect is greatest when the surface mean particle size is below 20μm (Capes, 1980; Iveson and Litster, 1998b). Particle size distribution will also effect ε_{min} due to the relationship between size distribution and packing density. The influence will be most marked for coarse powders which consolidate more easily.

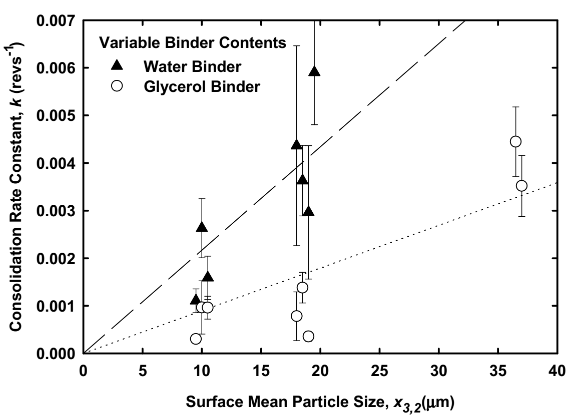
The effect of binder content is complex because it has opposite effects on inter-particle friction and viscous forces. Increasing binder content reduces inter-particle friction due to lubrication but increases viscous forces since there is more binder to be squeezed between particles as they rearrange. Consider Figure 4.12. When inter-particle friction dominates (e.g. low viscosity binders like water) increasing binder content increases ε_{min}. However, when viscous forces dominate (e.g. high-viscosity binders) increasing the amount of binder decreases ε_{min} . There is an intermediate binder viscosity at which the decrease in frictional resistance is balanced by the increase in viscous resistance. At this stage, ε_{min} is insensitive to binder content. *Thus, it is very difficult to predict the effect of changing binder content, even qualitatively!*

Increasing binder viscosity always decreases the rate of consolidation due to the increase in viscous forces. However, at low binder contents, granules with high viscosity binders consolidate to a lower ε_{min} (see figure 4.12a). This may also be due to the equilibrium between dilation and consolidation. The glycerol-bound granules are more able to resist dilation. Hence, although they consolidate more slowly, they can consolidate to a greater extent. Ritala *et al.* (1986) found similar results in high shear mixer granulation.

Lowering binder surface tension lowers a granule's resistance to dilation which shifts the dynamic equilibrium between consolidation and dilation towards a more porous state. Hence decreasing surface tension increases ε_{min} (see figure 4.12b). Thus granules with low surface tension binders consolidate more quickly, *but do not consolidate as far*.

Equation 4-5a predicts that increased process intensity (increased collision velocity) will increase the consolidation rate through its effect on the Stokes deformation number. There is limited good quality data to test the equation. However, in general, k increases and ε_{min} decreases with increasing intensity eg. high shear mixers compared to tumbling granulators.

The effect of process scale depends on the scale up rule for the granulator. In pan and drum granulators scaled using a Froude number criteria, collision velocities and consolidation rates increase with scale. For fluid beds, granule porosity increases with increasing bed depth. Many mixer granulator designs are not kept geometrically similar during scale up, making prediction difficult. In some cases, granule porosity increases on scale up. These scaling effects are discussed in more detail in the equipment chapters (chapter 7 to 11).



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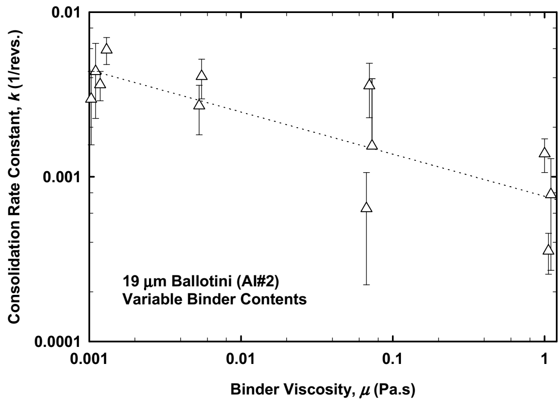

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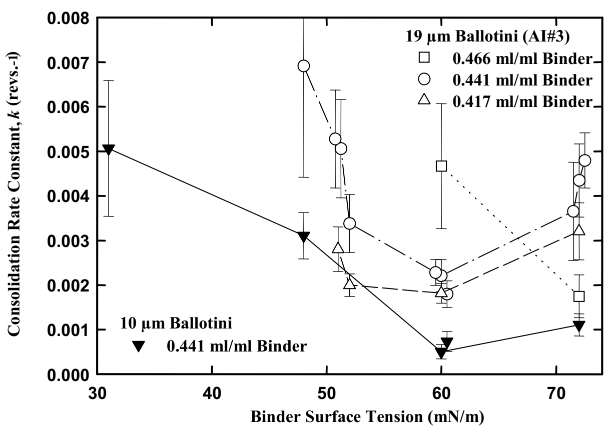


(c)

Figure 4.11. Effect of formulation properties on consolidation rate constant during laboratory drum granulation (a) particle size (b) binder viscosity (c) binder surface tension) (Iveson & Litster, 1998b).

4.2.3. Granule Consolidation with Simultaneous Drying

So far, we have implicitly assumed that the time for consolidation is the same as the time for granulation (batch time or residence time). Where granulation occurs with simultaneous drying eg. fluid beds and spouted beds, the time for consolidation is the *drying time*. This leads to a number of implications:

1. The granules produced have high porosities because the drying times are of the order of seconds compared to batch times of the order minutes.

2. Granules never reach ε_{min}. The granule porosity is dominated by kinetic effects ie. the consolidation rate constant k and time for consolidation (drying time).

3. The drying time is controlled largely by the liquid spray rate, the spray droplet size and the bed temperature. In general, any change to process conditions that decreases the drying time will increase the granule porosity.

The factors which effect k are the same, whether or not drying is occurring eg. using a viscous binder will lead to a more porous granule. However, it is important to note the formulation properties, especially binder viscosity will change as the granule dries and some time averaged properties should be used if possible (Ennis et al., 1991).

4.2.4. Theoretical Models for Consolidation

There are only two theoretical treatments of granule consolidation in the literature. Ouchiyama and Tanaka (1980) assumed granules were held together only by the capillary pressure of the binder. This pressure generates a normal force causing friction at inter-particle contacts. Considering how the coordination number of particles in the granular assembly increased when forces were applied and ignoring any dilation of the assembly, they derived an equation for the granule consolidation rate.

Their model predicts the dimensionless compaction rate, K_ε, is proportional to particle size and inversely proportional to liquid surface tension. Compaction rate is also proportional to the energy of granule collisions. The model predicts that the minimum porosity is related to the compaction rate by:

$$\frac{\varepsilon_{min}}{(1-\varepsilon_{min})^3} = \frac{1}{K_\varepsilon} \tag{4-6}$$

The model is consistent with the observed effect of binder surface tension and particle size on consolidation rate. However, the model predicts ε_{min} will decrease with decreasing surface tension, which is opposite to the observed effect. This discrepancy is caused by neglecting the effect of dilation during granule impacts.

The second theoretical treatment of granule consolidation was by Ennis *et al.* (1991) who considered only the effect of binder viscosity. By considering a one dimensional model of a granule as a string of particles joined by viscous liquid

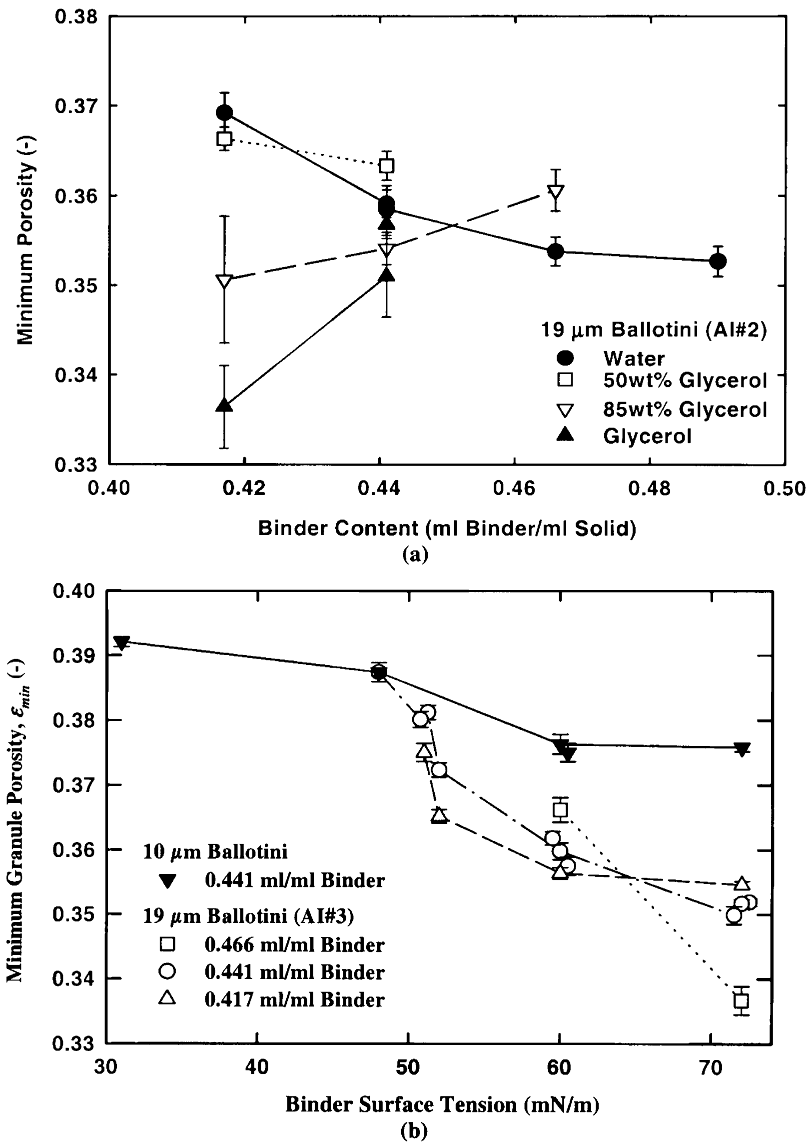


Figure 4.12. Effect of process parameters parameters on ε_{min} : (a) effect of binder content and viscosity; (b) effect of binder surface tension (Iveson and Litster, 1998b)

bridges, they found that the amount of consolidation per collision increased with increasing viscous Stokes number, St_v:

$$\frac{\Delta x}{h} = 1 - \exp(-St_v) \tag{4-7}$$

where

$$St_v = \frac{4\rho_g U_c d_g}{9\mu} \tag{4-7a}$$

This model correctly predicts the effect of binder viscosity to reduce the consolidation rate.

Both these models begin with the interaction between two particles in the assembly and try to derive the consolidation rate for the whole granule. Neither is comprehensive, as they each consider only one contribution to granule strength and deformation resistance. Their value is in understanding better the mechanisms by which consolidation occurs. We will return to Ennis's Stokes number model when discussing granule coalescence criteria.

4.3. Granule Coalescence Criteria for Near Elastic Particles

Having considered granule mechanics and consolidation, we move to study the complex area of granule growth. Different granulation systems show markedly different patterns of granule growth (see table 4-1). These various observations can all be explained by starting with the simple question. *When two granules collide in a granulator, will they stick (coalesce) or rebound?*

Let's begin by considering a simple model of two colliding granules in a granulator as a solid particle surrounded by a layer of binder liquid (figure 4.14). The granules have a diameter *d,* coefficient of restitution e_r and surface roughness (asperities) of height h_a. The binder liquid layer has a thickness h_b. The two granules approach each other with a relative collision velocity U_c.

When these granules approach to a distance $2h_b$ the liquid layers interact. For the granules to approach further, liquid must be squeezed out from between the granules. As this happens, some of the initial kinetic energy in the collision is dissipated through viscous friction in the liquid layer. The granules approach to a minimum distance $2h_a$. They then begin to rebound with more kinetic energy lost through viscous dissipation in liquid. Once the granules are $2h_b$ apart, the liquid layers will separate and rebound is complete.

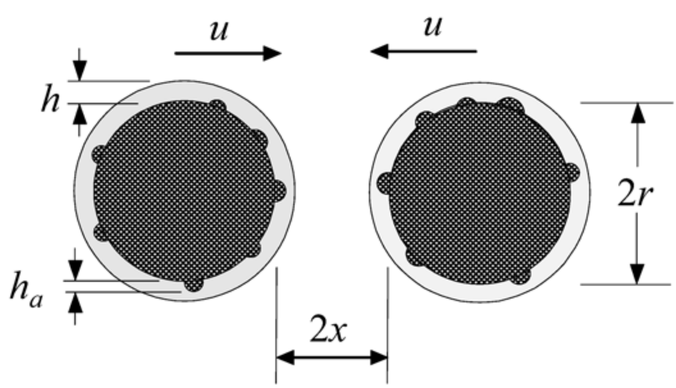


Figure 4.14. Two granules colliding – the basis for the coalescence/rebound criteria (Ennis et al., 1991)

The two granules will coalesce if their relative velocity becomes zero before this process is finished ie. granules will coalesce if the initial kinetic energy is too small to overcome the viscous lubrication resistance in the liquid layer. This is very similar to the lubrication problem in fluid mechanics. Ennis *et al.* (1991) solved this problem and showed that coalescence will occur if:

$$St_v < St^* \tag{4-8}$$

where the viscous Stokes Number St_v is given by eqn.4.7a and the critical Stokes Number St^* is:

$$St^* = \left(1 + \frac{1}{e_r}\right)\ln\left(\frac{h_b}{h_a}\right) \tag{4-9}$$

St_v is the viscous Stokes number, an important dimensionless group which represents the ratio of the initial collision kinetic energy to the energy dissipated by viscous lubrication forces. It is particularly sensitive to the liquid binder viscosity (a formulation property) and the characteristic collision velocity (a process parameter). The viscous Stokes number can vary over several orders of magnitude for different practical granulation systems.

4.3.1. Growth Regimes for near Elastic Granules

In a granulator, many collisions occur between granules. These collisions will have a range of collision velocities (and Stokes numbers) depending on their position in the granulator. Thus, we can define three types of growth that may occur based on the range of viscous Stokes numbers compared to the critical Stokes number (figure 4.15).

1. *Non-inertial growth* [$St_{v,\max} < St^*$]: The viscous Stokes number for all collisions in the granulator are less than the critical Stokes number. All collisions lead to sticking and growth by coalescence. In this regime, changes to process parameters will have little or no effect on the probability of coalescence.
2. *Inertial growth* [$St_{v,av} \approx St^*$]: Some collisions cause coalescence while others lead to rebound. There will be steady granule growth by coalescence. The extent and rate of growth will be sensitive to process parameters which will determine the proportion of collisions that lead to coalescence. Varying process parameters and formulation properties can push the system into either the non-inertial or coating regimes.
3. *Coating regime* [$St_{v,\min} > St^*$]: The kinetic energy in most or all collisions exceeds viscous dissipation in the liquid layer. There is no coalescence. Granule growth will only occur by the successive layering of new material in the liquid phase (melt, solution or slurry) onto the granule.

Changing formulation properties or process parameters will have a major effect on granule size distribution in the inertial regime because small changes may substantially increase or decrease the proportion of successful coalescence events. In contrast, these variables will have little effect on granule size in the non-inertial or coating regime unless the changes are large enough to shift the system into the inertial regime. It is important to establish the current operating regime, *before* deciding strategies for changing granule properties.

Another corollary of the growth regimes is that a batch granule may proceed through more than one regime as the granules grow. A granulation may begin in the non-inertial regime with small particles. As the granules grow, St_v will increase. Given enough time, the granules will eventually reach a maximum critical size where the system enters the coating regime and granule growth will stop. This maximum size will be influenced by the liquid viscosity and the collision velocity distribution in the granulator.

4.3.2. Estimating the Characteristic Collision Velocity

The characteristic collision velocity U_c is an important parameter in determining the viscous Stokes number. In any granulator there is a range of collision velocities. The collision velocity is a *relative* velocity ie. the velocity at which two granules move past each other in the granulator. While the maximum collision velocity will be of order of the granular flow velocity relative to the equipment walls or internals, the average collision velocity will relate to the granule flow gradient and may be much lower. Full velocity distributions are currently impossible to predict for real granule systems. Instead, we define some measures of the typical and maximum collision velocities in table 4-2 for common types of granulation equipment. These represent order of magnitude estimates only, but are useful because they relate the collision velocity to standard process parameters.

For fluidised beds, the motion of particles is dominated by the presence of bubbles in the fluidised bed and the collision velocity is derived by considering the relative velocity between two particles in the emulsion phase as a bubble passes near by (Ennis *et al.*, 1991). Thus, bubble size and velocity are the key process parameters to determine collision velocity. For spouted beds, the collision velocity is related to the jet velocity in the spout while for tumbling and mixer granulators, the rotation speed of the equipment or the impeller are the key parameters. Particle motion in the various granulator designs is discussed in more detail in the equipment specific chapters.

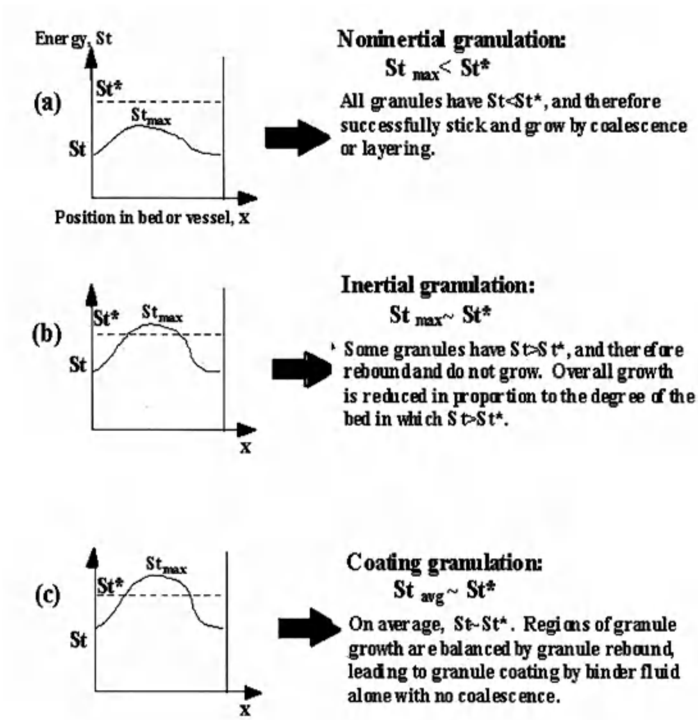


Figure 4.15. Growth regimes based on the Stokes number criterion

Table 4.2. Estimates of U_c for different granulation processes

Type of Granulator	Average U_c	Maximum U_c
Fluidised Beds	$\frac{6U_b d_g}{d_b}$	$\frac{6U_b d_g}{d_b \delta^2}$
Tumbling granulators	ωd	$\omega D_{drum} / 2$
Mixer granulators	$\omega_i d, \quad \omega_c d$	$\omega_i D_i / 2, \quad \omega_c D_c / 2$

4.3.3. Examples of Application of the Stokes Regime Analysis

Two examples of application of the Stokes Regime Analysis are given below. Further applications are discussed in Ennis et al. (1991).

Batch Fluidised Bed Granulation

Figure 4.16 shows growth of glass ballotini granules in a fluidised bed granulator using two different liquid binders at the same spray rate – Sodium carbomethyl cellulose (CMC-M) and polyvinylpyrolidone (PVP). These two binders have average viscosities during drying of 1.7P and 0.36P respectively. Figure 4.16 shows that for the first 5 hours of granulation, there is no difference in growth behaviour between the two systems, despite the difference in binder viscosity. This represents growth within the non-inertial regime (all collisions between wetted granules are successful). When the granule size reaches approximately 800μm, the PVP bound granule growth begins to slow indicating a transition to the inertial growth regime (only some collision are successful). Finally, the PVP granules level off at a maximum size of approximately 900μm showing transition to the coating regime where no granule collisions are successful. In contrast, the more viscous CMC-M granules grow steadily throughout the 8 hour experiment ie. they remain in the non-inertial regime for the whole experiment.

This example gives experimental support for the Stokes regime approach and also emphasises the role of formulation properties and operating parameters is very dependent on the growth regime. Growth was independent of binder viscosity while both systems remained in the non-inertial regime. However, the binder viscosity had a major effect on granule growth once the PVP granules entered the inertial and coating regimes, setting the maximum granule size for the PVP granules.

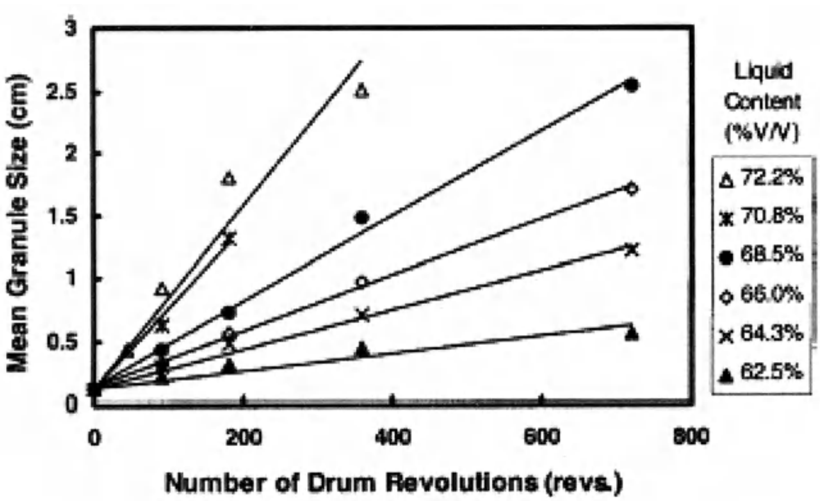


Figure 4.16. Growth of glass ballotini granules in a fluidised bed with binders of different viscosity (Ennis et al., 1991)

Drum granulation of fertilisers

The final example is the drum granulation of fertilisers. In the granulation of ammonium salt fertilisers, a newly formed salt solution is sprayed onto dried, recycled granules up to several millimetres in size. The salt solution coats the recycled granules which then grow by coalescence. Initial rapid growth occurs to a maximum granule size (non-inertial regime) followed in some cases by slower growth in the inertial regime.

Adetayo et al. (1995) found the size of granules formed after non-inertial growth at the same liquid content varied for three different fertilisers – di-ammonium phosphate (DAP), mono-ammonium phosphate (MAP) and ammonium sulphate (AS). They fitted a population balance model (chapter 7) to granule size distribution data and extracted the "extent of granulation" kt_{max} in the inertial regime. The model parameter kt_{max} is directly related to the mean granule size at the end of non-inertial growth:

$$d_{\max} = d_0 \exp(\frac{kt_{\max}}{6}) \tag{4-10}$$

Figure 4.17a shows the experimental data. DAP granules exhibited greater extent of granulation than either MAP or AS granules. These differences can be accounted for by the difference in properties of the different salt solutions. At the end of the non-inertial growth:

$$St_v = \frac{8\rho_g U_c d_{\max}}{9\mu} = St^* \tag{4-11}$$

Combining eqns. 4-10 and 4-11 gives:

$$kt_{\max} = 6\ln\left(\frac{St^* 9\mu}{8\rho_g U_c d_0}\right) \propto \ln\left(\frac{\mu}{\rho_g U_c d_0}\right) \tag{4-12}$$

When the extent of granulation is normalised for the effect variations in granule density and salt solution viscosity, results for the three fertilisers collapse onto a single line. Thus, the variation in behaviour between fertilisers is predicted from the simple Stokes analysis.

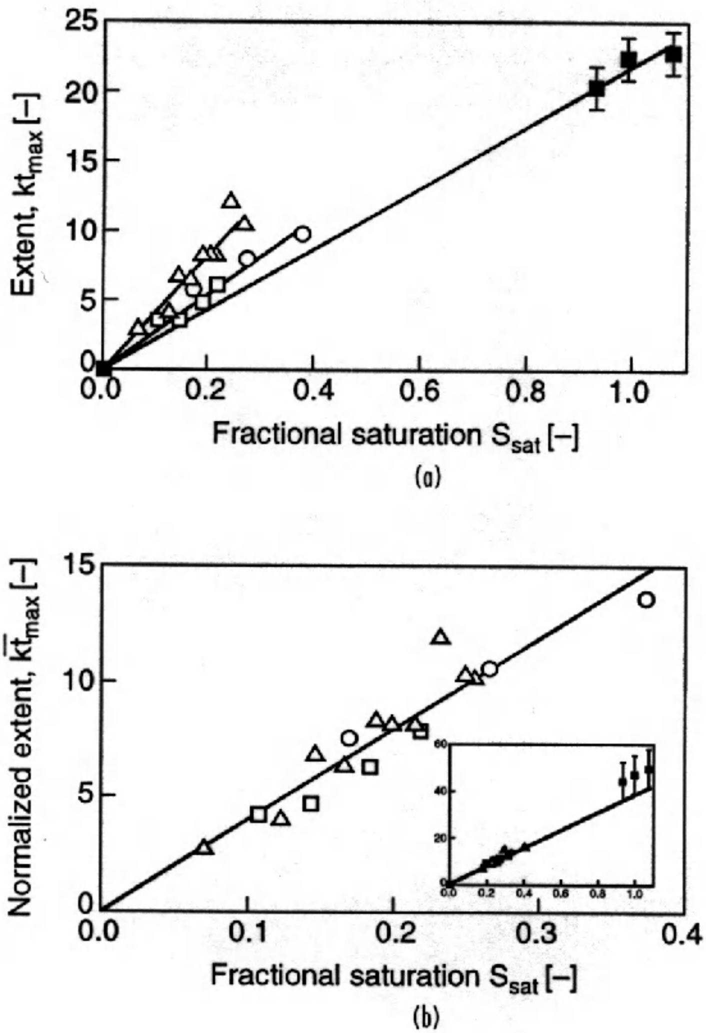


Figure 4.17. Drum granulation of fertilisers (a) extent of non-inertial granulation as a function of liquid content (b) results normalised for the effect of liquid viscosity and granule density (Adetayo et al., 1995).

4.3.4. Limitations of the Stokes Regime Analysis and Final Remarks

The examples above show that the Stokes regime analysis, despite its simplicity, is a powerful tool for understanding and quantifying granulation behaviour. However, it has some very important limitations.

The Stokes regime analysis is essentially a "ball bearing and honey" model that considers the granules as being nearly elastic and non-porous with all liquid present as a surface coating. In contrast, in many granulation processes:

1. The granules are porous with most of the liquid contained within the pores. The granules may be partially saturated and surface dry.
2. On collision, the porous granules may deform plastically to absorb much of the collision energy.

These systems show behaviour that cannot be predicted from the simple Stokes analysis. The Stokes analysis is best suited to systems where:

- The particles are large compared to the spray drop size so that particles are coated with a layer of liquid in the spray zone.
- Granulation occurs with simultaneous drying so that granules are relatively elastic.

These circumstances are met in a range of real applications including fluidised and spouted bed granulation and coating, drum granulation incorporating recycled dried granules (eg. the fertiliser case study above) and granulation of compacted, surface wet granules at the end of a granulation process. For these systems, the Stokes regime analysis gives at least semi-quantitative analysis of granulation behaviour.

For fine powder systems where nuclei are formed by drop penetration into a powder bed to form plastic, deformable granules, a more complete analysis incorporating granules mechanics is needed. This will be covered in the next section.

4.4. Granule Coalescence Criteria for Deformable, Porous Granules

4.4.1. Development of Coalescence Criteria

The mechanical properties and consolidation of porous, deformable granules have been discussed in sections 4.1.and 4.2. If a porous granule is over saturated with liquid, then it will have a liquid layer at the surface. Assuming that the liquid layer thickness is much smaller than the granule diameter ($h_0 << D$), then the liquid layer thickness is:

$$h_0 = \begin{cases} \dfrac{D(\nu - \varepsilon s^*)}{6} ; \nu > \varepsilon s^* \\ 0 ; \nu < \varepsilon s^* \end{cases} \tag{4-13}$$

Eqn. 4-13 allows us to relate the surface liquid layer to measurable system properties.

How do we handle the contribution of plastic deformation in the granule matrix? Ouchiyama and Tanaka (1975), considered surface-dry, deformable granules that collide without rebound. Kristensen *et al.* (1985) further extended the work of Ouchiyama and Tanaka by assuming that granule deformation for a given force was proportional to the ratio of critical strain and stress at failure measured during compression of bulk samples. Bond strength formed between granules was assumed to be the same as the bulk tensile strength of the granule. However, this approach neglects the importance of the liquid layer at the granule surface which the Stokes regime analysis above has shown to be extremely important. We will extend the granule collision model developed in section 4.3 using contact mechanics theory (Johnson, 1987) to account for the elastic and plastic deformation of the granules during collision. Full details of the derivation are given in Liu et al. (2000).

Consider a collision between two surface wet granules with a relative collision velocity $2u_0$ (see Figure 4.18). Assuming that the liquid layer thickness is larger than the granule surface roughness, the collision can be divided into three stages:

1. *Approach stage.* At a separation distance of $2h_0$ the liquid layers touch and merge (Figure 4.18(a)). This combined liquid layer will then be squeezed out as the granules approach, dissipating some of the kinetic energy of collision.
2. *Deformation stage*. When the separation distance reduces to $2h_a$ (the height of granule surface asperities) and the relative granule velocity to $2u_1$ (Figure 4.18(b)), the granules will begin to deform. The remaining kinetic energy is converted to stored elastic energy and dissipated by plastic deformation. When the relative collision velocity is reduced to zero, a contact area A^* is formed between the granules.
3. *Separation stage*. The granules then begin to rebound with an initial velocity of u_2 as the stored elastic energy is released (Figure 4.18(c)). Viscous dissipation in the surface liquid layer will again retard the granule movement. When the granules are separated to a distance $2h_0$ the liquid layers are assumed to separate and the granule rebound is complete leaving the granules with a velocity of u_3 (Figure 4.19(d)).

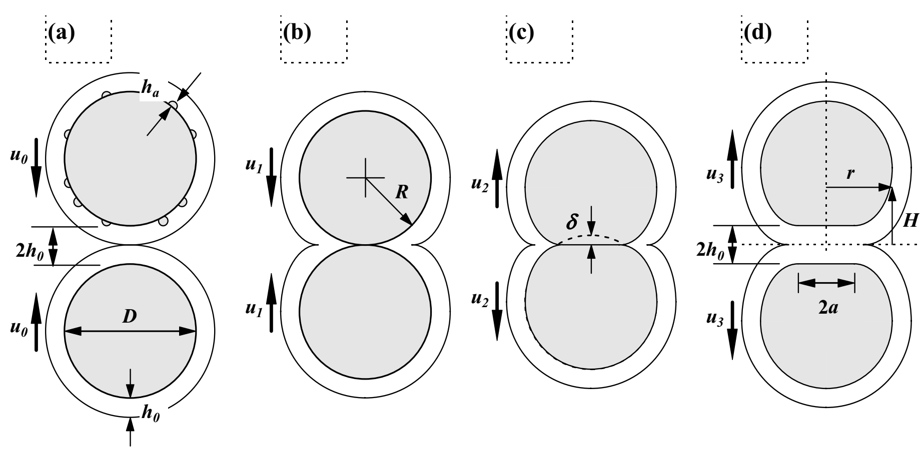


Figure 4.18. Collisions between surface wet, deformable granules (a) Approach (b) Deformation (c) Initial separation (d) Separation of liquid layers

The granules will coalesce if the collision kinetic energy is completely dissipated by viscous losses in the surface liquid layer and plastic deformation in the granule matrix i.e. the granule collision velocity reduces to zero in either the initial approach stage (type I coalescence) or the final separation stage (type II coalescence). Granules will rebound (no coalescence) if there is some remaining granule velocity at the end of the separation stage ($u_3 > 0$).

the approach stage, the model is similar to that for near elastic particles. The velocity when granules surfaces contact is:

$$u_1 = u_0\left[1 - \frac{1}{St_v}\ln\left(\frac{h_0}{h_a}\right)\right] \tag{4-14}$$

and the criterion for type coalescence is:

$$St_v < \ln\left(\frac{h_0}{h_a}\right) \tag{4-15}$$

From contact mechanics, we can calculate the initial rebound velocity:

$$u_2 \approx 2.46 u_0 \left(\frac{Y_d}{E^*}\right)^{1/2} \left(\frac{\widetilde{m}\, u_0^2}{2\widetilde{D}^3 Y_d}\right)^{-1/8} \left(1 - \frac{1}{\mathrm{St_v}}\ln\frac{h_0}{h_a}\right)^{3/4} \tag{4-16}$$

where:

$$St_{def} = \frac{\widetilde{m} u_0^2}{2\widetilde{D}^3 Y_d} \tag{4-17}$$

For equal sized granules, eqn. 4-17 reduces to the definition already given in section 4.2 (eqn. 4-4). St_{def} is the Stokes deformation number for the collision (Tardos et al., 1998, Iveson and Litster, 1998c) which gives a measure of the amount of plastic deformation of an equivalent surface dry granule. The permanent plastic deformation during collision is:

$$\delta'' = \delta^* - \delta' = \left(\frac{8}{3\pi}\right)^{1/2} \left(St_{def}\right)^{1/2} \widetilde{D}\left[1 - \frac{1}{St_v}\ln\left(\frac{h_0}{h_a}\right)\right] \left[1 - 7.36\left(\frac{Y_d}{E^*}\right)\left(St_{def}\right)^{-1/4}\left(1 - \frac{1}{St_v}\ln\left(\frac{h_0}{h_a}\right)\right)^{-1/2}\right] \tag{4-18}$$

Finally, the remaining velocity after rebound is:

$$u_3 = u_2 - \frac{\mu u_1^2 \widetilde{D}}{4 Y_d h_0^2}\left[\left(\frac{h_0^2}{h_a^2} - 1\right) + \frac{2h_0}{\delta''}\left(\frac{h_0}{h_a} - 1\right) + \frac{2h_0^2}{(\delta'')^2}\ln\left(\frac{h_0}{h_a}\right)\right] \left[1 - 7.36\left(\frac{Y_d}{E^*}\right)\left(St_{def}\right)^{-1/4}\left(1 - \frac{1}{St_v}\ln\left(\frac{h_0}{h_a}\right)\right)^{-1/2}\right]^2 \tag{4-19}$$

For granules to coalesce, u_3 must be less than or equal to zero. By setting $u_3 < 0$ and substituting for u_2 we have the condition for type II coalescence:

$$\left(\frac{Y_d}{E^*}\right)^{1/2}\left(St_{def}\right)^{-9/8} < \frac{\dfrac{0.172}{St_{vis}}\left(\dfrac{\tilde{D}}{h_0}\right)^2\left[1-\dfrac{1}{St_{vis}}\ln\left(\dfrac{h_0}{h_a}\right)\right]^{5/4}}{\left[\left(\dfrac{h_0{}^2}{h_a{}^2}-1\right)+\dfrac{2h_0}{\delta''}\left(\dfrac{h_0}{h_a}-1\right)+\dfrac{2h_0^2}{(\delta'')^2}\ln\left(\dfrac{h_0}{h_a}\right)\right]} \cdot\left[1-7.36\left(\frac{Y_d}{E^*}\right)\left(St_{def}\right)^{-1/4}\left(1-\frac{1}{St_v}\ln\left(\frac{h_0}{h_a}\right)\right)^{-1/2}\right]^2 \tag{4-20}$$

The main characteristics of type II coalescence is that granules are slowed to a halt during rebound, after their surfaces have contacted. Whether or not permanent plastic deformation has occurred depends on the elastic-plastic properties of the granule.

The liquid layer thickness h_0 can be written in terms of the granule size and relative saturation (eqn. 4-13). If the height of asperities on the granule surface h_a is assumed to be proportional to the diameter of the constituent particles making up the granule ($h_a = kD_p$ where k is less than or equal to ½) we have the following equations for the dimensionless groups:

$$\frac{\tilde{D}}{h_0} = \frac{6}{\nu - \varepsilon s^*} \tag{4-21}$$

$$\frac{h_0}{h_a} = \frac{\nu - \varepsilon s^*}{6k}\frac{\tilde{D}}{D_p} \tag{4-22}$$

Substituting eqns. (4-21) and (4-22) into eqn. (4-20) yields the condition for type II coalescence in terms of granule excess liquid and the diameters of the constituent particles. Similarly, the condition for type I coalescence (eqn. (6)) can be rewritten as:

$$St_v < \ln\left(\frac{\nu - \varepsilon s^*}{6k}\frac{\tilde{D}}{D_p}\right) \tag{4-23}$$

Eqns. (4-15) and (4-20) give us the conditions for Type I and II coalescence of deformable, surface wet granules in terms of measurable formulation properties and the characteristic collision velocity in the granulator.

ications of the Coalescence Criteria

Equations 4-15 and 4-20 show that granule coalescence depends on the following dimensionless parameters:

- the viscous Stokes number St_v which is the ratio of impact kinetic energy to viscous dissipation in the liquid layer;
- the Stokes deformation number St_{def} which is the ratio of impact kinetic energy to plastic deformation in the granule matrix;
- the ratio of plastic yield stress to elastic modulus, Y_d/E^*; (granule stiffness)
- the ratio of liquid layer thickness to surface asperity height h_0/h_a (or its equivalent the excess liquid content $v\text{-}\varepsilon s^*$) which gives the amount of liquid at the granule surface to aid coalescence; and
- the ratio of granule size to liquid layer thickness D/h_0 (or the related term $\tilde{D}/D_p$), which influences the relative importance of liquid layer to inertial effects.

Figure 4.19 is a plot of St_v vs. St_{def} showing the two coalescence criteria for two granules for given values of Y_d/E^*, h_0/h_a and $\tilde{D}/h_0$. This plot shows the complex nature of the granule impact. Eqns. 4-15 and 4-20 divide the plot into three regions: type I coalescence, type II coalescence and rebound. For type I coalescence, granules are stopped by viscous dissipation in the surface liquid layer before the granule surface touch. This occurs for all collisions below a critical value of St_v, independent of St_{def} since the mechanical properties of the granule do not come into play. Therefore, eqn. 4-15 is a horizontal line in Figure 4.20.

For type II coalescence (eqn. 4-20) the relationship is more complex. For weak, deformable granules (high St_{def}), type II coalescence may occur even at quite high St_v. There are two reasons for this. Firstly, a significant proportion of the impact kinetic energy is absorbed in plastic deformation of the granule matrix. Secondly, the viscous dissipation during the separation stage is strongly related to the contact area generated by plastic deformation ($F_{vis} \propto \left(\delta'' \tilde{D}\right)^2 = (a^*)^4$ or A^2). Hence, as more deformation occurs, the viscous forces during separation increase much more rapidly than the amount of stored elastic energy. As granule strength increases (decreasing St_{def}), the critical St_v for Type II coalescence approaches a minimum value given by eqn. 4-8 for perfectly elastic collisions.

In summary then, the three regions are: *rebound* (strong granules at large St_v); *type I coalescence* (small St_v independent of granules strength); and *type II coalescence* (weak granules at a wide range of St_v).

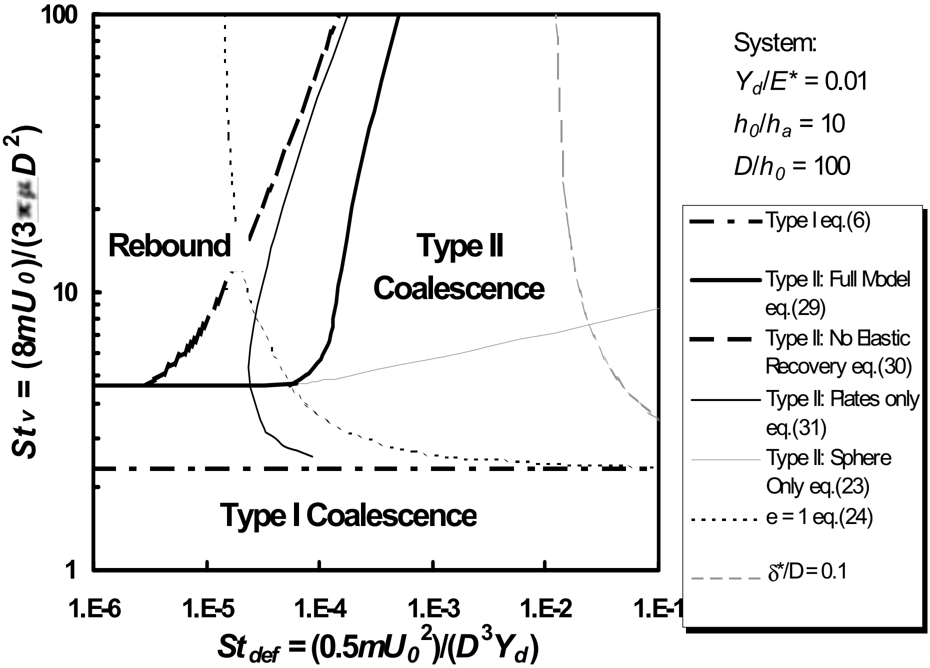


Figure 4.19. A plot of Stv vs. Stdef showing the boundaries for Type I and Type II coalescence (Liu et al., 2000)

4.4.3. Extension to Surface Dry Granules

It is implicit in this analysis that granules with any degree of elasticity will only coalesce if there is a liquid layer between them. This occurs automatically if the granules begin with a surface layer of liquid binder. However, it is possible for surface dry granule to coalesce if liquid is squeezed to the surface *during* the granule impact as a result of localised deformation and consolidation. This leads to type II coalescence. This is only likely if:

1. The liquid content is close to the critical value for a surface layer to appear;
2. There is significant contact area formed during the collision (high St_{def}); and
3. Liquid can move quickly into the contact zone (low viscosity binder and large granule pore size due to large primary particles).

These conditions correspond to a weak, nearly saturated deformable granule. Clearly, type I coalescence is impossible for surface dry granules. For type II coalescence, the derivation is similar except that there is no approach stage ($u_1 = u_0$) and during rebound the liquid layer is only present between the two flattened surfaces (Figure 4.20). Thus, eqn. (4-15) becomes:

$$\left(\frac{Y_d}{E^*}\right)^{1/2}\left(St_{def}\right)^{-9/8} < \frac{0.172}{St_{vis}}\left(\frac{\tilde{D}}{h_0}\right)^2\left(\frac{h_0^{\ 2}}{h_a^{\ 2}}-1\right)\left[1-7.36\left(\frac{Y_d}{E^*}\right)\left(St_{def}\right)^{-1/4}\right]^2 \tag{4-15a}$$

The permanent deformation δ'' is also different from eqn. (4-18):

$$\delta'' = \left(\frac{8}{3\pi}\right)^{1/2}\left(St_{def}\right)^{1/2}\tilde{D}\left[1-7.36\left(\frac{Y_d}{E^*}\right)\left(St_{def}\right)^{-1/4}\right] \tag{4-18a}$$

Figure 4.21 shows the coalescence predictions of eqn. 4-20a for two cases: h_0 constant and $h_0 = \delta''$. In both cases, the shape of the plot is similar. The curves asymptotically approaches a critical value of St_{def}. For all collisions where St_{def} is below this critical value, coalescence does not occur, regardless of St_v. This critical value occurs when there is no permanent plastic deformation, since there is then no liquid layer during rebound (in Figure 4.21 the critical value of St_{def} is 1.22 x 10^{-5} found by setting δ'' = 0 in eqn. 4-18a). For St_{def} above this critical value, St_v becomes significant, since it indicates the liquid viscosity.

Comparing the two different assumptions about h_0, assuming a constant value of h_0 gives a greater likelihood of coalescence since there is always a large amount of liquid present, whereas using h_0 = δ'' means that for low deformations, very little liquid is present. Currently, we have no good quantitative estimate for h_0 but expect it will be a function of the extent of granule deformation and the granule saturation (liquid content). The formation of bonds between surface dry granules during collision is the subject of ongoing research (Iveson and Page, 1998).

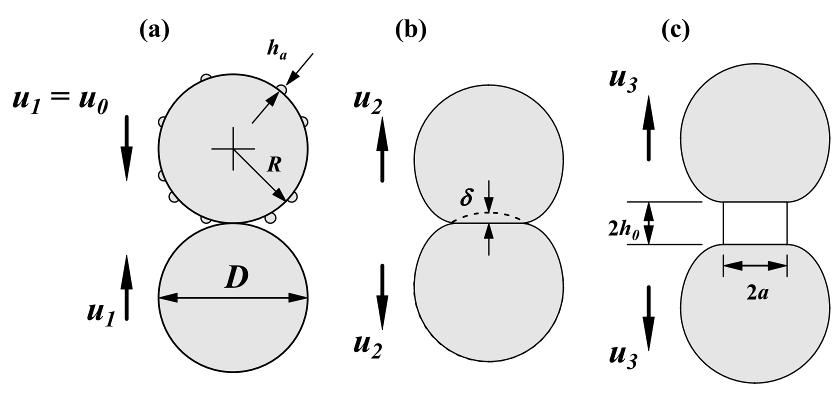


Figure 4.20. Model for collision of surface dry granules

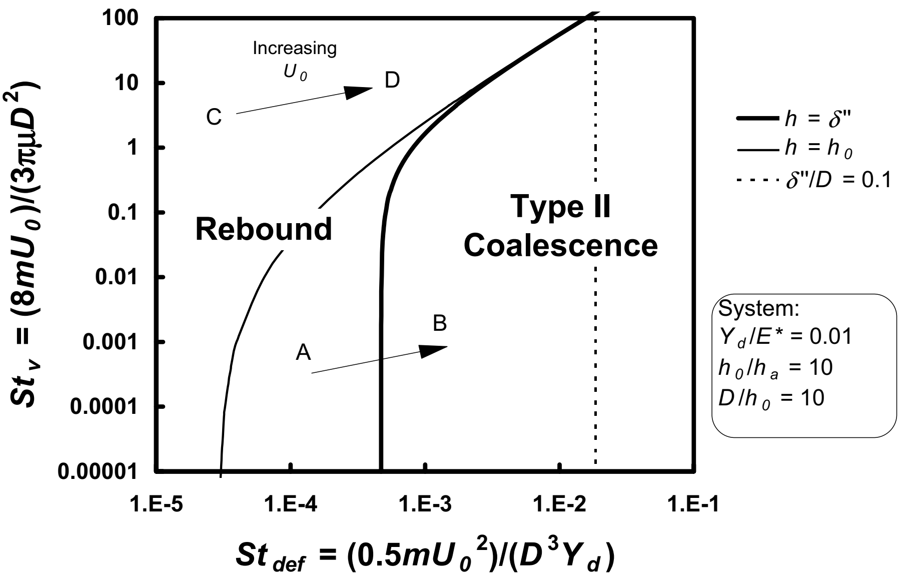


Figure 4.21. St_v vs. St_{def} showing model predictions for coalescence of surface dry granules (Liu et al., 2000)

4.4.4. Comparison with Laboratory Granulation Data

The above coalescence criteria should help explain observed phenomena in batch granulation data. Iveson *et al.* (1998a,b) measured dynamic yield strength, granule consolidation rate and granule growth behaviour for glass ballotini granulated with several liquid binders in a laboratory batch granulation drum. Table 4-3 lists the measured properties and derived dimensionless groups for their system. The Stokes numbers are calculated based on the peripheral drum speed $\omega D_{drum}/2$ which is the maximum collision velocity a granule is likely to experience in the drum (Adetayo *et al.*, 1993, Iveson *et al.*, 1998a). Iveson did not measure elastic modulus for the granules, but static measurements from other workers suggest that Y_d/E is of order 0.01 to 0.05 for these materials (e.g. Holm *et al.*, 1985). The water and glycerol binders cover viscosities ranging from 0.001 Pa.s to 1 Pa.s.

Figure 4.22 shows the plot of St_v vs. St_{def} calculated from eqns. 4-19 and 4-23 while figure 4.23 shows the growth behaviour for these systems. The two curves give the boundaries for type I and type II coalescence. The dimensionless parameters used in the calculations are listed in Table 4.4. The weak, water bound granules grew by steady growth where deformation is generally important (figure 4.23a). Most of these points are located in the type II coalescence region, with some points located in the rebound region. For the strong glycerol bound granules, St_v was much smaller and the systems were on the type I - type II transition boundary. These systems grew by induction growth once liquid was available at granule surfaces (figure 4.23b). Considering the complex collision velocity ution of granules in a rotating drum, the model predicts reasonably well how a particular system will behave.

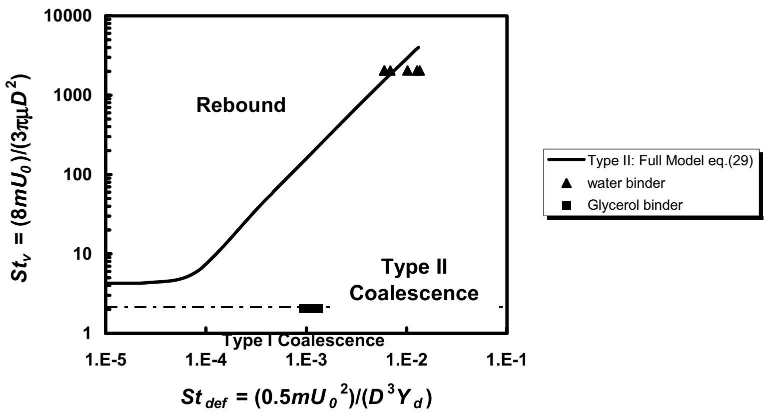


Figure 4.22. Experimental validation of the coalescence criteria granules (Liu et al., 2000)

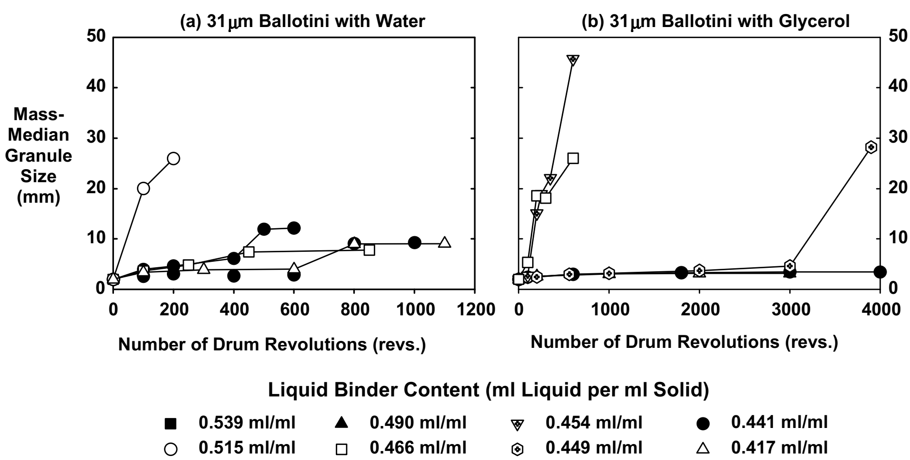


Figure 4.23. Growth behaviour of 31μm glass ballotini with water and glycerol binders

4.4.5. Summary

The deformable granule coalescence criteria developed here is a logical extension of the simpler Stokes analysis described in section 4.3. There is little data to quantitatively test this theory at present. However, we can make immediate use of the mechanistic understanding to develop a growth regime map for deformable granules (section 4.5 below).

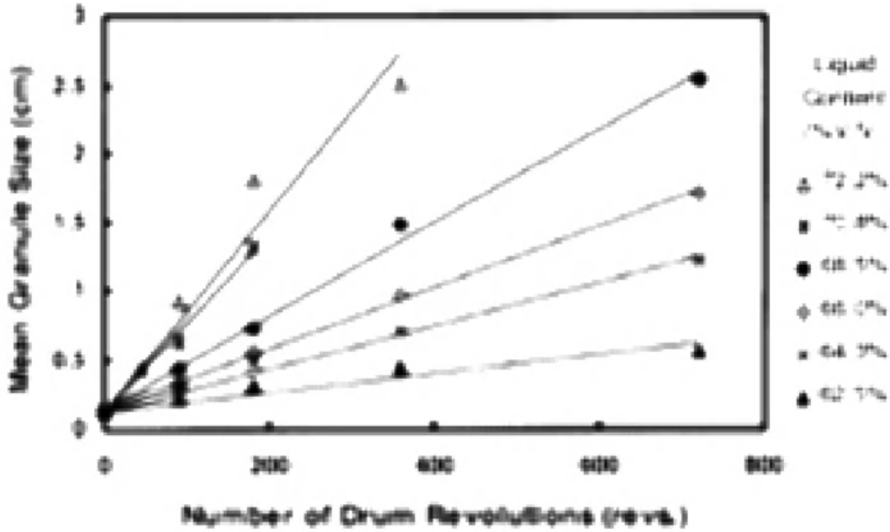
4.5. Granulation Behaviour for Deformable Granules: A Growth Regime Map

Table 4.1 shows that many different growth behaviours have been observed. Can these behaviours be explained in terms of our knowledge of granule consolidation and coalescence?

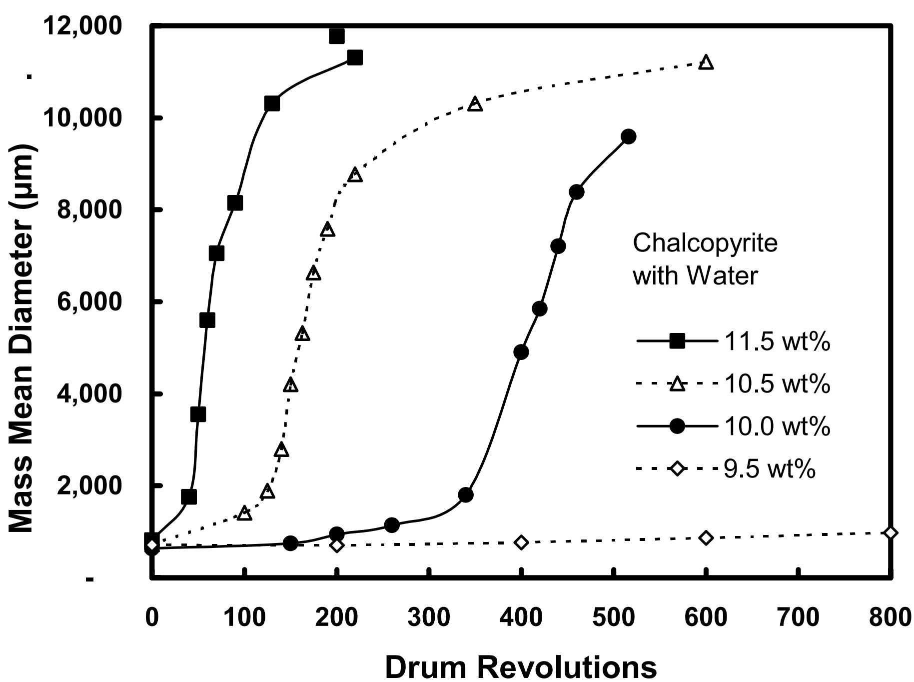
There are two broad classes of growth behaviour which can occur – either steady growth or induction time behaviour, illustrated schematically in figure 4.24. The type of growth that occurs depends on the deformability and rate of consolidation of the granules. Weak, deformable granules form a large area of contact during collision and absorb the impact kinetic energy through plastic deformation of the granule. Liquid binder may be squeezed into this contact zone. If this bond is strong enough to resist the separating forces within the granulator, then the pair of granules will be rounded into a new larger granule. This behaviour leads to a steady increase in granule size (steady growth) and is common in systems with coarse, narrowly-sized particles and low surface tension and/or low viscosity binder liquids (eg. Newitt and Conway-Jones, 1958; Capes and Danckwerts, 1965; Linkson *et al.*, 1973).

Strong, non-deformable, slowly-consolidating granules do not deform sufficiently during impact to form a strong contact bond. Pairs of colliding granules rebound and there is a period of little or no granule growth – the "induction" period (also called the "nuclei" region (Kapur, 1978) or "consolidation" period (Hoornaert, *et al.*, 1994)). During the induction time, the granules are surface dry. The length of this induction period decreases with increasing liquid content. If granules consolidate sufficiently, then liquid binder may eventually be squeezed to their surface. This surface liquid enables coalescence without the need for large amounts of deformation as predicted by the Stokes analysis. This triggers rapid granule growth until the granules become so large that the torque exerted on granule dumbbells prevents further coalescence. This class of behaviour is frequent in systems with fine particles and/or viscous binders (eg. Linkson *et al.*, 1973; Sastry *et al.*, 1977; Kapur, 1978; Hoornaert *et al.*, 1994). Figure 4.24 shows examples of both types of growth from the literature.

Figure 4.25 shows granulation results from glass ballotini in which the Stokes deformation number is deliberately varied by changing particle size and binder viscosity (Iveson and Litster, 1998a). Growth behaviour varies from steady growth to significant induction time behaviour.



(a) Drum granulation of 67μm silica with water (Newitt and Conway-Jones, 1958)



(b) Drum granulation of chalcopyrite with water (Waters et al., 2001)

Figure 4.24. Examples of (a) steady growth, and (b) induction behaviour in batch granulation

In summary, the growth behaviour of granules is determined by the granule deformation during collisions and the liquid content of the granules. We can therefore map the granulation behaviour in terms of two of the dimensionless group we have defined: the deformation number St_{def} (eqns. 4-4, 4-17) and the liquid saturation. As the liquid saturation varies with compaction, we use the maximum pore saturation as the characteristic value for the system:

$$s_{max} = \frac{w\rho_s(1-\varepsilon_{min})}{\rho_l \varepsilon_{min}} \tag{4-23}$$

Granule yield stress and density will also vary with compaction and St_{def} should also be calculated at the minimum porosity conditions.

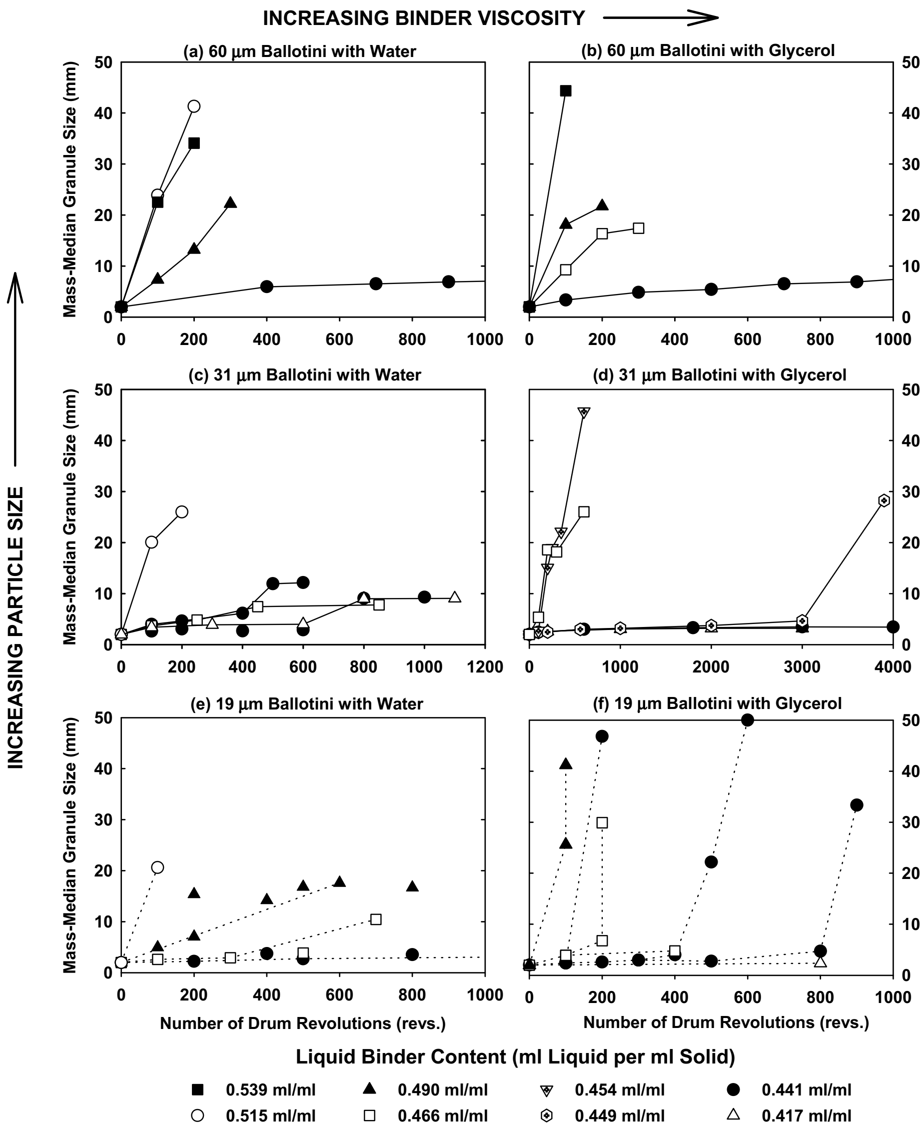


Figure 4.25. Mass-median size vs. granulation time for 19-, 31- and 60-µm ballotini granulated with water and glycerol binders across a range of liquid contents

The position of a system on a plot of St_{def} vs. s_{max} will characterise the type of growth behaviour observed:

- At very low liquid contents (low s_{max}), particles will either remain as a dry, free-flowing powder, or else will form nuclei due to van de Waals interactions, but not grow any further.
- At slightly higher liquid contents, either nuclei will form which will not grow, or if the system is extremely weak (high St_{def}) it will form a non-granular "crumb" material (Capes and Danckwerts, 1965; Tardos *et al.*, 1998).
- At medium levels of liquid content, if the granules are deformable and consolidate quickly (medium St_{def}), then they will grow steadily. If they are low-deformation, slowly-consolidating systems (low St_{def}) then they will exhibit induction-time behaviour.
- At high liquid contents ($s_{max} \rightarrow 1$), both fast and slow consolidating systems will grow rapidly.
- At very high liquid contents (s_{max}>1), a slurry or over-wetted mass will be formed.

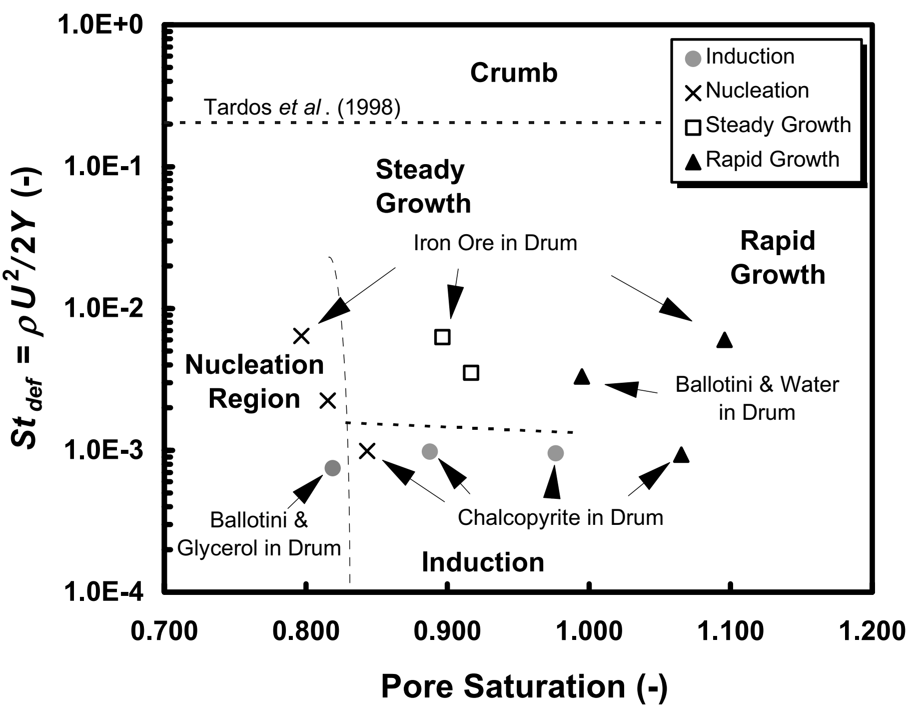


Figure 4.26. Growth regime map with data for granulation of a wide range of materials

The growth regime map is shown in figure 4.26 (Iveson et al., 2001). The map shows a wide range of literature data for drum granulation, including data from figures 4.24 and 4.25. The data fits the regime map reasonably well. The transition from nucleation to steady or induction growth occurred at pore saturations between 80% and 90%. Rapid growth occurred for all pore saturations greater than 100 %. The boundary between induction and steady growth occurred at St_{def} between 0.001 and 0.003. The steady growth results all lie well below the steady-crumb regime boundary of 0.2 measured by Tardos et al. (1997).

Figure 4.27 shows the regime map with the mixer experimental results added. In many cases mixer results do not fit the regime map as set for drum granulation. The most likely reason for the failure of the regime map to predict the granulation behaviour in the mixers is that, since St_{def} is proportional to impact velocity squared, any errors in estimating typical impact velocity will be greatly magnified. It is difficult to know what characteristic speed to use to characterise a mixer granulator. In a horizontal-axis, high shear Lödige mixer growth rate is sensitive to chopper speed but not to impeller (plough share) speed (Menon, 1996). The chopper tip-speed was chosen as the collision velocity to use in calculating St_{def}. when analysing the data of Hoornaert *et al.* (1998). However, these observations are not universal to all high shear mixers. In vertical axis mixers, several studies showed no effect of chopper speed on growth eg. (Knight (1993). For vertical axis mixers, collision velocity is probably closely related to impellor tip speed.

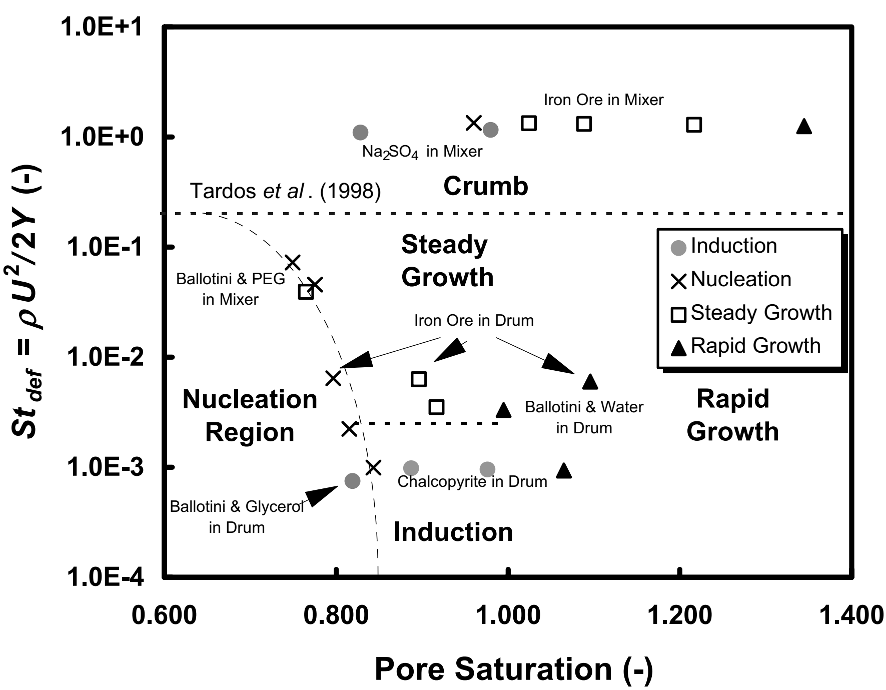


Figure 4.27. St_{def} vs pore saturation for drum and mixer granulation experiments with glass ballotini, iron ore, chalcopyrite and sodium sulphate powders (Iveson et al., 2000).

In summary, the regime map is a useful tool for predicting granule growth behaviour in tumbling granulators. There is still some way to go to extend it fully to the more complex behaviour in mixers.

he growth regime will influence profoundly approaches to scale up and control. If a powder-binder mixture lies in the induction growth region (strong, non-deformable granules), then provided its induction time is not exceeded, it will never grow rapidly. The granule size distribution will be controlled by the nucleation conditions. Layering growth could be promoted by spraying binder onto the surface of pre-existing granules, whereas smaller granules could be formed by spraying the binder onto the fresh feed to promote nucleation. Therefore the product size distribution will be strongly influenced by the positioning of the liquid nozzle(s) and powder feed, liquid droplet size distribution, powder wetting characteristics and the flow and mixing of material in the granulator.

This kind of formulation will be relatively easy to scale with respect to granule size distribution since there is a long period of little growth. However, care must be taken that the process agitation intensity does not increase significantly during scale up (eg. from a small to large pan). Such an increase may shift the formulation's behaviour from the induction region to the steady growth region, in which case scale up will be much more difficult.

The growth behaviour of powder-binder mixtures in the steady growth region (weak, easily deformable granules) is dominated by coalescence, not layering or nucleation. The rate of coalescence is very sensitive to binder content and scale of operation because these variables both affect the granule deformability. This type of formulation is difficult to scale-up with respect to granule size distribution. Granule size is controlled by the rate of liquid addition and is also a strong function of the residence time. On the other hand, porosity will be insensitive to residence time since granules will quickly reach their minimum porosity.

4.6. Control of Growth and Consolidation for Particle Design

The granule mean size, the breadth of the granule size distribution and the granule porosity (density) are all influenced by the way granules grow and consolidate. This section summarises the effect of process parameters and formulation properties on growth and consolidation behaviour, and hence granule attributes, based on the fundamental understanding of granule growth we have developed.

4.6.1. Elastic Granule Systems

Systems that can be treated as elastic (non-deformable) are:

- Granulation with simultaneous drying (fluidised beds, spouted beds, etc.)
- Systems where the maximum feed particle size is much larger than the spray drops (coating/layering processes, processes where feed includes recycled granules of order of the size of the desired product granules)

These systems are described by the Stokes regime analysis. It is important to establish in which regime the system in operating (non-inertial, inertial, coating) and whether the system traverses more than one regime during granulation. The effect of process parameters and formulation properties is encapsulated in the dimensionless groups St_v and St^*.

4.6.2. Deformable Granule Systems

Deformable granule systems have powder feeds wet by a spray whose drop size is large compared to the feed powder and where simultaneous drying does not occur. These systems are characterised primarily by the dimensionless groups s_{max} and St_{def} , with St_v having a secondary effect. The granulation regime map (figure 4.26) can be used to establish the growth regime and decide on scale up and control strategies.

Effect of formulation properties

The single most important formulation property is the liquid content. As liquid content increases, the granulation regime moves from nuclei only through steady growth or induction time to rapid growth and slurry formation. In the nuclei regime, slow controlled growth can be achieved by successive powder layering. The operating window for controlled growth by coalescence is often very small with growth rate and granule size increasing exponentially with liquid saturation (see figure 4.28) (Ritala et al., 1988, Litster and Waters, 1986). Thus, granule size is very sensitive to liquid content in this region.

The effect of other formulation properties should be viewed in terms of their effect on granule yield stress. Increasing binder viscosity increases granule dynamic strength and hence decreases St_{def} (see section 4.1.2). This decreases the rate of granule consolidation and increases the equilibrium granule porosity generally leading to a more porous granule (section 4.2). Increasing viscosity can move a system from the crumb region to steady growth (Keningley et al., 1997) or from steady growth to induction behaviour (Knight et al., 1998, see also figure 4.26). So the effect on growth *rate* and final granule size depends on the growth regime and the residence time in the granulator. This is emphasised by figure 4.29. In high shear mixer granulation, granules grew *faster* with the lower viscosity binder but reached a smaller final (equilibrium) granule size.

Increasing liquid viscosity or decreasing particle size both increase granule dynamic yield stress but have variable effects on minimum porosity (Iveson *et al.*, 1996, Iveson and Litster, 1998a, Iveson and Litster, 1998b). Hence, increasing binder viscosity and decreasing particle size will decrease St_{def} and should shift a formulation's behaviour towards the bottom of the regime map ie. from crumb to steady growth; and from steady growth to induction time behaviour.

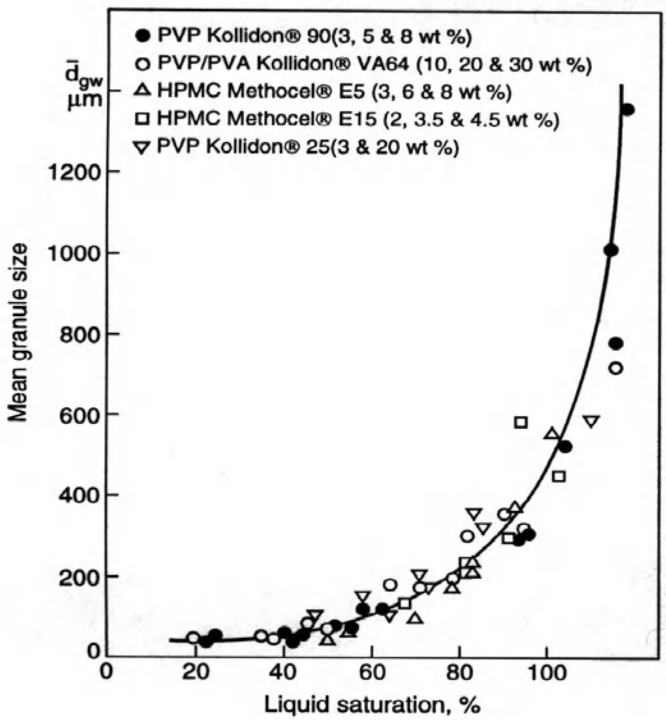
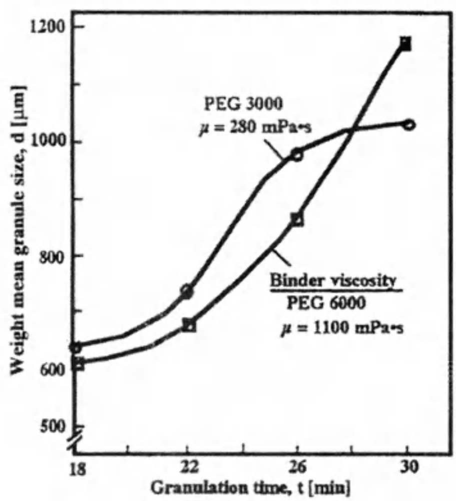
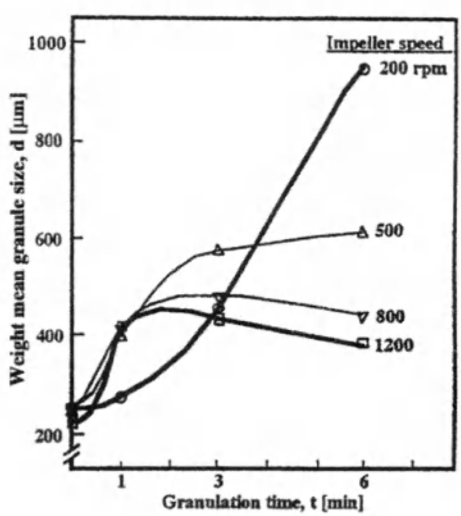


Figure 4.28. Effect of moisture as a percentage of granule saturation on mean granule diameter. Granulation of calcium hydrogen phosphate with aqueous binder solutions in a Fielder PMAT 25 VG high-shear mixer (Ritala et al., 1988)

Results in the literature support these predictions. Capes and Danckwerts (1965) found that decreasing particle size shifted behaviour from the crumb into the steady growth region. The increasing frequency of induction time behaviour with decreasing particle size can be seen by surveying the existing literature. Workers who studied coarse, narrowly-sized silica sands, did not observe induction behaviour (eg. Capes and Danckwerts, 1965; Newitt and Conway-Jones, 1958). Linkson *et al.* (1973) saw induction times of up to 200 revolutions with widely-sized, 43 µm silica sands. Induction times of over 1000 revolutions have been frequently observed with finely ground limestone powders (0.71 m^2/g) and hematite ores (0.295 m^2/g) (Kapur, 1978; Sastry *et al.*, 1977).



(a)



(b)

Figure 4.29. Granule diameter as a function of time for high-shear mixer granulation. (a) 10-liter vertical high-shear melt granulation of lactose with liquid loading of 15 wt % binder and impeller speed of 1400 rpm for two different viscosity grades of polyethylene glycol binders. (Schaefer et al., 1990). (b) 10-liter vertical high-shear mixer granulation of dicalcium phosphate with 15 wt % binder solution of PVP/PVA Kollidon® VA64, liquid loading of 16.8 wt % and chopper speed of 1000 rpm for varying impeller speed. (Schaefer et al., 52(9), 1990).

Decreasing binder surface tension has been found to decrease both dynamic yield stress and the extent of granule consolidation (Iveson and Litster, 1998a; Iveson and Litster, 1998b). Hence, lowering binder surface tension will increase St_{def} and decrease s_{max}. This will shift a formulation's behaviour towards the top-left of the regime map ie. from induction to steady growth; and from steady growth to crumb or nucleation behaviour. Capes and Danckwerts (1965) found that lowering binder surface tension did shift a material into the crumb region. However, no studies have been done to test whether lowering binder surface tension can shift a material's behaviour from induction to steady growth.

Effect of process parameters

Increasing process agitation intensity will increase the typical impact velocity and hence increase the deformation number. The consolidation models of Ouchiyama and Tanaka (1980) and Ennis *et al.* (1991) both predict that granule consolidation will increase with increasing frequency and energy of impacts. This has been confirmed experimentally in tumbling pans, high-shear mixers and agitated fluidised beds where it was found that increasing pan or impeller speed increased the initial rate of consolidation (Ouchiyama and Tanaka, 1980; Ritala *et al.*, 1986; Watano *et al.*, 1995). Likewise, the porosities of granules made in high-shear mixers are generally much lower than those made in tumbling drums and pans (eg. 20 to 40% measured by Kristensen *et al.* (1985) compared with 40 to 50% measured by Kapur and Fuerstenau (1969)). Hence, increasing process agitation intensity will increase the extent of granule consolidation and hence increase the maximum pore saturation.

Therefore, increasing agitation intensity should shift a system's behaviour towards the top-right of the proposed growth regime map ie. from nucleation to steady or induction growth; from induction to steady or rapid growth; and from steady growth to crumb, rapid or over-wetted growth (see Figure 4.31). There is some evidence for this in the literature. Poskart (1988) used a tumbling drum mounted on a spinning arm and found that increasing the centrifugal force from 1 to 10-50 g significantly increased the product density. At 50 g, destructive forces dominated and very few pellets were formed ("crumb" region). The process also became very sensitive to binder content and could easily "sludge" (ie. form an over-wetted mass or slurry). Similarly, Keningley *et al.* (1997) found that increasing impeller speed shifted a material's behaviour from the steady growth to the crumb region (their so-called "paste" region). However, no work is currently available to confirm whether increasing agitation intensity will also shift a material's behaviour from induction to steady growth as predicted by this regime map.

Table 4.3 gives a quick reference summary of the role of formulation properties and process parameters in controlling growth and consolidation.

Table 4.3. Controlling Growth and Consolidation in Granulation Processes

Typical changes in material or operating variables which maximize growth and consolidation	*Appropriate routes to alter variable through formulation changes*	*Appropriate routes to alter variable through process changes*
Rate of growth (low deformability): Increase rate of nuclei formation	Improve wetting properties. (See "Wetting" subsection.) Increase binder distribution.	Increase spray rate and number of drops.
Increase collision frequency		Increase mixer impeller or drum rotation speed or fluid-bed gas velocity.
Increase residence time		Increase batch time or lower feed rate.
Rate of growth (high deformability): Decrease binder viscosity	Decrease binder concentration or change binder. Decrease any diluents and polymers which act as thickeners.	Decrease operating temperature for systems with simultaneous drying. Otherwise increase temperature.
Increase agitation intensity		Increase mixer impeller or drum rotation speed or fluid-bed gas velocity.
Increase particle density Increase rate of nuclei formation, collision frequency and residence time, as above for low deformability systems		
Extent of growth: Increase binder viscosity	Increase binder concentration, change binder, or add diluents and polymers as thickeners.	Increase operating temperature for systems with simultaneous drying. Otherwise decrease temperature.
Decrease agitation intensity		Decrease mixer impeller or drum rotation speed or fluid-bed gas velocity.
Decrease particle density		
Increase liquid loading		Extent observed to increase linearly with moisture.
Rate of consolidation: Decrease binder viscosity	As above for high deformability systems. Particle size and friction strongly interact with binder viscosity to control consolidation. Feed particle size may be increased and fine tail of distribution removed.	As above for high deformability systems. In addition, increase compaction forces by increasing bed weight, or altering mixer impeller or fluid-bed distributor plate design.
Increase agitation intensity		
Increase particle density		
Increase particle size		Size is controlled in milling and particle formation.

ography

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4.8. Nomenclature

a	particle radius (m)
A_o	initial area of the sample (m^2)
A_1	the deformed contact area after impact (m^2)
Ca	Capillary number (-)
d_p	particle size (m)
d_b	volume equivalent bubble size (m)
d_g, D	granule size (m)
e	coefficient of restitution (-)
g	gravitational acceleration (m s^{-2})
h, h_b	liquid layer thickness (m)
h_a	height of surface asperities (m)
K_e	compaction rate (-)
m	particle mass (kg)
$N(t)$	total number of granules at time t
$n(v,t)$	number density of granules of size v at time t
s	liquid saturation of pores (-)
St_v	viscous Stokes number
St^*	critical Stokes number
Y	granule dynamic yield stress ((N m^{-2})
u	velocity (m s^{-1})
U_c	collision velocity (m s^{-1})
u_b	bubble rise velocity (m s^{-1})
w	effect mean particle mass during a collision (kg)
$\beta(u,v)$	coalescence rate constant for collisions between granules of volumes u and v
ε_{min}	minimum granule porosity (-)
ε_b	bed voidage (-)
ε	granule porosity (-)

	surface tension (N m)
δ	granule deformation (m)
ρ_g	granule density (kg m^{-3})
ρ_l	liquid density (kg m^{-3})
ρ_p	particle density (kg m^{-3})
σ_D	peak flow stress (N m^{-2})
σ_T	granule static tensile strength (N m^{-2})
μ	viscosity (Pa s)
ω	rotational speed (sec^{-1})

CHAPTER 5

BREAKAGE AND ATTRITION

This chapter considers the last of the three classes of granulation processes that control granule attributes – breakage and attrition. There are really two separate phenomena to consider here:

1. Breakage of *wet* granules in the granulator; and
2. Attrition or fracture of *dried* granules in the granulator, drier or in subsequent handling.

Breakage of wet granules will influence and may control the final granule size distribution, especially in high shear granulators. In some circumstances, breakage can be used to limit the maximum granule size or to help distribute a viscous binder. Attrition of dry granules leads to the generation of dusty fines. As the aim of most granulation processes is to remove fines, this is generally a disastrous situation to be avoided.

In this chapter, breakage of wet and dry granules are treated separately. We take the same approach as in chapter 3 and 4. Where possible, key formulation and granule properties are defined and the controlling groups or equations for the breakage processes are developed.

5.1. Breakage of Wet Granules

5.1.1. Experimental Observations

Few investigators have described or studied wet granule breakage in granulation processes. Some preferential growth mechanisms in tumbling granulation may involve attrition or breakage of weak granules (crushing and layering, abrasion transfer) (Sastry and Feurstenau, 1975). However, breakage is much more likely in higher intensity mixer and hybrid granulators. The limited work on wet granule breakage focuses on these processes.

Several studies show an increase in agitation intensity (increased impeller speed) reduces the final granule mean size in granulation experiments (Knight et al., 2000; Holm et al., 1983; Watano et al., 1995). For example, figure 5.1 shows median granule size from three scales of agitated fluid bed granulator decreases with increasing agitator tip speed. However, reduction in product size with increased agitation could also be explained by a reduction in the maximum granule size for coalescence (see chapter 4). So changes to granule size distribution, *on their own*, are insufficient evidence for wet granule breakage as a key mechanism for controlling granule properties.

However, wet granule breakage has been identified clearly in high shear mixer experiments by other means. Ramaker and co-workers (1998, Vonk et al., 1997) and Pearson et al. (1998) both used coloured tracer granules or liquid to identify breakage of wet granules. Pearson et al. added narrow size fractions of well formed tracer granules part way through a batch high shear granulation. Some of the tracer granules were broken, leaving coloured tracer fragments in smaller granule size fractions. Large tracer granules (>1mm) were more likely to be broken than smaller granules (figure 5.2). Knight et al (2000) showed mean granule size decreased after impeller speed was suddenly increased ugh a batch high shear mixer experiment. This was attributed to granule breakage.

Ramaker and co-workers added a coloured liquid at the start of the granulation process and observed the dispersion of the dye through a process of "destructive nucleation" where loosely bonded nuclei are broken down into smaller fragments via attrition or fragmentation (figure 5.3). The initial weak nuclei were quite large in these experiments (5mm diameter). We can view this process as simply a subset of breakage processes in the granulator. In fact, all binder distribution in the "mechanical dispersion" (chapter 3) is essentially a breakage process and should be treated as such.

In summary, wet granule breakage is potentially an important process affecting binder distribution and granule size in high intensity processes. Therefore it is important to establish the conditions under which breakage will occur.

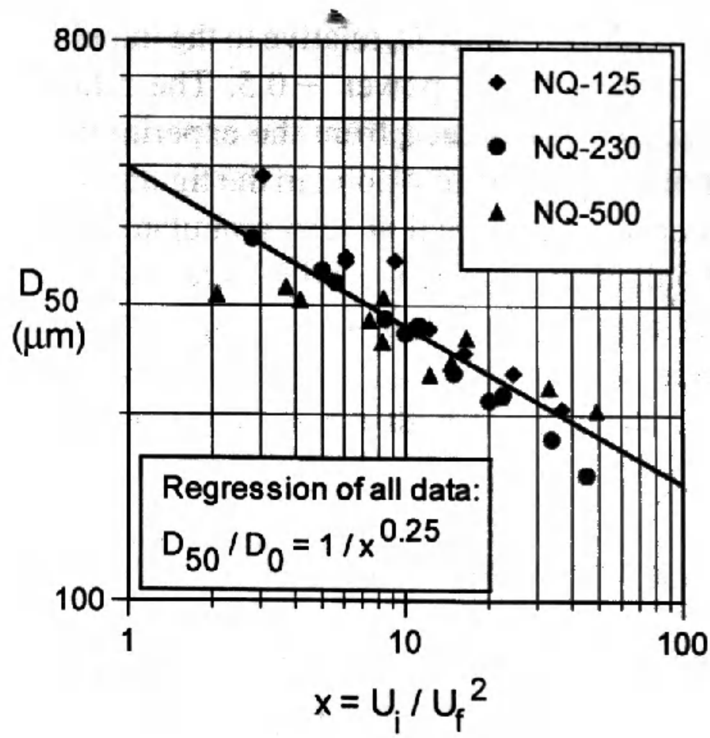


Figure 5.1. Effect of impeller speed on median particle size in an agitated fluid bed granulator (Tardos et al., 1997)

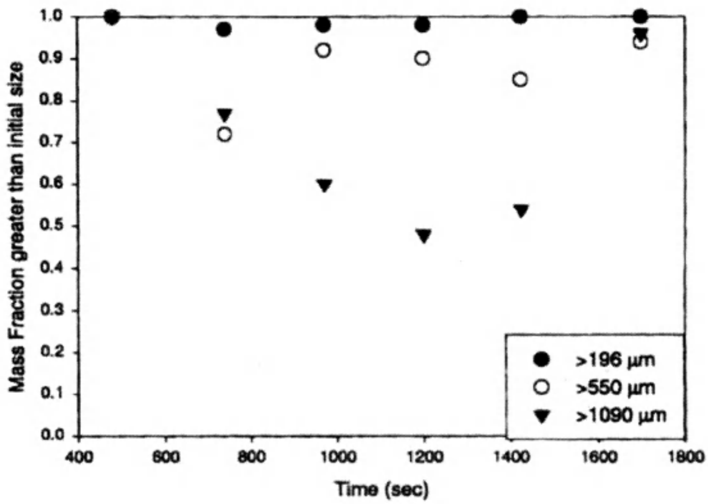


Figure 5.2. Breakage of tracer granules in high shear mixers: Effect of tracer granule size on mass fraction of unbroken granules (Pearson et al., 1998)

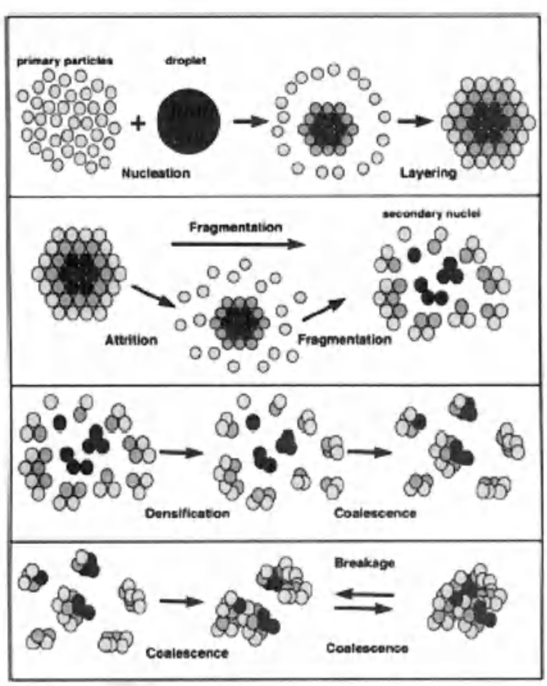


Figure 5.3. The destructive nucleation mechanism proposed by the Groningen group (Vonk et al., 1997)

cting Conditions for Breakage

There is very little quantitative theory or modelling available to predict conditions for breakage, or the effect of formulation properties on wet granule breakage. We present a criteria for breakage based on the concepts proposed by Tardos et al. (1997a,b) combined with the understanding of granule behaviour developed in chapter 4.

Consider that a granule will break if the applied kinetic energy during an impact exceeds the energy required for breakage. This analysis leads to a Stokes deformation number criteria for breakage:

$$St_{def} > St^{*}_{def} \tag{5-1}$$

where St_{def} is the Stokes deformation number as defined by eqn.4-14 and St^{*}_{def} is the critical value of Stokes number that must be exceeded for breakage to occur.

There are strong analogies to the development of the Stokes deformation number for granule deformation and growth (eqn. 4-4). It is likely the critical value for breakage will be greater than that for coalescence as granules may deform plastically at the impact point without breakage of the granule. Even this is an oversimplification. A purely plastic granule will smear rather than break when its yield stress is exceeded. At high impeller speeds such materials will coat the granulator wall or form a paste. Semi- brittle granules will break at high impact velocity giving a maximum stable granule size or a weak crumb. Thus, considerable information about the granule mechanical properties is needed to predict their behaviour. Note this yield behaviour should be measured at strain rates similar to those during impact in the granulator, not in static mechanical tests.

Note that the original work of Tardos et al. (1997a,b) proposed a more loosely defined characteristic stress than the dynamic yield stress in eqn.4-4 and considered breakage of granules by shear rather than impact. They postulate the granule will behave under shear as a Herchel-Buckley fluid ie.

$$\tau(\dot{\gamma}) = \tau_y + k\dot{\gamma}^n \tag{5-2}$$

Two simplifications were considered, neglecting either the apparent viscosity ($\tau(\dot{\gamma}) = \tau_y$) or the yield stress ($\tau(\dot{\gamma}) = k\dot{\gamma}^n$). In either case, the model predicts granules above a maximum size will break and this size is decreased with increasing shear rate.

Tardos and co-workers measured granule deformation and break up under shear in a novel constant shear fluidised bed granulator. Granules first elongated under shear and then broke at a Stokes deformation number of approximately 0.2.

Tardos's work was pioneering. However, in mixer granulators, granules may break on impact with the impeller or chopper, rather than in shear. In this case, the appropriate "critical stress" should be the dynamic yield stress measured under high strain rate conditions, as discussed in chapter 4.

Kenningley *et al.* (1997) developed a relationship for breakage (crumb, paste) or survival of granules in high shear mixer granulation by equating kinetic energy of impact to energy absorbed by plastic deformation of granules The granule yield strength was assumed to be due to viscous pressure loss for fluid flow between particles by Kozeny-Carman equation. The amount of strain (ε_m) was given by:

$$\varepsilon_m^2 = \frac{1}{540}\frac{\varepsilon^3}{(1-\varepsilon)^2}\frac{\rho u d_{3,2}}{\mu} = \frac{1}{540}\frac{\varepsilon^3}{(1-\varepsilon)^2} St_v \tag{5-3}$$

where $d_{3,2}$ is the Sauter mean size of the granules' constituent particles. Increasing Stv increases the amount of impact deformation. Above a critical value of ε_m (taken as 0.10 by Kenningley et al., 1997), granules will break. They showed reasonable agreement with their experimental data (see figure 5-4).

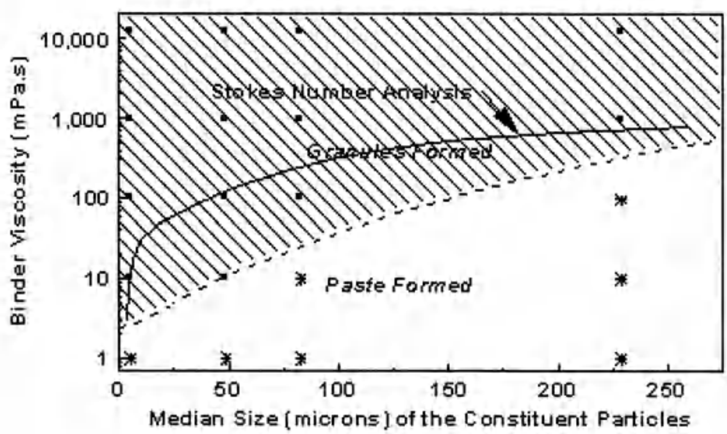


Figure 5.4. Variation of binder viscosity required to form granules with size of constituent particle (Source: Kenningley, et al. 1997).

Although, the fundamental basis for predicting breakage is incomplete, we can still use our limited knowledge for scale up. For breakage, the appropriate velocity for the Stokes deformation number is the *maximum* collision velocity a granule can experience with another granule or with part of the granulator equipment. For mixer granulators, this is clearly the impeller tip speed. Eqns 5-1 and 5-3 suggest breakage will increase with increasing tip speed. Figure 5-1 shows that the relationship between tip speed and granule mean size was the same for three different scales of agitated fluidised bed granulators, as we would expect if granule size were controlled by breakage processes.

5.1.3. Controlling Granule Attributes through Breakage

Controlling wet granule breakage gives the opportunity to give a narrow granule size distribution by growing granules up to a breakage limit (Tardos et al., 1997a; Knight et al., Mort and Tardos, 1999; Ramaker et al, 1998). This has been the driving force in the development of some newer granulator designs (Watano et al., 1995; Dencs and Ormos, 1993). It is important to note that size distribution control will also depend on the impact velocity *distribution* and turnover of granules through the high impact region (impeller or chopper). Granulators with broad impact velocity distributions and small, uncontrolled turnover through the high impact region are unlikely to ever yield narrow granule size distributions.

Controlling granule breakage for particle design is discussed further in section 5.3.

5.2. Attrition and Fracture of Dry Granules

Most granulation processes involve drying granules either simultaneously (fluidised bed and spouted bed granulators) or immediately after granulation in a separate drier. Attrition or fracture of the granules during granulation, drying or subsequent handling is generally undesirable. Therefore understanding the attrition processes and parameters which effect them are important.

There is limited fundamental work on the fracture of dry granules but we can draw on more general understanding of the fracture of brittle and semi-brittle materials. Adams and Briscoe (1987) provide good background in this area.

5.2.1. Fracture Properties of Dry Granules

From the point of view of breakage, we can consider a dry granule as a non-uniform physical composite rather than an agglomerate of primary particles. The composite possesses certain macroscopic mechanical properties including a yield stress. Instead of porosity, we see an inherent distribution of cracks and flaws. Dry granules fail in brittle or semi-brittle fashion i.e. they fail in tension by the propogation of pre-existing cracks which concentrate stress. Thus, the fracture stress may be much less than the inherent tensile strength of bonds between particles in the granule.

Consider a brittle material failing by crack propogation (figure 5.5). The tensile stress concentrates near the crack tip and is much higher than the applied stress. The fracture toughness of the granule K_c defines the elastic stress field in the granule ahead of the propogating crack and is given by (Lawn, 1975):

$$K_c = Y\sigma_f \sqrt{\pi c} \tag{5-4}$$

A simple elastic fracture model predicts an infinite stress at the crack tip. The elastic stress cannot exceed the yield stress of the material, leading to local yielding near the crack tip. To apply the simple framework of linear elastic fracture mechanics, the process zone size can be considered as an effective increase in crack length δ_c (Irwin, 1957). Fracture toughness is then given by:

$$K_c = Y\sigma_f \sqrt{\pi(c + \delta c)} \text{ with } \delta c \sim r_p \tag{5-5}$$

The process zone size is a measure of the yield stress or plasticity of the material in comparison to its brittleness. Yielding within the process zone may take place either plastically or by diffuse microcracking, depending on the brittleness of the material.

Crack propogation can also be viewed from an energy perspective. The critical strain energy release rate G_c due to crack propogation must exceed the rate of creation of surface energy for the crack to propogate (Griffith, 1920). G_c is the energy equivalent of fracture toughness. They are related by:

$$G_c = \frac{Y^2 \sigma_f^2 \pi c}{E} = \frac{K_c^2}{E} \tag{5-6}$$

These properties, together with the flaw distribution in the material dictate how it will break. Therefore, characterising these properties of a granule is an important prerequisite to determining its attrition and fracture behaviour.

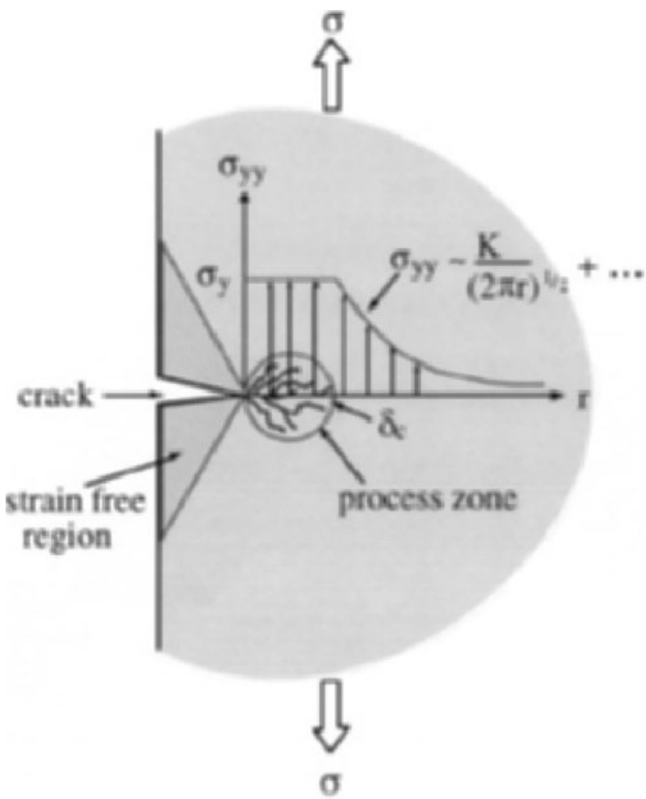


Figure 5.5. Fracture of a brittle material by crack propagation

5.2.2. Measurement of Fracture Properties

Granule breakage and attrition is an area in which rituals abound. However, ranking of granule breakage resistance by *ad hoc* tests may be test specific, and in the worst case differ from the ranking under actual conditions. Instead, standardised mechanical tests should be employed to minimise the effect of flaw and loading conditions under well defined geometries of internal stress, as described below.

To measure fracture properties reproducibly, very specific test geometry must be used since it is necessary to know the stress distribution at predefined induced cracks of known length. Three traditional methods are the three point bend test, indentation fracture testing and Hertzian contact compression between two spheres.

Figure 5.6 shows a schematic of a three point bend test. Firstly, formulation powder is premixed with liquid binder to the expected level for granulation. A series of bars, each of known crack length, are formed by compacting the moist premix in moulds containing a razor notch. The bars are then dried. The force displacement of the bars is then measured up to fracture in a 3 point bend test. K_c and δ_c are then determined by regressing the measured fracture stress against the known crack length (eqn.5-4). Figure 5.7 shows an example of the force displacement curve from a three point bend test on an agglomerate bar.

The main difficulty with the three point bend test for agglomerated materials is in the preparation of the bars. Results are sensitive to the way the bars are made and the bar structure will not match exactly the structure of a granule formed in, for example, a mixer granulator. Nevertheless, the technique been used successfully to study fundamentals of granule fracture, as well as for process troubleshooting (Seville and Ennis, 1990; Ennis and Sunshine, 1993).

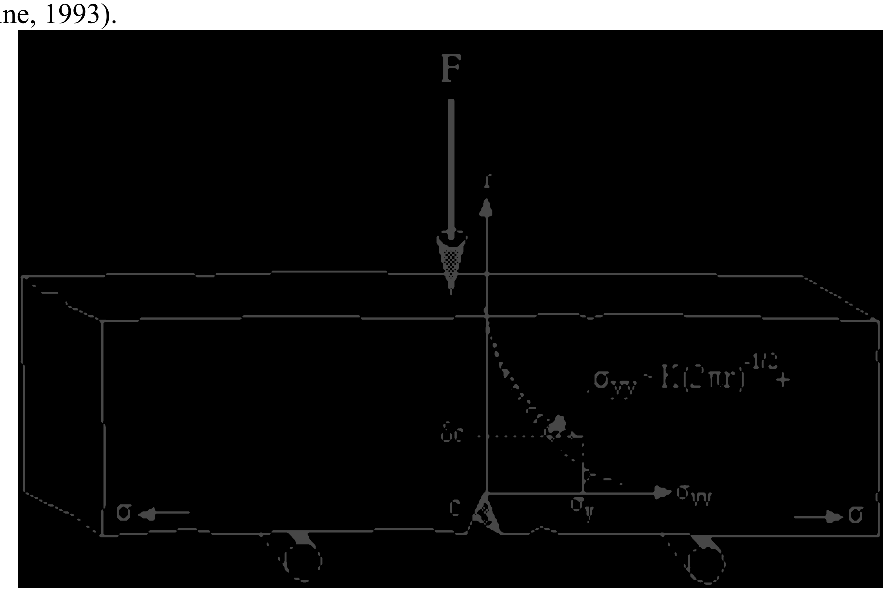


Figure 5.6. Schematic of 3 point bend test apparatus

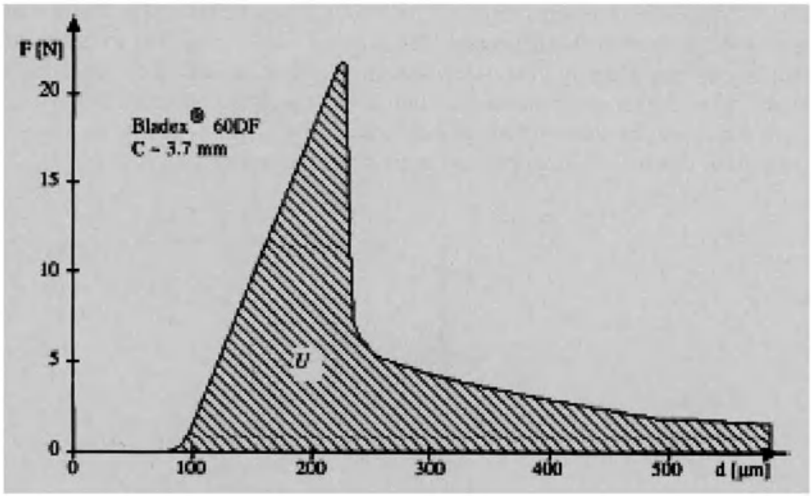


Figure 5.7. Three point bend test results: Typical force displacement curve for semi-stable fracture

Fracture toughness and hardness can also be determined from indentation tests (Johansonn and Ennis, 1994). An indent is made in a granule with known maximum force *F* (figure 5.8). The hardness is determined from the area of the residual plastic impression and the fracture toughness from the length of the cracks propogating from the indent as a function of load:

$$K_c = \beta \sqrt{\frac{E}{H}} \frac{F}{c^{3/2}} \text{ and } H \sim \frac{F}{A} \tag{5-7}$$

Figure 5.9 shows the plastic impression and radiating cracks from an indentation in an alumina catalyst particle. For this technique to be useful for agglomerates, the indentation must be large compared to the size of the feed particles but small compared to the size of the product granule. Careful presentation of the granule to the indentor is also important. This test does have the advantage that measurements can be made on real granules.

The heterogenous nature of agglomerates means they exhibit more scatter in mechanical testing than some other materials and consistent sample preparation is very important. Nevertheless, the key advantage of these tests remains that fundamental properties are measured and the influence of natural flaw distribution on the results is small.

Table 5-1 compares typical fracture properties of agglomerated materials. Fracture toughness K_c is seen to range from 0.01 to 0.06 MPa.m$^{1/2}$, less than that typical for polymers and ceramics, presumably due to the high agglomerate voidage. Critical strain energy release rates G_c from 1 to 200 J/m^2, typical for ceramics but less than that for polymers. Process zone sizes δc are seen to be large and of the order of 0.1-1 mm, values for polymers. Ceramics on the other hand typically have process zone sizes less than 1 μm. Critical displacements required for fracture may be estimated by the ratio G_c/E, which is an indication of the **brittleness** of the material. This value was of the order of 10^{-7} - 10^{-8} mm for polymer-glass agglomerates, similar to polymers, and of the order of 10^{-9} mm for herbicide bars, similar to ceramics. In summary, granulated materials behave similar to brittle ceramics which have small critical displacements and yield strains but also similar to ductile polymers which have large process or plastic zone sizes.

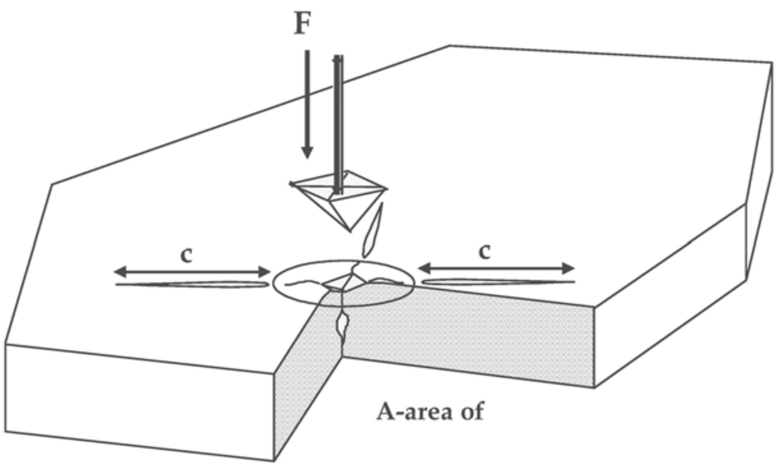


Figure 5.8. Schematic of indentation test for single particle or granules

Figure 5.9. Indentation in an alumina catalyst particle showing propogation of cracks

Table 5.1 Fracture Properties of Agglomerate Materials (Ennis and Sunshine, 1993)

Material	K_c *(Mpa.m$^{1/2}$)*	G_c *(J/m^2)*	*δc (μm)*	*E (MPa)*	G_c/E *(m)*
Bladex 60™†	0.070	3.0	340	567	5.29e-09
Bladex 90™†	0.014	0.96	82.7	191	5.00e-09
Glean™†	0.035	2.9	787	261	1.10e-08
Glean Aged™†	0.045	3.2	3510	465	6.98e-09
CMC-Na(M)‡	0.157	117.0	641	266	4.39e-07
Klucel GF‡	0.106	59.6	703	441	1.35e-07
PVP 360K‡	0.585	199.0	1450	1201	4.10e-08
CMC 2% 1kN‡	0.097	16.8	1360	410	5.28e-08
CMC 2% 5kN‡	0.087	21.1	1260	399	5.02e-08
CMC 5% 1kN‡	0.068	15.9	231	317	

† DuPont corn herbicides
‡ 50 μm glass beads with polymer binder

5.2.3. Empirical Crush Strength and Attrition Tests

There are a plethora of 'standard' empirical tests for measuring granule strength or attrition resistance. The details of these are industry, or even company specific. They fall into two broad categories:

- Crush strength tests
- Tests which measure size distribution after handling

Crush strength tests measure the fracture strength of individual granules or beds of granules under uniaxial compression. Although a compressive stress is applied, granules generally fail in tension via crack propogation.

During compaction of single particles or beds of particles, particles break in quasi-static compression. For a single particle, the extent of breakage will relate to the fracture toughness of the particles. For beds of particles, the transmission of stress through the bed is complex. The first inflection on the compaction curve for a granular bed in uniaxial compression, the consolidation starting stress P_c , has been related to the crushing strength of individual particles by several workers (Couroyer *et al.*, 1998; Naito *et al.,* 1998; Adams and McKeown, 1996; Mort *et al.*, 1995). Figure 5.10 shows an example of a compaction curve for spray dried alumina granules. However, compaction curves for granules that show brittle fracture are *similar* to those for plastic deformation so it is difficult *a priori* to extract granule mechanical properties from bed compaction tests. Most authors also compare bed compaction results to single granule crush strength, itself not a true property for brittle and semi-brittle fracture.

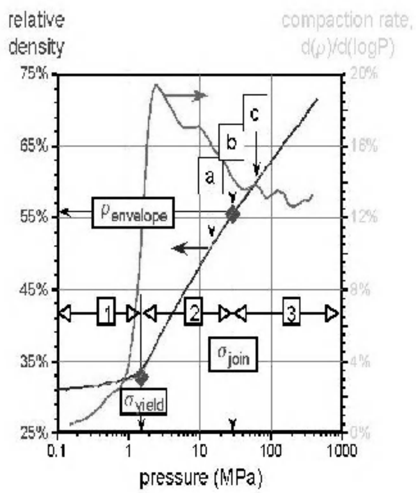


Figure 5.10. Typical compaction curve for spray dried alumina. σ_{yield} indicates the onset of either plastic deformation or fracture of granules (Mort et al., 1995)

Handling tests measure the size distribution of granules (typically the % “fines”) after a specific type of handling. Various handling methods are used, including drop tests, fluidised beds, pneumatic conveying loops, tumbling or sieving with metal balls and rotation in a shear cell.

It is important to realise that the mode of breakage in these empirical tests varies depending on the type of test (see section 5.2.4 below). These different modes of breakage have different functional dependence of fundamental material properties (fracture toughness, hardness) and flaw distribution. For this reason, breakage by one mode cannot be accurately predicted by a test base on a different breakage mode. In the worst case, different materials can rank differently in different tests, making even qualitative comparisons difficult. When empirical tests must be used, they should be chosen to mimic as closely as possible the expected handling conditions for the product in question.

5.2.4. Breakage Mechanisms for Dry Granules

The process zone plays a large role in determining the mechanism of granule breakage. Agglomerates with small process zones in comparison to granule size break by a brittle fracture mechanism into smaller fragments (figure 5.11a). This mechanism is called **fragmentation** or **fracture**.

However, for fracture to occur the granule must be able to concentrate enough elastic energy to propogate gross fracture during collision. This is harder to do as the process zone size increases. For well defined compacts under controlled stress testing conditions, it can be shown both theoretically and experimentally that fracture will only occur when the specimen size is significantly larger than the process zone size (Kendall, 1978; Puttick, 1980). For many agglomerate materials, the process zone size is of the order of the granule size (see Table 5.1). These granules will break by **wear, erosion** or **attrition** brought about by diffuse microcracking (figure 5.11b).

In contrast to fragmentation, these microcracking based processes lead to the generation of substantial amounts of fines. Thus, this class of breakage process is unfortunately both the most common and the most problematic.

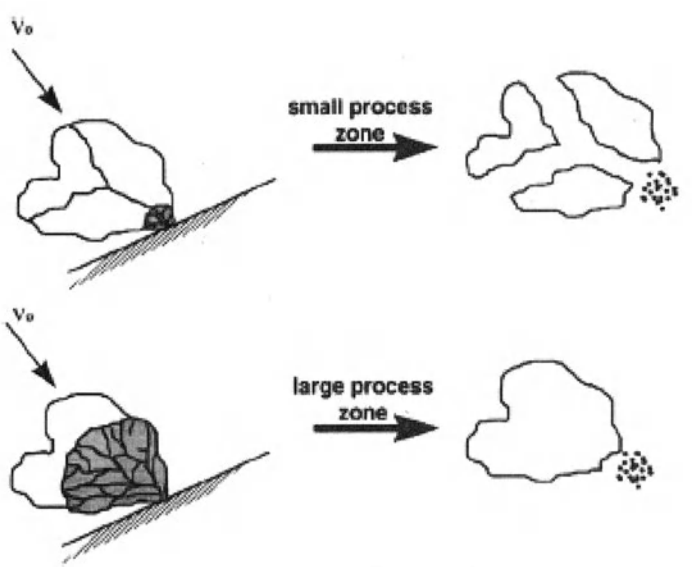


Figure 5.11. Schematic of breakage by (a) fracture, and (b) erosion/attrition depending on process zone size

There is relatively little good quality data relating attrition/erosion rates of granules and agglomerates to fundamental properties. However, we can make use of knowledge developed in the ceramics and tribology literature, as well as breakage of single particles.

The mode of stress application effects both the attrition rate and the functional dependence on particle properties. There are three classes of stress application: (i) wear and erosion, (ii) impact and (iii) compaction. For abrasive wear of agglomerates, the volumetric wear rate is (Evans and Wiltshire, 1976):

$$V = \frac{d_1^{1/2}}{A^{1/4} K_c^{3/4} H^{1/2}} P^{5/4} l \tag{5-8}$$

where d_l is indentor diameter, P is applied load, l is wear displacement of the indentor and A is apparent area of contact of the indentor with the surface. Note that the wear (erosion) rate is inversely dependent on both the fracture toughness and the hardness.

Experimental studies of the impact breakage of single crystals give a different dependence on material properties (Ghadiri et al. 1991; Yuregir et al 1987):

$$V \propto \frac{\rho_p u^2 d_p H}{K_c^2} \tag{5-9}$$

Impact attrition is more sensitive to fracture toughness than abrasive wear. In addition the effect of hardness is *the opposite* to that for wear, since hardness acts to concentrate stress for fracture during impact.

During compaction of single particles or beds of particles, particles break in quasi-static compression (section 5.2.3). Clear relationships with granule properties are not available but breakage rate will depend on fraction toughness (K_c), plasticity (G_c/E) and flaw distribution.

Different modes of breakage will dominate in various processing equipment and attrition tests, and there is a different dependence on agglomerate properties for the three modes of breakage. These differences emphasises the need for fundamental properties to be known in order to estimate attrition rates by different processes.

5.2.5. Case Study – Fluid Bed Attrition of Agglomerates

Within a fluid bed there are a large number of low velocity collisions between particles as they shear past each other. This process is analogous to abrasive wear. The number and relative velocity of the collisions depends on the number of bubbles in the bed and hence the excess gas velocity (U-U_{mf}). The applied pressure in a fluid bed depends on bed depth. Thus by analogy to the wear process described by eqn.5-8, we can write:

$$B_w = \frac{d_0^{1/2}}{K_c^{3/4} H^{1/2}} h_b^{5/4} (U - U_{mf}) \tag{5-10}$$

where h_b is the fluidised bed height and d_o is the distributor hole orifice size. We will now test the effect of granule properties on the attrition rate.

The materials shown in Table 5.1 were formed into bars and their wear rate against a rough surface measured (Ennis and Sunshine, 1993). Results are shown in figure 5.12a. The dependence on fracture toughness and hardness matches that predicted by eqn. 5-10. Figure 5.12b shows the attrition rate of granules of these materials in a fluid bed. The relationship between attrition rate and agglomerate properties is remarkably similar. This confirms that wear is a key mechanism of fines production in fluid beds and that the relative attrition rate can be accurately predicted if some fundamental mechanical properties of the agglomerate material are known.

5.3. Particle Design for Controlling Attrition and Breakage

Table 5.2 summarises typical changes in formulation and operating variables which can be used to minimise granule attrition and breakage. Granule attrition will increase with decreasing macroscopic fracture toughness, increasing flaw size and increasing intensity in handling. Figure 5.13 illustrates the point. The attrition rate for several formulations is a strong function of the granule density (porosity). The effect of porosity on macroscopic properties is two-fold. Firstly, macroscopic fracture toughness increases with decreasing porosity due to the increased number of bonds per unit volume and the increased strength of the bonds. Secondly, the maximum flaw size increases with porosity. However, even at the same porosity, there are very significant differences between the attrition resistance of different formulations. No amount of granule densification can completely overcome poor bonding between the particles.

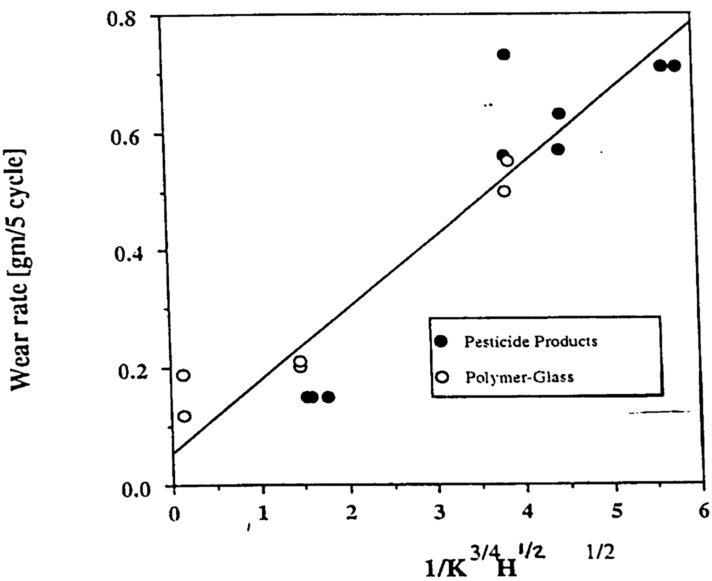
Attrition resistance can be improved by a combined effort in formulation design and process choice.

5.3.1. Effect of Process Variables

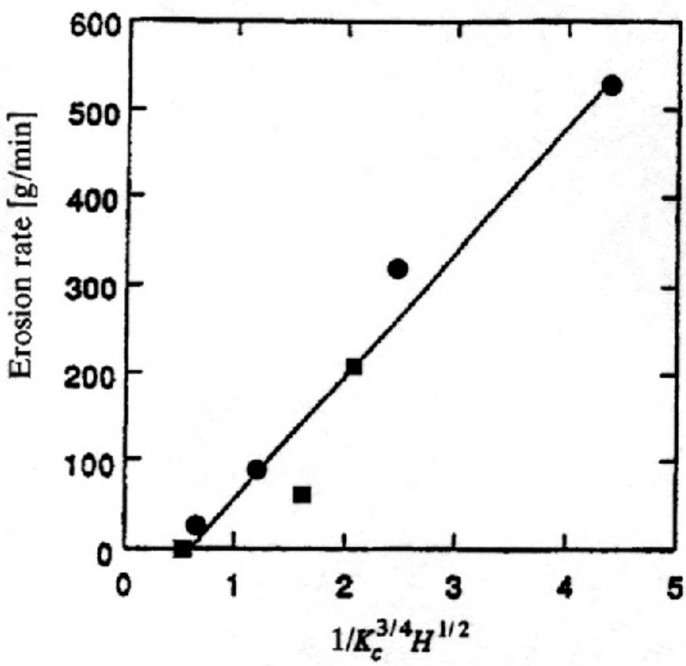
For granulation processes *without* simultaneous drying the effect of process intensity is quite clear. Granule attrition resistance is increased with increasing process intensity because the granule porosity is decreased (see chapter 4). Increasing residence time in the granulator may also decrease the granule porosity if the resistance of the *wet* granule to densification is large.

For granulators with simultaneous drying, the picture is much cloudier. Both wet and dry granules exist in the granulator. Increasing process intensity will both increase the density of wet granules (decreasing attrition) and increase the rate of attrition of dry granules in the granulator. Modifying bed temperature, spray rate or spray drop size to increase the drying (consolidation) time may be a more rewarding alternative in this case.

Note that the range of porosities that can be achieved for any one granulator is quite limited. Major changes to porosity (and strength) may require more intense processes such as high shear mixers or compaction processes.



(a)



(b)

Figure 5.12. Erosion rates of agglomerate materials (a) during wear test on formed bars; (b) attrition of granules in a fluidised bed

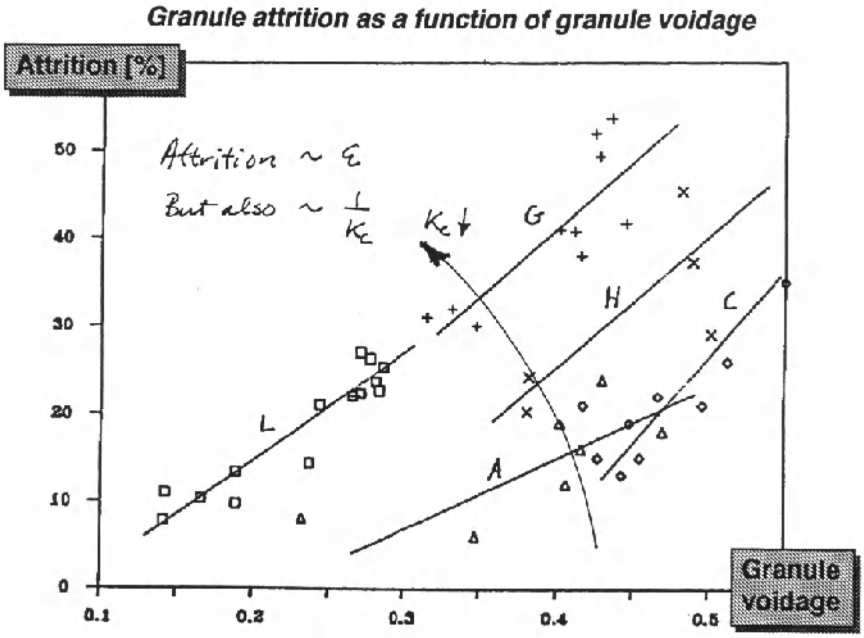


Figure 5.13. Effect of granule porosity on fluidised bed attrition of five agricultural chemical formulations

5.3.2. Effect of Formulation Properties

Both granule fracture toughness and hardness are strongly influenced by the compatibility of the binder with the primary particles and the elastic/plastic properties of the binder. If the binder has poor compatibility with the particles, adhesive failure will occur with the bond between the binder and the particle breaking. With high compatibility binders, cohesive failure (within the bond) may be the failure mechanism (figure 5.14). In principle, macroscopic granule properties can be derived from knowledge of the microscopic adhesive and cohesive failure modes. The microscopic fracture toughness of a bond between particles can be measured by nanoindentation techniques. Where polymeric binders are used, the adhesion between polymer and a surface similar to the particle can be measured by a peel test.

Feed powder size will also influence macroscopic properties. In general, reducing particle size increases attrition resistance as (a) the number of bonds per unit volume is increased, and (b) the size of pores (flaws) is smaller. Fine powder granules are more difficult to compact, however, so that the precise effect of feed powder size is very dependent on the consolidation processes in the granulation.

Figure 5.14. Cohesive failure in the Klucel bridge between glass spheres within a granule

Table 5.2. Controlling Breakage in Granulation Processes

Typical changes in material or operating variables which minimize breakage	*Appropriate routes to alter variable through formulation changes*	*Appropriate routes to alter variable through process changes*
Increase fracture toughness Maximize overall bond strength Minimize agglomerate voidage	Increase binder concentration or change binder. Bond strength strongly influenced by formulation and compatibility of binder with primary particles.	Decrease binder viscosity to increase agglomerate consolidation by altering process temperatures (usually decrease for systems with simultaneous drying). Increase bed-agitation intensity (e.g., increase impeller speed, increase bed height) to increase agglomerate consolidation. Increase granulation-residence time to increase agglomerate consolidation, but minimize drying time.
Increase hardness to reduce wear Minimize binder plasticity Minimize agglomerate voidage	Increase binder concentration or change binder. Binder plasticity strongly influenced by binder type.	See above effects which decrease agglomerate voidage.
Decrease hardness to reduce fragmentation Maximize binder plasticity Maximize agglomerate voidate	Change binder. Binder plasticity strongly influenced by binder type. Apply coating to alter surface hardness.	Reverse the above effects to increase agglomerate voidage.
Decrease load to reduce wear	Lower-formulation density.	Decrease bed-agitation and compaction forces (e.g., mixer impeller speed, fluid-bed height, bed weight, fluid-bed excess gas velocity, drum rotation speed).
Decrease contact displacement to reduce wear		Decrease contacting by lowering mixing and collision frequency (eg., mixer impeller speed, fluid-bed excess gas velocity, drum rotation speed).
Decrease impact velocity to reduce fragmentation	Lower-formulation density.	Decrease bed-agitation intensity (e.g., mixer impeller speed, fluid-bed excess gas velocity, drum rotation speed). Also strongly influenced by distributor-plate design in fluid-beds, or impeller and chopper design in mixers.

5.3.3. Controlling Granule Dispersibility

In some applications, the end use of granules requires them to be redispersed in water. In this case, granule need to be easily wetted (chapter 3). Once wet, granules need to rupture quickly under as additives in the formulation such as clays expand when wet. Clearly the wetting properties of the granule are important, but also the energy required to rupture the granule under internal stress G_c (eqn 5.6). Figure 5.15 shows (1) increasing granule porosity improves dispersibility, and (2) the ranking of formulations for dispersibility is almost the reverse of their ranking for attrition resistance (figure 5.13). Achieving a good ce between granule handling properties and their redispersion properties is tricky. In this respect, the role of wetting and swelling agents in the formulation of water dispersible granules is very important to achieve fast dispersing granules that still maintain reasonable dry attrition resistance for handling.

5.4. Concluding Comments

We are still "scratching the surface" in our understanding of the fundamentals of granule breakage. Further research work is needed and the key is understanding the fracture mechanics of both wet and dry granules. Nevertheless, we can still use our limited understanding for particle design and process scale up as has been shown by several examples in this chapter. Beware of empirical attrition tests. Use results from these tests with great care and try to match the test to the expected handling conditions for the granules.

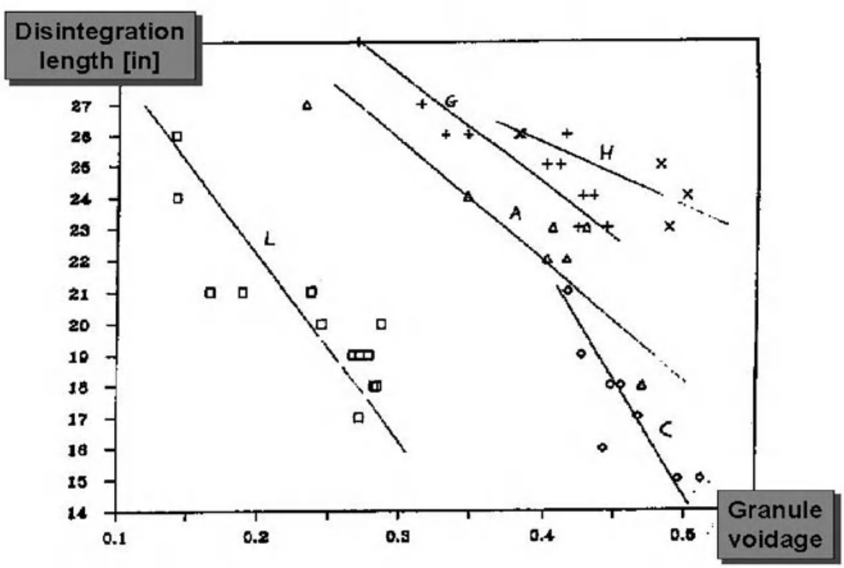


Figure 5.15. Dispersibility of five agricultural chemical formulations – effect of granule porosity

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5.6. Nomenclature

A indenter contact area (m^2)
B_w attrition rate ($kg\ s^{-1}$)
C crack length (m)
d_0 orifice diameter (m)
d_p particle diameter (m)
d_I indenter diameter (m)
d_{32} Sauter mean size of particles (m)
E Young's modulus (Pa)
H hardness (Pa)
h_b fluid bed height (m)
K_c Fracture toughness ($Pa.m^{1/2}$)
l wear displacement of indenter (m)

ied load (N)
r_p particle radius (m)
St^*_{def} critical Stokes deformation number (-)
St_{def} Stokes deformation number (-)
St_v Stokes viscous number (-)
u impact velocity (m s^{-1})
U fluidization velocity (m s^{-1})
U_{mf} minimum fluidization velocity (m s^{-1})
V volumetric attrition rate (m^3 s^{-1})
Y geometric factor (-)
ρ_p particle density (kg m^{-3})
τ shear stress (Pa)
τ_y yield stress (Pa)

CHAPTER 6

MATHEMATICAL MODELLING OF AGGLOMERATION PROCESSES

Figure 1.5 showed that a granulator can be analysed in several scales. So far, we have analysed important granulation rate processes at the particle and volume-of-powder scale (chapters 3 to 5). In this chapter, we describe the balance equations that describe the evolution of granule property distributions in the granulator ie. granulator scale analysis. A key tool in this analysis is the population balance.

The population balance (PB) is a rate equation which follows the change in number of particles of a given property(ies). The equation is analogous to the mass balance (continuity) equation for a chemical species which is familiar to chemical engineers. It includes convective (flow) terms, accumulation and a kinetic expression for each mechanism which changes the particle property similar to chemical reaction kinetics in reactor design (see figure 6.1). By combining the PB with overall mass and energy balances and information on mixing patterns in the granulator, we can apply traditional process engineering analysis to the granulation unit operation.

The population balance was introduced as a general equation for particulate systems independently by Hulbert and Katz (1964) and Randolph and Larsen (1962). It is a powerful tool with uses including:

- critical evaluation of data to determine controlling mechanisms
- in design, to predict the mean size and size distribution of product particles
- sensitivity analysis: to analyse quantitatively the effect of changes in operating conditions or feed variables on product quality
- optimisation and process control

Despite its power, the equation has been little used outside academic circles for attacking granulation problems. This contrasts markedly to crystallisation and grinding, where the population balance is a widely used design tool. There are two key historical reasons for this:

- the usefulness of the models is very dependent on knowledge of the kinetic parameters. These have been difficult to predict for granulation systems.
- solution to the equations is difficult, especially for coalescence problems.

These historical problems are now largely overcome so that the population balance has come of age as a tool for engineers and technologists dealing with granulation processes. In this chapter the population balance is derived, starting from the definitions of property distributions discussed in chapter 2. Rate expressions for the important granulation processes are developed and the rate parameters related to the quantitative understanding developed in chapters 3 to 5. Analytical and numerical solution techniques are briefly described and application case studies are presented

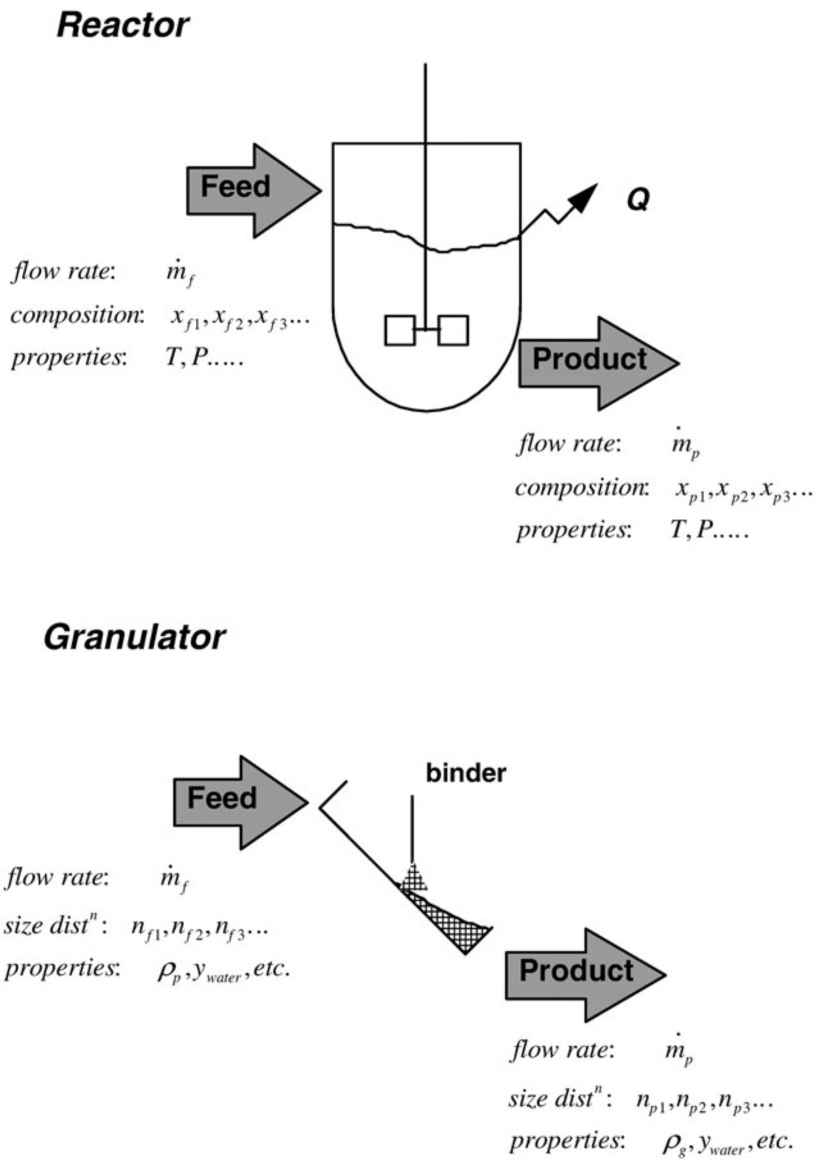


Figure 6.1. Analogy between a granulator and a chemical reactor

6.1. Derivation of the Macroscopic Population Balance

We start by deriving the macroscopic population balance for following the distribution of a single property, size.

Consider a well mixed granulator as shown in figure 6.2. We want to keep track of how the number of particles between sizes x and $x+dx$ in the process change with time. The number of particles of this size $n(x)dx$ within the process control volume V can change for a number of reasons. Let us divide these into three categories:

(a) Flows into and out of the control volume:

- flow of particles of size x into the process at a rate of $\dot{Q}_{in}n_{in}(x)dx$
- flow of particles of size x out of the process at a rate of $\dot{Q}_{ex}n_{ex}(x)dx$

(b) Sudden processes that generate or destroy particles of size x

- the sudden "birth" of new particles of size x at a rate of $V\dot{b}(x)dx$ eg. a new nuclei particle
- the sudden "death" of particles of size x at a rate of $V\dot{d}(x)dx$ eg. a particle of size x coalesces with another particle and "disappears"

(c) Differential growth processes eg. growth by layering

- a particle slightly smaller than size x grows to that size
- a particle of size x grows to give a particle that is slightly larger in size

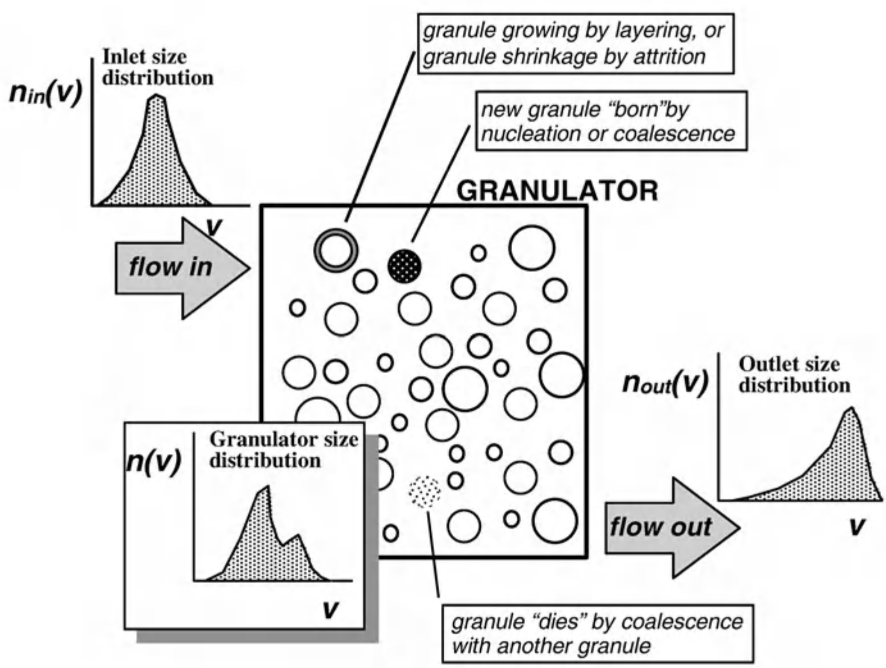


Figure 6.2. Changes to the particle size distribution in a particulate process

he growth term needs some explanation. It is a gradual (differential) process, not sudden like the birth and death terms. We can define the differential growth rate of an individual granule as:

$$\frac{dx}{dt} = G \tag{6-1}$$

We can think of growth as a convective process analogous to flow of a fluid down a pipe. Just as fluid will flow down the pipe at some velocity *v*, growing particles move along the size axis with a "velocity" *G* (see figure 6.3). The flux of particles into a size range between *x* and *x+dx* is from smaller sizes is $VnG\big|_x$ and the flux leaving the size range to larger sizes is $VnG\big|_{x+dx}$.

The rate of accumulation of particles in the size range between *x* and *x+dx* by all these processes (flow, sudden birth and death, growth) is then (figure 6.3):

$$\underset{(accumulat^n)}{\frac{\partial Vn(x,t)dx}{\partial t}} = \underset{(flow\ in\ and\ out)}{\dot{Q}_{in}n_{in}(x)dx - \dot{Q}_{ex}n_{ex}(x)dx} + \underset{(growth\ flux)}{VnG\big|_x - VnG\big|_{x+dx}} + \underset{(birth)}{V\dot{b}(x)dx} - \underset{(death)}{V\dot{d}(x)dx} \tag{6-2}$$

Dividing by *dx* and taking the limit as $dx \to 0$ gives:

$$\frac{\partial Vn(x,t)}{\partial t} = \dot{Q}_{in}n_{in}(x) - \dot{Q}_{ex}n_{ex}(x) - V\frac{\partial Gn(x,t)}{\partial x} + V\dot{b}(x) - V\dot{d}(x) \tag{6-3}$$

This is the ***one-dimensional macroscopic population balance*** for a control volume with the particular example of particle size as the property of interest. Note that it is written in terms of volumetric flows and un-normalised size distributions e.g. the inlet volumetric flow rate is $\dot{Q}_{in}\lfloor m^3 s^{-1}\rfloor$ and the inlet size distribution is $n_{in}(x)[no.m^{-4}]$.

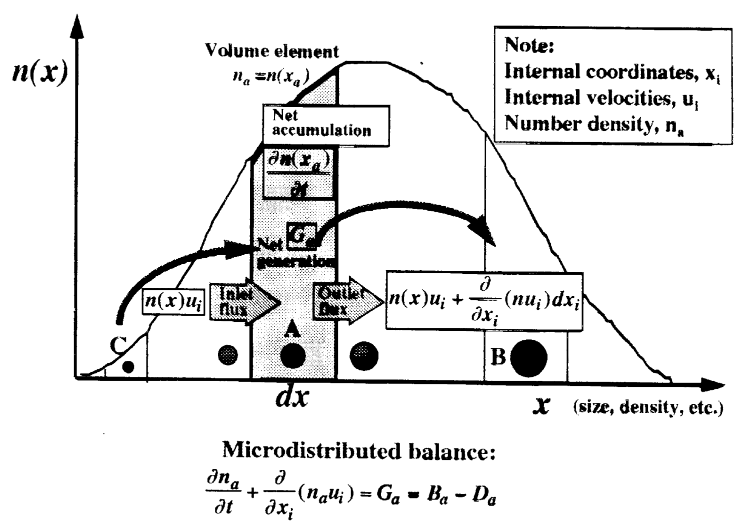


Figure 6.3. The influence of different granulation mechanisms on the granule size distribution

Equation 6-3 can be written in a number of equivalent forms:

- We can rewrite eqn. 6-3 in terms of number flows and normalised size distributions eg. the inlet number flow rate is $\dot{\upsilon}_{in}[no.s^{-1}]$ and the normalised inlet size distribution is $f_{in}(x)[m^{-1}]$. Thus $\dot{Q}_{in}n_{in}(x) = \dot{\upsilon}_{in}f_{in}(x)$ and the full population balance is:

$$\frac{\partial VN_T f(x,t)}{\partial t} = \dot{\upsilon}_{in}f_{in}(x) - \dot{\upsilon}_{ex}f_{ex}(x) - \frac{V\partial GN_T f(x,t)}{\partial x} + \dot{B}f_B(x) - \dot{D}f_D(x) \qquad (6\text{-}4)$$

- Equation 6-3 is written in terms of the linear size as the particle property. The balance is easily rewritten in terms of <u>any</u> property of the particle e.g. the distribution of particle volume $n(v)$, $f(v)$.

- Equation 6-3 is suitable for analysing well mixed vessels as it implicitly assumes particle properties are constant within the control volume. It is also possible to derive a much more general ***multi-dimensional microscopic population balance***. The microscopic balance is needed to examine processes where the particle property distribution varies with position in the vessel ie. distributed parameter systems. The adjective “multi-dimensional” applies because the equation can, in principle, be used to follow any combination of particle properties that have distributions (size, concentration, porosity, strength, etc.). This is the most general form of the population balance and is written:

$$\frac{\partial n}{\partial t} + \nabla.\bar{v}_e n + \nabla.\bar{v}_i n + \dot{d} - \dot{b} = 0 \qquad (6\text{-}5)$$

The particle distribution n is a function of the particle position with respect to its ***external*** co-ordinates (x,y,z) and its ***internal*** co-ordinates or properties which define the state of the particle (size, concentration, porosity, strength, etc.). $\bar{v}_e$ is a vector of the components of the particle external velocity in the $x,y,$ and z directions. $\bar{v}_i$ is the vector of velocity components with respect to the internal co-ordinates eg. G is the velocity component with respect to particle size as discussed above. The internal and external co-ordinates define the ***phase-space*** for the particle distribution n.

Equation 6-5 is the most general form of the population balance. It is next to useless as written. However, many very useful simplifications can be derived from this general equation including the macroscopic population balance (eqn. 6-3). Another important simplification is the microscopic population balance in one internal coordinate for a plug flow :

$$\frac{\partial n}{\partial t} + \frac{\partial un}{\partial l} + \frac{\partial Gn}{\partial x} - \dot{b} + \dot{d} = 0$$
$$where \;\; n = n(x,l,t) \qquad (6\text{-}6)$$

where u is the velocity through the vessel and l is the distance from the start of the vessel.

e population balance (eqns. 6-3 to 6-6) provides the mathematical framework for analysing all particulate systems. It is as important as the mass balance and the energy balance. In combination with appropriate kinetic expressions, these balances define the system for process engineering analysis. To make it useful for granulation processes we need to:

1. Write kinetic expressions for all key granulation rate processes.
2. Link the rate constants in these mechanisms to measurable formulation properties and process parameters.

6.1.1. Moments of the PB

Often, useful information on the granule size distribution can be obtained by taking the moments of the population balance. The *kth* moment of eqn. 6-4 is

$$\int_0^\infty x^k \left(eqn.6 - 3\right)dx \Rightarrow$$
$$\frac{dV\mu_k'}{dt} = \dot{Q}_{in}\mu_{k,in}' - \dot{Q}_{ex}\mu_{k,ex}' - \int_0^\infty x^k \frac{\partial VGn(x,t)}{\partial x}dx + V\mu_{k,B}' - V\mu_{k,D}' \tag{6-7}$$

The growth term can be simplified further if G is not a function of size. This is called ***size independent growth***. In this case, the growth term simplifies to:

$$\int_0^\infty x^k \frac{\partial Gn(x,t)}{\partial x}dx = VG\int_0^\infty x^k \frac{\partial n(x,t)}{\partial x}dx$$
$$= -VGk\mu_{k-1}' \tag{6-8}$$

Substituting into eqn. 6-7 gives:

$$\frac{dV\mu_k'}{dt} = \dot{Q}\mu_{k,\,in}' - \dot{Q}\mu_{k,\,ex}' + VGk\mu_{k-1}' + V\mu_{k,\,B}' - V\mu_{k,\,D}' \tag{6-9}$$

Note that eqn. 6-9 is an ODE and therefore much easier to solve than eqn. 6-4. Let us look at some specific examples:

- k=0: Eqn. 6-9 reduces to

$$\frac{dN_T V}{dt} = \dot{Q}_{in}\mu'_{0,in} - \dot{Q}_{ex}\mu'_{0,ex} + V\dot{B} - V\dot{D} = \dot{\upsilon}_{in} - \dot{\upsilon}_{ex} + V\dot{B} - V\dot{D} \tag{6-10}$$

This is the total number balance. As you would expect, the total number of particles in the control volume is affected by the numbers flowing in and out, dying and being born. Note that the total number of particles is not affected by particles growing.

- k=1

$$\frac{dN_T V \bar{x}_{1,0}}{dt} = \dot{\upsilon}_{in}\,\bar{x}_{1,0,in} - \dot{\upsilon}_{ex}\,\bar{x}_{1,0,ex} + N_T VG + V\dot{B}\,\bar{x}_{1,0,B} - V\dot{D}\,\bar{x}_{1,0,D} \tag{6-11}$$

Solution of eqn. 6-11 gives the rate of change of mean size with time.

- k=3: The third moment is important because the total mass of particles is $N_T\rho_p\alpha_v\mu_3$. Thus

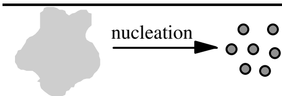
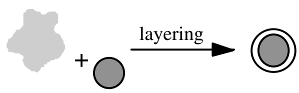
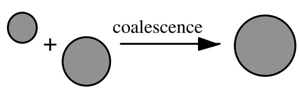
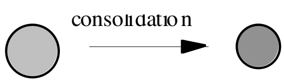
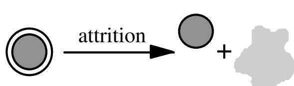
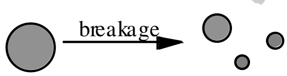
$$\frac{dM_T V}{dt} = \dot{m}_{in} - \dot{m}_{ex} + 3GN\rho_p\alpha_v\mu'_2 + \dot{m}_B - \dot{m}_D \tag{6-12}$$

Equation 6-12 is the mass balance for the particulate system.

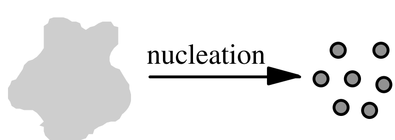
6.2. Kinetic Expressions for Granulation Rate Processes

Table 6.1 lists the granulation rate processes discussed in chapters 3 to 5 and their effect on granule size and porosity distributions. For PB modelling, "growth" as described in chapter 4 is divided into two mechanisms – layering (differential growth) and coalescence (birth and death terms). We now describe the kinetic expressions for each of these processes.

Table 6.1. Effect of granulation rate processes on granule size and porosity distributions

Mechanism	*Changes number of granules?*	*Changes mass of granules?*	*Change porosity of granules?*	*Discrete or differential?*
	yes	yes	no	discrete
	no	yes	no	differential
	yes	no	no	discrete
	no	no	yes	differential
	no	yes	no	differential
	yes	no	no	discrete

6.2.1. Nucleation



Nucleation is the formation of new granules from liquid or fine powder feed. Nucleation increases both the mass and number of the granules. For the case where new granules are produced by liquid feed which dries or solidifies, the nucleation rate is given by the new feed droplet size distribution n_s and the volumetric spray rate S:

$$V\dot{b}(v)_{nuc} = Sn_s(v) \tag{6-13}$$

Spray drying and prilling represent the extreme case in which no other rate processes are present and the granule size distribution is totally dominated by the drop size distribution. In granulation from melts or solution (common for fertilisers and inorganic chemicals), nucleation from the liquid feed may be present in combination with other processes.

In processes where new powder feed has a much smaller particle size than the smallest granular product, the feed powder can be considered as a "continuous phase" which can nucleate to form new granules (Sastry and Fuerstenau, 1975). The size of the nuclei is then related to nucleation regime. In the drop controlled regime, the size of the nuclei is larger than the droplet size and depends on the porosity of the powder bed and saturation of nuclei granule formed (see chapter 3). We can represent the nuclei size distribution as:

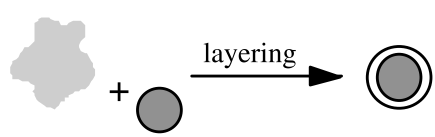
$$V\dot{b}(s\varepsilon v)_{nuc} = Sn_s(v) \tag{6-14}$$

The size of nuclei formed is a complex function of the powder and fluid properties (chapter 3). Nevertheless, there is a one to one correspondence to the drop size distribution generated in the spray zone. In fluid bed granulation of fine powders, final granule size distribution may be dominated by nucleation, and hence by the drop size distribution.

In the mechanically dispersed regime, the nuclei size distribution is harder to predict, but clearly will relate to the intensity of mechanical agitation ie. impeller or chopper speed.

Where the feed has a broad size distribution which overlaps the product size distribution, arbitrary division between the "powder" (continuous phase) and granules is neither sensible nor logical. This is often the case when granules are post treated (drying, sintering) **before** recycling e.g. fertiliser granulation, iron ore sinter feed granulation. In these cases, nucleation is rarely a significant mechanism.

6.2.2. Layering



Layering increases granule size and mass by the progressive coating of new material onto existing granules, but does not alter the number of granules in the system. As with nucleation, the new feed may be in liquid form (where there is simultaneous drying or cooling) or may be present as a fine powder. Where the feed is a powder, the process is sometimes called **pseudo-layerin**g or **snowballing**.

The key model parameter for growth by layering is the differential growth rate which can be written in terms of granule size $G(x)$ or granule volume $G^*(v)$. These two growth rates are related:

$$G^*(v) = 3\alpha_v x^2 G(x) \tag{6-15}$$

often reasonable to assume a linear growth rate $G(x)$ is independent of granule size. This implies that each granule has the same exposure to new feed material eg. equal exposure time in the liquid spray zone for liquid feeds. This will not be true if there is segregation on the basis of granule size or volume (Liu and Litster, 1993). Size-independent linear growth rate implies that the volumetric growth rate $G^*(v)$ is proportional to projected granule surface area:

$$G^*(v) \propto x^2 \propto v^{2/3} \tag{6-16}$$

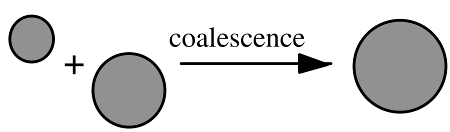
If all new feed goes to layered growth (no nucleation), the growth rate can be simply related via a mass balance to the volumetric feed rate:

$$\dot{V}_{feed} = (1-\varepsilon)\int_0^\infty G^*(v)n(v)dv \tag{6-17a}$$

$$\dot{V}_{feed} = (1-\varepsilon)\int_0^\infty 3\alpha_v x^2 G(x)n(x)dx \tag{6-17b}$$

where $\dot{V}_{feed}$ is the volumetric flow rate of new feed and ε is the granule porosity.

6.2.3. Coalescence



Coalescence is the formation of a new granule (birth term) from the coalescence of two smaller granules (death term). Coalescence decreases the number of granules and but does not change the total volume or mass of granules.

To model coalescence processes, it is convenient to use the volume based number distribution $n(v)$ because granule volume is additive in a binary coalescence event, whereas granule size is not. Consider the coalescence of two granules of volumes u and u-v to give a new granule of volume v. We assume the coalescence rate is proportional to the number density of granules of each size range and inversely proportional to the total number of granules (Kapur and Fuerstenau, 1968; Sastry, 1981). Thus, the rate of production new granules of size v as:

$$\left.\frac{dn(v)}{dt}\right|_{u,v-u} = \beta(u,v-u,t)\frac{n(u)n(v-u)}{N_T}du \tag{6-18}$$

$\beta(u,v\text{-}u,t)$ is the coalescence kernel or rate constant for the process. To obtain the total birth rate of granules of volume v , we integrate eqn. 6-18 over the range 0 to v:

$$\dot{b}(v) = \frac{1}{2N_t} \int_0^v \beta(u, v-u, t) n(u) n(v-u) du \tag{6-19}$$

The *1/2* in front of the integral term is to avoid double counting of combinations.

Similarly, we can write the death term for the disappearance of granules of volume *v* due to successful coalescence with granules of any other size:

$$\dot{d}(v) = \frac{1}{N_t} \int_0^\infty \beta(v, u, t) n(v) n(u) du \tag{6-20}$$

The key parameter is the coalescence kernel $\beta(u,v,t)$. The kernel dictates the overall rate of coalescence, as well as the effect of granule size on the coalescence rate. Several empirical kernels have been proposed and used for granule coalescence (table 6.2). The **order** of the kernel has a major effect on the shape and evolution of the granule size distribution eg. figure 6.4. In general, the higher the kernel order, the broader the granule size distribution that results.

All the kernels in table 6.2 are empirical, or semi-empirical and must be fitted to plant or laboratory data. The kernel proposed by Adetayo and Ennis (1998) is consistent with the granulation regime analysis described in section 4.4 and is therefore recommended:

$$\beta(u,v) = \begin{cases} k_o, w < w^* \\ 0, w > w^* \end{cases} \quad where \quad w = \frac{(uv)^a}{(u+v)^b} \tag{6-21}$$

where w^* is the **critical average granule volume** in a collision. Collisions lead to successful coalescence only if $w < w^*$. From the Stokes regime analysis for non-deformable systems *a=b=1* and w^* is given by:

$$w = \frac{(uv)}{(u+v)} = w^* = \frac{\pi}{6}\left(\frac{16\mu}{\rho u_0} St^*\right)^3 \tag{6-22}$$

For granule growth in the non-inertial regime where $St << St^*$, this kernel collapses to the simple random or size independent kernel $\beta = \beta_o$. Coalescence occurs only in the non-inertial regime and stops abruptly when $St_e = St^*(w = w^*)$.

Modelling coalescence where deformation is significant is more difficult. Adetayo and Ennis assumed the model parameters *a* and *b* would vary with granule deformability and allowed them to vary to fit granulation data. Note that to be dimensionally consistent, *2b-a=1*.

It should be possible, based on model for deformable granule collisions developed in section 4.5, to extend the coalescence kernel in eqn. 6-21 more rigorously. Liu and Litster (2002) have developed a physically based coalescence model for both deformable and non-deformable granules. Nevertheless, eqn. 6-21 gives a kernel based on the particle scale understanding developed in chapter 4 in which the effect of formulation properties and ss parameters is properly accounted and which will explain a wide range of experimental granulation data (see section 6.4.2 below).

Table 6.2. Coalescence Kernels for Granulation

Kernel	*Reference and Comments*
$\beta = \beta_o$	Kapur and Fuerstenau (1969), Size independent kernel.
$\beta = \beta_o \dfrac{(u+v)^a}{(uv)^b}$	Kapur, (1977) Preferential coalescence of limestone.
$\beta = \beta_o \dfrac{\left(u^{2/3} + v^{2/3}\right)}{1/u + 1/v}$	Sastry and Feurstenau (1975), Preferential balling of iron ore and limestone.
$\beta(u,v) = \begin{cases} \beta_o, w < w* \\ 0, w > w* \end{cases} \quad \mathrm{w} = \dfrac{(\mathrm{u}v)^{\mathrm{a}}}{(\mathrm{u}+v)^{\mathrm{b}}}$	Adetayo and Ennis (1997), Based on granulation regime analysis.

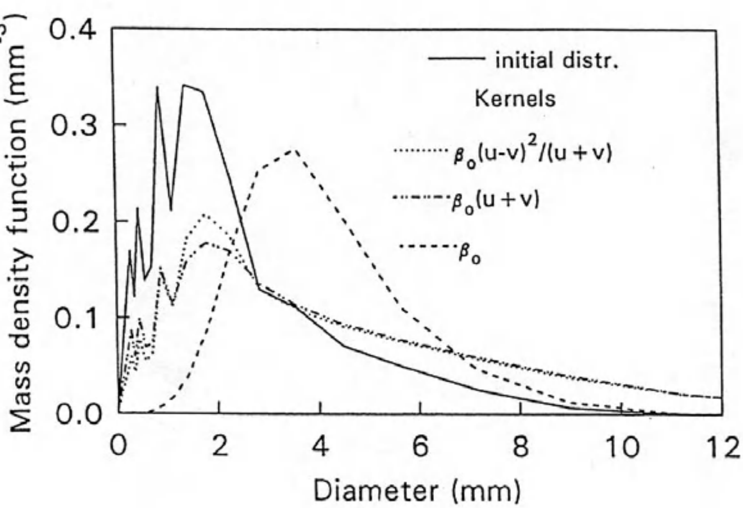
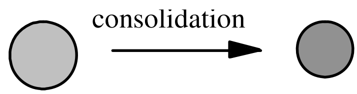


Figure 6.4. Effect of coalescence kernel on the shape of the granule size distribution. (Adetayo et al, 1995)

6.2.4. Consolidation



Consolidation causes a decrease in granule porosity with time. There is an accompanying slight decrease in granule size. This direct effect on granule size is usually negligible. However, the *indirect* effect on granule deformability and liquid saturation is very important. Granule saturation increases as the granule consolidates and the probability of coalescence is very sensitive to liquid saturation (chapter 4). For porous granules w^* in eqn. 6-21 will be a function of granule saturation, increasing suddenly as the granules become surface wet.

Ideally, we should write the PB for the two dimensional distribution of volume and porosity $n(v,\varepsilon)$ (Iveson, 2002). This is an area of on going research.

6.2.5. Attrition and Breakage

The wearing away of granule surface material by attrition is the direct opposite of layering. The key rate parameter is the attrition rate A which is essentially a negative growth rate. Attrition is an important mechanism when drying occurs simultaneously with granulation and granule velocities are high, *e.g.* fluidized beds and spouted beds. Based on the analysis presented in section 5.25, the attrition rate constant (a negative growth rate) for a fluidised bed can be related to bed conditions and formulation properties by the relationship:

$$A \propto \frac{d_o^{\frac{1}{2}}}{K_c^{\frac{3}{4}} H^{\frac{1}{2}}} h_b^{\frac{5}{4}} (U - U_{mf}) \tag{6-23}$$

For spouted beds, most attrition occurs in the spout and the attrition rate may be expressed as :

$$A \propto \frac{A_i U_i^3}{K_c} \tag{6-24}$$

where A_i and U_i are the inlet orifice area and gas velocity, respectively. Attrition rate also increases with increasing slurry feed rate (Liu and Litster, 1993).

Granule breakage by fragmentation is also possible, especially in high impact mixer granulators. In contrast to attrition, breakage is a discrete process, and results in birth and death terms in the PB. We model breakage as a first order rate process. There are two important parameters:

he first order breakage rate constant $k_b(v)$; and

2. The breakage function $b(u,v)$ which gives the number fraction of particles of size u to $u+du$ formed from the breakage of one granule of size v.

The birth and death terms for breakage are:

$$\dot{b}(v)_{br} = \int_{v}^{\infty} b(u,v) k_b(u) n(u) du \tag{6-25}$$

$$\dot{d}(v)_{br} = k_b(v) n(v) \tag{6-26}$$

Breakage is not the exact opposite to coalescence, because to model coalescence we assume only binary collisions, whereas as a breakage event may have more than two fragment particles. This is why two parameters are necessary for the breakage model.

There is relatively little quantitative work on granule breakage in granulators (see section 5.2). However, the population balance for breakage has been extensively used to model comminution in the mineral industry and this provides a good starting point for analysis (Austin and Rogers, 1985; Lynch , 1977).

Granule breakage will also occur in continuous granulation circuits where recycled oversize granules are crushed in a crusher eg. hammer mill or roll mill. These units can be modelled using the PB for breakage (see chapter 12).

6.2.6. The Complete Population Balance for Granulation

Table 6.3 summarises the PB expressions for all the granulation rate processes. The complete macroscopic PB for granulation in terms of the particle volume is:

$$\begin{aligned} \frac{\partial V n(v,t)}{\partial t} &= \dot{Q}_{in} n_{in}(v) - \dot{Q}_{ex} n_{ex}(v) - V \frac{\partial (G^* - A^*) n(v,t)}{\partial v} \\ &+ V(\dot{b}(v)_{nuc} + \dot{b}(v)_{coal} + \dot{b}(v)_{br} - \dot{d}(v)_{coal} - \dot{d}(v)_{br}) \end{aligned} \tag{6-27}$$

where $\dot{b}(v)_{coal}$, $\dot{b}(v)_{br}$, are given by eqns. 6-19, 6-20, 6-25 and 6-26.

Equation 6-27 is suitable for modelling a well mixed granulator. For plug flow granulators, the best starting point is the microscopic PB:

$$\begin{aligned} &\frac{\partial n(v,l,t)}{\partial t} + \frac{\partial u n(v,l,t)}{\partial l} + \frac{\partial (G^* - A^*) n(v,l,t)}{\partial v} \\ &- \dot{b}(v)_{nuc} - \dot{b}(v)_{coal} - \dot{b}(v)_{br} + \dot{d}(v)_{coal} + \dot{d}(v)_{br} = 0 \end{aligned} \tag{6-28}$$

The key model parameters in the PB for granulation are listed in table 6.3. At first glance, eqns. 6-27 and 6-28 look cumbersome and difficult with many parameters. However, in most cases only some of the rate processes are occurring in the granulator and the rate constants can be related to a fundamental understanding of physics of the process.

Section 6.4 gives examples of the use of the population balance in granulation applications, starting with very simple examples.

Table 6.3 Summary of PB expressions for granulation rate processes

	Macroscopic PB Expression	*Parameters*
nucleation	$V\dot{b}(v)_{nuc}$	$\dot{b}(v)_{nuc}$
growth (layering)	$-V\dfrac{\partial G^*(v).n(v,t)}{\partial v}$	$G^*(v)$
coalescence	$\dfrac{V}{2N_t}\int_0^v \beta(u,v-u,t)n(u,t)n(v-u,t)du$	$\beta(u,v,t)$
	$-\dfrac{V}{N_t}\int_0^\infty \beta(v,u,t)n(u,t)n(v,t)du$	
attrition	$-V\dfrac{\partial A(v).n(v,t)}{\partial v}$	$A(v)$
breakage	$V.\int_v^\infty b(u,v)k_b(v)n(u)du$	$b(u,v)$
	$-Vk_b(v)n(v)$	$k_b(v)$

6.3. Solution of the Population Balance Equation

6.3.1. Analytical Solutions

The population balance for granulation is a partial integro-differential equation. Solution of the PB is not trivial. Analytical solutions are available for only a limited number of special cases. Some of these of practical importance for granulation are summarized in table 6.4. For other analytical solutions, see Randolph and Larson (1991), Gelbart *et al.* (1990).

For batch granulation where the only growth mechanism is coalescence, at long times the size distribution may become **self-preserving**. The size distribution is self-preserving if the normalized size distributions $\varphi = \varphi(\eta)$ at long time are independent of mean size $\bar{v}$, or

$$\varphi = \varphi(\eta) \text{ only where}$$
$$\eta = v/\bar{v} \tag{6-29}$$
$$\bar{v} = \frac{\int_0^\infty v.n(v,t)dv}{N_T}$$

Table 6.4. Some Analytical Solutions to the Population Balance (Randolph and Larson, 1991, Gelbart et al., 1990)

Mixing state	*Mechanisms operating*	*Initial or inlet size distribution*	*Final or exit size distribution*
Batch	Layering only: G(x)=constant	Any initial size distribution, $n_0(x)$	$n(x) = n_0(x - \Delta x)$ where $\Delta x = Gt$
Continuous and well-mixed	Layering only: G(x)=constant	$n_{in}(x) = N_{in}\delta(x - x_{in})$	$n(x) = \frac{N_0 G}{\tau}\exp\left(-\frac{\tau(x - x_{in})}{G}\right)$
Batch	Coalescence only, size independent: $\beta(u,v) = \beta_o$	$n_o(v) = N_0\delta(v - v_o)$	$n(v) = \frac{N_0}{\bar{v}}\exp\left(-\frac{v}{\bar{v}}\right)$ where $\bar{v} = v_0 \exp\left(\frac{\beta_o t}{6}\right)$
Batch	Coalescence only, size independent: $\beta(u,v) = \beta_o$	$n_o(v) = \frac{N_0}{v_0}\exp\left(-\frac{v}{v_o}\right)$	$n(v) = \frac{4N_0}{v_0(N_0\beta_o t + 2)^2}\exp\left[\frac{-2\, v/v_0}{N_0\beta_o t + 2}\right]$

Analytical solutions for self-preserving growth do exist for some coalescence kernels (Sastry, 1978, Ramkrishna, 1985) eg. for the size independent kernel $\beta(u,v)=\beta_o$:

$$\varphi(\eta) = \exp(-\eta) \tag{6-30}$$

Self preserving size distributions are sometimes seen in practice (see figure 6.5). Roughly speaking, self-preserving growth implies that the spread of the size distribution increases in proportion to mean granule size, i.e. the spread is uniquely related to the mean of the distribution. The self preserving size distribution is independent of the initial size distribution.

Self preserving size distributions should be used with caution. Cumulative distribution plots against normalised size tend to disguise differences between distributions, rather than highlight them. The greater the number of distributions presented on a single plot, the easier it is to disguise trends! Frequency distribution plots are much more discerning.

Normalised moments should also be calculated and analysed for trends before committing to a "self preserving" solution.

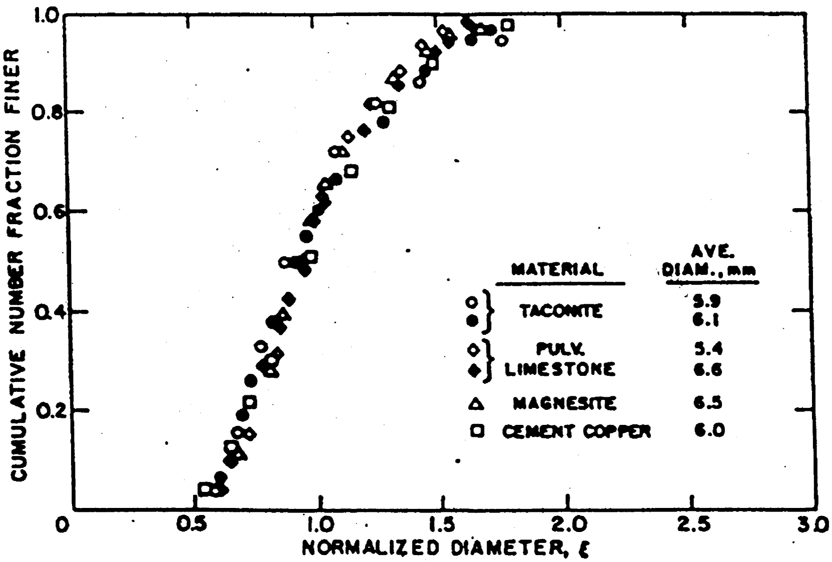


Figure 6.5. Self-preserving size distributions for batch coalescence in drum granulation, (Sastry, 1975)

6.3.2. Numerical Solutions

For many practical applications, numerical solutions to the population balance are necessary. Several numerical solution techniques have been proposed (Gelbard and Seinfeld, 1980; Batterham et al., 1981, Marchal et al., 1988 and Landgrebe and Pratsinis, 1990). It is usual to break the size range into discrete intervals and then solve the resulting series of ordinary differential equations. A geometric discretization reduces the number of size intervals (and equations) that are required eg. using a *qth* root of 2 series in granule volume:

$$v_j = 2^{1/q} v_{j-1} \tag{6-31}$$

where v_j is the volume of the largest granule in size interval j and q is an integer greater than or equal to 1. Using this discretisation, we convert eqn. 6-27 to a series of ODEs:

$$\frac{dN_i}{dt} = N_{in_i} - N_{e_i} + \left(\frac{dN_i}{dt}\right)_{growth} + \left(\frac{dN_i}{dt}\right)_{agg} + \left(\frac{dN_i}{dt}\right)_{break} \tag{6-32}$$

Litster *et al.* (1995) give the discretized terms for growth and coalescence,

$$
\begin{aligned}
\left(\frac{dN_i}{dt}\right)_{agg} = & \sum_{j=1}^{i-S(q)-1} \beta_{i-1,j} N_{i-1} N_j \frac{2^{(j-i+1)/q}}{2^{1/q}-1} \\
& + \sum_{k=2}^{q} \sum_{j=i-S(q-k+2)-k+1}^{i-S(q-k+1)-k} \beta_{i-k,j} N_{i-k} N_j \frac{2^{(j-i+1)/q} - 1 + 2^{-(k-1)/q}}{2^{1/q}-1} \\
& + \frac{1}{2} \beta_{i-q,i-q} N_{i-q}^2 \\
& + \sum_{k=2}^{q} \sum_{j=i-S(q-k+2)-k+2}^{i-S(q-k+1)-k+1} \beta_{i-k+1,j} N_{i-k+1} N_j \frac{-2^{(j-i)/q} + 2^{1/q} - 2^{-(k-1)/q}}{2^{1/q}-1} \\
& - \sum_{j=1}^{i-S(q)} \beta_{i,j} N_i N_j \frac{2^{(j-i)/q}}{2^{1/q}-1} \\
& - \sum_{j=i-S(q)+1}^{\infty} \beta_{i,j} N_i N_j
\end{aligned}
\qquad (6\text{-}33)
$$

$$
\left(\frac{dN_i}{dt}\right)_{growth} = \frac{2G}{(r+1)L_i}\left(\frac{r}{r^2-1} N_{i-1} + N_i - \frac{r}{r^2-1} N_{i+1}\right) \qquad (6\text{-}34)
$$

where

$$
r = 2^{1/q} \qquad (6\text{-}35)
$$

and q is an integral number. Hill and Ng (1995, 1996) give similar terms for breakage. The equation set generated by eqns. 6-31 to 6-35 can be solved numerically using standard numerical techniques for sets of ODEs. This discretisation is exactly correct for the zeroth and 3rd moments in particle size ie. the total number and total volume of particles. Figure 6.7 shows an example of the numerical solution of eqn. 6-32 in comparison with known analytical solution for one case study. Accuracy is increased (at the expense of computational time) by increasing the value of q. Given the uncertainty in laboratory and plant size distribution data used for parameter estimation or validation of such models, values of q in the range 1 to 4 are likely to be appropriate. For the special case of $q=1$ a number of the summation terms disappear to give a much simpler equation (Hounslow *et al.*, 1988). See the listed references for more details of the derivation of the discretised equations and further discussions of the accuracy of the solutions.

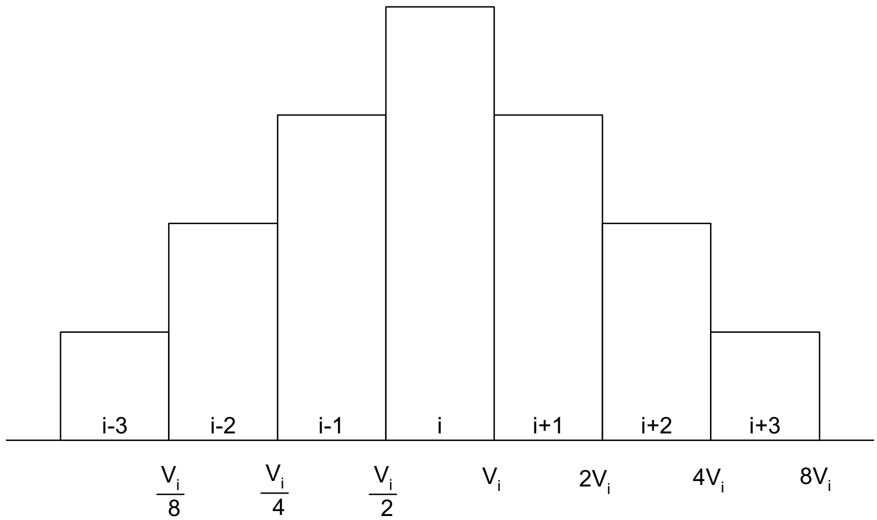


Figure 6.6. Geometric discretisation of the granule volume

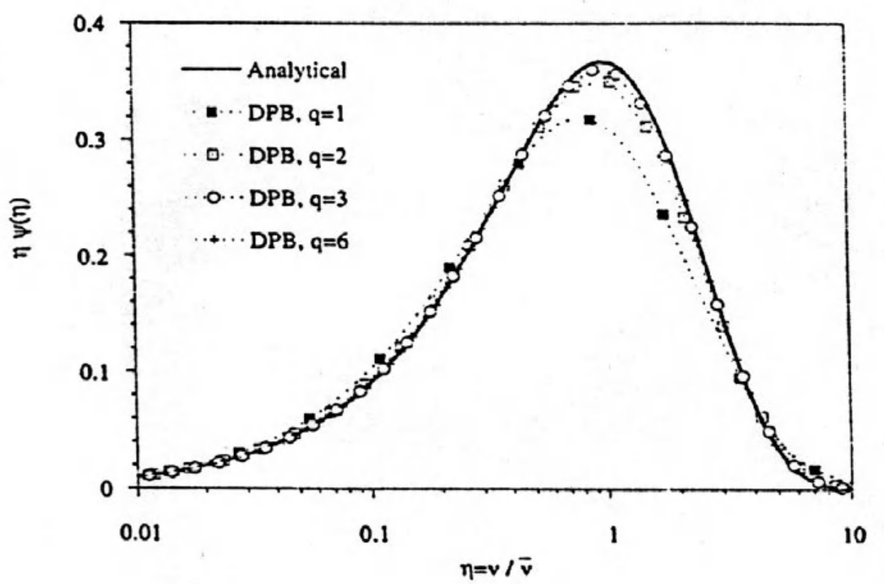


Figure 6.7. Comparison of numerical solution of the PB with a known analytical solution (Litster et al., 1995)

6.4. Example Applications of the Population Balance

In this section several examples of application of the PB for relatively simple case studies are shown.

6.4.1. Growth by Layering

Consider the simple case of growth by layering in a batch granulator. Examples of this simple application include "onion skin" layered growth of fertiliser granules from a melt or solution feed and coating of seed particles with a slurry or solution.

Assume all other granulation rate processes can be neglected. In this case, the PB in terms of size x, eqn. 6-27 reduces to:

$$\frac{\partial n(x,t)}{\partial x} = \frac{-\partial G(x)n(x,t)}{\partial t} \tag{6-36}$$

If the linear growth rate is size independent $G \neq G(x)$, the solution to eqn. 6-36 is a travelling wave equation with the granule size shifting forward in time with the shape of the granule size distribution unaltered (see table 6.4):

$$n(x) = n_o(x - \Delta x) \tag{6-37}$$

where $$\Delta x = \int_0^t G dt$$

Figure 6.8 shows an example of layered growth in drum granulation which shows this type of behaviour.

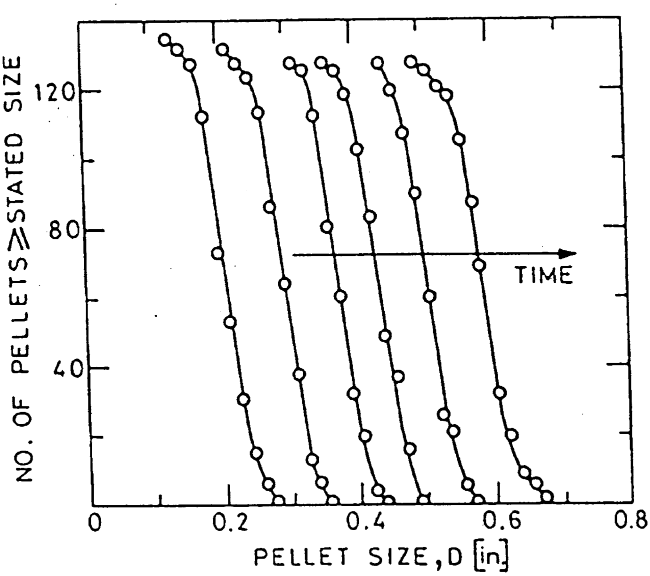


Figure 6.8. Batch drum growth of limestone pellets by layering with a linear size independent growth rate (Capes, 1967)

From a mass balance, the growth rate G can be directly related to rate of addition of new feed and the surface area of granules in the granulator (eqns. 6-16 and 6-17a).

Note that the mass growth rate is not size independent, but rather proportional to the particle surface area or 2/3 power of the particle mass. Figure 6.10 shows a comparison of experimental particle mass distributions after coating in a spouted bed with those predicted by solving eqns. 6-16, 6-17a, and 6-37. The model (with no fitting parameters) gives a good prediction of the coated particle mass and mass distribution.

The condition $G \neq G(x)$ only holds if each granule has equal exposure to the spray. Liu and Litster (1993) showed that larger particles were preferentially coated in a bottom sprayed spouted bed. This is probably because large particles spend more time in the spray zone due to differential acceleration effects (Litster, *et al.*, 1993). Figure 6.9 also shows model predictions for preferential growth. In this case, the differences are quite small because of the narrow initial size distribution. The effect will increase as the spread of the initial size distribution increases.

Particle segregation in the granulator can also cause preferential layering. This can be used to advantage if the nozzle is positioned to spray liquid feed onto the smaller particles. This approach is used in fluidised drum granulators (Litster and Sarwono, 1996). The PB is a useful way to predict the extent to which these approaches can be used to control the granule size distribution. Segregation in fluidised bed granulators also causes variation in granule size and coating thickness (Wnukowski and Setterwall, 1989; Maronga, 1998).

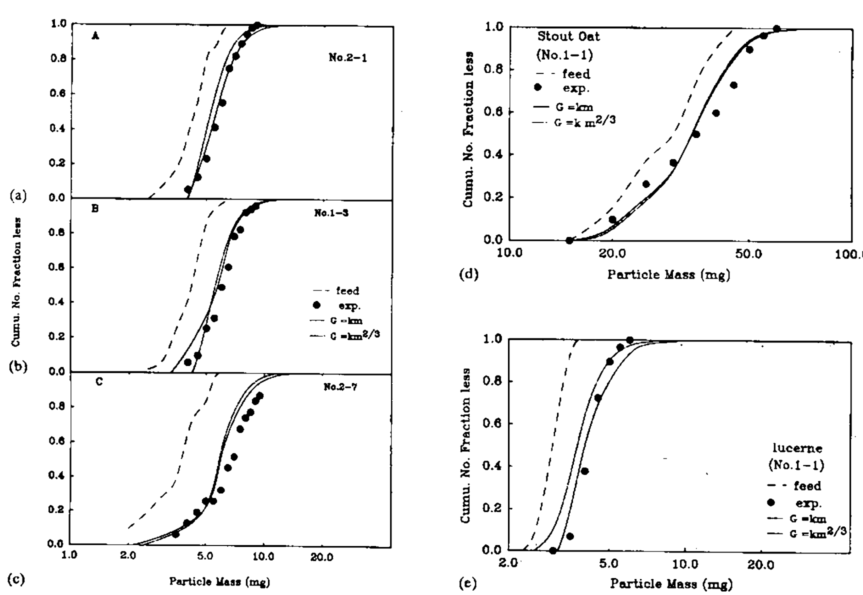


Figure 6.9. Comparison of experimental particle mass distributions with those calculated from the PB model for growth. Examples are for seeds coated with Monocalcium phosphate fertiliser (a), (b) and (c) rape seed, (d) stout oat, (e) lucern. (Liu and Litster, 1993)

. Growth by Coalescence

Now consider the case where the change in the granule size distribution is dominated by coalescence and other size change mechanisms can be neglected. For batch granulation, the PB becomes:

$$\frac{dn(v,t)}{dt} = \frac{1}{2N_T}\int_0^V \beta(v, v-u)\, n(v)n(v-u)du - \frac{1}{N_T}\int_0^\infty \beta(u,v)n(u)n(v)du \tag{6-40}$$

The coalescence kernel β is given by equation 6-21. Consider the case for non-inertial growth. Here $w < w^*$ for all collisions and $\beta(\mathrm{u,v}) = \beta_0$ ie. a size independent kernel. For this case, we can calculate the moments solution to the population balance. The zeroth moment of the PB is (Adetayo et al, 1995):

$$\frac{dN_T}{dt} = -\frac{1}{2}\beta_o N_T \tag{6-41}$$

$$\Rightarrow N_T = N_{T_0} \cdot \exp\left(-\frac{\beta_o t}{2}\right) \tag{6-42}$$

Thus, the number volume mean size $\bar{v}_{10}$ is given by:

$$\bar{v}_{10} = \bar{v}_{10,_0} \exp\left(\frac{\beta_o t}{2}\right) \tag{6-43}$$

Equations 6-41 to 6-43 show that the number of granules will decrease exponentially with time and the granule mean size will increase exponentially. Figure 6.10 shows an experimental example of exponential growth in the non-inertial regime.

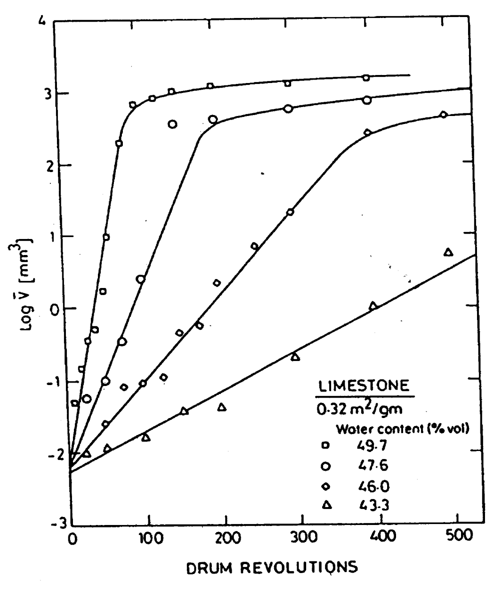


Figure 6.10. Batch drum growth of limestone by coalescence. Note granule size increases exponentially with time in the first stage of non-inertial growth (Kapur, 1978)

Figure 6.10 shows a sharp break point in the growth curve corresponding to the transition from inertial to non-inertial growth. This growth behaviour is well modelled via the Adetayo and Ennis kernel (eqn. 6-21), although numerical solution of the PB is now required. Figure 6.11 shows numerical solutions varying the coalescence rate constant β_o and the cut off size w^* which give very similar behaviour to the experimental data in figure 6.10.

However, this is not simply a data fitting exercise. The variation in w^* with formulation properties is predictable (eqn. 6-22). We have already demonstrated in section 4.3.4 how this combination of the population balance with growth physics was used to predict granulation behaviour of different fertiliser salts.

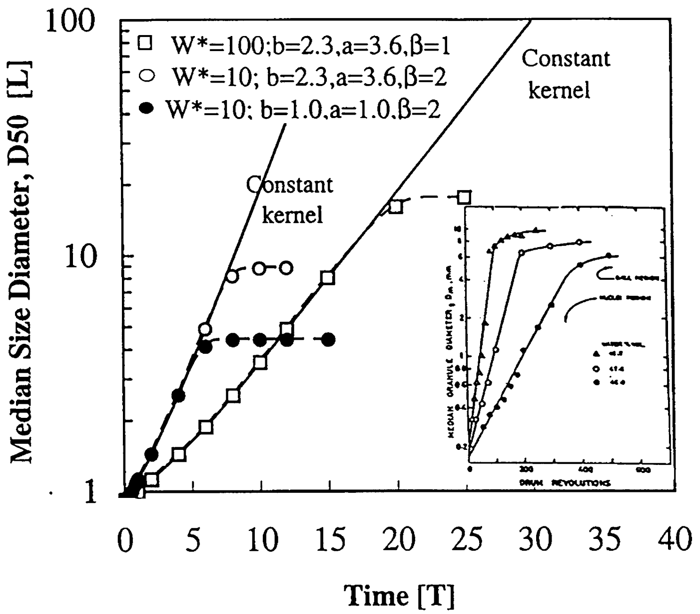


Figure 6.11. The effect of varying coalescence model parameters on mean granule size during non-inertial and inertial growth (Adetayo and Ennis, 1998)

Even for the simplest case of non-inertial growth, analytical solutions to the coalescence PB are only available for specific initial size distributions (see table 6.4). However, numerical solutions allow the shape of the granule size distribution to be easily tracked. If the initial size distribution is narrow, coalescence will tend to broaden the size distribution (see figure 6.12). In general, very narrow granule size distributions achievable by layered growth are impossible if coalescence is occurring.

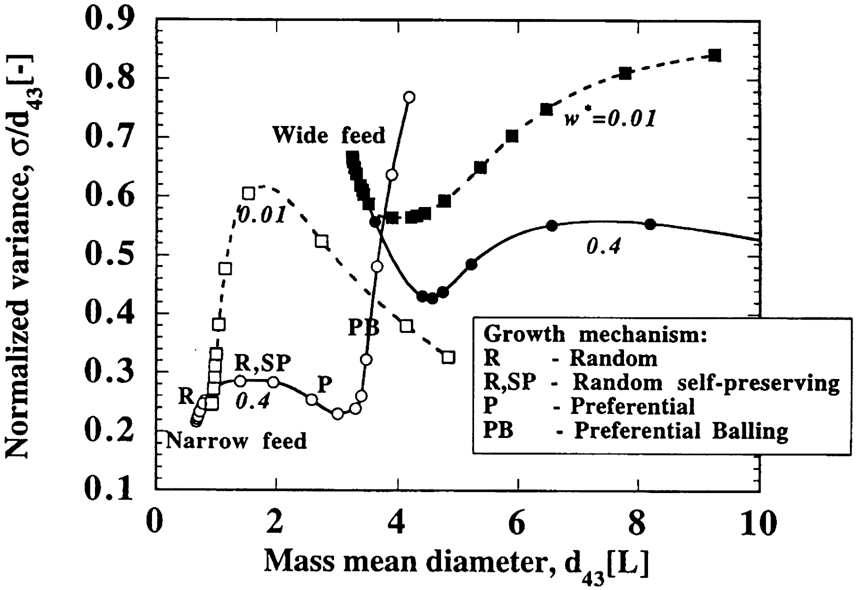


Figure 6.12. Changes to the shape of granule size distribution from different initial conditions (Adetayo and Ennis, 1997)

However, in continuous applications with recycle of crushed off spec granules, the initial size distribution may be very broad with large granules above the cut off size for successful coalescence. In this case, coalescence initially narrows the granule size distribution (see figure 6.13).

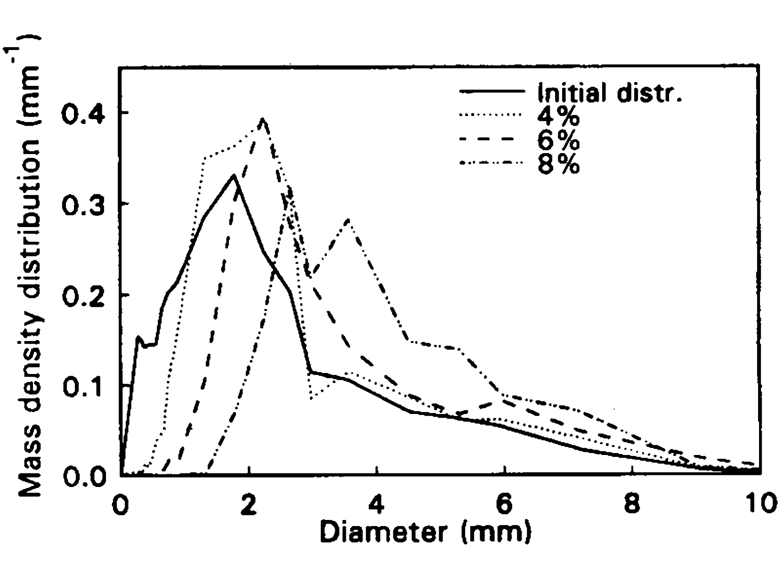


Figure 6.13. The narrowing of broad granule size distribution due to coalescence (Adetayo et al., 1993)

These changes can all be easily predicted by the PB for coalescence using the cut off kernel (Adetayo and Ennis, 1997).

6.4.3. Effect of Mixing in the Granulator

To this point, we have considered granulators in well mixed batch systems. For continuous granulators, the nature of mixing in the granulator has a profound effect on the granule property distributions in the product because of its influence on the residence time distribution of the granules. This is exactly analogous to the effect of mixing in chemical reactor design.

Consider, for example a process for growth by layering eg. urea granulation in a fluidised bed (figure 6.14). We have already derived the granule size distribution for a batch granulator (eqn.6-37). For a well mixed continuous granulator operating at steady state with representative overflow, the population balance (eqn. 6-27) simplifies to:

$$\begin{aligned} 0 &= \dot{Q}_{ex}\left(n_{in}(x) - n(x)\right) - VG\frac{\partial n(x,t)}{\partial x} \\ \Rightarrow G\frac{dn(x)}{dx} &= \frac{1}{\tau}\left(n_{in}(x) - n(x)\right) \quad where \quad \tau = \frac{V}{\dot{Q}_{ex}} \end{aligned} \tag{6-44}$$

Here τ is the mean residence time of a particle in the vessel. The total number balance (zeroth moment) is:

$$\dot{Q}_{in}N_{in} = \dot{Q}_{ex}N_T \tag{6-45}$$

Equation 6-44 is a linear ODE which is true for <u>any</u> inlet size distribution. Once $n_{in}(x)$ is specified, it can be solved numerically eg. by Runge-Kutta technique, or analytically by using the integration factor approach to give:

$$n(x)exp\left(\frac{x}{G\tau}\right) - n(0) = \frac{1}{G\tau}\int_0^x n_{in}(x)exp\left(\frac{x}{G\tau}\right)dx \tag{6-46}$$

For example, if the control volume is fed with particles which all have the same size, x_{in}, then $n_{in}(x) = N_{in}\delta(x - x_{in})$. Equation 6-44 or 6-46 is solved to give:

$$n(x) = \frac{N_{in}}{G\tau}exp\left[-\frac{x - x_{in}}{G\tau}\right] \quad for\ x > x_{in} \tag{6-47}$$

Compare this size distribution with that predicted for the equivalent batch solution (figure 6.15). For the continuous fluid bed, the product size distribution is exponential and greatly broadened from the inlet size distribution. The narrow granule size distributions from the batch granulator cannot be reproduced. This is a consequence of the exponential residence time distribution of particles in the control volume.

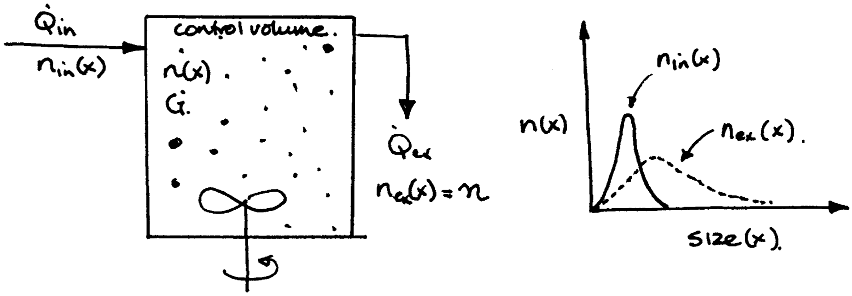


Figure 6.14. Granulation by layering in a continuous, well mixed granulator

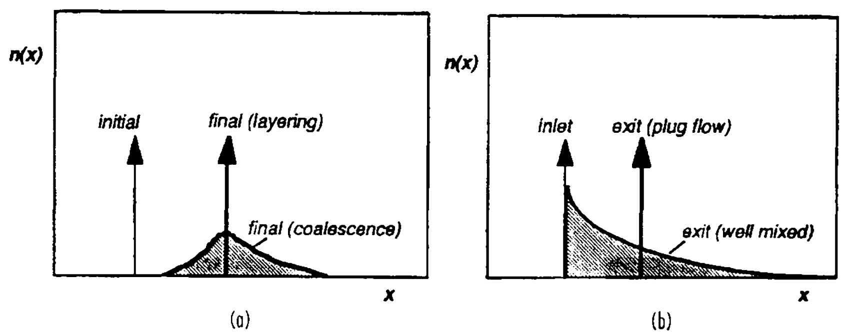


Figure 6.15. The effect of growth mechanism and mixing on product granule size distribution for (a) batch growth by layering or coalescence, and (b) layered growth in well-mixed or plug-flow granulators

This analysis is an example of the profound effect of mixing on product attributes and how the population balance can be used to quantify the effect. Table 6.5 lists some other mixing models that have been used for continuous granulators and particle coaters.

The multizone models (eg. Litster et al., 1993; Wnukowski and Ketterwell, 1989) divide the granulator into at least two zones – the spray zone and the rest of the granulator. The granule attributes are strongly affected by the circulation rate between the two zones and whether the residence time of a particle in the spray zone is dependent on the particle attributes (size and density). See the literature references from Table 6.5 for more details.

Table 6.5. Mixing Models For Continuous Granulators

Granulator	*Mixing Model*	*Reference*
Fluid bed	Well-mixed	
	Two-zone model	Wrukowski and Ketterwell, (1989)
	Three-zone model	Maronga and Wrukowski, (1997)
Spouted bed	Well-mixed	Liu and Litster, (1993)
	Two-zone model	Litster et al. (1993)
Drum	Plug flow	Adetayo et al., (1995)
Disc	Two well-mixed tanks in series with classified exit	Sastry and Loftus (1989)
	Well-mixed tank and plug flow in series with fines bypass	Ennis, Personal communication (1996)

6.5. Summary

This chapter has introduced quantitative process engineering analysis of granulators using the population balance. A series of examples have been given to demonstrate the power of the technique in critical evaluation of data, process design, sensitivity analysis and control.

This quantitative tool has come of age in analysis of granulation processes ***provided the kinetic parameters in the models are closely linked to the physical understanding of the processes described in chapters 2 to 5.***

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6.7. Nomenclature

A*	attrition rate interms of volume, m3. s-1
A_i	inlet orifice size of spouted bed, m
$\dot{B}$	birth rate, no.s^{-1}. m^{-3}
$\dot{b}(v)_{br}$	birth rate distribution due to breakage, no.s^{-1}.m^{-4}
$\dot{b}(v)_{nuc}$	birth rate distribution due to nucleation, no.s^{-1}.m^{-4}
$\dot{b}(v)_{coal}$	birth rate distribution due to coalescence,, no.s^{-1}.m^{-4}
$\dot{D}$	death rate, no.s^{-1}. m^{-3}
$\dot{d}$	death rate distribution, no.m^{-4}.s^{-1}
f	normalized number density, m^{-1}
f_{ex}	normalized number density of the exit , m^{-1}
f_{in}	normalized number density of the inlet, m^{-1}
G	growth rate in term of granule size, m.s^{-1}
G^*	growth rate in term of granule volume, m^3.s^{-1}
k_b	breakage rate constant
H	hardness, Pa
h_b	fluidized bed depth, m
K_c	granule fracture toughness, Pa.$m^{1/2}$
l	granule position in the granulator, m

n	number size distribution, no.m^{-4}
N_0	initial total numbers of particles per unit volume, no.m^{-3}
$n_0(x)$	initial size distribution, no.m^{-4}
n_{in}	number density of inlet, no.m^{-4}
n_{ex}	number density of the exit, no.m^{-4}
n_{out}	outlet number size distribution, no.m^{-4}
$n_s(v)$	spray drop side distribution, no.m^{-4}
N_i	number of particles per unit volume in size interval i, no.m^{-3}
$N_{in\ \mathrm{i}}$	rate of particles in size interval flowing in, no. m$^{-3.}$ s^{-1}
$N_{in\ \mathrm{i}}$	rate of particles in size interval flowing in, no. m$^{-3.}$ s^{-1}
$N_{in\ \mathrm{e}}$	rate of particles in size interval flowing out, no. m$^{-3.}$ s^{-1}
N_T	total number of particles no.m^{-3}
$\dot{Q}_{in}$	inlet volumetric flow rate of particles, m^3.s^{-1}
$\dot{Q}_{ex}$	exit volumetric flow rate of particles, m^3.s^{-1}
S	volumetric spray rate, m^3.s^{-1}
St	Stokes number
St^*	critical stokes number
t	time, s
U	fluidization velocity, m.s^{-1}
U_i	gas velocity at inlet orifice for spouted bed, m.s^{-1}
u	velocity, m.s^{-1}
v	particle volume, m^3
V	granulator volume, m^3
$v_{i,}$, v_j	largest volume of particles in interval *i,j*
$\dot{v}_{in}$	inlet number flow rate, no.s^{-1}
$\bar{v}$	average granule volume, m^3
$\bar{v}_i$	average granule volume of inlet, m^3
$\dot{V}_{feed}$	volumetric feed rate, m$^{3.}$ s^{-1}
$\dot{v}_{ex}$	inlet number flow rate, no.s^{-1}
w	granule size defined by eq. (6-22)
w^*	cut-off size, m
x	particle size, m
x_{in}	feed size, m

Greek

α_v	particle shape factor
ε	granule porosity
$\beta(u,v)$	coalescence kernel
β_0	coalescence rate constant , s^{-1}
μ	binder viscosity, Pa.s
μ_k	kth moment
$\mu_{K,in}$	kth moment of inlet particles
$\mu_{K,ex}$	kth moment of exit particles
ρ	binder density, kg m^{-3}
ρ_p	particle density, kg m^{-3}
η	dimensionless size
ρ_g	granule density, kg m^{-3}
τ	mean residence time of the particles in the vessel, s

CHAPTER 7

CLASSIFICATION AND CHOICE OF SIZE ENLARGEMENT EQUIPMENT

There is an enormous variety of size enlargement equipment available in the market place. In many industries, the choice of equipment and the expertise related to it has resided with the equipment vendor. This is not ideal from the practitioners point of view and unnecessary, given the current understanding of granulation science as described in chapters 2 to 6.

A better approach to equipment choice is to approach the decision from a **product design** perspective (section 1.3). This consists of two clear stages:

- Clear, quantitative definition of the desired product granule attributes.
- Choice of a combination of **equipment** and **formulation** that will achieve these attributes (see figure 7.1)

To use this approach, the engineer or technologist needs to have *both* a good understanding of both product use and sensible product characterisation; *and* a good understanding of the granulation processes that govern the product attributes and the formulation and equipment parameters that influence them.

There are many possible constraints that may also influence equipment choice including:

- The form of the active ingredient feed (dry powder, melt, slurry, solution)
- The need for a dry process for moisture sensitive materials
- Robustness of the process to handle wide ranging feeds
- The need for enclosure due to dust and fume issues
- The desired scale of operation
- The integration of the size enlargement equipment within the existing process plant
- Existing experience within the company on specific types of equipment
- Existing folklore within the company about specific types of the equipment.

In the past, the final two constraints may have often played the largest part in equipment choice. However, our improved quantitative knowledge of granulation processes makes such reliance unnecessary.

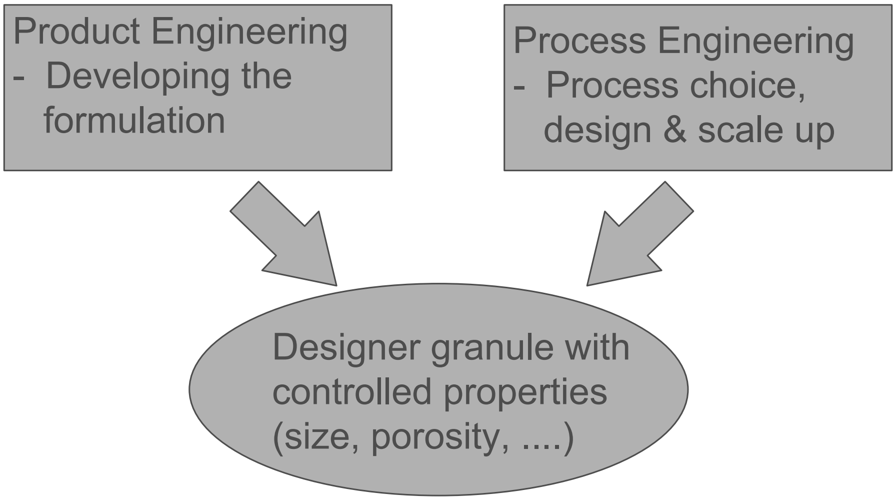

Figure 7.1. A product design approach to equipment selection and design

Table 7.1 lists a size enlargement processes with some broad comments on the range of granule size, morphology and porosity that can be achieved in different pieces of equipment. It is important to note that because both process parameters and formulation properties can be manipulated, there is no unique "best" choice of equipment for a given situation.

Below, size enlargement processes are classified and briefly introduced. Chapters 8 to 10 describe the different classes of equipment and their design, operation and scale up in more detail.

Table 7.1. Size enlargement methods and applications

Method	*Product size (mm)*	*Granule density*	*Scale of operation*	*Additional comments*	*Typical applications*
Tumbling granulators Drums Discs	0.5 to 20	Moderate	0.5-800 tph	Very spherical granules	Fertilizers, mineral ores, agricultural chemicals
Mixer granulators Continuous high shear Batch high shear	0.1 to 2	Low to high High	Up to 50 tph Up to 500 kg batch	Handles very cohesive materials well, both batch and continuous	Chemicals, detergents, clays, carbon black, pharmaceuticals, ceramics
Fluidized granulators Fluidized beds Spouted beds	0.1 to 2	Low (agglomerated) Moderate (layered)	100-900 kg batch 50 tph continuous	Flexible, relatively easy to scale, difficult for cohesive powders, good for coating applications	Continuous: fertilizers, inorganic salts, detergents Batch: pharmaceuticals, agricultural chemicals, nuclear wastes
Centrifugal granulators	0.3 to 3	Moderate to high	Up to 200 kg batch	Powder layering and coating applications	Pharmaceuticals, agricultural chemicals
Spray methods Spray drying	0.05 to 0.5	Low		Morphology of spray dried powders can vary widely	Instant foods, dyes, detergents, ceramics
Prilling	0.7 to 2	Moderate			
Pressure compaction Extrusion Roll presses Tablet press Molding press Pellet mill	>0.5 >1 10	High to very high	Up to 100 tph	Strongest bonding	Mineral ores, cement clinker, ceramics, inorganic chemicals

ssification of Size Enlargement Equipment

Figure 7.2 shows the range of granulation and compaction processes as a function of the applied stresses (granulation) or pressure (compaction). The applied stresses vary over several orders of magnitude. Granulation processes vary from low to medium level applied stress. The presence of liquid as a binder is essential for granule growth and green strength. The granules produced are of low to medium density. Compaction processes rely on pressure to increase agglomerate density and give sufficient compact strength without a liquid binder. The density of these compacts is very high.

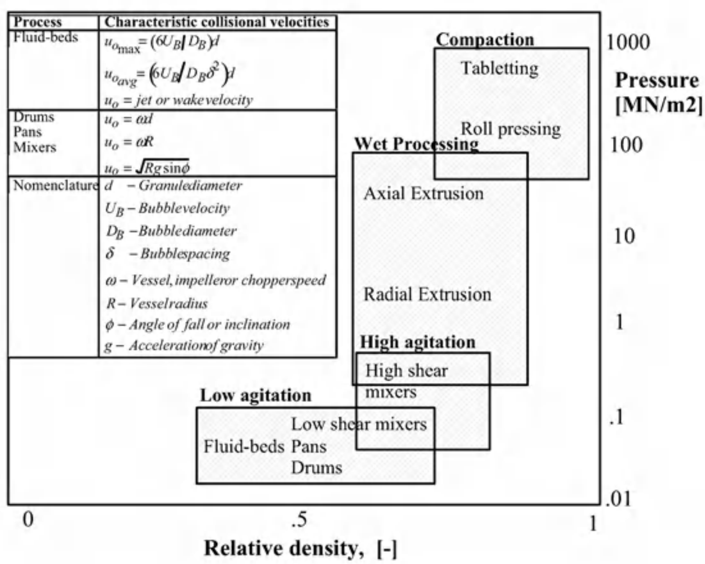

Figure 7.2. Classification of granulation and compaction processes by relative density of the product

For more detailed analysis, we divide the granulation and compaction processes into four groups as follows. Thermal processes, spraying drying and prilling are beyond the scope of this book.

Tumbling granulators (Chapter 8)

Tumbling granulators (drums, pans, discs) produce low to medium density, very spherical granules in the size range 0.5 to 20mm. They have the highest throughput of any granulation process. The balance between granule nucleation, compaction and growth determines product size distribution and density. Tumbling granulators usually operate with continuous feed and sometimes with very high recycle ratios.

Fluidised granulator-driers (Chapter 9)

Fluidised granulator driers (fluidised beds and spouted beds) use air to provide solids circulation and mixing. The particle velocities and applied stress are of the same order as tumbling granulators. However, the fluidised granulators act simultaneously as driers.

This leads to the production of porous, non-spherical granules from fine powder feed. Fluidised granulators give the lowest density agglomerates of any of the processing equipment. Fluidised beds and spouted beds are also useful for coating larger particles or to produce sperical layered granules from solution or melt feeds.

Mixer granulators (Chapter 10)

The term mixer granulator denotes equipment in which the wall of the equipment are fixed and powder motion is induced using a rotating impeller or ploughs. If fact, the term mixer granulator covers a huge range of equipment designs and applied stresses. Generalised design approaches are therefore difficult. Mixer granulators are robust enough to handle a wide variety of powder and liquid properties in the formulation. In high shear mixers, granule breakage may have a significant effect on final granule attributes. High shear mixers produce the highest density granules of any granulation equipment.

Compaction processes

Extrusion processes can operate wet or dry to produce a very narrowly sized, fairly dense product. Many designs are available but all operate on the principle of forcing the powder in a plastic state through a die, perforated plate or screen. The material undergoes considerable shear in the equipment and operation and product attributes are strongly influenced by the frictional interaction between the powder phase and the wall. For wet extrusion the rheology of the paste is also important.

Compaction processes, including roll presses and tablet presses are confined pressure devices in which material is directly consolidated between two surfaces. The processes operate dry and exert the highest applied force of any size enlargement device to give the highest density product. Successful operation depends on good transmission of the applied force through the powder and the development of strong interparticle bonds. Thus, results are very sensitive to powder flow and mechanical properties.

In the following chapters each of these classes of granulation equipment is described in more detail. The fundamentals described in chapters 2 to 6 are applied to the rational design, scale up and operation of the equipment wherever possible.

7.2. Future Directions in Granulation Process Equipment

Granulation processes have been used for many years. Equipment developments in the last 15 years have tended to focus on hybrid granulators which incorporate the features of more than one class of granulator eg. drum granulators with simultaneous drying or fluidised beds incorporating mechanical mixing. These designs add flexibility. However, the numerous processes occurring simultaneously and the sensitivity to changes in operating conditions or feed formulation make rational design and scale up difficult.

If granulation is treated as particle design, then formulation parameters and/or the process equipment can be chosen *a priori* to force one mechanism to be controlling. The logical extension of this thinking is the design of new granulators which physically separate the granulation processes and base design on the new fundamental understanding. Designs based on this approach are beginning to appear:

- Fluidised bed granulators with new nozzle designs and operation (Schaafsma et al, 2000)
- The nucleation only granulator (Wildeboer, 2002)
- Two stage continuous mixer granulators for detergents (Mort, 2001)
- Staged drum granulators (Wildeboer, 2002)

This *modular* approach, based on fundamental understanding with simple scale up rules and predictable product attributes, is the likely way of the future. Potential benefits in reduced pilot plant testing and improved speed to market of new products may be very substantial.

7.3. Bibliography

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CHAPTER 8

TUMBLING GRANULATION

8.1. Introduction

Tumbling granulators and coaters include discs, drums, pans and a range of similar equipment. In tumbling granulators, particles are set in motion by the tumbling action caused by the balance between gravity and centrifugal forces. These granulators have the following characteristics:

- Product granule size is in the range 2 to 20 mm. Tumbling granulators are not suitable for producing very small granules.

- Tumbling granulators are good for producing high density "balls" or pellets. It is more difficult to produce high porosity agglomerates.

- Tumbling equipment can also be used for coating relatively large (group D) particles.

- Discs and drums generally operate continuously and can be up to 4 m in diameter. They are capable of very large throughputs (up to 100 tonne/hr) and are therefore extensively used in mineral processing and fertiliser granulation.

In this chapter the operation of different types of tumbling granulators is described focussing on particle motion and mixing, operating variables and scale up. The role of the various granulation processes in tumbling granulators is discussed in some detail.

8.2. Disc Granulators

Figure 8.1 shows a typical disc or pan granulator. It consists of a tilted rotating disc with a rim to hold the tumbling granules. Powder feed are continuously fed to the disc, typically at the edge of the rotating granular bed. Liquid binder is added through a series of single fluid nozzles distributed across the face of the bed (figure 8.2). A coating of feed powder builds up on the disc to form a false base on which the mass rolls. The thickness of this layer is controlled by scrapers or ploughs. The tumbling motion causes segregation and size classification which is central to the disc operation.

Variations on the simple disc design include:

- An outer reroll ring which promotes granule coating and densification without further growth
- Multistepped side walls
- A pan in the form of a truncated cone.

ranule Hold Up, Mixing and Segregation on the Disc

A key operating parameter for the disc is the critical speed. This is defined as the speed at which a granule is just held stationary on the rim of the disc by centripetal forces alone (see figure 8.3):

$$N_c = \sqrt{\frac{g \sin \beta}{2\pi^2 D}} \tag{8-1}$$

Discs are typically operated at between 50 and 75% of critical speed with the angle β between 45 and 55°. If the speed is too low, the particle mass will slide against the disc instead of tumbling. If the speed is too high particles are pinned to the rim or may be thrown off the disc. Optimum drum speed is also influenced by the powder flow properties of the feed material.

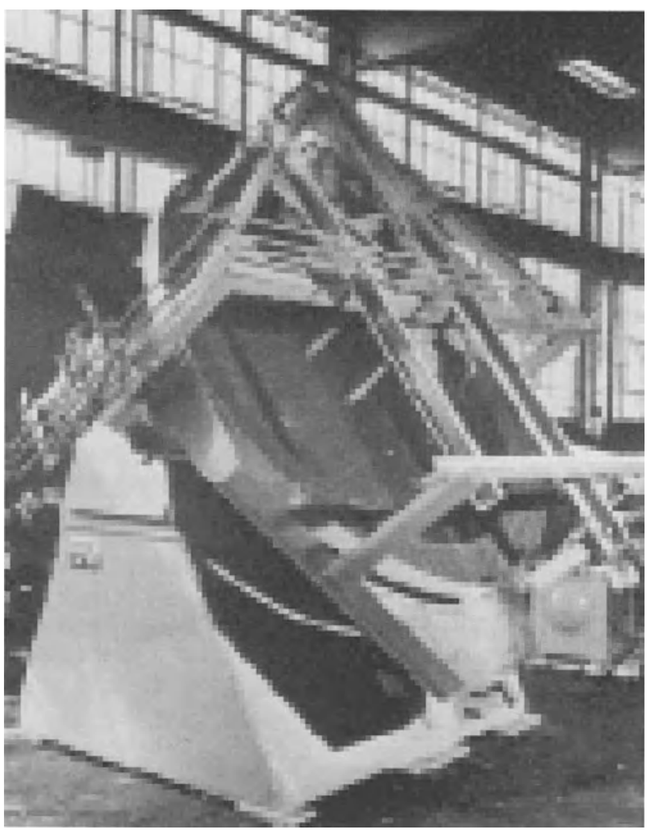

Figure 8.1. A typical disc granulator (Capes, 1985)

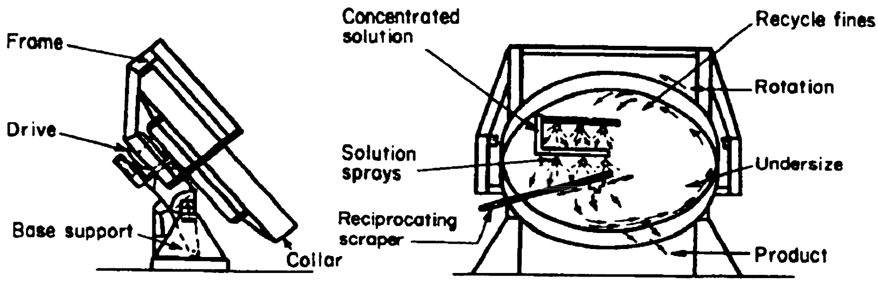

Figure 8.2. Disc granulator details (Sherrington and Oliver, 1981)

Frictional forces between a granule and the disc hold smaller particles longer as they are carried up the disc. The small particles travel higher on the disc and then roll underneath the larger pellets in the eye of the disc (see figure 8.4). Thus the larger well formed granules sit on a bed of new nuclei and ungranulated powder. This process leads to a natural size classification on the disc with only large granules or pellets overflowing the eye of the disc.

As a result of this natural size classification, disc granulators rarely need to recycle undersize and oversize granules which is a distinct advantage over drum granulators. The segregation of granules on the disc also allows the balance of granulation processes to be controlled by adjusting the position of the powder feeder and liquid spray nozzles.

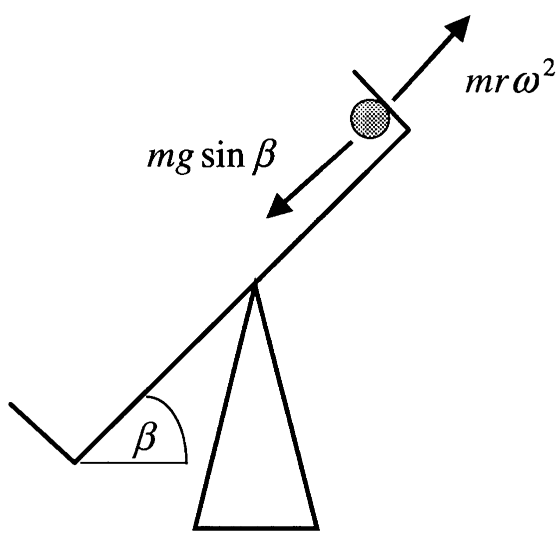

Figure 8.3. Force balance on a granule with the disc operating at critical speed

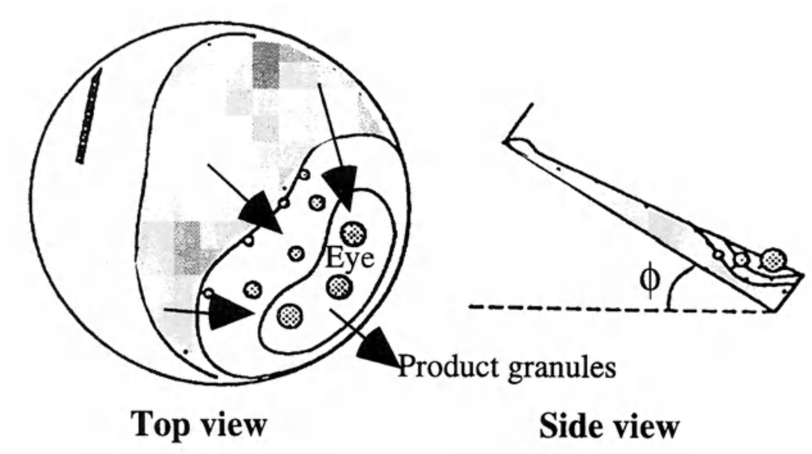

Figure 8.4. Granule segregation on a disc granulator illustrating a size classified granular bed sitting on ungranulated feed powder

The total holdup of granules on the disc can be adjusted by varying operating parameters. Hold up will increase with:

- decreasing disc angle at the same fraction of critical speed;
- increasing disc speed: granules are carried higher on the disc and cover a large fraction of the disc area; and
- increasing moisture content: granules are stickier and therefore are carried higher on the disc.

The total hold up determines the average residence time of granules on the disc. Typical granule residence times are in the range 1 to 2 minutes. However, the mixing patterns (residence time distribution) are also important in determining the product granule properties. The simplest residence time distribution models are continuous stirred tank reactor (CSTR) and the plug flow reactor (PFR). Figure 8.5a shows that a disc granulator lies somewhere between these two extremes. The residence time distribution moves closer to plug flow as the angle of the disc is increased. Figure 8.5b shows a possible two stage model for mixing on the disc. The first stage is a CSTR corresponding to feed powder spread over a wide disc area. The second stage is a PFR corresponding to the well classified tumbling flow in the eye of the disc.

Increasing the disc angle narrows the residence time distribution and moves it towards ideal plug flow. The residence time distribution has a marked effect on granule size distribution and structure (see chapter 6).

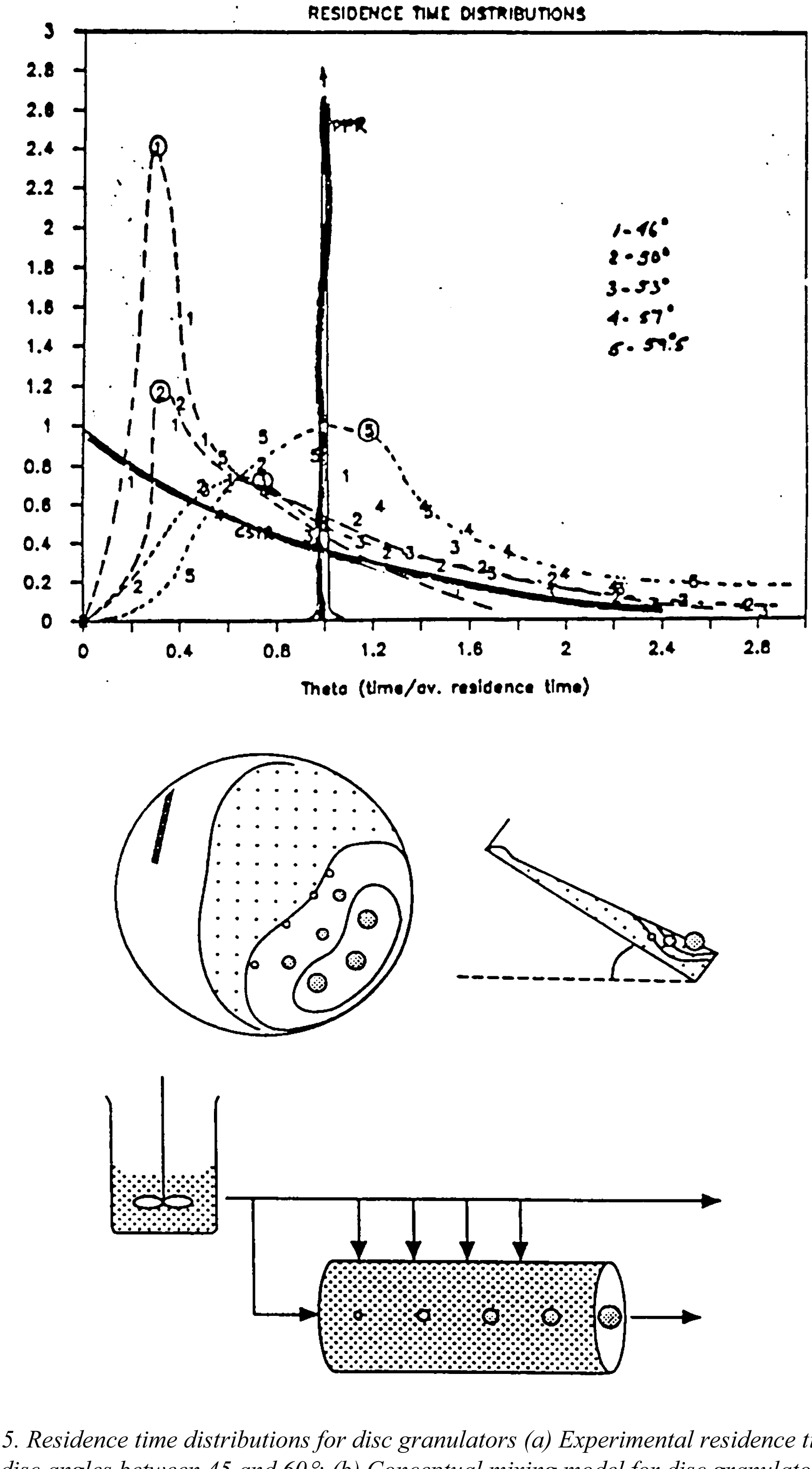

Figure 8.5. Residence time distributions for disc granulators (a) Experimental residence time data at disc angles between 45 and 60°; (b) Conceptual mixing model for disc granulators

Granulators

Drum granulators are extensively used in the mineral processing and fertiliser industries. They are capable of very large capacity (see Table 8.1). In contrast to disc granulators, there is no size classification at the output. Drum granulators often run with large recycles of undersize and crushed oversize granules.

A granulation drum consists of an inclined cylinder which may be open ended or fitted with annular retaining rings (see figure 8.6). Feeds may be either premoistened by mixers to form granule nuclei, or liquid may be sprayed onto the tumbling bed via nozzles or distributor pipe systems. Scrapers are often, but not always, used to limit build up on the drum wall.

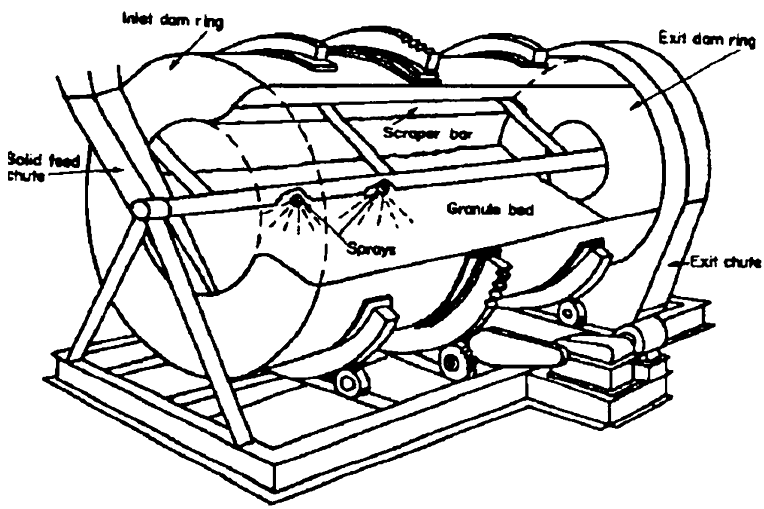

Figure 8.6. Typical configuration for a continuous drum granulator (Sherrington and Oliver, 1981)

Table 8.1. Characteristics of large scale granulation drums (Capes, 1980)

Diameter, (ft)	*Length, (ft)*	*Power, (hp)*	*Speed, (rpm)*	*Approximate capacity, (ton/hr)**
Fertilizer granulation				
5	10	15	10-17	7.5
6	12	25	9-16	10
7	14	30	9-15	20
8	14	60	20-14	25
8	16	75	20-14	40
10	20	150	7-12	50
Iron-ore balling				
9	31	60	12-14	54
10	31	60	12-14	65
12	33	75	10	98

*Capacity excludes recycle. Actual drum throughput may be much higher.

8.3.1. Granule Hold Up, Mixing and Segregation

In contrast to discs, there is no output size classification in drum granulators and high recycle rates of off-size product are common. The desired granule motion is a tumbling, cascading motion. Equation 8.1 can be used to calculate the drum critical speed with $\beta = 90^{\circ}$. Drums are operated at a smaller fraction of critical speed than discs, typically 30 to 50% N_c. If the drum speed is too low, the granular bed will slide, rather than cascade. High drum speeds give unwanted cataracting of the charge (similar to the motion of the balls during ball milling). Spray blow through and wall build up are more likely. Figure 8.7 illustrates different powder flow patterns in the drum.

The drum pitch angle is simply to assist the flow of the charge through the drum. Typical drum pitch angles vary from 0 to 10°.

Hold up in the drum is between 10 and 20% of the drum volume. Holdup and mean residence time are controlled by the drum length which is typically 2 to 10 times the drum diameter. Holdup in the drum is increased by reducing the pitch angle or providing a lip at the drum exit. Residence times of 2 to 5 minutes are common.

Residence time distributions are less complex than in disc granulation. As a first approximation, granules can be considered to flow through the drum in plug flow, although some back mixing is common.

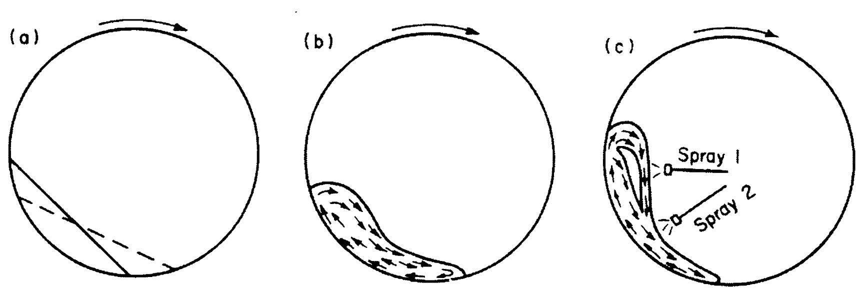

Figure 8.7. Powder flow patterns in drum granulators (a) sliding (b) cascading (c) onset of cataracting

8.3.2. Granulation Drum Circuits

Granulation circuits often involve large recycle streams (2:1 to 5:1 are common in fertiliser and iron ore circuits). The role of the recycle stream in circuit stability and control is critical. Surging and limit cycle behaviour is common. Surges alter instantaneous moisture content and size distribution in the drum (see figure 8.8). Modern control techniques with on-line moisture and size analysis can overcome these effects.

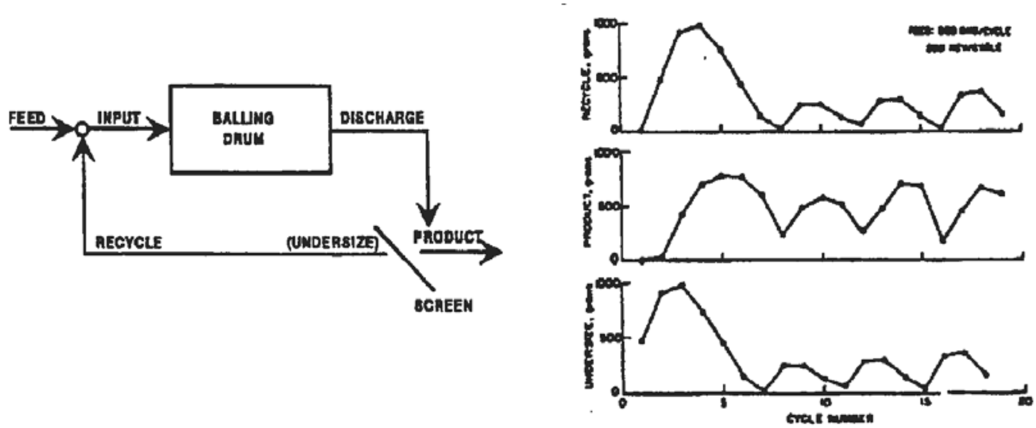

Figure 8.8. Simulation of surging behaviour in iron ore balling circuits (Sastry, 1978)

8.4. Granulation Rate Processes

Granulation rate processes have been discussed generically in depth in part 1 of this book. This section highlights the application of the quantitative analysis of these processes to drum and disc granulators. Readers are referred to the earlier chapters for more detailed discussion and especially to tables 3-3, 4-5 and 5-3 for trouble shooting.

The important rate processes that occur in tumbling granulators are nucleation, growth by layering or coalescence, and consolidation. Breakage and attrition are typically not important as applied stresses and impact velocities are low and there is no simultaneous drying.

Granulation processes are extremely complicated in tumbling granulators for several reasons:

- Granules remain wet and can deform and consolidate. The behaviour of a granule is therefore a function of its history.
- Different granulation behaviour is observed for broad and narrow size distribution feeds.
- There is often complex competition between growth mechanisms that may interact strongly with powder flow patterns.

8.4.1. Nucleation

Nuclei (new granules) can be formed by primary nucleation from liquid feed drops, or by secondary nucleation generated by the scraper bar. Where the liquid drop size is large compared to the feed powder size, the nuclei distribution will reflect the drop size distribution.

For drop controlled nucleation, the spray flux ψ_a should be less than 0.2. To calculate ψ_a (eqn.3-27), the powder surface flux is required. In a tumbling granulator, the powder surface velocity is of order of the peripheral speed of the drum wall or disc rim *DN*. Provided the spray nozzles are well characterised, ψ_a can be estimated without difficulty and can, in principle, be kept in the desired range by careful choice of spray nozzle(s) and liquid spray rate.

Drop controlled nucleation also requires small drop penetration times which are primarily controlled by feed formulation properties (eqn.3-27) rather than equipment parameters. As tumbling granulators have low applied stress, liquid distribution and nucleation in the mechanical dispersion regime are impossible. Thus, tumbling granulators are unsuitable, in isolation, for very fine or poorly wetting powders, highly viscous liquid binders and for mixing feeds with very different moisture contents eg. dry feed and filter cake. For these types of materials a high shear mixer granulator, or a tumbling granulator in combination with a premixer for nucleation/binder dispersion is a much better option.

In drum granulation, the flow of particles into the drum is often dominated by recycled granules. These granules are usually dried or sintered before recycle and are therefore hard and not deformable. The size range of the recycle granules is very broad, often containing particles as large as several millimetres in size. In this case, the spray drop size is often significantly smaller than many of the particles and the role of nucleation is limited. For fertiliser granulation from liquid (melt or solution) feed, the recycle represents the *only* particulate feed to the drum and nucleation can be neglected. This is in contrast to discs and pans, where the balance between nucleation of fine powder feed and layering of powder onto existing granules can profoundly effect the product size distribution.

8.4.2. Consolidation

Consolidation of the granules tumbling in the drum or disc is an important process. It directly determines the granule density (porosity). Consolidation can proceed over extended times and achieve high granule densities as there is no in situ drying to stop the consolidation process. Consolidation also has important indirect effects on other granulation processes by changing the amount of liquid needed to saturate the granule and changing the granulation deformability through "strain hardening".

Section 4.2 gave an extensive discussion of granule consolidation. Granule porosity decreases towards a minimum value with time in the granulator (figure 4.9):

$$\frac{\varepsilon - \varepsilon_{\min}}{\varepsilon_0 - \varepsilon_{\min}} = \exp(-kN) \qquad (4\text{-}3)$$

The rate of consolidation is a function of the Stokes number:

$$\frac{\varepsilon - \varepsilon_{\min}}{\varepsilon_0 - \varepsilon_{\min}} = \exp(-kN) \qquad (4\text{-}3)$$

$$k = \beta_c \exp(a.St_{def}) \qquad (4\text{-}5a)$$

which incorporates effects of both formulation properties and the effective collision velocity in the granulator. The minimum porosity ε_{min} is a more complex function of binder viscosity, interparticle friction and surface forces (see section 4.2).

reasing drum size and/or speed will increase the degree of consolidation by increasing the energy of collision. The maximum collision velocity for consolidation is proportional to the peripheral speed of the drum or disc:

$$U_c \approx \omega D / 2 \tag{8-2}$$

Thus, consolidation will increase with both drum/disc size and speed. This leads to some implications for scale up (see section 8.4.5).

8.4.3. Granule Growth by Coalescence

The fundamentals of granule growth by coalescence is covered in depth in sections 4.3 to 4.6. Coalescence is usually a key rate process in tumbling granulation. There is no simultaneous drying and relative collision velocities are fairly low, leading to a high probability of successful coalescence. Depending on the circumstances, the existing models for either near elastic collisions (section 4.3) or deformable granule collisions (sections 4.4 and 4.5) may be applicable.

In both models, the effective collision velocity is a key parameter. The relative velocity between two adjacent granules in the tumbling granulator varies with position. The maximum collision velocity will be of order the peripheral speed of the drum or disc (eqn.8-2), and we assume that this velocity is the key to determine successful coalescence. Thus, this single collision velocity is used to be representative of surface velocities (for nucleation calculations) and collision velocities (for consolidation and coalescence calculations).

Deformable granule coalescence

Fine powder feeds exhibit deformable granule growth in tumbling granulators. This is typically the case for disc granulators and sometimes in drum granulators. A range of growth behaviours are observed depending on the deformability of the granules (figures 8.9, 4.25). The behaviour is predictable if the Stokes deformation number (eqn.4-3) and fraction granule saturation (eqn.4-23) can be estimated. Figure 8.10 shows the growth regime map for deformable granules. The regime map has been well validated for tumbling granulators (Iveson and Litster, 1998; Iveson et al., 2001).

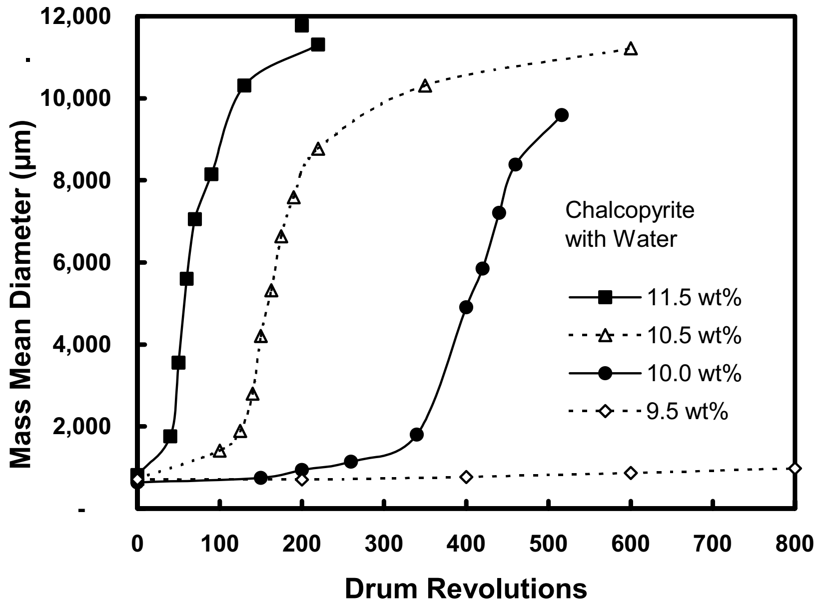

Figure 8.9. Examples of growth behaviour in tumbling granulators: Induction growth behaviour of copper concentrate (Wauters et al., 1999)

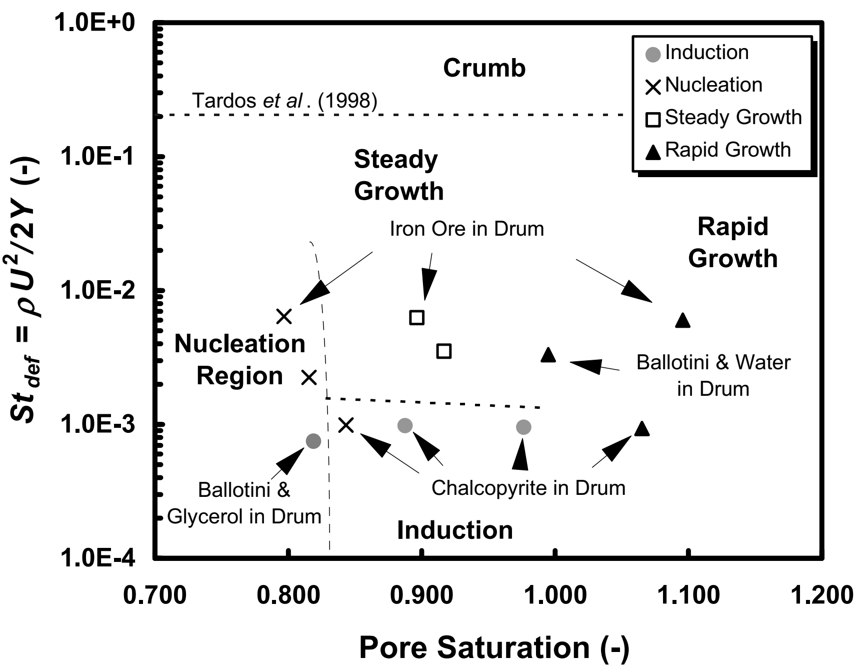

Figure 8.10. Growth regime map for drum granulators (Iveson et al., 2000)

ubstantial granule growth (steady growth or induction growth) occurs within a fairly narrow range of moisture contents (85 to 110% pore saturation). Thus, an initial estimate of moisture requirements can be made by calculating the amount of liquid needed to saturate the granule pores (eqn.4-23). Capes (1980) provides a rule of thumb based on this equation:

$$w = \frac{\varepsilon\rho_l}{\varepsilon\rho_l + (1-\varepsilon)\rho_s}$$

$$w = \frac{1}{1+1.85\left(\rho_s/\rho_l\right)};\, d_p < 30\mu m \qquad (8\text{-}3)$$

$$w = \frac{1}{1+2.17\left(\rho_s/\rho_l\right)};\, d_p > 30\mu m$$

Table 8.2 gives typical moisture requirements for a range of common materials. It is important to realise that eqn. 8-3 and Table 8.2 are only suitable for preliminary mass balance requirements for liquid binders with similar properties to water. Product granule size distributions and growth characteristics are extremely sensitive to liquid content with the narrow operating range (figures 4.29, 8.9).

For detailed discussion of the effect of other formulation properties on granule attributes, see section 4.6.

Table 8.2. Typical moisture requirements for the granulation of some common materials in tumbling granulators

Raw material	*Approximate size of raw material, less than indicated mesh*	*Moisture content of balled product, wt % H_2O*
Precipitated calcium carbonate	200	29.5-32.1
Hydrated lime	325	25.7-26.6
Pulverized coal	48	20.8-22.1
Calcined ammonium metavanadate	200	20.9-21.8
Lead-zinc concentrate	20	6.9-7.2
Iron pyrite calcine	100	12.2-12.8
Specular hematite concentrate	150	8.0-10.0
Taconite concentrate	150	8.7-10.1
Magnetic concentrate	325	9.8-10.2
Direct-shipping open-pit iron ores	10	10.3-10.9
Underground iron ore	¼ in.	10.4-10.7
Basic oxygen converter fume	1μ	9.2-9.6
Raw cement meal	150	13.0-13.9
Fly ash	150	24.9-25.8
Fly ash-sewage sludge composite	150	25.7-27.1
Fly ash-clay slurry composite	150	22.4-24.9
Coal-limestone composite	100	21.3-22.8
Coal-iron ore composite	48	12.8-13.9
Iron ore-limestone composite	100	9.7-10.9
Coal-iron ore-limestone composite	14	13.3-14.8

Elastic particle coalescence

For drum granulation systems where dry granules with a very broad distribution (up to several mm in size) are recycled, the Ennis model for elastic particle coalescence is appropriate (section 4.3). Metallurgical ore sinter feed granulation and fertiliser granulation are two good examples of this case. In these systems, fines are removed up to some critical size after addition of the liquid binder, tending to narrow the very broad size distribution of the recycle granules (Adetayo and Ennis, 1995). The critical size is a function of both moisture content and viscosity as predicted by the Stokes regime analysis (see section 4.3.4 and eqns. 4-10 to 4-13). Typically, this elastic particle growth is fast. In some cases, the granules formed through this process may be deformable enough to continue to grow at a slower rate.

8.4.4. Layering

Here we can consider two types of layering. True layering occurs when the feed to the drum is a solution or melt e.g. fertiliser granulation. The liquid feed layers onto recycled granules. Unless there is simultaneous drying, this is rarely the dominant growth mechanism in drum granulation systems.

However, we can also consider the addition of fine powder feed to the outside of tumbling granules as a layering process. This process is important:

- for the addition of fine powder feed to recycled well formed granules in drum granulation or balling circuits.
- for disc granulators and pelletisers

In each case, layering will compete with nuclei formation and coalescence as growth mechanisms. Layering is favoured by a high concentration of pellets to new feed; positioning liquid feed to spray onto existing granules, not new feed; and positioning the powder feed to fall onto tumbling granules.

Layered growth leads to a smaller number of larger, denser granules with a narrower size distribution than growth by coalescence. Figure 8.11 shows an example of the effect of spray position on disc granulation of a powder. Spraying near the top of the disc onto granules about to coated with feed powder encourages layering and gives large, narrowly distributed granules.

This balancing of layering and nucleation is just one example of the control of the granule attributes through controlling the rate processes during granulation. In general, feed size distributions may either narrow or broaden during granulation depending the initial feed size distribution, mixing in the granulator and the balance between nucleation, layering, deformable and elastic coalescence. Population balance models (see chapter 6) are the ideal vehicle for quantifying and predicting these effects (see, for example, figure 6.12 and section 6.4.2).

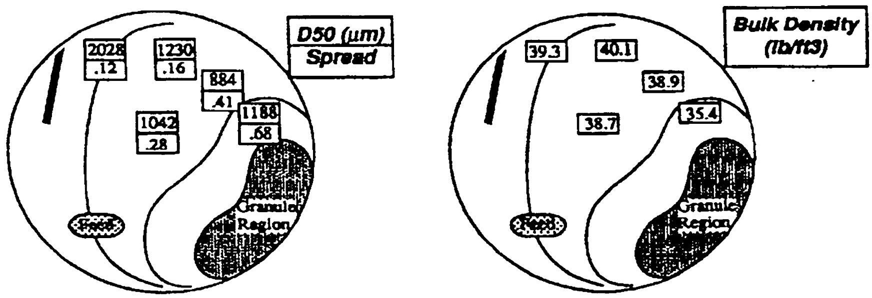

Figure 8.11. The influence of nozzle position on granule size distribution and density during disc granulation

8.5. Scale Up and Operation of Tumbling Granulators

Typical dimensions and scale up rules for discs and drums are shown in Table 8.3. These data are suitable for quick order of magnitude calculations. For drum granulators, power consumption and capacity are related to drum volume, allowing greater throughputs than are feasible with discs, where capacity scales with disc area (see table 8.3).

Table 8.3. Dimensions and scale up rules for disc and drum granulators

Inclined Disc	*Drum*
Throughput or capacity: $Q \propto D^2$ (area)	$Q \propto LD^2$ (10-20% of volume)
Power consumption: $P \propto D^2$ (area)	$P \propto LD^2$ (volume)
Speed: 50-75% of N_c	30-50% of N_c
Angle: 40-70° from vertical (45-55° most common)	0-10° from horizontal
Height or Length: 20% of D (10-30% of D range)	2 to 5 D

For both discs and drums, it is usual to maintain a constant fraction of critical speed during scale up to keep powder flow patterns and mixing regimes similar. This is a constant Froude number scale up rule:

$$Fr = \frac{DN^2}{g} = const.$$
$$\Rightarrow \quad N \propto D^{-0.5} \tag{8-4}$$

For a constant Froude number, the peripheral speed of the drum or disc increases with scale:

$$ND \propto D^{0.5} \tag{8.5}$$

As both surface powder velocity and effective collision velocity scale with peripheral speed, there are a number of implications for effect of scale up on granulation processes.

For wetting and binder distribution, we require the dimensionless spray flux ψ to remain low. For a similar nozzle and spray area, the powder flux increases with $D^{0.5}$. However, as disc capacity (therefore liquid spray flux) scales with D^2, a larger spray area on the powder surface is needed to keep ψ constant. This is most easily achieved by increasing the number of spray nozzles.

Due to the increased collision velocities, large scale granulators tend to produce higher density granules when operating within the nucleation or induction growth regimes. In this regime, granule size does not change with scale provided the induction time is not exceeded. Instead, granule size distribution is controlled by the balance between nucleation and layering.

For operation in the steady growth regime, increasing scale may increase granule size and changes are very sensitive to moisture content.

In general, changes to granule attributes with scale in tumbling granulators cannot be predicted by simple rules of thumb, but require careful characterisation of the formulation and knowledge of the controlling rate processes (see chapters 3 and 4).

Due to the ability to manipulate the balance of growth mechanisms and the inherent size classification, disc granulators are far more flexible to operate than drums. However, steady operation can be difficult and operator sensitive. Some operational issues include:

- *Controlled solid feeding:* Gravimetric, rather than volumetric, control of solids feed is desirable due to the sensitivity of operation to moisture content and solids flow pattern.
- *Control of liquid spray rate and distribution:* Pressure nozzles need to be maintained regularly and the pressure monitored to avoid dripping nozzles. Liquid flow rate should be tied to the solid feed flowrate (feed forward control). The number and position of nozzles can be used as a secondary control parameter to optimise granule attributes through varying the balance of granule growth mechanisms.
- *Powder flow on the disc:* Many parameters effect the powder flow on the disc including variations in feed powder flow properties, disc angle and speed, and the positioning of the powder feed onto the disc. Good powder cascading on the disc is important to avoid spray blow through and reduce variability in product quality.

On-line measurement of both moisture content and granule size distribution is now quite feasible, making automatic control of disc and drum granulators possible and desirable. Nevertheless, the industrial application of automatic control to tumbling granulators is still in its infancy.

8.6. Variations on a Theme

Some modified drum granulator designs incorporate internals to modify the granule flow pattern and allow the drum as a simultaneous granulator/drier with granule growth predominately by layering. These designs include falling curtain granulator and the fluidised drum granulator (Sarwono and Litster, 1994a). In both these designs the granules do not cascade in the drum. Instead, they are lifted to the top of the drum and fall in a curtain past the liquid spray nozzle. In the fluidised drum granulator, the granules cross an air table where they are warmed and dried before falling past the spray nozzle.

In many respects, these granulator/driers have more in common with fluidised granulators than other tumbling granulators. There is currently only limited application of these granulators in industry. Analysis of the operation of the fluidised drum granulator is considered by Sarwono and Litster (1994b).

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8.8. Nomenclature

d_p constituent particle size (mm)
D disc diameter (m)
Fr Froude number
g gravity (m.s-2)
k proportional constant (s.rev-1)
m granule mass (g)
N rotational speed (rev.s-1)
N_c critical rotational speed (rev.s-1)
r disc radius (m)
St_{def} Stokes deformation number

U_c collision velocity (m.s-1)
w moisture content (%)
β disk angle
βc proportional constant (s. rev-1)
ε granule porosity
ε_{min} minimum granule porosity
ε_0 initial granule porosity
ω drum peripheral speed (s-1)
Ψa spray flux
ρ_l liquid binder density (kg. m-3)
ρ_s solid density (kg. m-3)

CHAPTER 9

FLUIDISED GRANULATORS

In fluidised granulators (fluidised beds and spouted beds), the particles are set in motion by air, rather than by mechanical agitation. Areas of application include fertilisers, industrial chemicals, agricultural chemicals, pharmaceutical granulation and a range of coating processes. Fluidised granulators produce either high porosity agglomerates (from powder feeds) or high strength layered granules (from liquid feeds).

Figure 9.1 shows a typical production size batch fluid bed granulator. The air handling unit dehumidifies and heats the inlet air. The heated air enters the fluid bed through a distributor which also supports the particle bed. Liquid binder is sprayed through an air atomising nozzle located above, in or below the bed. Bag filters or cyclones are needed to remove dust from the exit air. Other fluidisation gases such as nitrogen are sometimes used in place of air if there are potential explosion hazards.

Continuous fluid bed granulators are used in the fertiliser industry. Near size granules are recycled to control the granule size distribution. Dust is not recycled directly, but first remelted or slurried in the liquid feed. Continuous fluid bed granulators usually operate with layering as the dominant growth mechanism.

The advantages and disadvantages of fluidised systems when compared to other systems are shown in Table 9.1.

In this chapter we describe fluidised bed granulators in detail covering the hydrodynamics, mass and energy balances, granulation processes, modelling, design and operation. At the end of the chapter, related equipment including draft tube designs and spouted beds are covered separately.

Table 9.1. Advantages and Disadvantages of Fluid-Bed Granulators

Advantages include:	Disadvantages include:
• High volumetric intensity • Simultaneous granulation and drying removes the need for an additional drier • Mechanically simple • Can easily produce small (minus 1 mm) porous granules from powder feeds • Can produce high strength layered granules with a very tight size distribution from liquid feeds • Very good heat and mass transfer rates • Robust and sensitive with regard to the impact of operating variables on product quality	• Defluidization due to quenching (excessive growth) if the operating variables are not controlled well • High operating costs for air handling • Attrition rates may be high and dust recovery equipment is essential • Poor operation with fine, cohesive powders

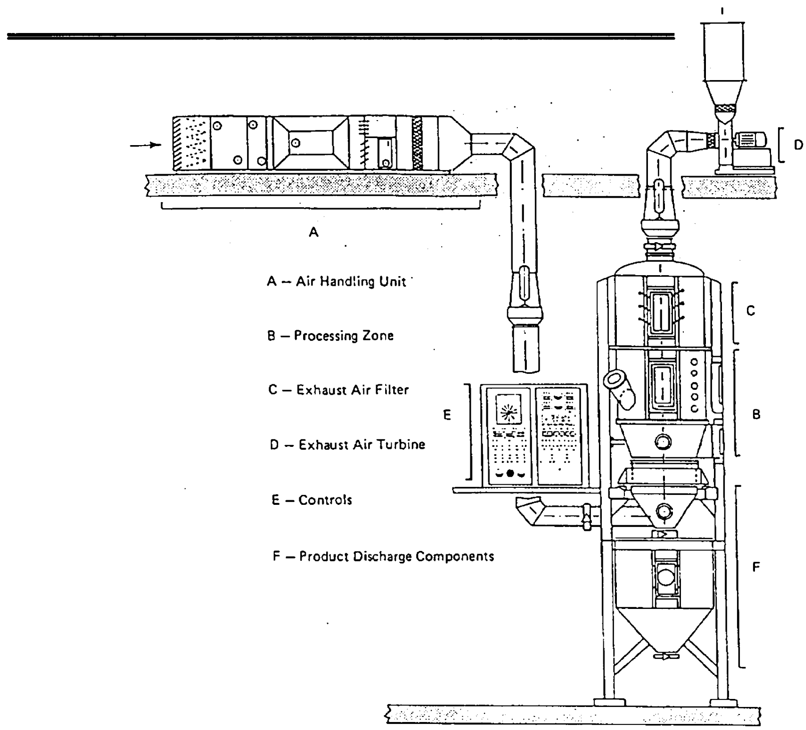


Figure 9.1. Fluid-bed granulator for batch processing of powder feeds (Ghebre-Sellasie, 1989)

9.1. Hydrodynamics of Fluidised Beds

In order to design or operate a fluidised bed granulator, the basic hydrodynamics of fluidised systems needs to be understood. Figure 9.2 shows the essential elements of any fluidised bed granulator. The fluidising gas enters the bed at the bottom through a grid or distributor. The gas passes up through a bed of solid, causing it to "fluidise". This gas/solid mixture behaves much like a liquid of similar bulk density e.g. the fluidised mixture will flow through an opening and pressure increases linearly with depth (see figure 9.3).

Above the fluidised bed, a freeboard section is needed to remove particles thrown up from the bed by the bursting of gas bubbles at the bed surface. In addition, cyclones or bag filters are necessary to separate fines elutriated with the exit gas.

In this section we will summarise the basis aspects of fluid bed hydrodynamics only. For a more comprehensive and detailed analysis of the hydrodynamics of fluidised systems see Kunii and Levenspiel (1991).

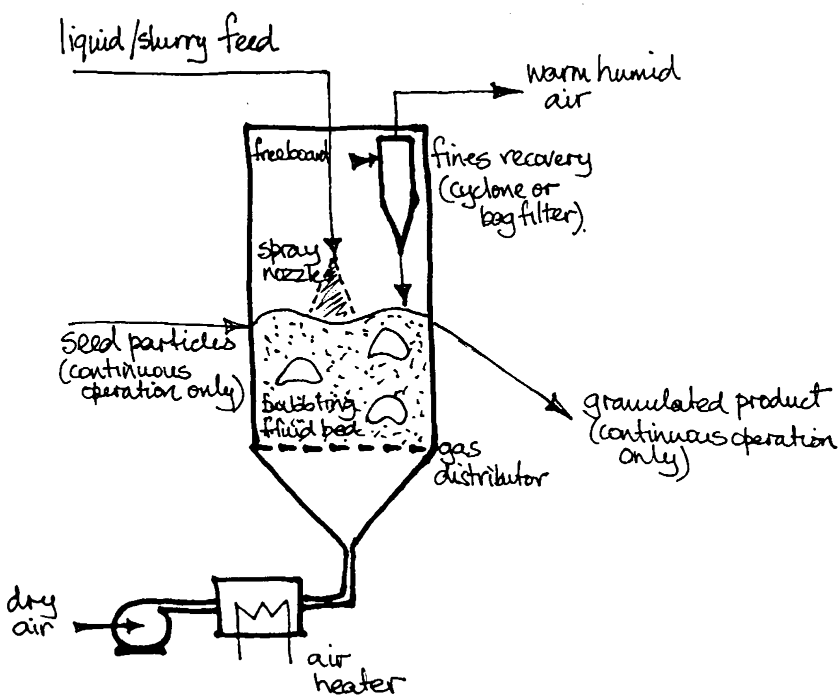


Figure 9.2. Elements of a fluidised bed granulator

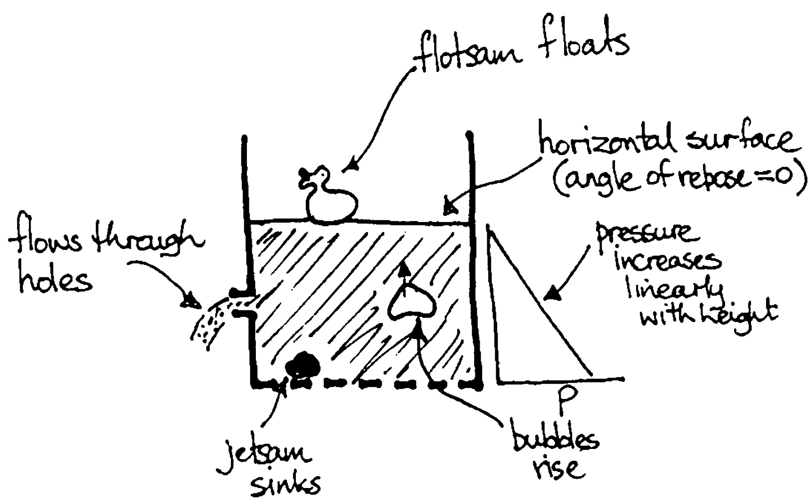


Figure 9.3. Liquid like properties of fluidised beds

9.1.1. Minimum Fluidisation Velocity and Pressure Drop

Consider a packed bed of particles with gas flowing up through the bed (figure 9.4). Initially the particles support each other and behave as a bulk solid. The pressure drop through the bed is equal to the drag force of the gas on the bed. As gas velocity is increased, a point is reached where the gas drag force just equals the weight of the particles in the bed. The particles no longer need to support each other and the bed fluidises. At this point the pressure drop across the fluid bed is equal to the bed weight ie.:

$$\Delta P_{fb} = \rho_s (1-\varepsilon) g H \tag{9-1}$$

If the velocity is increased further, the pressure remains constant (equal to the bed weight) and the bed expands. The bed may expand smoothly (particulate fluidisation) or by the appearance of gas bubbles (bubbling fluidisation).

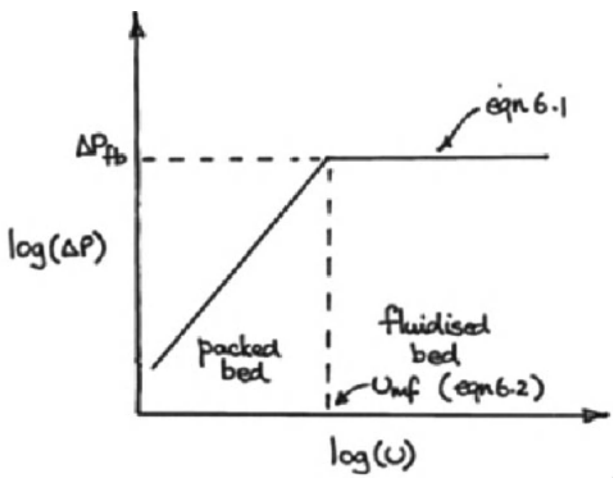


Figure 9.4. Pressure drop - velocity relationship for packed and fluidised beds

The bed becomes fluidised when the superficial gas velocity reaches the minimum fluidisation velocity u_{mf}. The minimum fluidisation velocity can be calculated from the Wen and Yu correlation:

$$\frac{d_v u_{mf} \rho_s}{\mu} = \left[33.7^2 + \frac{0.0408 d_v^3 \rho_g (\rho_s - \rho_g) g}{\mu^2} \right]^{1/2} - 33.7 \tag{9-2}$$

This correlation is valid for a wide range of fluid and particle properties to an accuracy of ±30%. The equation is written in terms of the volume equivalent particle size d_v. For a wide size distribution, the specific surface mean size $\bar{x}_{32}$ should be used as the measure of particle size (see section 2.15).

9.1.2. Types of Fluidisation Behaviour

As gas velocity is increased above u_{mf}, a range of fluidisation behaviour may be observed. Immediately after fluidising, the bed may expand smoothly (particulate fluidisation). At the minimum bubbling velocity u_{mb}, bubbles will form in the bed. After this point, all additional gas added to the bed will form bubbles. If the fluidised bed is narrow, the bubbles may fill the tube, and slugging occurs. If the gas velocity is increased till greater than the terminal settling velocity of the particles, turbulent or fast fluidisation behaviour occurs. In these high velocity regimes, there is no clearly defined bed surface and a significant proportion of particles are in the freeboard. There is a high rate of elutriation from the bed and particles must be captured and recirculated to maintain particles in the bed. Finally, at very high velocity, a lean phase conveying regime exists. Particles are distributed uniformly throughout the tube at very low concentration.

The fluidisation behaviour depends on powder properties as well as gas velocity. Geldart (1973) classified powders based on their size and the density difference between the particles and the gas (see figure 9.5).

- Type A powders (aeratable) are easily fluidised. These powders show some particulate expansion when fluidised before bubbling takes over at higher velocities i.e. $u_{mb} > u_{mf}$.
- Type B powders (sandy) also fluidise easily but do not show any particulate expansion. All excess gas above minimum fluidisation goes into bubbles i.e. $u_{mb} = u_{mf}$.
- Type C powders are cohesive and fluidise poorly. Attempts to fluidise beds of cohesive powders lead to severe channelling and a sudden but unpredictable change to the fast fluidisation or lean phase transport regime.
- Type D particles (spoutable) are very large. High velocities are needed to fluidise these materials and slugging behaviour occurs unless the bed diameter is quite large. These materials lend themselves to spouted bed processing.

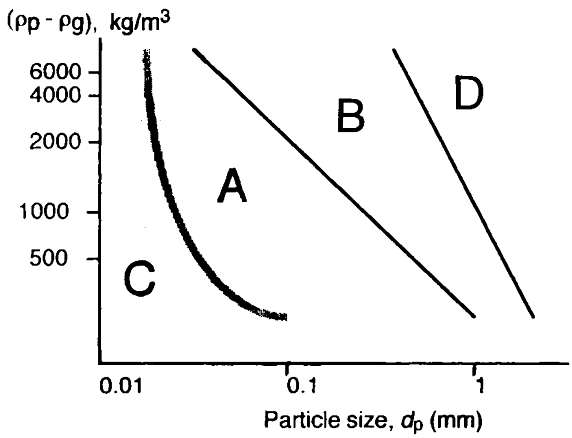


Figure 9.5. Geldart classification of powders (fluidisation in air)

Figure 9.6 summarises the fluidising behaviour for the different types of powder. Product granules from a fluidised bed granulator are typically either group B or group D particles and granulators are most likely to operate in the bubbling regime with gas velocity in the range $1.5u_{mf} < u_g < 5u_{mf}$. However, the batch granulation may start with a bed of group A powder whose fluidisation characteristics change dramatically as granule growth occurs by coalescence. For such systems, the initial fluidisation velocity may be many times u_{mf} and gas velocity may need to be increased continuously through the batch time to maintain a fluidised state.

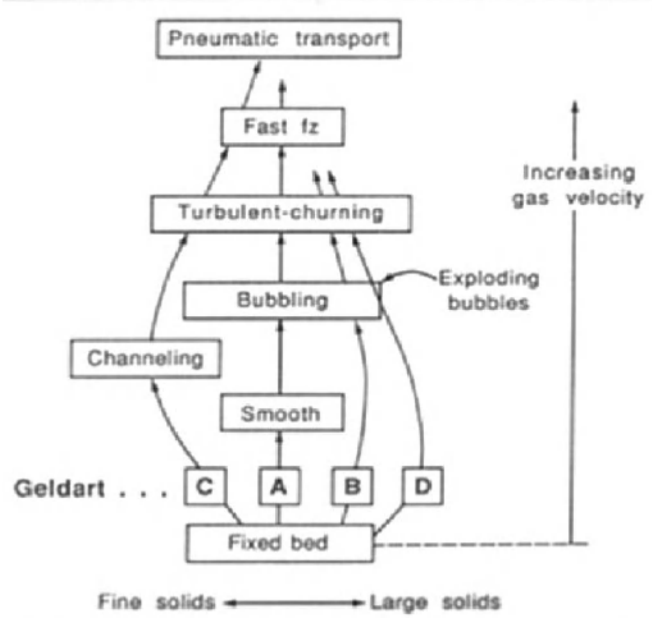


Figure 9.6. Summary of fluidisation behaviour of different powders (Kunii & Levenspiel, 1991)

9.1.3. The Role of Bubbles in Fluidised Beds

As the bubbling bed regime is the most common for granulation systems, it is worth a more detailed analysis. Consider the fluidisation of a group B powder. We can explain the behaviour of the fluidised bed by a two phase model (see figure 9.7). These two phases are the dense (emulsion) phase and the bubbles. The gas velocity and voidage in the dense phase remain at minimum fluidisation conditions. All excess gas above minimum fluidisation is present as bubbles. In summary:

$$\varepsilon_d = \varepsilon_{mf} \tag{9-3a}$$

$$\varepsilon_b = 1.0 \tag{9-3b}$$

$$Q_d = \mu_{mf} A \tag{9-3c}$$

$$Q_b = \left(u_g - u_{mf}\right)A \tag{9-3d}$$

Based on this model, the expansion equation for the fluidised bed is (Rhodes, 1998):

$$\frac{H - H_{mf}}{H_{mf}} = \frac{\varepsilon - \varepsilon_{mf}}{1-\varepsilon} = \frac{u_g - u_{mf}}{0.71\sqrt{g d_b}} \tag{9-4}$$

where d_b is the effective bubble size in the bed.

Bubbles play an extremely important role in the motion of particles in the fluidised bed. Solids are carried in the wakes behind rising bubbles. This causes the excellent solids mixing in bubbling fluidised beds. These solids may be thrown into the freeboard when the bubble bursts at the bed surface (figure 9.8). The relative velocity between particles in the bed (which helps determine rates of attrition and coalescence) is related to the bubble velocity. Thus, we expect all these phenomena to be related to the bubble size d_b and the bubble frequency i.e. $u_g - u_{mf}$.

A number of correlations exist for estimating bubble size in a fluidised bed (Kunii and Levenspiel, 1991). In general, bubble sizes are fairly small for group A particles (2 to 5 cm). For group B particles, bubbles grow with bed height to large size and can easily form slugs in narrow diameter beds.

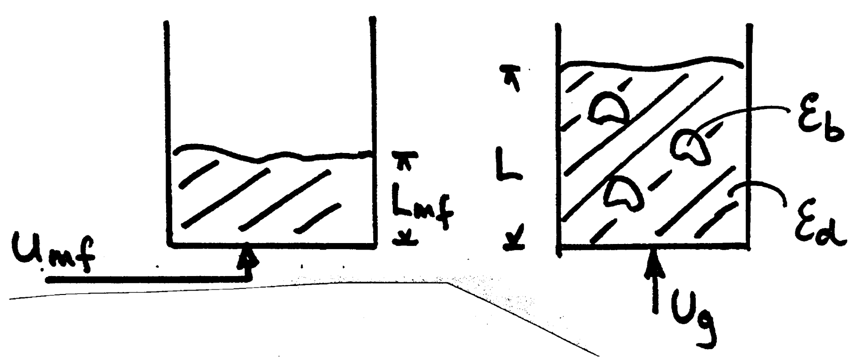
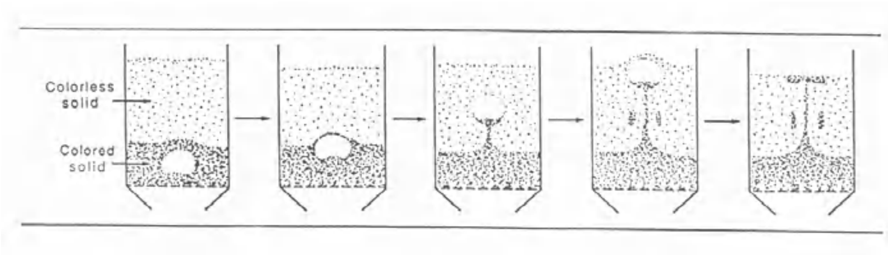
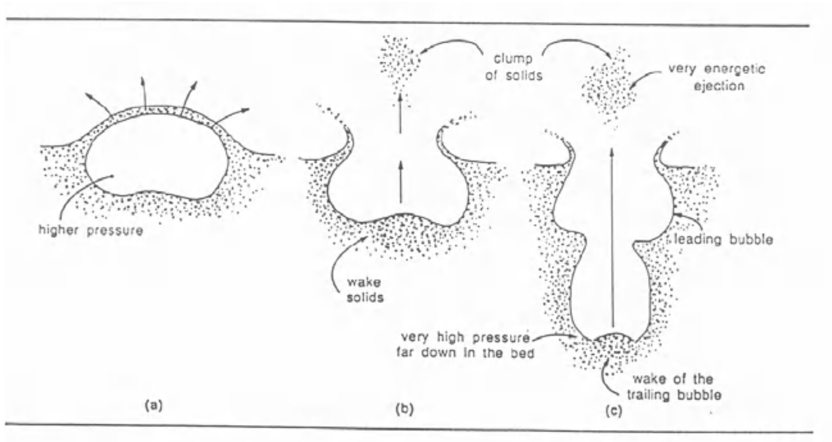


Figure 9.7. Schematic of the two phase model for bubbling fluidise beds



(a)



(b)

Figure 9.8. Effect of bubbles on (a) solids mixing and (b) solids entrainment (Kunii and Levenspiel, 1991)

9.2. Mass and Energy Balances

A fluidised bed granulator acts simultaneously as a drier. It therefore has operational limits that are defined by its ability to evaporate solvent from the liquid or slurry feed. Consider the schematic of a fluidised bed granulator in figure 9.2. Let us denote the streams flowing in and out of the granulator as (1) liquid/slurry feed, (2) seed particles feed, (3) warm air entering the bed, (4) granulated product leaving the bed, and (5) warm, humid air leaving the bed. $\dot{Q}_1$ and $\dot{Q}_2$ are the heat input to the air heater and directly to the fluidised bed respectively. (Direct heating is not shown in figure 9.2).

For a batch operation, the mass balance for the solvent (usually water) is:

$$\dot{M}_1 X_{w1} + \dot{M}_3 X_{w3} = \frac{dM_b X_{wb}}{dt} + \dot{M}_5 X_{w5} \tag{9-5}$$

If we neglect the small amount of water in the bed granules, eqn. (9-5) simplifies to:

$$\dot{M}_1 X_{w1} + \dot{M}_3 X_{w3} = \dot{M}_5 X_{w5} \tag{9-6}$$

Similarly, for steady state continuous operation, the solvent mass balance is:

$$\dot{M}_1 X_{w1} + \dot{M}_2 X_{w2} + \dot{M}_3 X_{w3} = \dot{M}_4 X_{w4} + \dot{M}_5 X_{w5} \tag{9-7}$$

The energy balance for a batch granulator is:

$$\dot{Q}_2 + \dot{M}_1 h_1 + \dot{M}_3 h_3 = \frac{dM_b h_b}{dt} + \dot{m}_5 h_5 \tag{9-8}$$

Similarly, the energy balance for steady state continuous operation is:

$$\dot{Q}_2 + \dot{M}_1 h_1 + \dot{M}_2 h_2 + \dot{M}_3 h_3 = \dot{M}_4 h_4 + \dot{M}_5 h_5 \tag{9-9}$$

Often, the energy required to heat the granules is small compared to that required to evaporate the solvent. If this is the case, both eqns. (9-8) and (9-9) can be approximated as:

$$\dot{M}_1 x_{w1} \lambda \approx \dot{M}_3 C_{p,air} (T_3 - T_5) + \dot{Q}_2 \tag{9-10}$$

Note that both the exit gas and particles from the fluid bed have the same temperature as the granules in the bed i.e. $T_5 = T_4 = T_b$. This is due to the very good mixing properties of fluid beds and limits the thermal efficiency that can be achieved. The thermal efficiency can be improved by supplying some of the heat indirectly through the vessel walls but this is rarely done in practice.

There are both mass and energy balance limits on the maximum liquid feed rate:

- Firstly, the exit gas humidity cannot exceed the saturation humidity in the gas at the exit temperature i.e. $x_{w5} < x_{w,sat}(T_b)$. Once the exit air is saturated, no more liquid can be removed from the bed.
- Secondly, the energy required to evaporate the liquid cannot exceed that available from external heating and the incoming gas (eqn. 9-10). In practice, external heating is rarely used in granulators so that the liquid feed rate is limited by the maximum inlet gas temperature and therefore the capacity of the air heater.

If either of these limits are exceeded, liquid will accumulate in the bed causing bed collapse. It is possible that uncontrolled agglomerate formation could collapse the bed well below the mass and energy balance limits to liquid feed rate. So these limits do not give a conservative design. They simply represent two constraints that must always be met.

9.3. Granulation Mechanisms and Rate Processes

Figure 9.9 is a schematic representation of the important granulation processes occurring in a fluidised bed granulator. These processes are wetting, layering, nucleation, coalescence, consolidation and attrition. Note that much of action occurs in or near the spray zone. In this section, the effect of feed properties and operating variables on each of these processes is discussed in some detail. We will draw on the fundamental understanding of these processes developed in previous chapters and apply it specifically to the fluidised bed.

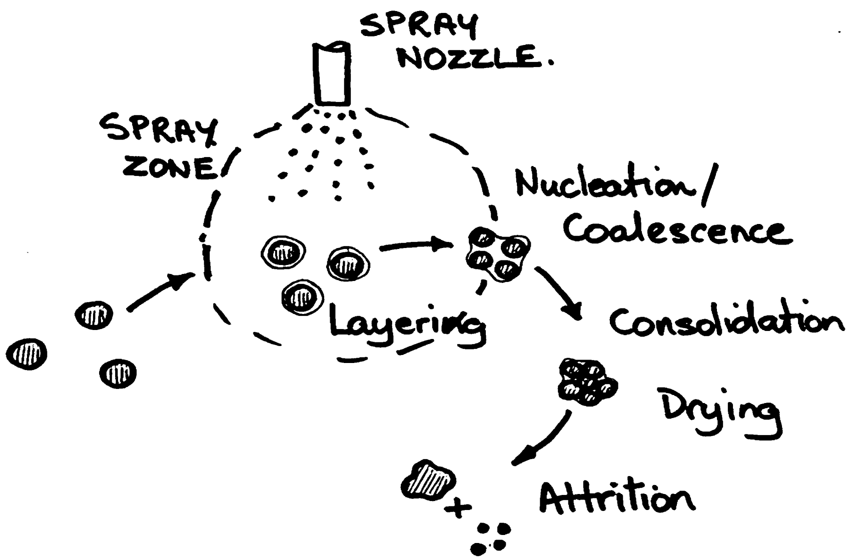


Figure 9.9. Important granulation processes in the fluidised bed

9.3.1. Nucleation

Nucleation is an important process when the droplet size from the atomising nozzle is large compared to the particle size in the bed. In this case, the nuclei formation rate and size distribution are directly related to the droplet size distribution. Nucleation is important when the initial particle size in the bed is very small (group A or C powders).

Ideally, the droplet size should be small and the droplet size distribution narrow. Large liquid drops cause the formation of large nuclei which dry slowly. These nuclei can settle out of the fluid bed causing quenching (defluidisation). To achieve small drops with a narrow size distribution, two fluid atomising nozzles are almost always employed.

Powder wetting also influences nuclei size distribution, although this effect is secondary compared to the droplet size distribution. Nuclei size will increase with increasing adhesive tension $\gamma^{lv}\cos\theta$ (see figure 9.10).

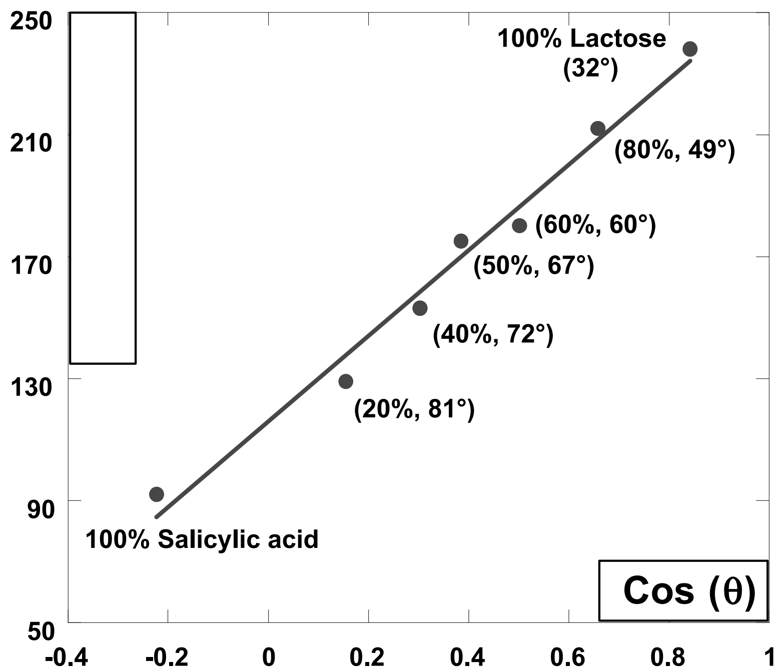
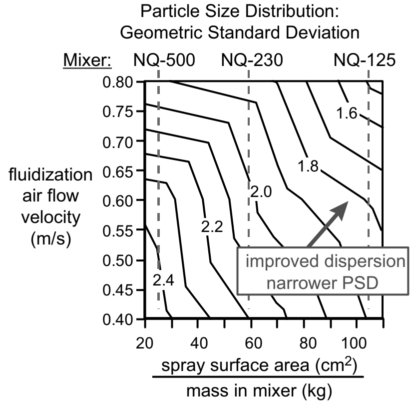


Figure 9.10. The influence of contact angle on nuclei size formed in fluid-bed granulation of lactose/salicylic acid mixtures (Aulton & Banks, 1979)

Where the spray is above the fluidised bed, the dimensionless spray flux Ψ_a can be used to estimate the maximum liquid spray rate to avoid overwetting (eqn. 3-27). To apply to the fluid bed, the powder flux through the spray zone $\dot{A}$ is needed. This is equivalent to the surface renewal rate and can be estimated from standard fluid bed correlations.

Recently, several different workers have demonstrated how correct design of the nozzle spray rate relative to bed surface removal can produce narrow granule size distribution (Akkermans *et al*. 1998; Schaafsma *et al*. 1999; Schaafsma *et al*. 2000; Tardos *et al*. 1997). For example, figure 9.11 shows granule size distribution spread decreases with increased spray area at constant spray rate.



Agglomeration in an Agitated Fluidized Bed

	Mixer Type		
	NQ-125	NQ-230	NQ-500
Powder feed (kg)	0.36	2.23	22.9
Spray area (cm^2)	38.5	132.7	594
Spray area / Feed	107	60	26

Figure 9.11. Geometric standard deviation of granule size in an agitated fluid-bed granulator as a function of gas fluidisation velocity and binder dispersion (measured using spray surface area to mass in mixer). Figure from Mort et al. (1998). Data from Tardos et al. (1997).

9.3.2. Coalescence

If a particle is significantly larger than the liquid droplet size, when it traverses the spray zone, droplets which collide with the particle will form a liquid layer around it. Nucleation is no longer an important mechanism for forming agglomerates. Instead, agglomeration must occur by coalescence of surface wet particles. Growth by coalescence is relatively fast, or order 100 μm/hr (Nienow and Rowe, 1985).

When two surface wet particles collide they will coalesce if they meet the Stokes number criterion discussed in chapter 4 i.e. coalescence will occur if the viscous dissipation in the liquid layers exceeds the kinetic energy of collision. Reviewing the Stokes criterion:

$$St_v < St_v^* \Rightarrow non-inertial\ regime$$
$$St_v \approx St_v^* \Rightarrow inertial\ regime$$
$$St_v < St_v^* \Rightarrow \text{coating}\ regime$$
$$where$$
$$St_v = \frac{4\rho_s V d_v}{9\mu};\text{and } St_v^* = \left(1+\frac{1}{e}\right)\ln\left(\frac{h}{h_a}\right) \tag{9-11}$$

From these equations, we expect gas velocity, binder viscosity and liquid feed rate to affect the extent of coalescence in the bed ***but only if the bed is operating in the inertial regime***. For example, if the bed is made up of very small particles so that $St_v << St_v^*$, doubling the gas velocity will not reduce the probability of coalescence at all. Instead, the rate of coalescence may actually increase due to an increase in particle collision frequency. This comment should be born in mind in the following discussion.

Motion of particles in the bed is dictated by the movement of bubbles. By looking at the velocity gradients in the dense phase of a fluidised bed we can estimate the maximum collision velocity between particles (Ennis *et al*, 1991). As might be expected, the maximum velocity is found where the nose of a bubble is pushing throught the surrounding dense phase. The particle collision velocity is related to the velocity and size of bubbles in the bed:

$$V \approx \frac{6u_B d_v}{d_B} \tag{9-12}$$

It is now clear why it is important to understand the two phase model for a bubbling fluidised bed. Bubble size and rise velocity are related to the excess gas velocity (eqn. 9-4.) so that particle collision velocity should increase with gas velocity. Thus increasing gas velocity should increase the maximum granule size that can be achieved by coalescence. Figure 9.12 demonstrates this effect for the granulation of glass beads with a CMC binder. The granules initially grow at similar rates. However, with the higher excess gas velocity, the inertial regime is reached at a granule size of approximately 500μm i.e. $St_v \approx St_v^*$. At the lower excess gas velocity, the particle collision velocity and therefore the viscous Stokes number are smaller. Granules continue to grow to greater than 600μm in size.

Increasing binder viscosity has a similar effect. Figure 9.13 shows results for granulation of glass beads with a low viscosity binder (CMC-V) and a high viscosity binder (PVP 360K). Note that initially the two systems behave identically. It is not until the inertial regime is reached for the PVP system at about 900μm, that the effect of viscosity is seen. Note that simultaneous drying means that the binder viscosity is changing with time. There will be a spread of viscous Stokes numbers for particles with different extents of drying. If possible, the binder viscosity during drying should be characterised.

Increasing liquid feed rate increases the average liquid layer thickness picked up by a particle in the spray zone. This will increase the critical Stokes number (eqn. 9-11) and therefore the probability of coalescence. Increasing the liquid feed rate also increases the mal-distribution of liquid in the bed. Therefore increasing liquid flow rate increases both the mean granule size and the spread of the granule size distribution.

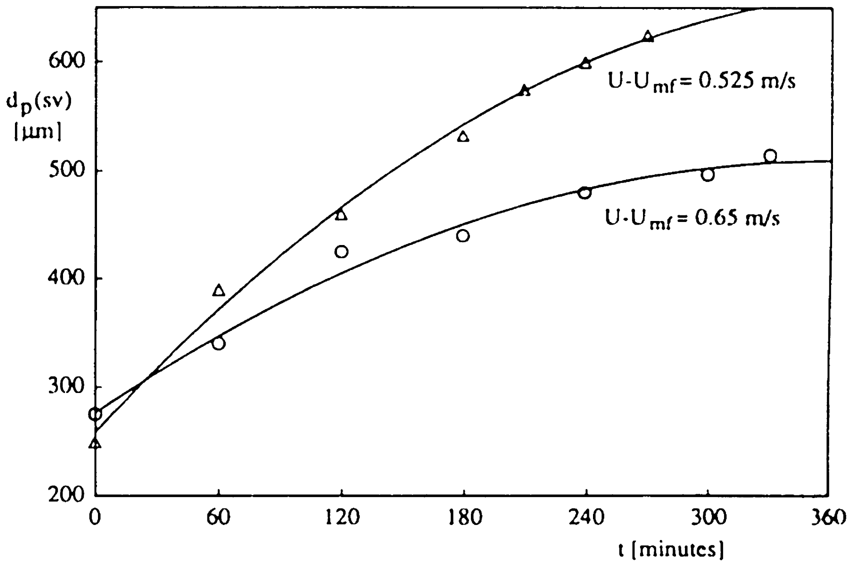


Figure 9.12. Effect of gas velocity on growth by coalescence (Smith & Nienow, 1983)

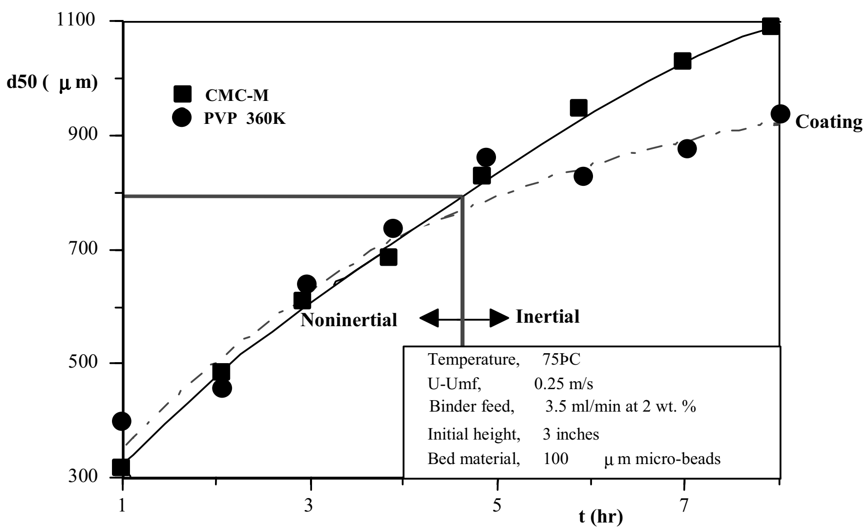


Figure 9.13. An illustration of the regimes of granule growth for fluid-bed granulation of 50 μm glass microbeads with two wt. percent PVP and Na-CMC polymeric binder solutions. The PVP binder exhibits all three regimes, whereas the CMC remains nearly entirely within noninertial growth. (Ennis et al., 1991)

9.3.3. *Layering*

If conditions are not suitable for granule growth by nucleation or coalescence, layering remains the only mechanism available to increase particle size. Layering is the dominant growth mechanism in the coating regime ($St_v >> St_v^*$) and is therefore most important for beds of large granules (group B or D) and low binder viscosity e.g. ammonium nitrate granulation, urea prill fattening and coating applications. Layered growth gives dense, strong granules with a very narrow size distribution. However, growth rates are an order of magnitude lower than growth by coalescence, typically 5 to 10μm/hr.

At a first approximation, all particles in the bed will have the same linear growth rate i.e. each particle will receive the same layer thickness. Note that this implies that the largest particles in the bed get more of the mass of the liquid feed due to their larger surface area.

Variation in layer thickness could be due to segregation of particles in the bed or the stochastic nature of particle circulation between the spray zone and the rest of the bed. The variation in layer thickness will increase with increasing liquid feed rate. However, provided layering remains the growth mechanism, granule size distributions will remain narrow when compared to those produced by coalescence.

The bed population balance can be used to predict the granule size distribution by layering (see Chapter 6).

9.3.4. *Consolidation*

Consolidation has a strong effect on the final porosity of granules produced by coalescence. Therefore, consolidation is also important in determining a range of porosity related properties including strength, bulk density and dispersibility.

In fluidised bed granulators, consolidation will only occur over a short time scale. Once the granule is dry, no further consolidation occurs. For this reason, granules from fluidised beds are often more porous than granules from tumbling or mixer granulators where no drying occurs. This, of course, is only true where coalescence is the dominant mechanism of growth.

The consolidation is due to collisions with other granules in the bed. The degree of consolidation decreases with the Stokes Deformation number according to equation 4-5a. If a highly viscous binder is used, granules will not consolidate much and energy of the collision will be dissipated within a few particle diameters of the agglomerate surface. This leads to high porosity granules and shell formation ie. a low porosity shell at the granule surface protecting a more porous core. This non-uniformity in granule porosity may affect other granule properties eg. dispersion.

The extent of consolidation will increase with the number of collisions ie. with consolidation time. In the fluidised bed, consolidation time is dependent on the rate of drying of the granules. Drying time can be increased by increasing the liquid feed rate or by decreasing the bed temperature.

9.3.5. Attrition

Due to the relatively high velocities in fluidised beds, attrition may be a significant mechanism to generate fines which are elutriated from the bed. The attrition rate depends on both the magnitude of the applied force and the strength of the granules. Granules will attrit due to collisions with other granules and bed walls and internals. The motion of granules in the bed is set by the bubble motion and therefore varies with excess gas velocity.

Attrition rate is inversely proportional to granule strength which is closely related to granule porosity (see section 5.2.4 and 5.2.5).

Where uncontrolled granule growth (dry quenching) is a problem, jet grinding is sometimes employed to limit the maximum granule size. The jet velocity is much higher than other velocities in the bed, leading to much increased attrition rates as well breakage of larger granules.

9.3.6. Summary

The effect of operating variables on granule properties is very complex. These effects are summarised in Table 9.2. "Rules of Thumb" for operation depend very much on knowing the dominant granulation mechanisms i.e. the operating regime. They should therefore be treated with great caution. However, understanding the micromechanics allows more rational and quantitative analysis to be used.

Table 9.2. Effect Of Variables On Fluidized-Bed Granulation

Operating or material variable	*Effect of increasing variable*
Liquid feed or spray rate	Increase size and spread of granule size distribution Increase granule density and strength Increase chance of defluidization due to quenching
Liquid droplet size	Increase size and spread of granule size distribution
Gas velocity	Increase attrition and elutriation rates (major effect) Decrease coalescence for inertial growth No effect on coalescence for noninertial growth Increase granule consolidation and density
Bed height	Increase granule density and strength
Bed temperature	Decrease granule density and strength
Binder viscosity	Increase coalescence for inertial growth No effect on coalescence for noninertial growth Decrease granule density
Particle or granule size	Decrease chance of coalescence Increase required gas velocity to maintain fluidization

9.4. Scale up of Fluidised Bed Granulators

Fluidised bed granulators are more easily scaled than tumbling or mixer granulators. Granule size distribution and density will not change greatly on scale up, provided the following rules are considered.

- The pilot plant fluid bed should be at least 0.3 meters in diameter so that bubbling rather than slugging fluidization behavior occurs. As the bubbling phenomena are so important to coalescence and attrition, bubbles of similar size and frequency are needed in the pilot and full scale plant.
- Where possible, scale up maintaining the fluidised bed height constant. Granule density and attrition resistance increase linearly with operating bed height whereas the rate of granule dispersion decreases. If bed height has to be increased on scale up, agitation intensity should also be increased.
- Provided bed height is maintained, gas throughput and solids capacity will scale with bed cross sectional area. Mass and energy balance limits (eqn. 9-10) should be checked on scale up.
- The spray zone is critical to the operation of a fluid bed as a granulator. The type of spray nozzle, droplet size distribution and relative nozzle position should be unchanged on scale up. The dimensionless spray flux ψ_a should be used to choose the number and position of spray nozzles in the bed.

Figure 9.14 shows a flexible pilot plant design for testing new products in either batch or continuous mode.

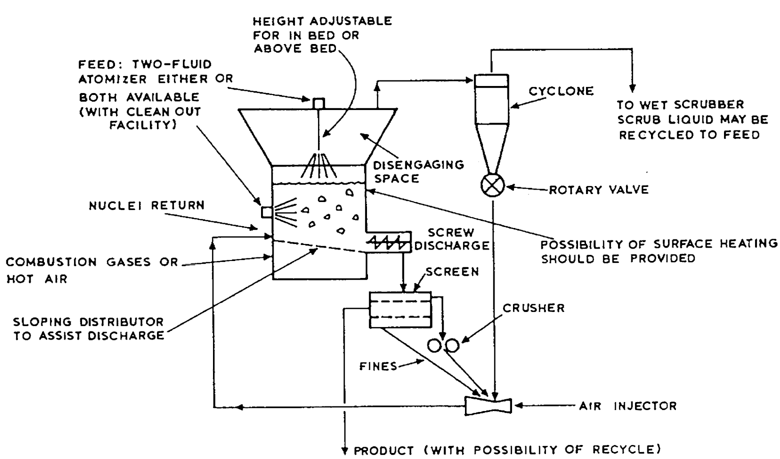


Figure 9.14. Fluidized-bed pilot plant (Nienow and Rowe, 1985)

9.5. Operation of Fluidised Bed Granulators

Fluidised bed granulators are sometimes fraught with operating difficulties. These are often associated with either the nozzle and spray zone, or defluidisation (quenching) phenomena.

Problems with the liquid feed nozzle include caking outside the nozzle and clogging inside the nozzle. Caking is caused by wet particles or drops blowing back and sticking to the outside of the nozzle. The cake may eventually intrude on the spray and prevent good liquid distribution or large lumps may fall into the bed and sink to the bottom. Caking is largely a problem where the nozzle is above the bed surface. If the nozzle is submerged into the bed, fast capture of the liquid droplets as well as scouring of the nozzle by particles prevent caking.

Blockages within the nozzle are always a potential danger, especially if a slurry feed is used. The problem is minimised by keeping the nozzle design as simple as possible. In situ cleaning or easy removal of the nozzle is essential to cope with blockages and caking.

Defluidisation is caused by the formation of very large agglomerates in the bed. These agglomerates settle out at the bottom of the bed causing chanelling, poor gas distribution, and reduced mixing in the bed which promote conditions for further agglomerate formation. The cycle continues and shut down inevitably follows. Early detection is therefore important to prevent this positive feedback leading to disaster.

Two causes of defluidisation are identified: wet quenching and dry quenching. Wet quenching is the formation of large wet agglomerates. It is caused by high liquid spray rates, large liquid droplets or dripping nozzles. The problem can generally be fixed by adjusting the liquid feed rate or the nozzle operation.

Dry quenching is the formation of large stable agglomerates. It occurs if growth by coalescence dominates even at large granule size or due to caking at the nozzle or bed internals. Dry quenching is more difficult to combat than wet quenching. One approach is the use of jet grinding within the bed to limit the maximum size of agglomerates.

Early detection of quenching is very important. Quenching is detected by careful monitoring of the bed temperature just above the distributor plate. If the bed is well fluidised, temperature gradients will be negligible and the temperature above the distributor will be similar to the average bed temperature. Sudden changes in temperature low in the bed indicate poor fluidisation. An increase or decrease in temperature indicate dry quenching and wet quenching respectively.

Control of fluidised beds is accomplished primarily by monitoring bed temperature, as well as granule size and density. Temperature is controlled best by adjusting the liquid feed rate. For batch granulation, the fluidizing air velocity must be increased during the batch. Bed pressure fluctuations can be used to monitor the quality of fluidization and to indicate when gas velocity increases are required. In addition, intermittent sampling systems may be employed with on-line size analyzers to monitor granule size. There is no simple heuristic to control the final granule size distribution (batch) or exit granule size distribution (continuous). A good knowledge of the granulation processes combined with model based control schemes can be used to fix the batch time (batch), or adjust the seed granule recycle (continuous) to maintain product quality.

9.6. Draft Tube and Spouted Bed Coaters

Numerous modifications to the simple fluid bed design are possible. Placing a central draft tube in the fluidised bed forces a more regular circulation of solids through the coating zone. The commonly used Wurster coater employs this technique (see figure 9.15). These systems are commonly used for layered growth of large granules and for coating applications. The Wurster coater is a bottom sprayed system and the gas distribution is varied between the inside and outside of the draft tube to encourage the circulation pattern. (see table 9.3)

In a spouted bed, no attempt is made to uniformly distribute the gas entering the bed. Instead, all air enters through a single central orifice at the bottom of a conical section. The bed forms two regions:

- a low density, high velocity spout through which most of the air travels; and
- a high density, moving bed annular region through which air percolates at a low velocity.

Granules entrained in the spout are carried to the bed surface and rain down on the annulus. Particles move downward through the annulus until they are entrained in the spout again. Figure 9.16 shows the elements of a spouted bed granulator or coater.

The spouted bed has many of the elements of a draft tube fluidised bed. Table 9.4 compares the spouted bed with the more common simple fluidised bed.

Table 9.3. Sizes and Capacities of Wurster Coaters

Bed diameter, inches	*Batch size, kg*
7	3-5
9	7-10
12	12-20
18	35-55
24	95-125
32	200-275
46	400-575

Table 9.4. Comparison between spouted beds and fluidised beds

Similarities to fluid beds	*Differences from fluid beds*
• gas causes particle motion and mixing • simultaneous granulation and drying • too high liquid rate causes spout collapse, equivalent to wet quenching • high attrition, elutriation rates • exceed minimum spouting velocity to operate • mass, energy balance limitations	• very good for group D particles • growth by layering only, therefore good for coating applications • more controlled particle circulation • better at handling non-spherical particles • does not easily scale past 2m diameter

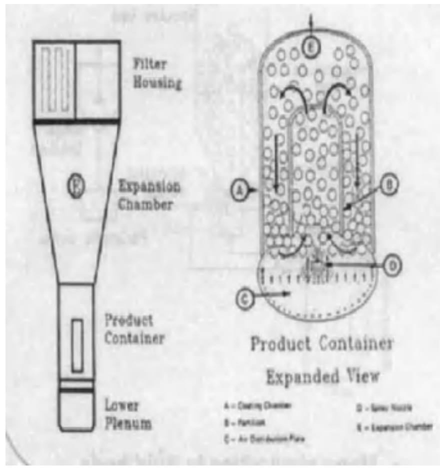


Figure 9.15. Schematic of Wurster draft tube coater

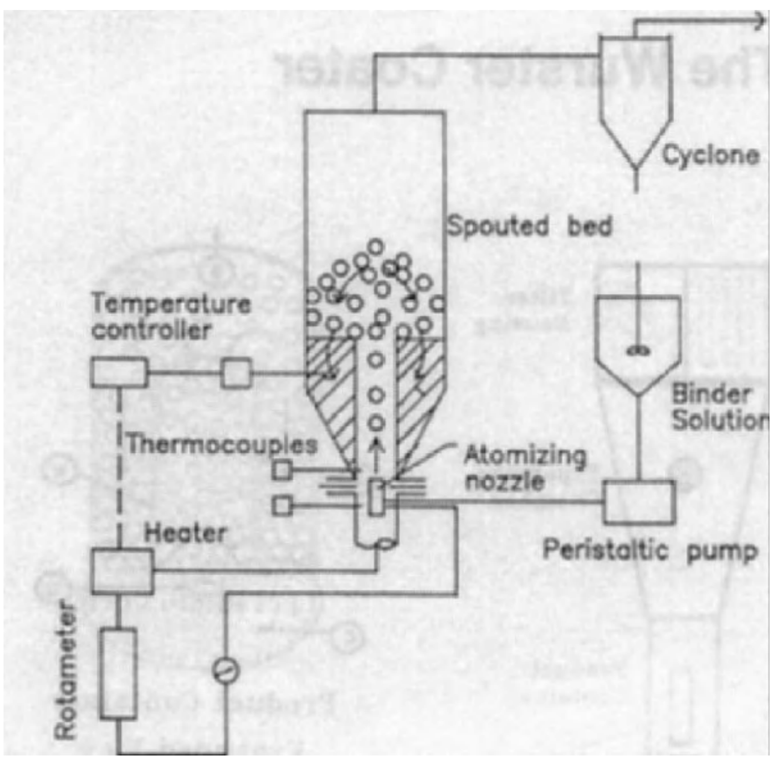


Figure 9.16. Elements of a spouted bed granulator or coater

Spouted beds have most application for Group D particles. These particles are difficult to fluidise uniformly. In contrast, they produce very stable spouted beds using less air than is required for the equivalent fluidised bed. Mathur and Epstein (1974) give details on spouted bed hydrodynamics.

In a spouted bed, granules grow exclusively by layering. Gas and particle velocities in the spout are very high and particle sizes are large. This leads to high Stokes numbers (eqn. 9-11) and operation in the coating regime. For this reason, spouted beds are often used for coating. For both bottom and top sprayed coating, the growth of large granules is favoured over smaller granules, though for different reasons. For bottom sprayed coating, the coating mass is proportional to the particle mass (see figure 9.17).

$$\textit{attrition}\ \text{rate}\,\alpha\,\frac{\mathrm{A_i}u_i^3}{K_c} \tag{9-13}$$

where A_i is the area of the inlet orifice and u_i is the velocity of gas through the orifice. Figure 9.18 shows the attrition rate as a function of inlet gas kinetic energy for seeds coated with tricalcium phosphate.

During granulation or coating operation, spouted beds are subject to similar mass and energy balance limitations to fluidised beds. However, the maximum liquid feed rate may be limited to below this level by spout collapse or ratholing. Both these phenomena relate to agglomerate formation in different regions in the bed. Figure 9.19 show the maximum liquid rate for seed coating applications limited by spout collapse. In this case, agglomerates form in the coating zone at the bottom of the spout causing the spout to collapse. Agglomerates form when the liquid layer around particles in the spray zone is thick enough for transition from the coating regime to non-inertial regime (eqn. 9-11). Experimentally, the maximum liquid rate was typically 20 to 90% of that required to saturate the exit air and found to increase with increasing gas velocity, bed temperature and decreasing binder viscosity (Liu & Litster, 1993a).

Ratholing occurs if the coating is not completely dried as the granule traverses the spout and fountain regions and is still sticky while the particle are in the annulus. The annular region is a moving packed bed where particle relative velocities are very low. If particles in the annulus are sticky, a cohesive mass forms and particles will no longer fall into the spout. Instead a stable rathole forms with no particle circulation.

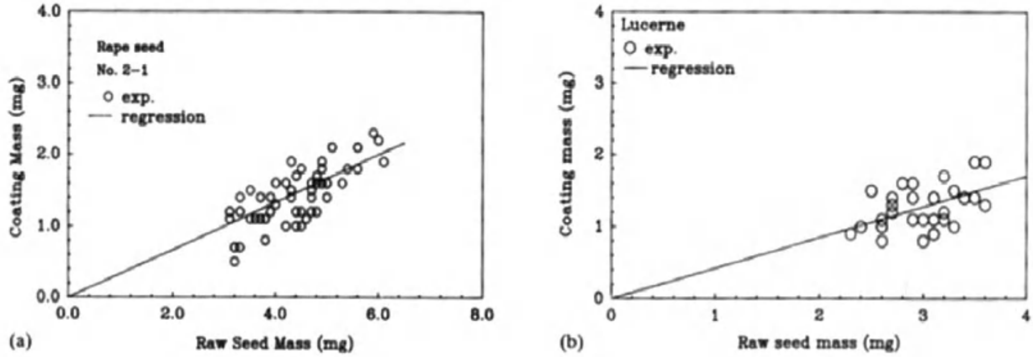


Figure 9.17. Relationship between particle mass and seed mass for seeds coated in a spouted bed (Liu & Litster, 1993b)

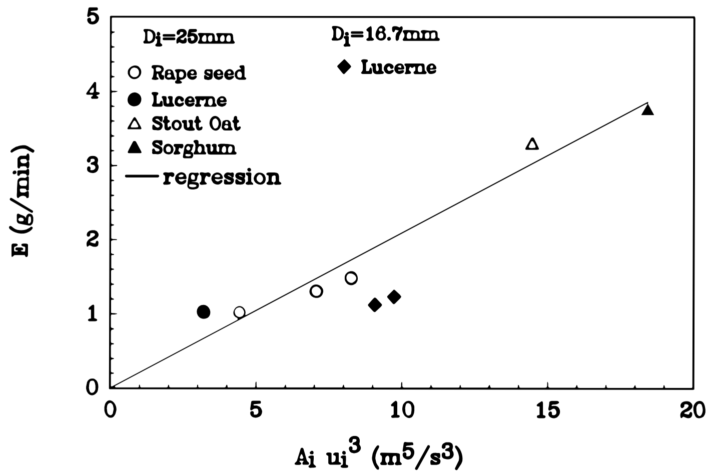


Figure 9.18. Attrition rates for coated seeds in a spouted beds (Liu & Litster, 1993a)

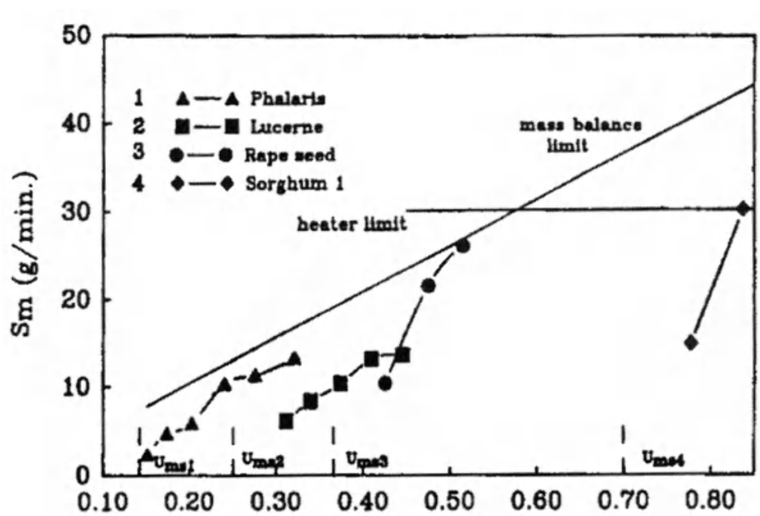


Figure 9.19. Maximum liquid rate for a spouted bed coater as a function of gas velocity and particle type (Liu and Litster 1993a)

9.7. Concluding Comments

Fluidised bed and their related techniques are well established as devices for both granulation and coating applications. Correct use of the technique requires a good understanding of both fluid bed hydrodynamics **and** granulation processes in the bed. With this understanding reliable design and scale up is quite possible.

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9.9. Nomenclature

A	bed cross sectional area (m^2)
A_i	inlet orifice area (m^2)
C_A	specific heat of air ($J\ kg^{-1}\ K^{-1}$)
C_v	specific heat of water vapour ($J\ kg^{-1}\ K^{-1}$)
C_F	specific heat of wet feed ($J\ kg^{-1}\ K^{-1}$)
C_s	specific heat of granular product ($J\ kg^{-1}\ K^{-1}$)
d_v	volume equivalent particle size (m)
d_b	volume equivalent bubble size (m)
e	coefficient of restitution
g	gravitational acceleration (ms^{-2})
H	fluid bed height (m)
H_{mf}	minimum fluidisation bed height (m)
h	liquid layer thickness (m)
h_a	height of surface asperities (m)
J_o	exit gas absolute humidity

J_{sat}	equilibrium saturation humidity of exit gas
K_c	fracture toughness
Q_d	volumetric flow of gas through dense phase ($m^3\ s^{-1}$)
Q_b	volumetric flow of gas through bubble phase ($m^3\ s^{-1}$)
q_w	external heat input (W)
q_L	heat losses (W)
St_v	viscous Stokes number
St_v^*	critical Stokes number
T_b	fluid bed temperature (K)
T_F	wet feed temperature (K)
T_{Ao}	exit air temperature (K)
T_{Ai}	inlet air temperature (K)
T_s	granular product temperature (K)
u_{mf}	minimum fluidisation velocity ($m\ s^{-1}$)
u_g	superficial gas velocity ($m\ s^{-1}$)
u_i	gas velocity through inlet orifice ($m\ s^{-1}$)
u_b	bubble rise velocity ($m\ s^{-1}$)
V	particle collision velocity ($m\ s^{-1}$)
W_F	mass flow rate of wet feed (kgs^{-1})
W_A	mass flow rate of inlet air (kgs^{-1})
x_s	mass fraction solid in wet feed
ΔP_{fb}	pressure drop across the fluid bed (Pa)
ε	fluid bed average voidage (volume fraction of fluid)
ε_{mf}	minimum fluidisation voidage
ε_d	dense phase voidage
ε_b	bubble phase voidage
ρ_s	particle density ($kg\ m^3$)
ρ_g	gas density ($kg\ m^3$)
μ	viscosity (Pa s)
λ	latent heat of vaporisation of water ($J\ kg^{-1}$)

CHAPTER 10

MIXER GRANULATORS

10.1. Introduction

Mixer granulators use a mechanical impeller to cause powder motion, while the granulator as a whole is usually fixed in space. Liquid binder is sprayed (or sometimes poured) onto the moving powder bed. Figure 10.1 shows a schematic of the elements of a typical batch high shear mixer granulator. In fact, there are an enormous range of geometries and designs over mixer granulators covering batch and continuous configurations and with a very wide range of agitation intensities (shear rates). In many cases, the designs are modifications of those for mixer/blenders in which granulation was noted as a desirable side effect.

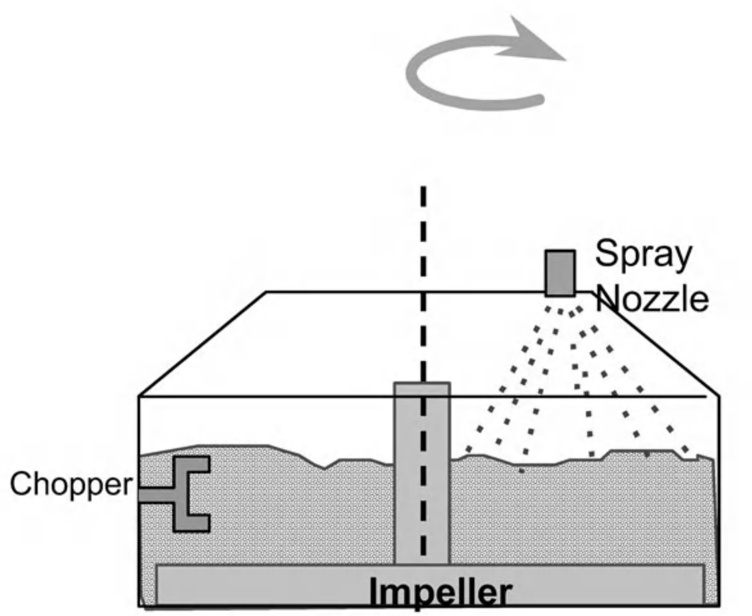


Figure 10.1. Key elements of a vertical shaft batch mixer granulator

Mixer granulators find wide application for pharmaceuticals, clays and ceramics, detergents and agricultural chemicals. Table 10.1 lists some of the advantages and disadvantages of mixer granulators. Mixer granulators will handle a wider range of feed materials than tumbling or fluidised granulators including very fine cohesive powders and very viscous liquid binders. This is due to the ability of the impeller or chopper to break up clumps of material and mechanically disperse the liquid binder. However, the complex flow patterns, wide range of shear rates and simultaneous influence of all three rate processes make control of product attributes in mixer granulation very difficult. Mixer granulators tend to make smaller, denser, less spherical granules than their tumbling counterparts.

Table 10.1. Advantages and disadvantages of Mixer granulators

Advantages	*Disadvantages*
Can process cohesive powders and spread viscous binders.	High capital, maintenance and operating costs.
Will generally produce a product (more robust than other process).	Granules are less spherical than tumbling granulators.
Only process to make small (less than 1mm) dense granules.	Most complex of granulators. Difficult to scale and maintain product attributes.

10.2. Description of Mixer Granulation Equipment

The geometry of tumbling and fluidised bed granulators is reasonably standard. However, there is an extremely wide range of mixer granulation equipment. In many cases, different industries have preferred designs and suppliers for largely historical rather than technical reasons. In many cases, designs have come from mixers or blenders used originally in the food and mineral processing industries. The equipment can be divided, somewhat arbitrarily, into high shear and low shear mixers. High shear mixers are now much more common and will be discussed in some detail. Low shear mixers will be described more briefly.

10.2.1. High Shear Mixer Granulators

In high shear mixers, the impeller and/or chopper rotate at moderate to high speed generating high shear rates and impact velocities in some region of the mixer. Depending on the design and powder flow patterns, this region may be only small proportion of the whole mixer.

Batch high shear mixers can be further divided into three categories (see figure 10.2):

- Vertical shaft: Fielder, Diosna, Gral
- Horizontal shaft: Lodige
- Eccentric shaft: Eirich

In these mixers, the impeller blade/ploughs rotates at speeds of between 60 and 800 rpm depending on scale and design with impeller tip speeds up to of order 10 ms^{-1}. Most designs incorporate an off centre chopper rotating at very high speed (500 to 3500 rpm). These mixer granulators range in scale from 10 litres to 1200 litres and typical batch times are of the order of 5 minutes.

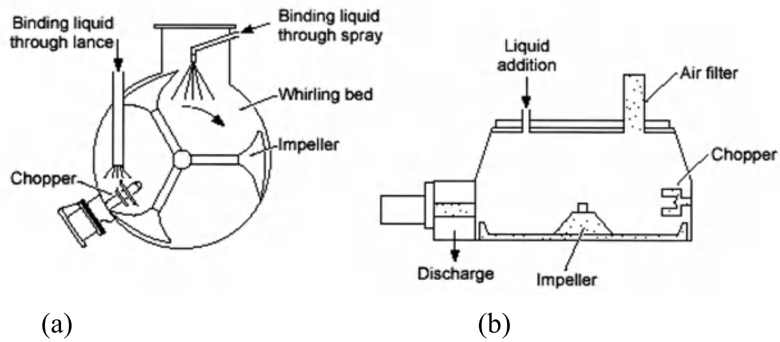


Figure 10.2. Schematic of some designs of high shear batch mixer granulators (a) horizontal shaft; (b) vertical shaft

Batch high shear mixer granulators are used extensively in the pharmaceutical industry where they are valued of their robustness and ease of enclosure (Kristensen, 1988, Schaefer, 1988). Often dry mixing of ingredients and granulation take place in the same piece of equipment. Figure 10.3 shows detail of different designs of impellers and choppers from Diosna granulators. Variations in equipment, impeller and chopper geometry cause powder flow patterns and shear rates to vary enormously. For this reason, great caution should be exercised in transferring empirical knowledge or folk lore from one design of mixer to another. For example, the effect of the chopper on granule attributes is relatively small for Fielder and Diosna designs (Litster *et al.*, 2002). In contrast, in the Lodige design, the plough shear mixing tools push a large proportion of the powder mass through the relatively large chopper region and the effect of the chopper is more pronounced (Iveson et al., 2000).

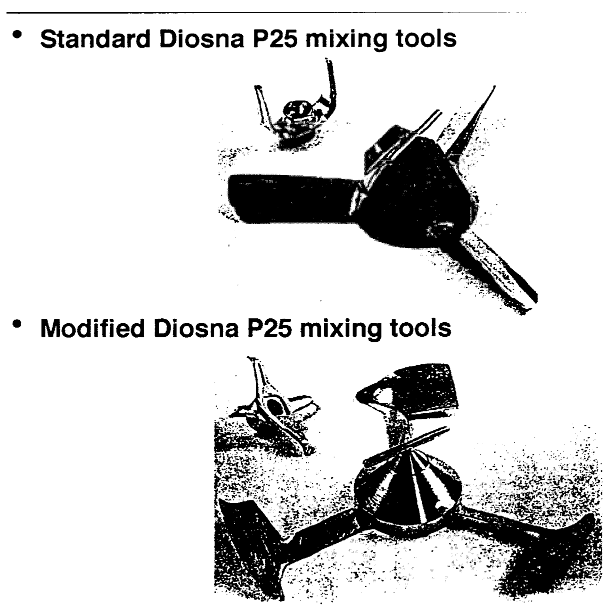


Figure 10.3. Detail of impeller and chopper design from a Diosna mixer granulator

he Eirich mixer granulator is a unique design. Both the impeller blades and chopper rotate on eccentrically mounted vertical shafts (see figure 10.4). In addition, the bowl rotates giving a hybrid effect with some features of a tumbling granulator. This type of granulator gives more spherical granules than those achieved in other batch high shear mixers. The Eirich granulator is commonly used for granulation of ceramics and clays, although its use in other industries is limited.

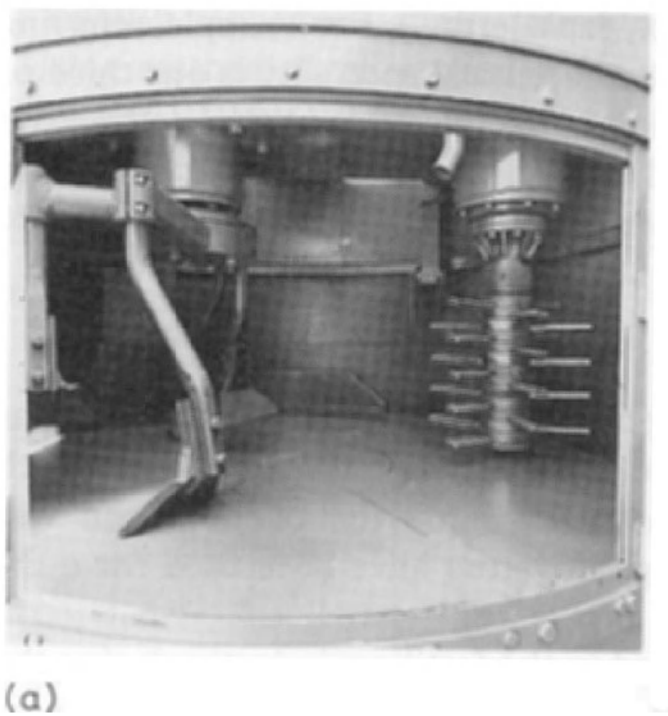


Figure 10.4. Detail of impeller blades and chopper in an Eirich Mixer (Pietsch, 1992)

There are a number of design of continuous high shear mixer granulators for medium to high capacity processes. Most of these designs have blades or pins mounted on a vertical shaft which rotates at high speed. The action of the blades promotes powder mixing and binder dispersion as well as controlling hold up by the rate at which feed material is transported from the feed to the discharge end of the granulator. Typical residence times are of the order of minutes. An example is the peg granulator used for granulation of ceramic clays and carbon black (see Figure 10.5). Full scale granulators have a diameter of 1.5m, the shaft rotates at 225rpm and the capacity is of order 15 tonne/hour.

Lodige makes continues granulators with similar design to the batch granulators described above (see Figure 10.2). This design of granulator is common in the manufacture of detergents.

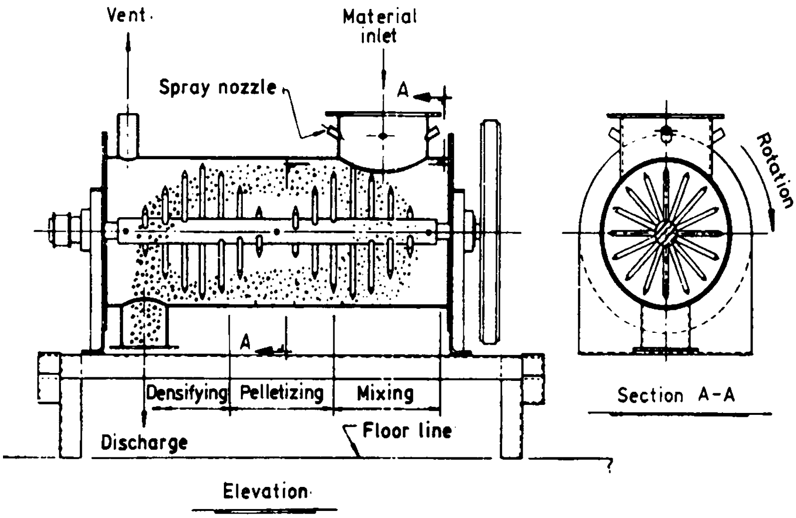


Figure 10.5. Schematic of a peg granulator (Capes, 1985)

The Shugi Fleximix granulator (figure 10.6) is a continuous high shear granulator with blades mounted on a vertical shaft that spin at extremely high speed (1000 to 3500 rpm). Residence time in the granulator is only a few seconds during which intimate mixing of a sprayed liquid binder with even fine cohesive powders is achieved. However, there is no time for densification and growth. The “granules” produced are irregular, fluffy and relatively low density. They will continue to grow and densify with further handling. Usually the product from a Shugi mixer drops directly into a fluidised bed drier. Shugi mixers have capacities up to 50 tones per hour with power requirements up to 100kW.

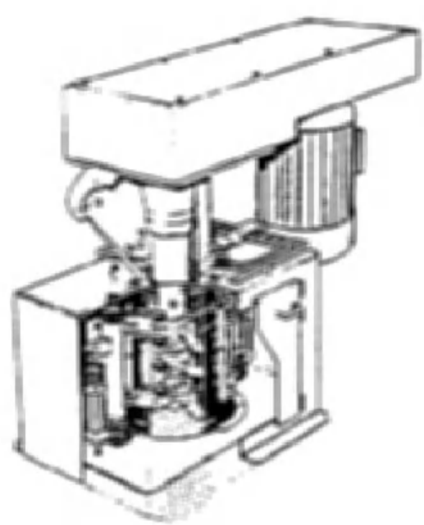


Figure 10.6. The Shugi Flexomix vertical high shear granualtor (Courtesy of Hosikawa Micron Australia)

ow Shear Mixer Granulators

Pug mills and paddle mills are used for both batch and continuous operation (figure 10.7). These devices have horizontal troughs in which central shafts with attached mixing blades rotate relatively slowly (<100rpm). Both single and double trough designs are used. These mills have largely been replaced by tumbling granulators in fertiliser and metallurgical applications. However, they are still used commonly as a pre-mixing step for blending very different raw materials eg. dry powder and wet filter cake.

Batch planatary mixers were very common in pharmaceutical manufacture. A typical unit has a batch size of 100 to 200 kg and a power input of 10 to 20 kW. Mixing times in these granulators are quite long (20 to 40 minutes) and they have been largely replaced by high shear batch granulators.

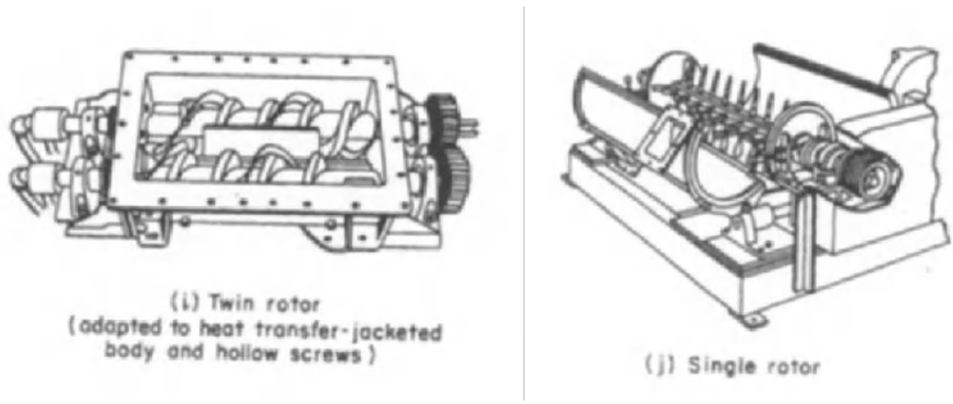


Figure 10.7. Twin rotor and single rotor pug mills (Perry and Green, 1997)

10.3. Powder Flow in Mixer Granulators

Powder flow in mixer granulators is induced by transfer of momentum from the impeller to the powder through a combination of (a) friction and shear, and (b) inertia. Clearly, the powder flow characteristics will have a significant impact on the granulation processes and therefore the product granule attributes. However, predicting the powder flow patterns is problematic for two reasons:

1. The dynamics of flow of fine, unfluidised powders is poorly understood. Powder flow in mixers falls into the intermediate powder flow regime. There are currently no generally accepted constitutive model for the flow of powders in this regime. Therefore, there is no equivalent in powder mixing to the use of CFD models to study flow in liquid mixers.
2. Powder flow is very sensitive to equipment geometry, which varies widely for different mixer granulator designs.

This means that the impact of mixer geometry, impeller speed and powder properties on the powder flow patterns in the mixer cannot be predicted *a priori*. Nevertheless, visualisation studies give some useful information on powder flow in mixers. We will look at some results from studies for two types of mixers – a vertical shaft mixer (Fielder) and a horizontal shaft mixer (Lodige).

10.3.1. Powder Flow in a Vertical Shaft Mixer

In a vertical shaft Fielder mixer granulator, two flow regimes are observed (Litster *et al.*, 2002). At low speeds, the powder is displaced vertically as the impeller blade passes underneath leading to slow, bumpy powder motion in the tangential direction. There is almost no vertical turnover of the powder bed (see figure 10.8a).

At higher impeller speed, there is a transition to "roping" behaviour (see figure 10.8b). Material from the bottom of the powder bed is forced up the vessel wall and tumbles down the angle of the bed surface towards the centre of the bowl. There is both good rotation of the bed and good vertical turn over.

At the transition impeller speed the bed is very unstable. The bed pushes out and up the wall of the bowl before collapsing (avalanching) back towards the centre of the bowl.

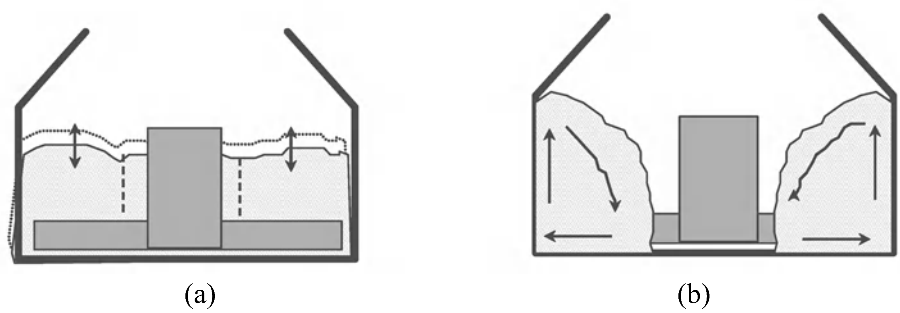


Figure 10.8. Flow regimes in Fielder mixer granulators (a) bumping at low speed; (b) roping at high speed.

Figure 10.9 shows powder surface velocity data from a 25 l Fielder mixer granulator. In the bumping flow regime, the powder surface velocity increases in proportion to the impeller speed. There is a break point in the curve corresponding to the change in regime from bumping to roping. In the roping regime, the surface velocity stabilises and is less sensitive to impeller speed. Note that in this case, the measured surface velocities were in the range 0.2 to 1.2 ms^{-1} which is an order of magnitude lower than the impeller tip speed.

The transition from bumping to roping is due to the change in the balance between the powder rotational inertia, which tends to push the powder to the outside of the bowl, and gravity. This balance between rotational inertia and gravity is given by the Froude Number:

$$Fr = \frac{DN^2}{g} \qquad 10.1$$

where D is the impeller diameter, N is the impeller speed and g is the gravitational acceleration. The critical Froude number for the bumping/roping transition Fr_c will increase ncreasing dimensionless bed height (H/D) and decrease as powder cohesion increases. Thus, Fr_c decreases as binder liquid is sprayed onto the dry powder.

Ramaker et al. (1998) observed a similar toroidal flow pattern. Impeller blade design also has a significant impact of bed turnover rate and Fr_c (Schaefer, 1997). Knight et. al. (2000) also studied flow in the toroidal (roping) regime and showed the dimensionless torque was a function of dimensional bed height and Froude Number.

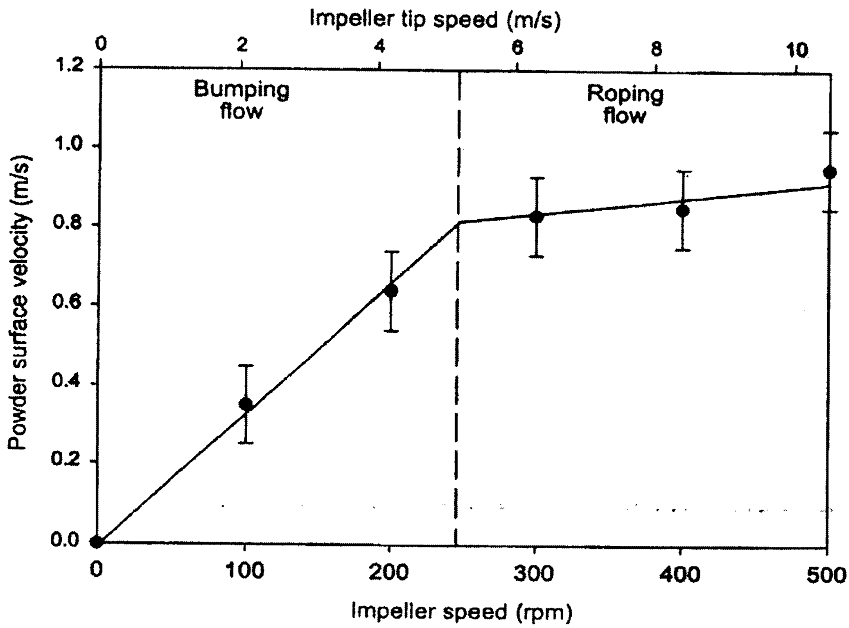


Figure 10.9. Powder surface velocities as a function of impeller tip speed (dry lactose powder in a 25 l Fielder granulator) (Litster et. al., 2002)

10.3.2. Powder Flow in Horizontal Axis Lodige Granulator

In a Lodige mixer, the powder flow pattern is quite different (Forrest *et al.,* 2001; Jones, 1997). Figure 10.10 illustrates the flow of the powder bed in the region of a plough shear as it rotates through one revolution. These results are from position emission particle tracking (PEPT) studies. PEPT is the currently the most powerful experimental tool for studying powder flow in mixers and granulators.

Granules in the red region are pushed by the central blade 2. Blue represents the region falling under gravity and granules pushed by adjacent blades 1 and 3 are marked in yellow. Within each of these regions, there is little relative motion between the granules. However, at the interface between regions, granules collide with high relative velocity and create a slip zone between the flow regions. The cycle behaviour observed as the plough sweeps through the powder bed is evident at all practical plough speeds. There is no relatively smooth flow regime analogous to roping in the simpler geometry vertical shaft mixers.

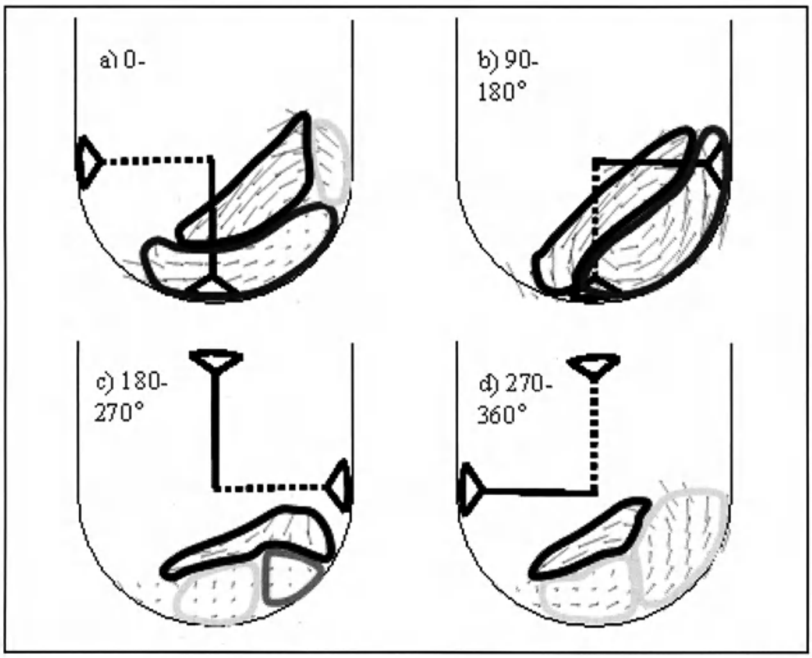


Figure 10.10. Flow regions and velocity vectors for powder flow in a Lodige mixer during granulation (Forrest et al., 2001)

For a dry, freeflowing powder, average particle tangential velocity scales directly with blade tip speed. For a wet granulating powder, however, the granule velocity is much less sensitive to tip speed. Furthermore, the powder velocity changes as the extent of granulation increases (see figure 10.11), presumably due to changes in granule density and surface wetness due to consolidation.

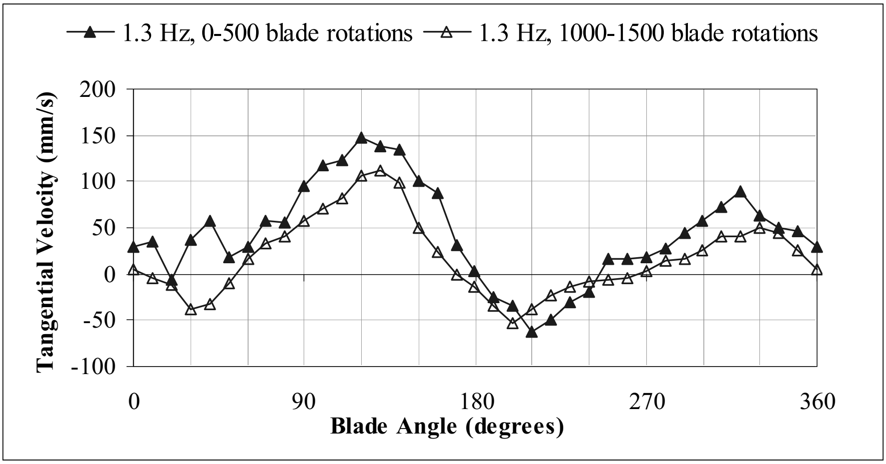


Figure 10.11. Tangential velocity of calcium carbonate granulated with aqueous glycerol solution in a Lodige mixer granulator as a function of granulation time. Results show velocity as a function of blade position early in the granulation (0-500 rotations) and late in the granulation (1000-1500 rotations) (Forest et. al., 2001)

10.4. Granulation Rate Processes in Mixer Granulators

Mixers are the most complicated of all granulators for predicting product attributes because

- All three classes of rate processes can be important in mixer granulation.
- A very wide range of shear rates and impact velocities occur in the mixer.
- Powder flow patterns are difficult to understand and predict.

To understand the impact of changing process parameters or formulation properties on generating granule attributes in the mixer granulator, the first step is to identify which rate process is dominating in your granulation. This can be done by applying the understanding we have developed in earlier chapters combined with carefully planned experiments.

10.4.1. Wetting and Nucleation

In pharmaceutical granulation, where significant growth and consolidation of granules is not required, wetting and nucleation is probably the key rate process. For good nucleation and wetting, operation should be in the drop controlled regime (see chapter 3) with low drop penetration time and low dimensionless spray flux. Operation in this regime gives better liquid distribution and narrower granule size distributions. Figure 10.12 gives an example of the spread of the granule size distribution in a mixer granulator a function of position on the nucleation regime map. The narrowest size distributions are achieved in the drop controlled regime (bottom left hand corner).

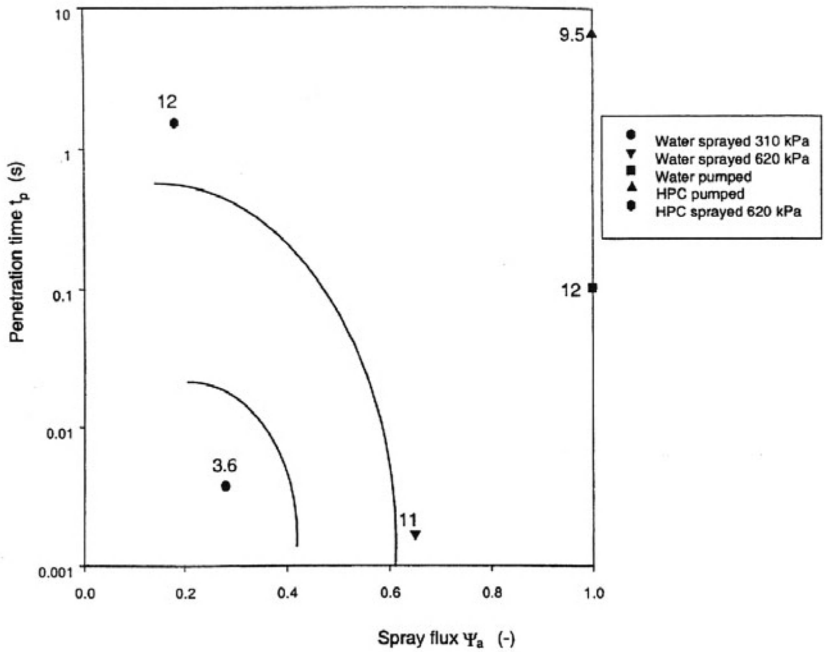


Figure 10.12. Nucleation regime map for lactose granulation in a 25 l Fielder mixer at 15% liquid content (Hapgood et al., 2003)

Furthermore, good turn over of the bed is required so that new powder is continuously presented to the spray zone. This implies operation in roping regime for Fielder mixers. A case study for calculation of drop penetration time and spray flux in a pharmaceutical mixer granulation is given below.

: Wetting and nucleation for lactose granulation in a 25 l Fielder granulator.

Consider some data from a typical laboratory scale mixer granulator in which a lactose based formation is granulated with (a) water, and (b) HPC solution. In which nucleation regime are operating?

The equations needed are given in chapter 3. For calculating the drop penetration time we have:

$$t_p = 1.35 \frac{V_{drop}^{2/3}}{R_{eff}\,\varepsilon_{eff}^2} \frac{\mu}{\gamma_{LV}.\cos\theta} \qquad \text{3-23}$$

$$\varepsilon_{eff} = \varepsilon_{tap}(1-\varepsilon_{macrovoids}) = \varepsilon_{tap}(1-\varepsilon+\varepsilon_{tap}) \qquad \text{3-21}$$

$$R_{eff} = \frac{\phi d_{32}\varepsilon_{eff}}{3(1-\varepsilon_{eff})} \qquad \text{3-22}$$

Thus we can calculate drop penetration time from the following data:

Powder: Lactose
Spray liquid: Water
3.5% HPC solution
Nozzle type: Spray systems TP650017 @ 310 kPa for water
Spray systems TP650017 @ 620 kPa for HPC solution

Parameter	Water	HPC solution
Drop size (μm)	96	240
Drop Volume (m^3)	$4.6*10^{-13}$	$72.4*10^{-13}$
Loose packed voidage	0.61	0.61
Tapped voidage	0.39	0.39
Effective voidage	0.304	0.304
Powder mean size (μm)	18	18
Effective pore size (μm)	2.6	2.6
Adhesive tension (Nm^{-1})	0.033	0.036
Liquid viscosity (Pa s)	0.001	0.1
Drop penetration time (s)	**0.0008**	**0.43**

Clearly, drop penetration time is low enough for granulation with low viscosity water. When using the higher viscosity HPC solution, drop penetration time increases 500 times, suggesting wetting in of these drops could be an issue.

We can continue to calculate the dimensionless spray flux for operation in a 25 l Fielder granulator for operation with water spray. The mixer operates with 6kg charge of dry lactose operating at 500 rpm. We have a spray time of 5 minutes and require a final granule saturation of 80%.

To calculate required liquid content:

$$w = \frac{s\rho_l \varepsilon_{\min}}{\rho_s (1 - \varepsilon_{\min})} \qquad \text{4-23}$$

The required liquid flow rate is therefore:

$$\dot{V} = \frac{wM_p}{\rho_l t} \qquad \text{10-2}$$

The powder flux is:

$$\dot{A} = Wv_p \qquad \text{10-3}$$

Therefore the dimensionless spray flux is given by:

$$\Psi_a = \frac{3\dot{V}}{2\dot{A}d_d} \qquad \text{3-27}$$

Using these equations, the key operating parameters can be summarised as follows:

Parameter	**Value**
Drop size (μm)	96
εmin	0.3
w	0.22
$\dot{V}$ (m3/s)	4.5*10-6
Spray width (m)	0.13
Powder surface velocity (m/s)	1
$\dot{A}\,(m^2/s)$	0.13
Ψ_a	0.52

Thus, we are operating *outside* the drop controlled regime. To get closer to the drop controlled regime, the spray rate could be decreased and/or multiple nozzles could be used in the mixer.

10.4.2. Consolidation and Growth

The techniques developed in chapter 4 can be used to analyse granule growth and consolidation in mixer granulators. Granule impact velocities in the region of the impeller and chopper are higher than in other granulator design leading to increased granule deformation and sometimes breakage. This reflects in high values of the Stokes deformation number St_{def}, high rates of consolidation and relatively dense granules.

It is still difficult to predict quantitative growth behaviour in mixer granulators because:

- There is a very wide range of collision velocities in different parts of the granulator leading to a wide distribution of St_{def}; and
- The granule yield stress is strain rate dependent, further complicating the correct estimation of St_{def}.

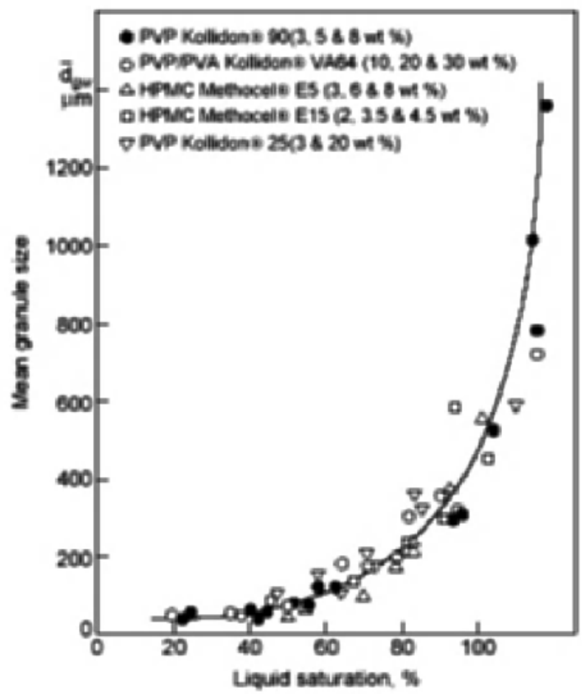


Figure 10.13. Sensitivity of granule growth to liquid saturation for the granulation. Granulation of calcium hydrogen phosphate with aqueous binder solutions in a 25ℓ Fielder granulator (Ritala et. al., 1988)

For these reasons, the growth regime map (figure 4.26) cannot be simply applied to mixer granulation. More sophisticated models are needed, probably considering powder flow in the mixer in at least two regions (near impeller and far from impeller). These models need to build on the type of powder flow studies described in section 10.3 above.

Nevertheless, we can explain many aspects of granule growth behaviour in mixers based on the understanding developed in chapter 4. Firstly, granule growth is very sensitive to moisture content and granule growth rates increase exponentially in the region of 80 to 100% saturation (see figure 10.13).

Our growth regime analysis can also explain complex growth behaviour sometimes observed in mixer granulators. For example, consider the growth behaviour shown in figure 10.14. In this case, granules grow faster when a low viscosity binder is used but granules made with a high viscosity binder reach a higher stable maximum size. The higher viscosity binder increases the granule strength (yield stress). Thus the high viscosity binder system will be in the induction growth regime and there will be a delay before there is substantial granule growth. On the other hand, once the granule is consolidated and the viscous binder is squeezed to the granule surface, it will promote successful coalescence between even large granules (high St_v).

Figure 10.14 shows similar complex behaviour as impeller speed is changed. Increasing impeller speed increases St_{def} and therefore the rate of consolidation and growth. However, increased impeller speed also limits the maximum granule size for successful collisions (high St_v). Thus the lowest impeller speed ultimately leads to the largest granule size. The contribution of the impeller to granule breakage will also contribute to this size limiting effect.

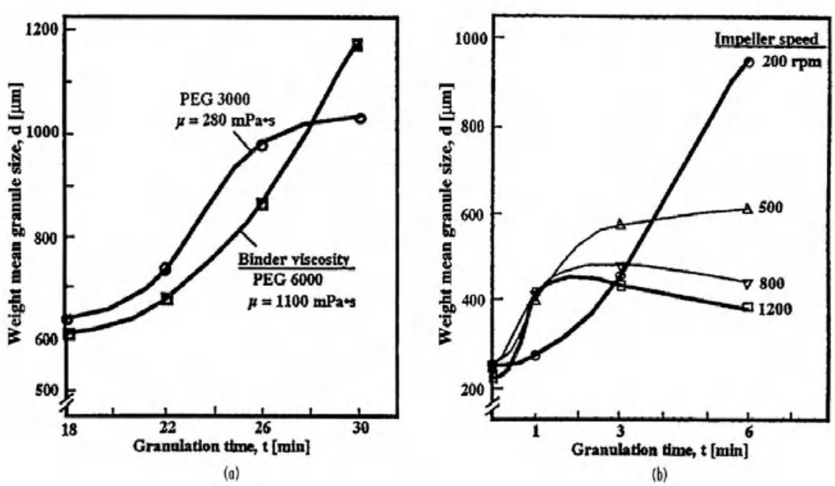


Figure 10.14. Variations in granule growth rate and extent of growth with changes to (a) binder viscosity, and (b) impeller speed

10.4.3. Granule Breakage

Mixer granulators are the ***only*** granulators in which significant breakage of wet granules occurs. In fact, for the dispersion of viscous binders in fine powders (outside the drop controlled nucleation regime) breakage of large wet granules/clumps is essential for producing granulated product.

chapter 5, examples of evidence of this breakage are given (see figures 10.15 and 10.16). We are still some way from a granule breakage regime map and predictive models but we can make some key statements:

- Extent of granule breakage will increase as the granule strength (yield stress) decreases and will increase with increasing impeller tip speed. The Stokes deformation number St_{def} is an important controlling group using the impeller tip speed as the collision velocity.
- Breakage also depends on the mode of granule failure. Semi-brittle failure leads to production of smaller granule fragments. Purely plastic deformation will not lead to granule breakage as such, but to smearing of the granule on surfaces in the granulator and perhaps formation of a dough or paste.
- Breakage rate decreases with increasing granule density because of its effect on granule strength. In a batch granulation, this means breakage rates will decrease as granules consolidate.
- Breakage rate increases with increasing granule size.

In principle, a balance between granule growth in the body of the mixer and granule breakage in the impeller region can lead to a narrow granule size distribution near a equilibrium granule size (Tardos et al., 1997; Vonk *et al.*, 1997). The equilibrium granule size could be manipulated by varying granule strength and impeller speed. To achieve this delicate balance consistently, much better quantitative models of behaviour in mixer granulators are necessary.

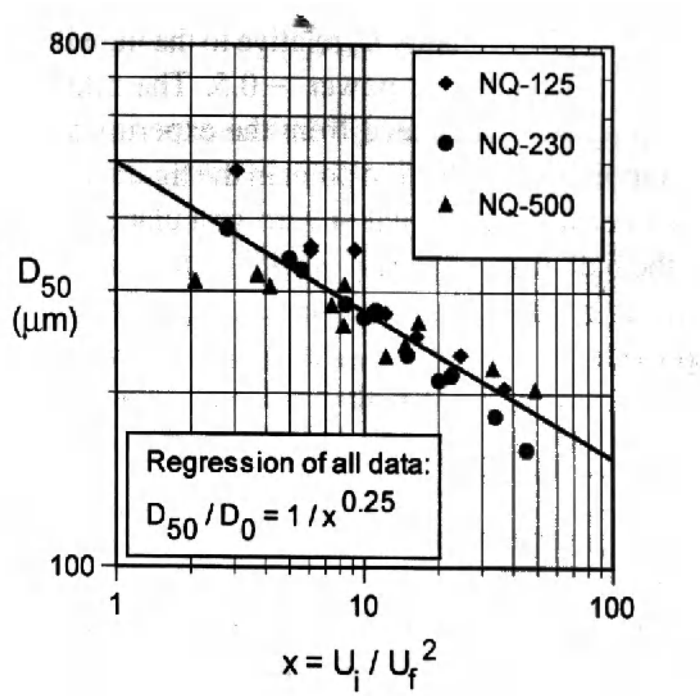


Figure 10.15. Effect of impeller speed on median particle size in an agitated fluid bed granulator (Tardos et al., 1997)

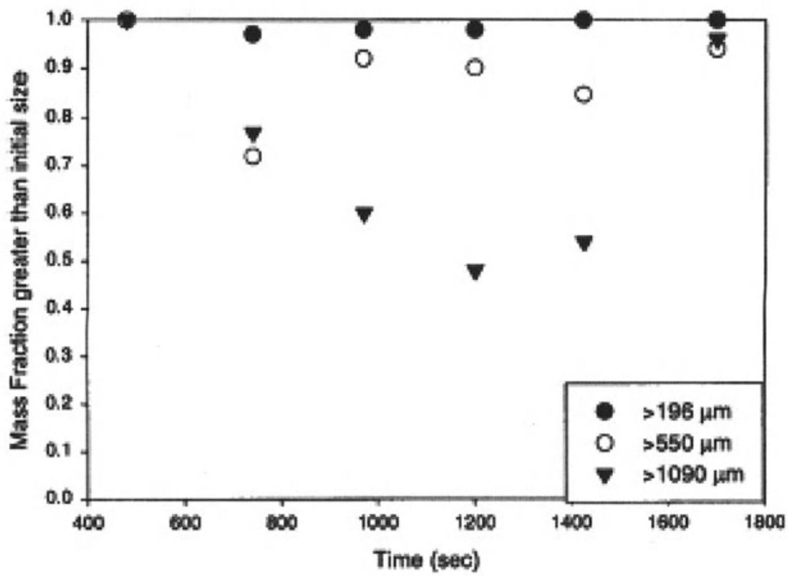


Figure 10.16. Breakage of tracer granules in high shear mixers. Effect of granule size on selection for breakage (Pearson et al.,1998)

10.5. Scale Up of Mixer Granulators

Scale up of mixer granulators is difficult and currently there are no universally applicable scale up rules. There are several good reasons for this. Firstly, the geometry of mixer granulators varies so much, scale up rules are likely to be highly equipment design specific (see section 10.2). Secondly, the very first rule in approaching scale up is to keep equipment geometrically similar during scale up. In fact, geometric similarity is the exception, rather than the rule, in common mixer granulator design (Kristensen *et al*., 1987; Schaefer, 1997; Landin *et al*., 1996). Users should be careful to look for changes in bowl geometry, impeller geometry and relative size of the chopper. [In some designs, the absolute size of the chopper remains unchanged as the granulator volume increases 10 to 50 times. Thus, chopper effects are much more significant on the small scale.]

Finally, scale up maintaining all granule attributes constant is very difficult because several granulation rate processes are occurring simultaneously in the granulator. Each of these will vary differently with equipment parameters. A scale up rule that maintains good binder distribution (keep the granule size distribution under control) may also lead to significant changes in granule density and strength. Often, scale up leads to poor liquid distribution, higher porosity granules and broader granule size distributions.

In attempting to define the best scaling rule for any application and equipment design it is important to define up front (1) the most important granule attribute to be maintained, and (2) the most significant granulation process that is occurring.

ill look at the approach to scale up of mixer granulators through a case study for vertical shaft mixers eg. Fielder or Diosna designs.

10.5.1. Scale Up of a Vertical Shaft Mixer

Consider a vertical shaft mixer similar to that shown in fig.10.1. Litster *et al.* (2002) discuss scaling rules for this type of mixer and their approach will be followed here. The first step in scale up is to ***maintain similar powder flow patterns***. Powder flow patterns in such mixers were described in section 11.3.1. Clearly, the roping regime is the most desirable for good granulation. The powder surface velocity and bed turn over are much higher in this regime leading to improved liquid distribution and nucleation. To maintain a roping regime, the impeller Froude number must remain above Fr_c as the process is scaled. This is achieved most easily through a constant Froude number scaling rule that varies impeller speed from laboratory to pilot to full scale in order to keep Fr constant:

$$\frac{N_2}{N_1} = \sqrt{\frac{D_1}{D_2}} \tag{10-4}$$

In addition, the dimensionless bed height should also be kept constant, ie. The same fraction of the bowl is filled at all scales:

$$\frac{H_2}{H_1} = \frac{D_2}{D_1} \tag{10-5}$$

where H is the bed height. These are similar scaling rule to that used for tumbling granulators (see chapter 8). Mixer granulators have been more commonly scaled using constant tip speed or constant relative swept volume (Kristensen *et al.*, 1986; Schaefer *et al.*, 1992). Maintaining constant impeller tip speeds leads to the scaling rule:

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \tag{10-6}$$

This scale up rule leads to Fr decreasing as scale increases. Combined with common practice of overfilling full scale granulators, this approach to scaling can often lead a change in operating regime from roping to bumping on scale up.

The constant swept volume approach to scale up was introduced partly to account for variations in geometry on scale up. The relative swept volume is defined as:

$$\dot{V}_R = \frac{\dot{V}_{imp}}{V_{mixer}} \tag{10-7}$$

On scale up, then:

$$\dot{V}_{R,1} = \dot{V}_{R,2} \tag{10-8}$$

This approach is useful for comparing granulators where geometry changes with scale. For geometrically similar granulators, eqn.10-8 is equivalent to scale up with constant tip speed (eqn.10.6).

To maintain equivalent liquid distribution and nucleation, the dimensionless spray flux Ψ_a (eqn.3-27) should be kept constant on scale up. If drop size from the full scale nozzle is similar to the small scale, this implies:

$$\frac{\dot{V}_2}{\dot{A}_2} = \frac{\dot{V}_1}{\dot{A}_1} \quad \Rightarrow \quad \frac{\dot{V}_2}{\dot{V}_1} = \frac{\dot{A}_2}{\dot{A}_1} \tag{10-9}$$

A common scale up approach is to keep the same total spray time and still use a single nozzle at large scale. Thus, $\dot{V}$ is proportional to D^3. Despite the fact that the powder area flux will increase slightly with scale, this approach generally leads to a substantial increase in dimensionless spray flux. To keep dimensionless spray flux constant, multiple spray nozzles and/or longer spray times should be used at the large scale.

Consolidation, growth and breakage processes are controlled by St_{def}. In a mixer granulator, the maximum collision velocity for a granule will be of the order of the impeller tip speed. To maintain constant St_{def}, the impeller tip speed should be kept constant (eqn.10.6). If a constant *Fr* rule is used:

$$\frac{St_{def,2}}{St_{def,1}} = \frac{U_{c,2}^2}{U_{c,1}^2} = \frac{N_2^2 D_2^2}{N_1^2 D_1^2} = \frac{D_2}{D_1} \tag{10-10}$$

Thus, St_{def} increases with scale. This will lead to an increase in the maximum achievable granule density and a decrease in the maximum achievable particle size. The *actual* granule density and size may depend also on the kinetics of consolidation and growth and are difficult to predict without more sophisticated quantitative modelling (see section 10.4.3 and figure 10.14b above). Thus, the variation in St_{def} with scale potentially leads to changes in granule attributes that are difficult to predict.

In summary, mixer granulator scale up is complex. Scale up with constant Froude number will ensure good quality powder flow and mixing in the granulator, but may cause unwanted changes to granule density or size. To achieve good binder distribution, Ψ_a should be kept constant on scale up. This is likely to mean multiple spray nozzles at the large scale. Scale up with constant tip speed is possible, provided that at large scale $Fr > Fr_c$. Conflicting scale up goals lead us to consider different operating strategies at large scale eg. beginning the granulation with high impeller speed (constant *Fr*) to induce good dry powder turn over and later reducing the impeller speed to control granule density or size.

Scale up of a lactose granulation from 25l to 300l

A lactose based granulation in a 25l granulator has given granules with acceptable properties. This granulation is to be scaled to 300 l using the following rules and heuristics:

- Keep *Fr* constant
- Keep spray time constant
- Spray from a single nozzle at large scale

How do Ψ_a and St_{def} change on scale up? What are the implications from granulation rate processes at full scale?

The operating conditions for the 25l granulator are summarised below:

Parameter	**Value**
Nominal volume (l)	25
Powder charge (kg)	5
Impeller speed (rpm)	330
Spray time (min)	8
Drop size (μm)	100
ε_{min}	0.3
w	0.15
$\dot{V}$ (m^3/s)	$1.6*10^{-6}$
Spray width (m)	0.13
Powder surface velocity (m/s)	0.85
Ψ_a	**0.22**

Scaling to 300l granulation:

$$D_2 / D_1 = 12^{1/3}$$

Assuming geometric similarity:

$$N_2 = (D_1 / D_2)^{0.5} N_1 = 218 rpm$$

Keeping *Fr* constant:

$$W_2 = (D_2 / D_1) W_1 = 0.3 m$$

Assume spray width scales with impeller diameter:

$$v_2 = (D_2 N_2 / D_1 N_1) v_1 = 1.28 \text{ ms}^{-1}$$

Powder surface velocity scales with tip speed:

$$\dot{V}_2 = 12 \dot{V}_1$$

Keeping spray time constant with one nozzle:

Thus, the dimensionless spray flux at 300 l is:

$$\psi_{a,2} = \frac{3\dot{V}_2}{2W_2 v_2 d} = \frac{3\left(12\dot{V}_1\right)}{2.\left(12^{1/3}W_1\right)12^{1/6} v_1 d} = 3.41\psi_{a,1} = 0.75$$

There has been a substantial increase in Ψ_a on scale up taking the granulation from nearly drop controlled into the mechanical dispersion regime. This could result in a much broader granule size distribution at large scale. A similar spray flux could be achieved by using an array of 4 nozzles spaced at 90º intervals around the granulator (all positioned so the spray fan is a right angles to the direction of powder flow).

We can't calculate the value of St_{def} because the dynamic yield stress Y for the lactose/binder system is not given. However, if we neglect changes in Y due to the larger strain rate, then St_{def} will increase as:

$$St_{def,2} = \frac{U_{c,2}^2}{U_{c,2}^1} St_{def,1} = \frac{\left(D_2 N_2\right)^2}{\left(D_1 N_1\right)^2} St_{def,1} = 2.3 St_{def,1}$$

There is a significant increase in St_{def} with scale up that could impact on the granule density and maximum size. It is not possible to scale with constant St_{def} while simultaneously maintaining constant Fr. A scale up summary data is:

Parameter	**25l**	**300l**
Nominal volume (l)	25	300
Powder charge (kg)	5	60
Impeller speed (rpm)	330	218
Spray time (min)	8	8
Drop size (μm)	100	100
ε_{min}	0.3	0.3
w	0.15	0.15
$\dot{V}$ (m^3/s)	1.6*10^{-6}	19.2*10^{-6}
Spray width (m)	0.13	0.3
Powder surface velocity (m/s)	0.85	1.28
Ψ_a	**0.22**	**0.75**
$St_{def} / St_{def,25l}$	**1**	**2.3**

tation and Control in Mixers

In line and in process measurement of granule size distributions is now possible with recent developments in particle sizing technology. Mort *et al.*(2003) reported the use of an image analysis based technique for size measurement in continuous mixer granulation of detergents. Fig.11.17 gives an example measured size distributions from this imaging system. The detergent granulation plant described by Mort consisted of two stages of mixer granulation in series (see fig.11.18). Parameters from the size distribution leaving the second mixer are used to adjust the liquid feed rate to that mixer, while in line measurement of granule bulk density is used to adjust the liquid feed rate to mixer 1.

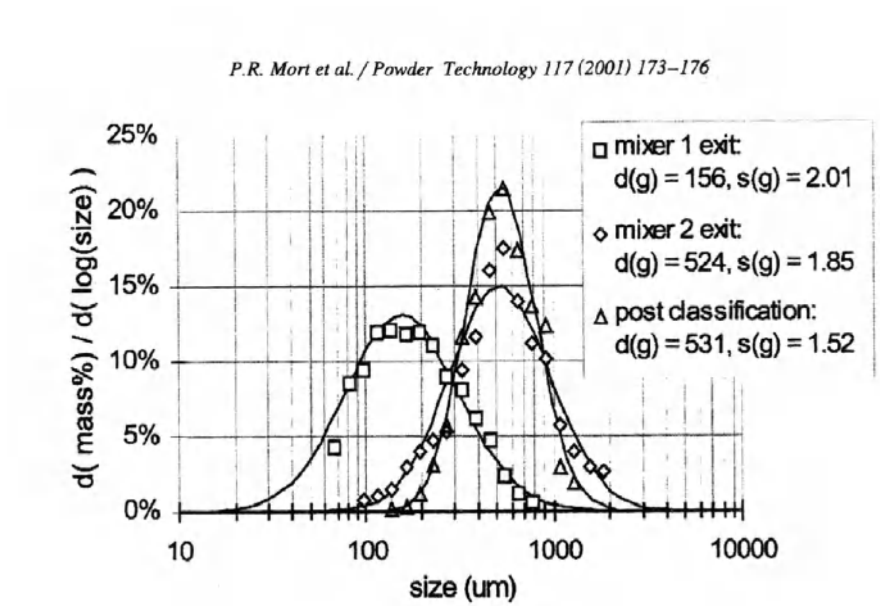


Figure 10.17. Examples of size distribution profiles from in-line imaging for continuous mixer granulation of detergents (Mort et al., 2003)

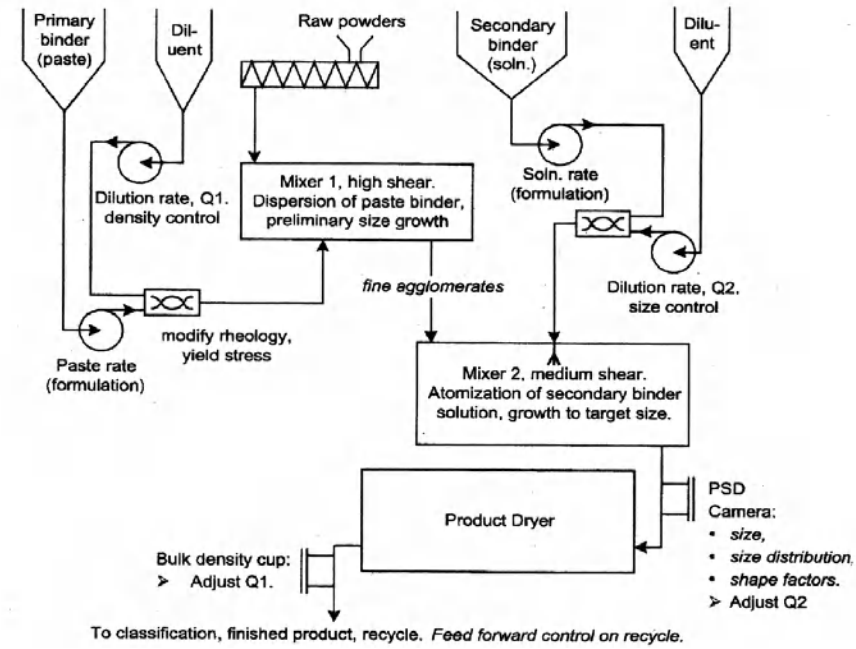


Figure 10.18. Simplified detergent granulation process flow diagram with control handles for granule size (Q1) and density (Q2) (Mort et al., 2003)

In process measurement of particle size distributions has also been used successfully to monitor and control batch mixer granulation. Watano (2003) used an imaging probe in an agitated fluid bed granulator (fig. 10.19) and was able to track changes in the granule size distribution with time very successfully.

We should add some cautionary notes, however. In all published cases where image analysis or light diffraction techniques have been successful, measurements are taken on granules in free fall or in a fluidised state. In most batch mixer granulators, the wet mass is not fluidised but behaves as a loosely packed moving bed. Such systems push the limits of existing technology.

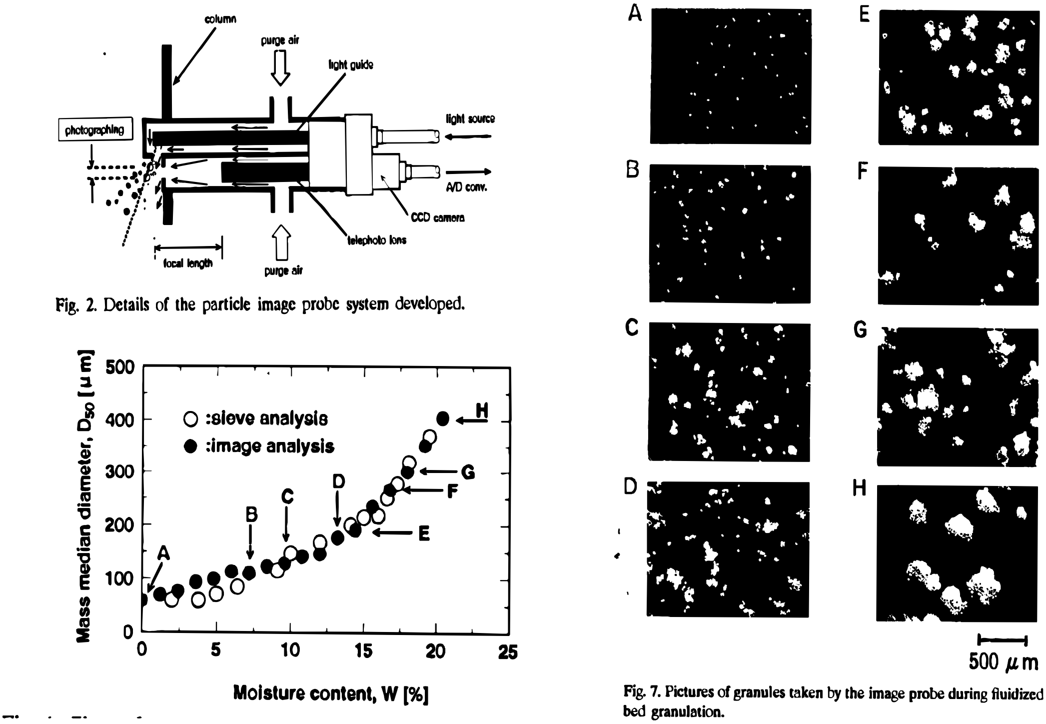
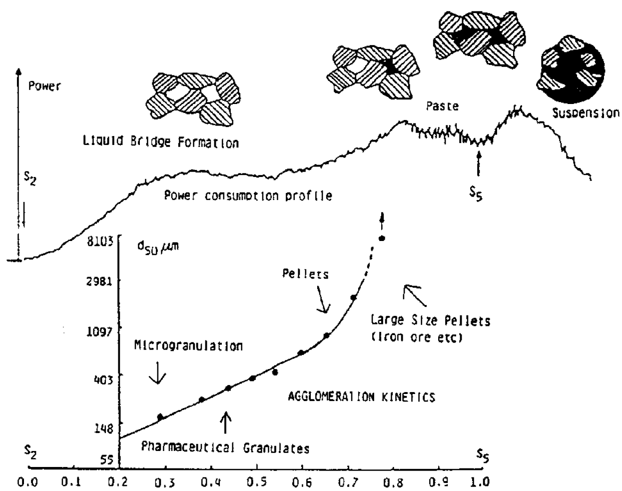
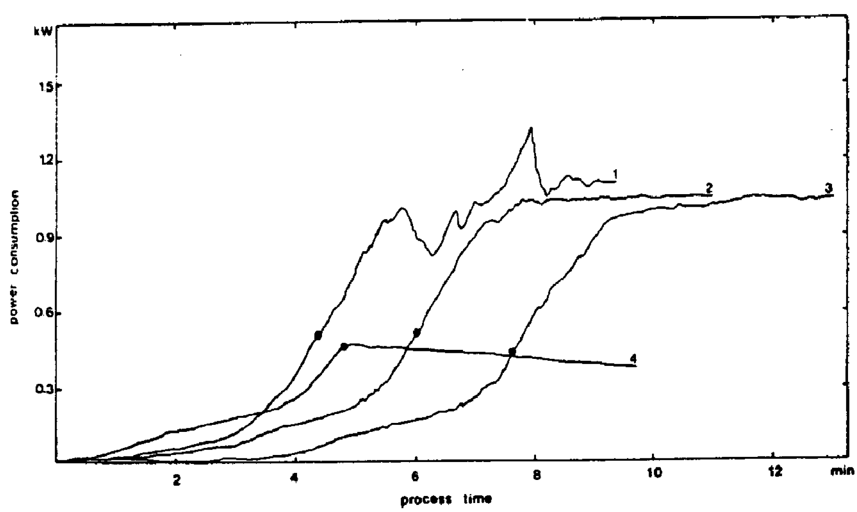


Figure 10.19. In process granule size measurement in an agitated fluid bed granulator [Watano, 2003]

For this reason, more indirect measures are commonly used for end point control in batch mixer granulation. The techniques measure specific power consumption or impeller torque as the wet mass granulates. The torque (or power) reflect the transfer of momentum to the powder mass and therefore is directly related to the wet mass "rheology". As granulation occurs, the powder rheology changes, giving a characteristic power vs time curve for a given granulation (see figure 10.20). Typically, power consumption increases as liquid is added because the powder becomes more cohesive and more effectively transfers momentum into the powder mass. As granules densify and change behaviour, the powder consumption levels off.



(a)



(b)

Figure 10.20. Power consumption measurements for end point control (a) Variations in power consumption as granulation proceeds (b) Characteristic power curves vary for different formulations

This approach was first used by Leuenberger *et al.* (1979) and has been studied extensively. It provides a good tool for simple endpoint control where there is little change in the formulation. However, again we should make some cautionary comments. Essentially torque or power measurements use the mixer as a crude rheometer. In chapter 4, we looks extensively at granule mechanics and showed that they depend on:

- Liquid content
- Granule density
- Powder size distribution
- Powder morphology
- Liquid properties and powder-liquid interactions

If there are batch to batch variations in powder or binder properties, these will change maximum power consumption and characteristic power curve for a granulation. Thus, the link between power consumption and a granule attribute of interest, eg. mean size, is very tenuous.

10.7. Summary

Mixer granulators are the most robust of all granulators, handling the widest range of feed powders and liquid binders. However, they are also the most complex in terms of the powder flow field and the numbers of rate processes that can impact on granule attributes. Be wary of general rules of thumb for scale up and operation due to this complexity, as well as the enormous variety of geometries that are available.

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10.9. Nomenclature

$\dot{A}$ Area flux of powder through the spray zone (m^2s^{-1})
D impeller diameter (m)
d_d liquid drop size (m)
d_{32} specific surface mean particle size (m)
Fr Froude number (-)
g gravitational acceleration (N/m^2)
H powder bed height in granulator (m)
M_p Mass of powder in granulator (kg)
N impeller speed (s^{-1})
Q flow rate of liquid into powder bed (m^3s^{-1})
R_{eff} effective pore radius (m)
s liquid saturation
St_{def} Stokes deformation number (-)
St_v viscous Stokes number (-)
t time (s)
t_p drop penetration time (s)
U_c collision velocity (ms^{-1})
$\dot{V}$ volumetric spray rate (m^3s^{-1})
V_d drop volume (m^3)
$\dot{V}_{imp}$ rate of swept volume of impeller (m^3s^{-1})
$\dot{V}_R$ relative swept volume of impeller (s^{-1})
V_{mixer} mixer volume (m^3)
w liquid content (kg/kg dry powder)
γ_{lv} liquid-vapour interfacial energy ($N\ m^{-1}$)
ε loose packed powder bed voidage (-)
ε_{ett} effective powder bed voidage (-)
ε_{tap} tapped powder bed voidage (-)
ε_{min} minimum granule porosity (-)
θ contact angle (°)
ψ_a dimensionless spray flux (-)
φ sphericity (-)
μ Viscosity (Pa.s)
ρ_l fluid density ($kg\ m^{-3}$)
ρ_s solid density ($kg\ m^{-3}$)

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